

Mixed Holomorphic–Topological Strings

Hamiltonian fields, chiral operations, and reconstruction

Raez Lorgat

26 July 2026

ABSTRACT

Hamiltonian fields on a holomorphic symplectic plane act on matrix boundary observables. Their traces provide one comparison with bulk operations. Recovering the full boundary action also requires the Lie brackets, associative products, and the homotopies relating them. We develop finite pointed Hamiltonian models and specify the maps needed for their mixed holomorphic–topological realization.

The pointed Hamiltonians are formal functions vanishing to second order, so their vector fields fix the Darboux point. Their finite jet quotients are unimodular Lie algebras. The corresponding finite cotangent models satisfy the quantum master equation for the constant measure. Relative antisymmetrization compares their Rees enveloping cochains with the adjoint Chevalley complex over truncated and formal quantization coefficients. For quadratic Hamiltonians the decisive associative relation is

$$e \star f + f \star e + \frac{1}{2}h \star h = -\frac{3}{8}\hbar^2, \quad e = q^2/2, \quad f = -p^2/2, \quad h = -qp,$$

where the Moyal product has Poisson normalization $\{q, p\} = 1$.

These finite constructions distinguish the chosen Hamiltonian algebra from its scalar trace image. Their extension to continuous jets, chiral collisions, and physical bulk observables requires compatible coefficient maps, support conditions, and quantum operations. The open–closed comparison must also retain its cyclic pairing and sewing data.

HAMILTONIAN OPERATIONS AND BOUNDARY RECONSTRUCTION

A Hamiltonian function determines a vector field through the symplectic form. At a matrix boundary it also determines an invariant trace. Comparing these two operations requires their actual brackets and differentials. A scalar trace by itself does not define an action on all boundary-changing operators.

The finite model begins at a fixed formal symplectic point. Hamiltonians in the square of its maximal ideal have vector fields that vanish there, and truncation by polynomial degree gives finite Lie algebras. Linear Hamiltonians instead move the point. They belong to the separately defined affine Hamiltonian algebra. The distinction matters when a brane position is allowed to vary.

The finite cotangent theory and its Rees cochains make the first comparison explicit. The oscillator relation above then shows that the associative algebra carries data beyond its Lie generators. Scalar reduction and an augmentation produce stable traces in a commutative target. Passing from these targets to a completed geometric centre requires a compatible action on modules, morphisms, and higher compositions.

For a physical mixed theory, define the bulk first from fields, an action, a BV pairing, boundary conditions, and a quantization on the chosen support domain. Define its boundary category and the object of permitted coherent local actions independently. The comparison problem asks for a map from bulk observables in a chosen collar to those boundary actions, preserving disjoint products, moving insertions, and collision operations. Source descent, target descent, and equivalence of the resulting objects are separate mathematical assertions.

Open–closed amplitudes require additional cyclic maps and pairings. Transverse closure and holomorphic sewing use different geometries and parameters. The matrix and finite Hamiltonian calculations provide inputs to these comparisons; the full quantum and global reconstruction requires their stated geometric and analytic extensions.

NOTATION AND NORMALIZATION

The ground field has characteristic zero. Stable categories are presentable and idempotent-complete. Tensor products preserve colimits separately. Completions are separated inverse limits of finite weight or proper-height quotients. The Igusa normalization is

$$\Delta_5^\theta = 64D_5, \quad \chi_{10} = D_5^2,$$

with D_5 monic. Physical charge is additive; Gram degree is carried through its polarization cocycle. Bar, factorization, and Hopf coproducts retain distinct notation and distinct geometric meanings.

CONTENTS

ABSTRACT	iii
HAMILTONIAN OPERATIONS AND BOUNDARY RECONSTRUCTION	v
NOTATION AND NORMALIZATION	vii
CONTENTS	vii
HAMILTONIAN FIELDS AND THE TRACE SECTOR	i
Mixed Holomorphic–Topological Deligne Theory	3
BOOK ENTRANCE: MIXED HOLOMORPHIC–TOPOLOGICAL DELIGNE THEORY	3
I Finite Hamiltonian geometry	5
1 THE FORMAL SYMPLECTIC PLANE	7
1.1 Hamiltonians and the Poisson bracket	7
1.2 Unimodularity	8
2 THE SHIFTED COTANGENT THEORY	9
2.1 The Lie algebra and its functions	9
2.2 The BV action	9
3 THE UNIVERSAL BOUNDARY BRANE	ii
3.1 Polarization	ii
II Relative Hochschild comparison	13
4 HOCHSCHILD COCHAINS OF THE UNIVERSAL BRANE	15
4.1 The coefficient ring and the Chevalley differential	15
4.2 The formal coefficient limit	16
5 THE KONTSEVICH–DUFLO MAP	19
5.1 Linear Poisson formality	19

5.2	The boundary action	19
6	THE FORMAL HAMILTONIAN LIMIT	21
6.1	Weight completion and comparison data	21
III	The finite-jet laboratory	23
7	THE FIRST THREE HAMILTONIAN JETS	25
7.1	Quadratic Hamiltonians	25
7.2	Cubic Hamiltonians	25
7.3	The first nonlinear layer	25
8	MOYAL QUANTIZATION IN COORDINATES	27
8.1	The bidifferential formula	27
8.2	Weyl generators	28
9	CHEVALLEY COORDINATES AND THE POISSON DIFFERENTIAL	29
9.1	The $\mathfrak{sl}_2$ invariant	29
10	RESIDUE DUALITY AND THE FORMAL CURRENT ALGEBRA	31
10.1	The continuous dual	31
11	ALL-ARITY OPEN–CLOSED OPERATIONS	33
11.1	Hochschild braces	33
11.2	The first mixed identities	33
12	FINITE CHAN–PATON RANKS	35
12.1	Rank one	35
12.2	Rank two and Cayley–Hamilton reduction	35
12.3	Stable passage	35
IV	Global holomorphic symplectic descent	37
13	FEDOSOV DESCENT	39
13.1	The Weyl algebroid	39
14	FACTORIZATION AND DESCENT	41
14.1	The algebraic mixed-HT realization	41
14.2	BV realization	41
V	Matrix branes and trace geometry	43
15	THE DERIVED COMMUTING STACK	45
16	NECKLACE HAMILTONIANS	47
17	STABLE TRACES	49
18	THE TWO-BRANE CONE	51
VI	Wheel corrections and exact quantum structure	53
19	THE DUFLO WHEEL	55
20	THE LOCAL CENTRE WITH COMPARISON DATA	57
21	CYCLIC CLASSES, STABLE TRACES, AND LIE HOMOLOGY	59
22	SYNTHESIS	61

VII	Technical appendices to the preceding book	63
23	SIGNS IN THE SHIFTED COTANGENT COMPLEX	65
24	A DIRECT JACOBI CHECK FOR HAMILTONIAN JETS	67
25	BIBLIOGRAPHIC NOTE	69
	Ordered and Associative Chiral Geometry	71
	BOOK ENTRANCE: ORDERED AND ASSOCIATIVE CHIRAL GEOMETRY	71
VIII	The invariant object	73
26	TWO DIRECTIONS OF LOCALITY	75
26.1	The local experiment	75
26.2	Carrier and operation	75
27	THE RAN COEFFICIENT CATEGORY	77
27.1	The fixed six-functor setting	77
27.2	Factorization coalgebras	78
28	THE TWO NONCOMMUTATIVE CHIRAL OBJECTS	81
28.1	Transverse order	81
28.2	Associativity inside chiral convolution	81
28.3	Formal-disc realization	82
28.4	Three separation laws	82
29	TRANSVERSELY ORDERED CHIRAL ALGEBRAS	83
29.1	Product field theories	83
30	BLOCK-ORDERED OPERATIONS	85
30.1	Mixed forests	85
IX	Bar, cobar, and the boundary state	87
31	THREE BAR CONSTRUCTIONS	89
31.1	Realization and factorization	90
32	THE INTERNAL ORDERED BAR	93
32.1	The bar comparison on separated configurations	93
32.2	Three coproducts	94
32.3	Interval factorization homology	95
33	COBAR AND TWISTING COCHAINS	97
34	COMPLETE BAR-COBAR RECONSTRUCTION	99
34.1	Quadratic recognition	99
35	MODULES, COMODULES, AND MORITA THEORY	101
36	DERIVED CENTRE AND BULK ACTION	103
37	CYCLIC CLOSURE AND MODULAR SEWING	105
X	Exact noncommutative models	107
38	FREE AND SQUARE-ZERO PAIRS	109
39	THE QUANTUM PLANE	III

40	QUIVER PATH ALGEBRAS	113
41	WEYL AND CLIFFORD DEFORMATIONS	115
42	ASSOCIATIVE CHIRAL ALGEBRAS ON THE FORMAL DISC	117
43	ZAMOLODCHIKOV–FADDEEV EXCHANGE	119
44	THE RTT YANGIAN OF $\mathfrak{gl}_2$	121
45	THE HALL ALGEBRA OF THE A_2 QUIVER	123
46	MATRIX TRACES AND THE NECKLACE BRACKET	125
XI	The nonlocal ladder	127
47	SPECTRAL EXCHANGE AS AN ASSOCIATIVE CHIRAL OPERATION	129
47.1	The coefficient field and the exchange operator	129
47.2	A diagonal exchange model	129
48	RTT AND REFLECTION ALGEBRAS	131
48.1	The RTT bialgebra	131
48.2	Boundary exchange	131
49	QUANTUM VERTEX REALIZATION	133
49.1	Two expansion regions	133
49.2	The Yangian field	133
50	SHUFFLE REALIZATION AND THE POSITIVE CURRENT ALGEBRA	135
50.1	Pairwise kernel	135
50.2	Coproduct and double	135
51	THE LADDER IN ONE DIAGRAM	137
XII	Field theory and master equations	139
52	A FINITE COTANGENT BF MODEL	141
53	BRST AND ORDERED REDUCTION	143
54	STOKES AND ALGEBRAIC IDENTITIES	145
55	TWO EDGE COLORS AND TWO LOOP OPERATORS	147
56	TYPED ANOMALIES	149
XIII	Calabi–Yau and automorphic interfaces	151
57	CALABI–YAU CATEGORIES	153
58	HALL AND BPS ALGEBRAS	155
59	THE IGUSA DENOMINATOR AS A BAR CHARACTER	157
60	FINITE-HEIGHT CURRENTS AND AN EXACT PHYSICAL MODEL	159
61	CHARACTER, BRACKET, PARITY, AND ORIENTATION	161
62	EXACT DEPENDENCY THEOREM	163
63	SYNTHESIS	165
XIV	Technical appendices to the preceding book	167
64	SIGN CALCULATION FOR THE LENGTH-THREE BAR	169
65	RATIONAL R -MATRIX VERIFICATION	171

66	CARRIERS AND COMPARISON MAPS	173
67	ENVELOPING BARS AND PLANAR PATH SEQUENCES	175
67.1	Chevalley comparison	175
67.2	The Virasoro row	177
67.3	Planar path sequences	178
68	THE OBSTRUCTION CONSTITUTION	181
	The Igusa Square Root	183
	BOOK ENTRANCE: THE IGUSA SQUARE ROOT	183
XV	The automorphic determinant	185
69	THE PROTECTED JACOBI INPUT	187
69.1	The K_3 elliptic genus	187
69.2	The virtual half-index	187
70	THE BORCHERDS PRODUCT	189
70.1	The effective chamber	189
70.2	The theta constant normalization	189
71	DIVISOR, CHARACTER, AND THE MEANING OF THE SQUARE ROOT	191
71.1	Automorphic section, orientation line, and Pfaffian operator	191
XVI	The denominator algebra	193
72	THE LORENTZIAN ROOT LATTICE	195
72.1	Real simple roots and Weyl vector	195
73	THE DENOMINATOR ALGEBRA	197
74	THE ORDERED BAR AS THE DENOMINATOR COMPLEX	199
75	FINITE-HEIGHT CURRENTS	201
75.1	Classical local theory and exact quantum zero modes	201
XVII	The compact scalar character	203
76	REDUCED CURVE COUNTING ON $K_3 \times E$	205
77	EXTENSIONS OVER THE ELLIPTIC CLASSIFYING STACK	207
78	ADDITIVE PHYSICAL CHARGE AND QUADRATIC AUTOMORPHIC DEGREE	209
78.1	The polarized central extension	209
79	POINT-LOCAL AND WRAPPED SECTORS	211
XVIII	The Borchers algebra and its quantum group	213
80	THE LOCAL-WRAPPED ROOT ALGEBRA	215
81	THE FORMAL IGUSA QUANTUM GROUP	217
82	A HOLOMORPHIC-TOPOLOGICAL BF REALIZATION	219
83	THE DETERMINANT AND FREDHOLM SQUARE-ROOT OPERATORS	221
83.1	Trace-class realization on the root Hilbert superspace	221
84	WHAT THE AUTOMORPHIC CHARACTER DETERMINES	223

XIX	The universal Borchers Hall and three-Calabi–Yau models	225
85	FINITE SIMPLE-ROOT WINDOWS	227
86	THE REAL-ROOT SEED WINDOW	229
86.1	The first Ginzburg differential	229
87	HALL REALIZATION OF THE POSITIVE ALGEBRA	231
87.1	Parity and the quantized superdatum	231
88	THE THREE-CALABI–YAU COMPLETION	233
89	HALL STATES AND PROTECTED SUPERSTATES	235
90	EXTENSION DETERMINANTS AND CRITICAL ORIENTATIONS	237
91	THE COLORED ELLIPTIC RAN SPACE	239
92	PHYSICAL CHARGE AND GRAM DEGREE	241
93	THE UNIVERSAL PROTECTED FIELD THEORY	243
94	ABSOLUTE CONVERGENCE OF THE VIRTUAL ROOT OPERATOR	245
95	THE VIRTUAL ROOT FREDHOLM OPERATOR	247
95.1	Automorphic orientation	247
96	THE UNIVERSAL IGUSA PROTECTED EFFECTIVE THEORY	249
97	THE PRIMITIVE COMPARISON AND ITS EXTENSIONS	251
98	THE EIGHT DIAGONAL-DIVISOR FORMS	253
99	ALGEBRAIC AND PHYSICAL CLOSURE	255
XX	Technical appendices to the preceding book	257
100	EXACT THETA-SERIES CALCULATION	259
101	THE ROOT REFLECTION MATRICES	261
102	AUTOMORPHIC, HOMOLOGICAL, AND HALL COMPARISONS	263
	Calabi–Yau Quantum Groups	265
	BOOK ENTRANCE: CALABI–YAU QUANTUM GROUPS	265
XXI	Categorical and field-theoretic foundations	267
103	CALABI–YAU CATEGORIES	269
103.1	Smoothness, properness, and the Calabi–Yau class	269
104	LOCAL BV THEORIES AND FACTORIZATION OBSERVABLES	271
104.1	Cyclic elliptic local Lie algebras	271
104.2	Boundary and defect restriction	271
105	THREE CURVE-LEVEL ALGEBRAIC OUTPUTS	273
106	HOLOMORPHIC CHERN–SIMONS THEORY	275
107	PUSHFORWARD TO A CURVE	277
108	THE DERIVED CENTRE AND THE PHYSICAL BULK	279
XXII	Hall and BPS algebras	281
109	EXTENSION CORRESPONDENCES	283
110	ORIENTATION DATA AND THE THREEFOLD HALL PRODUCT	285

111	THE GEOMETRIC E -EQUIVARIANT HALL SOURCE	287
112	LOCAL AND WRAPPED GEOMETRY INSIDE THE HALL ALGEBRA	289
113	COHOMOLOGICAL REFINEMENT AND THE GRAM CHARACTER	291
114	BPS PRIMITIVES	293
115	THE INTEGRAL HALL–BORCHERDS AMALGAM	295
116	COPRODUCTS, PAIRINGS, AND DOUBLES	297
117	ORDERED CHIRAL BIALGEBRAS	299
XXIII	Exact model cases	303
118	THE A_2 HALL ALGEBRA	305
119	THE TRIPLED JORDAN QUIVER AND $\mathbb{C}^3$	307
120	SURFACE COHAS AND ADE YANGIANS	309
121	COULOMB BRANCHES AND SHIFTED YANGIANS	311
XXIV	K_3 symmetries	313
122	THE MUKAI LATTICE	315
123	THE NAKAJIMA HEISENBERG ALGEBRA	317
124	THE ORTHOGONAL LLV ALGEBRA	319
125	LOCAL ADE YANGIANS	321
126	THE K_3 SYMMETRY TRICHOTOMY	323
127	K_3 CORRESPONDENCES ON FIXED COEFFICIENTS	325
128	THE ALGEBRA GENERATED BY GEOMETRIC ACTIONS	327
129	TENSOR RECONSTRUCTION AND GEOMETRIC KERNELS	329
XXV	The exact $K_3 \times E$ protected completion	331
130	THE PROTECTED INPUT	333
131	CYCLIC OPEN-STRING OPERATIONS	335
132	AN ALGEBRAIC OPEN–CLOSED COMPARISON	337
133	THE EXACT SCALAR AND FIRST-ORDER OBSERVABLES	339
134	PHYSICAL MEANING	341
135	PRIMITIVE COMPARISONS AND THEIR ALGEBRAIC EXTENSIONS	343
136	FINITE QUIVERS AND CALABI–YAU COMPLETIONS	345
XXVI	Quantum groups, exchange, and representation theory	347
137	FROM POSITIVE HALF TO FULL QUANTUM GROUP	349
138	TANAKA RECONSTRUCTION	351
139	KZ AND RATIONAL EXCHANGE	353
140	THE CENTRE AND THE DOUBLE	355
XXVII	A compact six-dimensional protected source	357
141	HOLOMORPHIC COTANGENT BF THEORY ON $K_3 \times E$	359

142	TREE EXACTNESS AND THE QUANTUM MASTER EQUATION	361
143	CLASSICAL REDUCTION ALONG K_3	363
144	PROTECTED STATES AND THE IGUSA TRACE	365
145	THE COMPACT-SOURCE COMPARISON	367
XXVIII	Physics	369
146	HOLOMORPHIC–TOPOLOGICAL BOUNDARY THEORIES	371
147	SIX-DIMENSIONAL HCS ON $K_3 \times E$	373
148	OMEGA BACKGROUNDS	375
149	ANOMALIES AS TYPED CLASSES	377
XXIX	Automorphic forms	379
150	BORCHERDS PRODUCTS AS CHARACTERS	381
151	THE UNIVERSAL BORCHERDS WEIGHT FORMULA	383
152	THE GRITSENKO–CLÉRY ATLAS	385
153	SCALARIZATION HIERARCHY	387
XXX	The unified theorem system	389
154	THE FOUNDATIONAL THEOREM	391
155	LANDSCAPE OF ESTABLISHED REALIZATIONS	393
156	CONSTRUCTED COMPARISONS AND THEIR GEOMETRIC EXTENSIONS	395
157	ACTIONS AND COMPARISON MAPS	397
158	COMPARISON DATA FOR CALABI–YAU CONSTRUCTIONS	399
XXXI	Technical appendices to the preceding book	401
159	TYPED DATA TABLE	403
160	THE THREE K_3 BRACKETS	405
161	EXACT ARITHMETIC CHECKS	407
	Universal Chiral BV and Einstein Completion	409
	COHERENT BOUNDARY ACTIONS AND BV FIELDS	409
XXXII	The full mixed boundary datum	411
162	THREE PRODUCTS AND ONE INTERFACE	413
163	COMPATIBLE MODULES AND THE MIXED CENTRE	415
164	RELATIVE INDUCTION FROM A CHIRAL ALGEBRA	417
XXXIII	The deformation algebra and its BV envelope	419
165	THE CENTRE-EXTENDED DEFORMATION ALGEBRA	421
	165.1 Field equations and Bianchi identities	421
166	SHIFTED COTANGENT TRANSGRESSION	423
167	INTRINSIC ALGEBRAIC QUANTIZATION	425

168	PERTURBATIVE FACTORIZATION REALIZATION	427
XXXIV	Boundary duality, exchange, and reduction	429
169	THE KOSZUL–MORITA SQUARE	431
170	PLANAR EXCHANGE AND TANAKA RECONSTRUCTION	433
171	BRST REDUCTION	435
XXXV	Conformal covariance and Einstein completion	437
172	PROJECTIVE CONNECTIONS AND THE VIRASORO TANGENT COMPLEX	439
173	THREE-DIMENSIONAL GRAVITY AS A FLAT CARTAN THEORY	441
174	FOUR EQUIVALENT LINEAR COMPLEXES	443
175	BROWN–HENNEAUX AND THE PROJECTIVE BOUNDARY	445
176	EINSTEIN COMPLETION OF A CONFORMAL PAIR	447
177	HIGHER-SPIN COMPLETION	449
XXXVI	Exact gravitational complexes and quantum tests	451
178	THE FOUR COMPLEXES IN COORDINATES	453
178.1	Flat adjoint complex	453
178.2	Einstein–Cartan complex	453
178.3	Metric complex	453
178.4	Calabi complex	454
179	BOUNDARY BFV THEORY AND THE VIRASORO MOMENT MAP	455
180	MATTER STRESS TENSOR AND THE EINSTEIN MOMENT MAP	457
181	ONE-LOOP GRAVITY AND ANALYTIC TORSION	459
182	BTZ, CARDY, AND THE AREA LAW	461
183	HIGHER-SPIN HOLONOMY AND BOUNDARY CHARGES	463
XXXVII	Anomalies, Hall sources, and automorphic completion	465
184	FOUR NATIVE ANOMALY CLASSES	467
185	HALL AND CALABI–YAU SOURCES	469
185.1	Mixed realization of the amalgam	469
186	IGUSA ROOT STATES AND THE MIXED-BOUNDARY COMPARISON	471
187	THE CARRIER LATTICE	473
188	SIXTEEN STRUCTURAL LAWS	475
189	COHERENT ACTIONS AND GRAVITATIONAL RECONSTRUCTION	477
XXXVIII	Technical appendices to the preceding book	479
190	THE CHERN–SIMONS NORMALIZATION	481
191	THE PRINCIPAL EMBEDDING COUNT	483
192	LOW-ARITY MIXED SIGNS	485

HAMILTONIAN FIELDS AND THE TRACE SECTOR

A Dirac brane probe on $\mathbb{R}_{\text{top}}^2 \times \mathbb{C}_{\text{hol}}^2$ leads to a local comparison between Hamiltonian BF fields and boundary observables. Pointed Hamiltonian jets, shifted cotangent fields, and brane algebras on the formal symplectic plane give finite models for this problem. Pointed Hamiltonians vanish to second order and fix the chosen point. The affine Hamiltonian algebra also contains translations and is a separate input when the brane moves. The trace sector asks which Hamiltonian operations can be detected by invariant functions of matrix representations.

An ordinary trace retains the scalar dimension of a representation. Scalar reduction and a chosen augmentation remove that rank dependence and define compatible reduced traces. With the stated adic completions, the resulting stable map is a continuous homomorphism of commutative algebras. Lie and Poisson compatibility for full traces uses a separately specified bracket identity. Reduced Chevalley and polyvector operations must be compared on their actual carriers, with their differentials and topologies fixed.

The relative Rees comparison and finite formality calculations provide local algebraic inputs. A nonzero Poisson differential requires the corresponding twisted formality data, and a completed trace-sector comparison requires compatibility with the selected transition maps. Holomorphic chiral operations and mixed open–closed operations require their own operadic refinements. The global field-theoretic problem is to extend these local comparisons through descent and quantization while preserving the brane action, pairing, and anomaly data.

The physical bulk and the geometric centre are independently specified. A bulk action on a chosen collar must preserve the boundary operations and their higher compatibility maps. An equivalence criterion applies after that action is constructed. Cyclic amplitudes further require a trace domain, a pairing, and comparison maps for the two sewing geometries.

BOOK ENTRANCE: MIXED HOLOMORPHIC–TOPOLOGICAL DELIGNE THEORY

*Hamiltonian jets, shifted cotangent fields, universal branes, Duflo quantization, and the all-arity
open–closed map.*

BOOK I

Finite Hamiltonian geometry

THE FORMAL SYMPLECTIC PLANE

1.1 HAMILTONIANS AND THE POISSON BRACKET

Let $V = \mathbb{C}^2$ with coordinates (z_1, z_2) and symplectic form

$$\omega = dz_1 \wedge dz_2.$$

Let $\mathfrak{m} = (z_1, z_2)$. The pointed formal Hamiltonian Lie algebra is

$$\mathfrak{h} = \mathfrak{m}^2 \subset \mathbb{C}[[z_1, z_2]], \quad \{f, g\} = \partial_{z_1} f \partial_{z_2} g - \partial_{z_2} f \partial_{z_1} g.$$

It consists of Hamiltonians whose vector fields fix the chosen Darboux point. The shifted order filtration $F^p \mathfrak{h} = \mathfrak{m}^{p+2}$ is complete and obeys

$$\{F^p \mathfrak{h}, F^q \mathfrak{h}\} \subset F^{p+q} \mathfrak{h}.$$

The affine Hamiltonian algebra $\mathfrak{h}^{\text{aff}} = \mathbb{C}[[z_1, z_2]]/\mathbb{C}$ also contains translations. Its linear sector is adjoined separately when a moving brane or center-of-mass coordinate is required.

Write $R_r = \text{Sym}^r(V^\vee)$ for the homogeneous polynomials of degree r . The bracket obeys

$$\{R_r, R_s\} \subset R_{r+s-2}.$$

For $d \geq 2$ define the finite jet quotient

$$\mathfrak{h}_{\leq d} = \bigoplus_{r=2}^d R_r,$$

where every bracket component of degree greater than d is set equal to zero.

Theorem 1.1 (Finite Hamiltonian jet algebra). *The truncated bracket makes $\mathfrak{h}_{\leq d}$ a finite-dimensional Lie algebra. Its dimension is*

$$\dim \mathfrak{h}_{\leq d} = \frac{d(d+3)}{2} - 2.$$

The quadratic part is

$$R_2 \cong \mathfrak{sl}_2,$$

and each R_r is the irreducible $\mathfrak{sl}_2$ -module $\text{Sym}^r(\mathbb{C}^2)$.

Proof. The Jacobiator of the polynomial Poisson bracket vanishes before truncation. Projection to degrees at most d preserves this identity because the discarded terms form a Lie ideal. The dimension is

$$\sum_{r=2}^d (r+1) = \frac{d(d+3)}{2} - 2.$$

Quadratic Hamiltonians act linearly on V and preserve ω ; this identifies R_2 with $\mathfrak{sp}(V) = \mathfrak{sl}_2$. The standard action on homogeneous polynomials gives the final statement. $\square$

1.2 UNIMODULARITY

Theorem 1.2 (Unimodularity of every finite jet). *For every $d \geq 2$ and every $x \in \mathfrak{h}_{\leq d}$,*

$$\mathrm{Tr}_{\mathfrak{h}_{\leq d}}(\mathrm{ad}_x) = 0.$$

Proof. A homogeneous Hamiltonian of degree greater than two raises polynomial degree. Its adjoint action is strictly upper triangular in the decomposition $\bigoplus_{r=2}^d R_r$, hence has trace zero. A quadratic Hamiltonian belongs to $\mathfrak{sl}_2$. Its action on each finite-dimensional $\mathfrak{sl}_2$ -module R_r has trace zero. Linearity gives the result. $\square$

Corollary 1.3 (Canonical finite BV measure). *The linear odd symplectic space $T^*[-1]\mathfrak{h}_{\leq d}[1]$ has a translation-invariant Berezinian for which the cotangent BF Hamiltonian satisfies the quantum master equation.*

Proof. Choose dual linear coordinates on the shifted cotangent space and use their constant Berezinian. The classical term vanishes by Jacobi, while the BV divergence is the adjoint trace. Unimodularity makes that trace zero. $\square$

THE SHIFTED COTANGENT THEORY

2.1 THE LIE ALGEBRA AND ITS FUNCTIONS

Put

$$\mathfrak{g}_d = T^*[-1]\mathfrak{h}_{\leq d} = \mathfrak{h}_{\leq d} \ltimes \mathfrak{h}_{\leq d}^\vee[-1],$$

where $\mathfrak{h}_{\leq d}$ acts on the dual by the coadjoint action and the shifted dual is abelian.

Proposition 2.1 (Chevalley–Poisson identification). *There is a canonical isomorphism of commutative dg algebras*

$$C_{\text{CE}}^*(\mathfrak{g}_d) \cong C^*(\mathfrak{h}_{\leq d}, \text{Sym}(\mathfrak{h}_{\leq d})).$$

The differential on the right is the Chevalley differential for the adjoint action on $\text{Sym}(\mathfrak{h}_{\leq d})$. Under the identification $\text{Sym}(\mathfrak{h}_{\leq d}) = \mathcal{O}(\mathfrak{h}_{\leq d}^)$, it is the linear Poisson differential.*

Proof. The dual of $\mathfrak{h}_{\leq d}^\vee[-1]$ shifted by $[-1]$ is $\mathfrak{h}_{\leq d}$ in degree zero. Chevalley cochains therefore contain polynomial functions on $\mathfrak{h}_{\leq d}^*$ and exterior ghosts for $\mathfrak{h}_{\leq d}$. The semidirect bracket produces the coadjoint action, which is the Hamiltonian action of the linear Poisson tensor. $\square$

2.2 THE BV ACTION

Choose dual bases e_a and e^a . Let c^a be the ghost coordinates and b_a the cotangent coordinates. The finite action is

$$S_d = \frac{1}{2} b_c f^c_{ab} c^a c^b.$$

The canonical BV bracket satisfies

$$\frac{1}{2} \{S_d, S_d\} = 0$$

by Jacobi.

Theorem 2.2 (Exact finite quantum master equation). *With the translation-invariant BV Laplacian,*

$$\Delta S_d = \sum_b \text{Tr}(\text{ad } e_b) c^b = 0.$$

Consequently

$$\frac{1}{2} \{S_d, S_d\} + \hbar \Delta S_d = 0$$

for every d .

Proof. Contracting one field and one antifield in the cubic Hamiltonian gives the divergence of the adjoint vector field. The preceding unimodularity theorem makes every coefficient vanish. $\square$

CHAPTER 3

THE UNIVERSAL BOUNDARY BRANE

3.1 POLARIZATION

Place the cotangent theory on a half-plane. The standard Lagrangian boundary condition fixes the cotangent field and leaves the $\mathfrak{h}_{\leq d}$ field free. Boundary observables form the Rees enveloping algebra

$$U_{\hbar}(\mathfrak{h}_{\leq d}) = T(\mathfrak{h}_{\leq d})[[\hbar]] / \langle xy - yx - \hbar[x, y] \rangle.$$

The Poincaré–Birkhoff–Witt symmetrization gives a filtered vector-space isomorphism

$$\mathrm{sym}_{\hbar} : \mathrm{Sym}(\mathfrak{h}_{\leq d})[[\hbar]] \xrightarrow{\sim} U_{\hbar}(\mathfrak{h}_{\leq d}).$$

Theorem 3.1 (Boundary quantization). *The factorization algebra of boundary observables of the finite cotangent theory is locally constant and its value on an interval is $U_{\hbar}(\mathfrak{h}_{\leq d})$. Fusion of adjacent intervals is multiplication in the enveloping algebra.*

Proof. Canonical quantization turns the linear boundary phase space $\mathfrak{h}_{\leq d}^*$ into the Gutt star algebra. For a linear Poisson bracket the Gutt product is transported by PBW to $U_{\hbar}(\mathfrak{h}_{\leq d})$. Excision identifies ordered interval fusion with multiplication. $\square$

BOOK II

Relative Hochschild comparison

HOCHSCHILD COCHAINS OF THE UNIVERSAL BRANE

4.1 THE COEFFICIENT RING AND THE CHEVALLEY DIFFERENTIAL

The Rees relation scales both the Lie bracket and its adjoint action. Its Chevalley differential must retain the same parameter. For example, take the quadratic jet $\mathbf{I} = \mathfrak{sl}_2$ with $[h, e] = 2e$. Modulo $\hbar$, the enveloping algebra is commutative, so the Hochschild differential of e vanishes. The unscaled Chevalley differential instead satisfies $(d_{\text{CE}}e)(h) = 2e$.

Let $\mathbf{I}$ be a finite-dimensional complex Lie algebra. For $N \geq 0$, put

$$R_N = \mathbb{C}[\hbar]/(\hbar^{N+1}), \quad \mathbf{I}_N = R_N \otimes_{\mathbb{C}} \mathbf{I}, \quad [x, y]_N = \hbar[x, y],$$

and

$$A_N = U_{R_N}(\mathbf{I}_N) = T_{R_N}(\mathbf{I}_N)/(xy - yx - \hbar[x, y]).$$

Here N denotes the order in $\hbar$, independently of a Chan–Paton rank. Write $\bar{A}_N$ for the augmentation ideal and $S = \text{Sym}_{\mathbb{C}} \mathbf{I}$. PBW symmetrization is the R_N -module isomorphism

$$\text{sym}_N : S \otimes R_N \longrightarrow A_N, \quad x_1 \cdots x_r \longmapsto \frac{1}{r!} \sum_{\sigma \in S_r} x_{\sigma(1)} \cdots x_{\sigma(r)}.$$

The source has its ordinary symmetric product. Use normalized relative Hochschild cochains

$$C_{\text{HH}}^n(A_N/R_N, A_N) = \text{Hom}_{R_N}(\bar{A}_N^{\otimes_{R_N} n}, A_N)$$

with differential

$$\begin{aligned} (bF)(a_1, \dots, a_{n+1}) &= a_1 F(a_2, \dots, a_{n+1}) \\ &+ \sum_{i=1}^n (-1)^i F(a_1, \dots, a_i a_{i+1}, \dots, a_{n+1}) \\ &+ (-1)^{n+1} F(a_1, \dots, a_n) a_{n+1}. \end{aligned}$$

Let d_{CE} act on $C^n(\mathbf{I}, S) = \text{Hom}_{\mathbb{C}}(\Lambda^n \mathbf{I}, S)$ by the adjoint Chevalley differential.

Theorem 4.1 (Relative Rees comparison). *Antisymmetrization defines a quasi-isomorphism of R_N -complexes*

$$\text{Alt}_N : C_{\text{HH}}^{\bullet}(A_N/R_N, A_N) \longrightarrow (C^{\bullet}(\mathbf{I}, S) \otimes R_N, \hbar d_{\text{CE}}),$$

given by

$$(\text{Alt}_N F)(x_1, \dots, x_n) = \text{sym}_N^{-1} \sum_{\sigma \in S_n} \text{sgn}(\sigma) F(x_{\sigma(1)}, \dots, x_{\sigma(n)}).$$

These maps commute with coefficient reduction in N .

Proof. Write $A = A_N$, $R = R_N$, and $A^e = A \otimes_R A^{\text{op}}$. In homological degree n , set

$$K_n = A \otimes_R \Lambda_R^n \mathbf{I}_N \otimes_R A.$$

Its augmentation is multiplication. For $X = x_1 \wedge \cdots \wedge x_n$, let $X_{\widehat{i}}$ and $X_{\widehat{ij}}$ omit the indicated entries without reordering. Define

$$\begin{aligned} \partial(a \otimes X \otimes b) &= \sum_i (-1)^{i+1} (ax_i \otimes X_{\widehat{i}} \otimes b - a \otimes X_{\widehat{i}} \otimes x_i b) \\ &\quad + \hbar \sum_{i < j} (-1)^{i+j} a \otimes [x_i, x_j] \wedge X_{\widehat{ij}} \otimes b. \end{aligned}$$

The normalized relative bimodule bar has terms $\mathcal{B}_n = A \otimes_R \tilde{A}^{\otimes n} \otimes_R A$. Define

$$\iota_n(a \otimes X \otimes b) = \sum_{\sigma \in \mathcal{S}_n} \text{sgn}(\sigma) a [x_{\sigma(1)} | \cdots | x_{\sigma(n)}] b.$$

The outer bar faces give the first sum in ∂ . Pair adjacent transpositions in the internal faces and use $xy - yx = \hbar[x, y]$ to obtain the second sum. For two inputs this reads

$$\begin{aligned} b(\iota[x|y]_1 - \iota[y|x]_1) &= x[y]_1 - y[x]_1 + \iota[x]y - \iota[y]x \\ &\quad - \hbar \iota[[x, y]]_1. \end{aligned}$$

Thus $b\iota = \iota\partial$. PBW makes $\mathbf{I}_N$ a free summand of $\tilde{A}$. Since every factorial is invertible in R , ι is injective. The identity $b^2 = 0$ then gives $\partial^2 = 0$.

Give K_n filtration degree equal to its outer PBW degrees plus n . The associated graded is the diagonal Koszul complex of $\text{Sym}_R(\mathbf{I}_N)$. Its differential uses differences of left and right polynomial generators. A linear change of variables makes these independent polynomial variables. They form a regular sequence even when R has nilpotents. For an augmented cycle, lift a primitive of its highest-degree symbol and subtract its differential. The filtration degree decreases, so this procedure terminates. Hence $K_\bullet \rightarrow A$ is a free bimodule resolution. The relative bar is also free over A^e , with its usual unit-insertion contraction. The map ι lifts the identity of A . Projectivity constructs its chain homotopy inverse degree by degree. Applying $\text{Hom}_{A^e}(-, A)$ gives a cochain homotopy equivalence.

The induced differential on $\text{Hom}_R(\Lambda_R^n \mathbf{I}_N, A)$ is

$$\begin{aligned} (\partial_N \phi)(x_1, \dots, x_{n+1}) &= \sum_i (-1)^{i+1} [x_i, \phi(x_1, \dots, \widehat{x}_i, \dots, x_{n+1})]_A \\ &\quad + \hbar \sum_{i < j} (-1)^{i+j} \phi([x_i, x_j], x_1, \dots, \widehat{x}_i, \dots, \widehat{x}_j, \dots, x_{n+1}). \end{aligned}$$

Commuting x past the factors of a symmetrized monomial gives

$$[x, \text{sym}_N(f)]_A = \hbar \text{sym}_N(\text{ad}_x f).$$

Both sums therefore become $\hbar$ times the ordinary Chevalley differential. Precomposition with ι gives Alt_N . All formulas commute with coefficient reduction. $\square$

4.2 THE FORMAL COEFFICIENT LIMIT

Put $R = \mathbb{C}[[\hbar]]$ and $A = \varprojlim_N A_N$. PBW identifies its underlying module with $S[[\hbar]]$. Each coefficient of $\hbar$ is a polynomial. Define

$$A^{\widehat{\otimes}_{R^n}} = \varprojlim_N A_N^{\otimes_{R_N} n}, \quad C_{\text{HH,cont}}^\bullet(A/R, A) = \varprojlim_N C_{\text{HH}}^\bullet(A_N/R_N, A_N).$$

Equivalently, these are continuous R -multilinear normalized cochains on the completed relative tensors.

Corollary 4.2 (Formal relative Rees comparison). *The compatible antisymmetrizations induce a quasi-isomorphism*

$$C_{\text{HH,cont}}^\bullet(A/R, A) \longrightarrow (C^\bullet(\mathbf{I}, S)[[\hbar]], \hbar d_{\text{CE}}).$$

Proof. Coefficient reduction is surjective in each cochain degree. For Hochschild cochains, lift outputs on the PBW input basis. The Chevalley exterior input has finite rank, so its outputs also lift. The comparison cones are acyclic and their transition maps are degreewise surjective. The long exact sequence shows that every transition kernel is acyclic. Choose a primitive of a compatible cycle at level zero. Lift it one level and correct its differential by a primitive in the transition kernel. Iteration gives compatible primitives at every order. The inverse-limit cone is therefore acyclic. $\square$

The coefficient ring is part of this result. Absolute $\mathbb{C}$ -linear cochains also detect variations of the coefficient parameter. For $\mathbf{I} = \mathbf{o}$, the continuous absolute derivation $\partial_{\hbar}$ on R has no relative counterpart. For nonabelian $\mathbf{I}$, the associated graded for the $\hbar$ filtration has zero differential, whereas d_{CE} need not vanish.

THE KONTSEVICH–DUFLO MAP

5.1 LINEAR POISSON FORMALITY

For $\mathfrak{l} = \mathfrak{h}_{\leq d}$, put $A_d = U_{\hbar}(\mathfrak{h}_{\leq d})$ and

$$\mathcal{P}_d = (C^*(\mathfrak{h}_{\leq d}, \text{Sym } \mathfrak{h}_{\leq d})[[\hbar]], \hbar d_{\text{CE}}).$$

This is the polynomial polyvector complex on $\mathfrak{h}_{\leq d}^*$ with differential $[\hbar\pi_d, -]$ for the linear Poisson tensor π_d . The relative comparison gives $C_{\text{HH,cont}}^\bullet(A_d/R, A_d) \rightarrow \mathcal{P}_d$ as a map of complexes. Higher operations require a formality map with specified operadic structure.

Fix Kontsevich’s formality morphism $\mathcal{U}$ on the affine space $\mathfrak{h}_{\leq d}^*$. Its twist by $\hbar\pi_d$ defines a star product $\star_d$ and an L_∞ quasi-isomorphism on the degree-shifted complexes. Let $T_d : (\text{Sym } \mathfrak{h}_{\leq d}[[\hbar]], \star_d) \rightarrow A_d$ be the algebra identification fixing the linear generators. This identification includes the equivalence between the Kontsevich and PBW products. PBW symmetrization alone identifies the underlying modules. The first Taylor coefficient of the twisted comparison is

$$\Phi_d(v) = (T_d)_* \sum_{r \geq 0} \frac{1}{r!} \underbrace{\mathcal{U}_{r+1}(\hbar\pi_d, \dots, \hbar\pi_d, v)}_r.$$

It has source $\mathcal{P}_d$ and target $C_{\text{HH,cont}}^\bullet(A_d/R, A_d)$. At each order in $\hbar$, polynomial inputs have polynomial outputs. Its reduction modulo $\hbar$ is the Hochschild–Kostant–Rosenberg map, with the normalized antisymmetrization convention. That map is a quasi-isomorphism for the polynomial algebra. The complete $\hbar$ filtration then proves that Φ_d is a quasi-isomorphism. The higher Taylor coefficients supply the L_∞ identities after shifting by [1]. These statements use the affine formality construction [51, Sections 6 and 8.3].

On invariant polynomials, the standard Duflo normalization gives

$$\text{Duf}_\hbar = \text{sym}_\hbar \circ j^{1/2}(\hbar d), \quad j^{1/2}(x) = \det_{\mathfrak{h}_{\leq d}} \left(\frac{\sinh(\text{ad}_x/2)}{\text{ad}_x/2} \right)^{1/2}.$$

It is an algebra isomorphism from $(\text{Sym } \mathfrak{h}_{\leq d})^{\mathfrak{h}_{\leq d}}[[\hbar]]$ to $Z(A_d)$, by the Duflo theorem [51, Section 8.3.4]. The square root has constant term one. This assertion concerns invariant degree-zero cohomology. It does not identify antisymmetrization with a homotopy- E_2 map.

5.2 THE BOUNDARY ACTION

The brace operad acts on $(C_{\text{HH,cont}}^\bullet(A_d/R, A_d), A_d)$. Its closed color is the relative derived centre. A homotopy- E_2 refinement means a morphism or zigzag

$$\tilde{\Phi}_d : \mathcal{P}_d \simeq C_{\text{HH,cont}}^\bullet(A_d/R, A_d)$$

in complete R -linear homotopy- E_2 algebras, with the chosen Gerstenhaber structure on $\mathcal{P}_d$. A compatible Swiss-cheese refinement also specifies the mixed operations on the pair and their coherence homotopies.

Corollary 5.1 (Open–closed structure with a refinement). *Given these refinements, the pair $(\mathcal{P}_d, A_d)$ has the transported homotopy Swiss-cheese action. Its open product is the enveloping product. The specified closed and mixed Taylor coefficients determine its closed bracket and boundary action.*

Proof. Transport the two-colored operations through the supplied equivalence of complete operadic algebras. The coherence homotopies are part of this equivalence. $\square$

THE FORMAL HAMILTONIAN LIMIT

6.1 WEIGHT COMPLETION AND COMPARISON DATA

The Hamiltonian Lie algebra and its boundary algebra have quotient systems

$$\mathfrak{h} = \varprojlim_d \mathfrak{h}_{\leq d}, \quad \widehat{U}_{\hbar}(\mathfrak{h}) = \varprojlim_d U_{\hbar}(\mathfrak{h}_{\leq d}).$$

A Lie quotient does not by itself define a covariant map between Chevalley cochains with adjoint coefficients. Nor does an algebra quotient define a map between all Hochschild cochains with values in the algebra itself. A cochain must descend through the input quotient and the output quotient. Consequently the coefficient limit of Corollary 4.2 is distinct from this jet limit.

A *compatible jet comparison* consists of the following data. Choose complete R -linear cochain models P_d, H_d with transition maps, levelwise comparisons $P_d \simeq \mathcal{P}_d$ and $H_d \simeq C_{\text{HH,cont}}^{\bullet}(A_d/R, A_d)$, and compatible quasi-isomorphisms $P_d \rightarrow H_d$. Require degreewise surjective transitions, or an explicit derived inverse-limit construction with the same exactness property. Specify identifications

$$\varprojlim_d P_d \simeq \mathcal{P}_{\infty}, \quad \varprojlim_d H_d \simeq C_{\text{HH,cont}}^{\bullet}(\widehat{U}_{\hbar}(\mathfrak{h})/R, \widehat{U}_{\hbar}(\mathfrak{h})),$$

where $\mathcal{P}_{\infty}$ is the proposed continuous Hamiltonian polyvector complex with differential $[\hbar\pi, -]$. The topology on the right combines the stated Hamiltonian weight completion with the $\hbar$ completion. The identifications must use this same topology. For higher operations, also supply compatible homotopy- E_2 and Swiss-cheese refinements whose operations converge in that topology.

Theorem 6.1 (Continuous comparison from compatible jets). *A compatible jet comparison induces a continuous quasi-isomorphism*

$$\Phi_{\infty} : \mathcal{P}_{\infty} \xrightarrow{\sim} C_{\text{HH,cont}}^{\bullet}(\widehat{U}_{\hbar}(\mathfrak{h})/R, \widehat{U}_{\hbar}(\mathfrak{h})).$$

Compatible homotopy- E_2 and Swiss-cheese refinements give the corresponding complete open-closed operations.

Proof. For surjective transitions, the cones of the levelwise maps are acyclic inverse systems with surjective maps in each degree. The primitive-lifting argument of Corollary 4.2 proves exactness of their limit. For a derived inverse limit, use its invariance under levelwise equivalences. Apply the supplied identifications of the two limits. Compatibility and convergence of the operadic maps give the final assertion. $\square$

The finite PBW comparison constructs the relative centre at each fixed jet. A continuous field-theory comparison additionally requires the jet datum and a map from the specified field observables to $\mathcal{P}_{\infty}$. Universal graph formulas alone do not provide the cochain transition maps.

Principle 6.2 (Universal brane principle). The boundary algebra $\widehat{U}_h(\mathfrak{h})$ is the universal formal Hamiltonian brane. Finite matrix branes are representation-theoretic images of this algebra. Its complete relative derived centre is the algebraic target for bulk observables. A field-theory identification requires the continuous comparison data stated above.

BOOK **III**

The finite-jet laboratory

THE FIRST THREE HAMILTONIAN JETS

7.1 QUADRATIC HAMILTONIANS

Set

$$e = \frac{1}{2}z_1^2, \quad f = -\frac{1}{2}z_2^2, \quad h = -z_1z_2.$$

A direct differentiation gives

$$\{h, e\} = 2e, \quad \{h, f\} = -2f, \quad \{e, f\} = h.$$

Thus $\mathfrak{h}_{\leq 2} = R_2$ is $\mathfrak{sl}_2$ with its standard Chevalley basis. The finite cotangent theory for $d = 2$ is the linear BV theory of $T^*[-1]\mathfrak{sl}_2$ and its boundary algebra is $U_h(\mathfrak{sl}_2)$.

7.2 CUBIC HAMILTONIANS

Let

$$v_j = z_1^{3-j} z_2^j, \quad 0 \leq j \leq 3.$$

Then

$$\{h, v_j\} = (3 - 2j)v_j, \quad \{e, v_j\} = jv_{j-1}, \quad \{f, v_j\} = (3 - j)v_{j+1},$$

with the boundary terms understood as zero. Hence R_3 is the irreducible highest-weight-three module.

Proposition 7.1 (The cubic jet algebra). *The degree-three jet algebra is the semidirect product*

$$\mathfrak{h}_{\leq 3} \cong \mathfrak{sl}_2 \ltimes \text{Sym}^3(\mathbb{C}^2),$$

where the second summand is abelian.

Proof. The bracket of two cubic polynomials has degree four and vanishes in the degree-three quotient. The bracket with a quadratic polynomial is the displayed $\mathfrak{sl}_2$ action. $\square$

7.3 THE FIRST NONLINEAR LAYER

Write $w_k = z_1^{4-k} z_2^k$, $0 \leq k \leq 4$. In $\mathfrak{h}_{\leq 4}$ the cubic modes satisfy

$$\{v_i, v_j\} = 3(j - i)w_{i+j-1}, \tag{7.1}$$

whenever the index lies between 0 and 4; the coefficient makes every remaining case vanish.

Calculation 7.2 (Two brackets).

$$\{z_1^3, z_2^3\} = 9z_1^2 z_2^2 = 9w_2, \quad \{z_1^2 z_2, z_1 z_2^2\} = 3z_1^2 z_2^2 = 3w_2.$$

The two independent cubic pairs reach the same quartic channel with amplitudes in the ratio 3 : 1.

Proof. Differentiate each monomial in the two symplectic coordinates and substitute in the Poisson formula. The first pair gives $3z_1^2 \cdot 3z_2^2$. The second gives $2z_1z_2 \cdot 2z_1z_2 - z_1^2 \cdot z_2^2 = 3z_1^2z_2^2$. $\square$

Theorem 7.3 (The quartic jet presentation). *The Lie algebra $\mathfrak{h}_{\leq 4}$ is generated by $e, f, h, v_0, v_1, v_2, v_3$. Its relations are the $\mathfrak{sl}_2$ relations, the irreducible action on the v_j , formula (7.1), and the induced $\mathfrak{sl}_2$ action on R_4 . The quartic subspace is abelian in the quotient.*

Proof. Quadratic and cubic generators produce every quartic monomial through (7.1). Brackets involving a quartic and a cubic have degree five, and brackets of two quartics have degree six, so both vanish after projection. Jacobi follows from the polynomial Poisson bracket before projection. $\square$

MOYAL QUANTIZATION IN COORDINATES

8.1 THE BIDIFFERENTIAL FORMULA

For polynomials on $\mathbb{C}^2$, the Moyal product is

$$f \star g = \sum_{n \geq 0} \frac{1}{n!} \left(\frac{\hbar}{2} \right)^n \sum_{r=0}^n (-1)^r \binom{n}{r} (\partial_1^{n-r} \partial_2^r f) (\partial_1^r \partial_2^{n-r} g). \quad (8.1)$$

Every polynomial pair gives a finite sum.

Theorem 8.1 (Quadratic exactness). *For polynomials f, g of degree at most two,*

$$[f, g]_\star = \hbar \{f, g\}.$$

The quadratic generators therefore define a Lie representation of the Rees-scaled $\mathfrak{sl}_2$. Their associative algebra has an additional Casimir relation.

Proof. The commutator retains the odd powers of the Poisson bidifferential. Derivatives of order three vanish on quadratics, so the first-order term is the entire commutator. $\square$

Proposition 8.2 (The oscillator quotient). *Let $R_N = \mathbb{C}[\hbar]/(\hbar^{N+1})$ and let $W_N = R_N[q, p]$ have the Moyal product for $\{q, p\} = 1$. Let U_N have generators E, F, H with relations*

$$[H, E] = 2\hbar E, \quad [H, F] = -2\hbar F, \quad [E, F] = \hbar H.$$

Set $\Omega = EF + FE + H^2/2$. The assignments

$$E \mapsto e = q^2/2, \quad F \mapsto f = -p^2/2, \quad H \mapsto h = -qp$$

induce an algebra isomorphism

$$U_N/(\Omega + 3\hbar^2/8) \xrightarrow{\sim} W_N^{(q,p) \mapsto (-q,-p)}.$$

These compatible isomorphisms also hold after $\hbar$ -adic completion. If linear Hamiltonians are retained, the generated associative algebra is the full Weyl algebra.

Proof. Quadratic exactness proves the three commutator relations. The associative products are

$$\begin{aligned} e \star f &= -q^2 p^2/4 - \hbar q p/2 - \hbar^2/8, \\ f \star e &= -q^2 p^2/4 + \hbar q p/2 - \hbar^2/8, \\ h \star h &= q^2 p^2 - \hbar^2/4. \end{aligned}$$

Thus Ω maps to $-3\hbar^2/8$. Its commutator with each of E, F, H vanishes by the defining relations. Give these generators PBW degree one. Give an even monomial $q^a p^b$ degree $(a+b)/2$. The leading symbol of $\Omega + 3\hbar^2/8$ is

$C = 2EF + H^2/2$. It is a non-zero-divisor in $R_N[E, F, H]$, since $2C$ is monic in H . Centrality and this property give

$$\text{gr}(U_N/(\Omega + 3\hbar^2/8)) = R_N[E, F, H]/(C).$$

Indeed, every element of the two-sided ideal is a multiple of the central generator. Its leading symbol is the product of the two leading symbols. The quotient has basis $E^a H^b F^c$ with $a, c \geq 0$ and $b = 0, 1$. Their leading symbols map to unit multiples of the distinct monomials $q^{2a+b} p^{2c+b}$, precisely the even monomials. Hence the associated-graded map is an isomorphism. Lifting highest-degree terms proves surjectivity and comparing them proves injectivity. Every element at fixed N has finite polynomial degree. The formulas commute with reduction in N , so their inverse limit gives the formal isomorphism. Finally q, p generate $\mathcal{W}_N$ and topologically generate its formal limit. $\square$

The primitive coproduct on U_N does not descend to this quotient. Modulo $\hbar$, it sends the classical relation to

$$\Delta(C) = C \otimes 1 + 1 \otimes C + 2E \otimes F + 2F \otimes E + H \otimes H.$$

The cross-term is nonzero in the tensor square of $\mathbb{C}[E, F, H]/(C)$. For example, evaluation at $(E, F, H) = (1, 0, 0)$ in the first factor and $(0, 1, 0)$ in the second gives 2. Thus the oscillator quotient is an associative algebra, with no coproduct inherited from these primitive generators.

Calculation 8.3 (Cubic correction). For $f = z_1^3$ and $g = z_2^3$,

$$f \star g - g \star f = 9\hbar z_1^2 z_2^2 + \frac{3}{2}\hbar^3.$$

The first term quantizes the quartic Hamiltonian channel. The constant is the first wheel-free ordering correction visible beyond the quadratic Lie algebra.

Proof. In (8.1), the odd orders are $n = 1$ and $n = 3$. The first contributes $9\hbar z_1^2 z_2^2$. The third contributes $2(\hbar/2)^3(3!) = 3\hbar^3/2$. $\square$

8.2 WEYL GENERATORS

With the convention $\{q, p\} = 1$, the symbols satisfy $[q, p]_\star = \hbar$. Their Weyl-ordered differential operators are

$$\widehat{q} = q, \quad \widehat{p} = -\hbar \partial_q, \quad [\widehat{q}, \widehat{p}] = \hbar.$$

The abstract Weyl algebra is defined by this commutator relation. Weyl ordering transports its multiplication to the Moyal product. The differential-operator representation realizes these relations over $\mathbb{C}[\hbar]$ and its formal completion. The finite quotient argument above uses the abstract R_N -algebra, so it does not require a faithful differential-operator representation at nilpotent $\hbar$.

CHEVALLEY COORDINATES AND THE POISSON DIFFERENTIAL

Choose a basis e_a of $\mathfrak{h}_{\leq d}$ with structure constants $[e_a, e_b] = f^c_{ab} e_c$. Let c^a be the odd Chevalley coordinate and let x_a be the even coordinate on $\mathfrak{h}_{\leq d}^*$. Then

$$d_{\text{CE}} c^c = -\frac{1}{2} f^c_{ab} c^a c^b, \quad d_{\text{CE}} x_a = f^c_{ba} c^b x_c. \quad (9.1)$$

The first equation encodes the Lie bracket. The second is the adjoint action on $\text{Sym}(\mathfrak{h}_{\leq d})$, viewed as functions on the dual.

Proposition 9.1 (Hamiltonian interpretation). *For a polynomial $F(x)$,*

$$d_{\text{CE}} F = c^a \{x_a, F\}_{\text{lin}}.$$

Consequently $d_{\text{CE}}^2 = 0$ is the Jacobi identity for the linear Poisson tensor.

Proof. The linear Poisson bracket is $\{x_a, x_b\} = f^c_{ab} x_c$. The Leibniz rule gives (9.1) on every polynomial. The square is one half of the Schouten bracket of the Poisson tensor with itself. $\square$

9.1 THE $\mathfrak{sl}_2$ INVARIANT

For the quadratic jet, identify the coordinate functions with e, f, h . The invariant

$$q = 2ef + \frac{1}{2}h^2$$

is d_{CE} -closed. Its formal Duflo image is $\Omega + \hbar^2/2$, where $\Omega = ef + fe + h^2/2$ in the Rees enveloping algebra. This is the degree-zero invariant comparison. Under the oscillator quotient of Proposition 8.2, its image is $\hbar^2/8$.

RESIDUE DUALITY AND THE FORMAL CURRENT ALGEBRA

10.1 THE CONTINUOUS DUAL

Let

$$\rho_{ab} = z_1^{-a-1} z_2^{-b-1} dz_1 dz_2, \quad a, b \geq 0.$$

The Grothendieck residue pairing is

$$\text{Res}_0(z_1^i z_2^j \rho_{ab}) = \delta_{ia} \delta_{jb}.$$

The continuous dual of $\mathbb{C}[[z_1, z_2]]$ is the direct sum of the ρ_{ab} . The continuous dual of the pointed algebra $\mathfrak{h} = \mathfrak{m}^2$ is the span of the modes with $a + b \geq 2$. The two modes of total degree one pair with the translation sector of $\mathfrak{h}^{\text{aff}}$.

Proposition 10.1 (Coadjoint current formula). *For $f = z_1^i z_2^j$ and ρ_{ab} ,*

$$\text{ad}_f^*(\rho_{ab})(g) = -\rho_{ab}(\{f, g\}).$$

In the residue basis this is a finite linear combination of modes of bidegree

$$(a - i + 1, b - j + 1).$$

Proof. The definition is the coadjoint action. The Poisson bracket lowers the total degree of $f g$ by two, so residue conservation fixes the displayed bidegree. The Poisson formula contains two derivative terms. $\square$

The shifted cotangent current algebra

$$\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}_{\text{cont}}^\vee[-1]$$

is therefore a complete topological Lie algebra with a continuous invariant pairing. Its finite jet quotients are precisely the $\mathfrak{g}_d$ used above.

ALL-ARITY OPEN–CLOSED OPERATIONS

II.1 HOCHSCHILD BRACES

For Hochschild cochains $F \in C^p(A, A)$ and $G_i \in C^{q_i}(A, A)$, the brace

$$F\{G_1, \dots, G_r\}$$

is the signed sum over ordered insertions of the G_i into the inputs of F . The brace identities state that a nested insertion is the sum over all ways of distributing the inner insertions among the outer slots.

Theorem II.1 (Brace realization with a refinement). *For $A = A_d$, supply the homotopy- E_2 and mixed refinements of Corollary 5.1. They identify the selected homotopy Gerstenhaber operations on $\mathcal{P}_d$ with the relative Hochschild centre operations. The chosen mixed refinement also identifies their action on the boundary algebra.*

Proof. The relative Hochschild complex has its brace operations. The supplied operadic equivalence preserves their coherent homotopies. An equivalence of complexes alone would not imply this conclusion. $\square$

II.2 THE FIRST MIXED IDENTITIES

A relative Hochschild 1-cocycle D of an algebra concentrated in degree zero is an R -linear derivation:

$$D(ab) = D(a)b + aD(b).$$

For two such cocycles D, E , their Gerstenhaber bracket acts by

$$[D, E](a) = D(E(a)) - E(D(a)).$$

For a graded algebra and homogeneous derivations, these formulas become

$$\begin{aligned} D(ab) &= D(a)b + (-1)^{|D||a|}aD(b), \\ [D, E](a) &= D(E(a)) - (-1)^{|D||E|}E(D(a)). \end{aligned}$$

Higher Hochschild cochains act through multilinear evaluations and braces. They need not act by derivations or carry a coproduct defining a diagonal action. A compatible mixed refinement transports these operations, including their homotopies.

Corollary II.2 (Continuous open–closed structure with compatible jets). *Under the jet comparison and mixed refinement hypotheses of Theorem 6.1, the pair $(\mathcal{P}_\infty, \widehat{U}_h(\mathfrak{h}))$ has the corresponding complete homotopy Swiss-cheese action. Its closed color is equivalent to the continuous relative derived centre.*

Proof. The supplied finite operations commute with the transition maps and converge in the specified completion. Their limit acts on the two limit objects. The comparison theorem identifies the closed complex with continuous relative Hochschild cochains. $\square$

FINITE CHAN–PATON RANKS

12.1 RANK ONE

For $N = 1$, the commuting equation is automatic. The boundary coordinate algebra is $\mathbb{C}[x, y]$, and

$$J_1(f) = f(x, y).$$

The necklace bracket reduces to the ordinary Poisson bracket. Restriction to $f \in \mathfrak{m}^2$ gives the defining pointed-Hamiltonian representation. The two linear Hamiltonians act by translations and recover the affine center-of-mass extension.

12.2 RANK TWO AND CAYLEY–HAMILTON REDUCTION

For a 2×2 matrix X ,

$$X^2 - (\text{Tr } X)X + (\det X)I = 0.$$

Hence every trace containing two consecutive powers of X reduces to traces with X -degree at most one after adjoining $\text{Tr } X$ and $\det X$. The same statement holds for Y . On the commuting locus, simultaneous diagonalization identifies the invariant algebra with the symmetric polynomials in two ordered pairs.

Proposition 12.1 (First finite-rank trace relation). *For commuting 2×2 matrices,*

$$p_{20} = p_{10}^2 - 2 \det X, \quad p_{02} = p_{01}^2 - 2 \det Y,$$

and

$$p_{11}^2 - p_{20}p_{02} = -(x_1y_2 - x_2y_1)^2.$$

The stable generators remain independent until a relation forced by the rank is reached.

Proof. Write the joint eigenvalues as (x_1, y_1) and (x_2, y_2) and expand both sides. □

12.3 STABLE PASSAGE

The transition from rank N to rank $N + 1$ adds a zero eigenvalue. Each fixed weighted degree stabilizes once N exceeds that degree. The inverse limit is therefore the completed multisymmetric algebra. The stable trace is an exact quotient of the universal centre on the polynomial trace sector.

BOOK **IV**

Global holomorphic symplectic descent

FEDOSOV DESCENT

13.1 THE WEYL ALGEBROID

All Hochschild cochains and Morita centres in this chapter are continuous and relative to $R = \mathbb{C}[[\hbar]]$. The symplectic globalization uses its own sheafwise formality and Fedosov resolution data, independently of the finite Hamiltonian comparison. Let (S, ω) be a holomorphic symplectic surface. The bundle of formal Weyl algebras has fibre the completed Weyl algebra of a formal symplectic plane. A holomorphic symplectic connection and a Fedosov one-form define a flat connection

$$D_F = \nabla + \frac{1}{\hbar}[\gamma, -]$$

whose flat sections form a deformation-quantization algebroid $\mathcal{W}_{S, \hbar}$.

Theorem 13.1 (Globalized formality on a holomorphic symplectic surface). *Fix a stable formality morphism and a compatible homotopy- E_2 refinement. There is a sheafwise homotopy- E_2 quasi-isomorphism*

$$\Phi_S : (\mathrm{PV}_S[[\hbar]], [\hbar\pi, -]) \xrightarrow{\sim} C_{\mathrm{HH}}^\bullet(\mathcal{W}_{S, \hbar}, \mathcal{W}_{S, \hbar}),$$

where $\pi = \omega^{-1}$. A change of holomorphic symplectic connection or Fedosov one-form gives a gauge-equivalent resolution and a homotopic quasi-isomorphism. Holomorphic symplectomorphisms act equivariantly on the resulting homotopy class.

Proof. Fedosov resolutions replace both sheaves by acyclic sheaves of formal fibrewise polyvectors and polydifferential operators. The fixed covariant formality morphism acts fibrewise and intertwines the Fedosov differentials. A change of connection or Fedosov one-form is a gauge transformation in the pronilpotent resolution and therefore gives a homotopy. $\square$

Corollary 13.2 (Global mixed-HT centre). *The deformation-quantization category $\mathrm{Perf}(\mathcal{W}_{S, \hbar})$ has Morita centre*

$$Z_{\mathrm{Mor}}(\mathrm{Perf}(\mathcal{W}_{S, \hbar})) \simeq \mathbb{R}\Gamma(S, \mathrm{PV}_S[[\hbar]], [\hbar\pi, -]).$$

For a $K3$ surface this computes the symplectic Poisson centre in the stated sheafwise model. A Hamiltonian BF identification additionally requires a comparison from its Chevalley observables to this symplectic polyvector complex.

Proof. The Morita centre of perfect modules over an algebroid is the derived global section complex of its Hochschild cochains. Apply the globalized formality equivalence and then derived global sections. A $K3$ surface is holomorphic symplectic, so the same sheafwise calculation applies whenever the chosen globalization data are available. $\square$

FACTORIZATION AND DESCENT

14.1 THE ALGEBRAIC MIXED-HT REALIZATION

The sheaf

$$\mathcal{P}_S = (\mathrm{PV}_S[[\hbar]], [\hbar\pi, -])$$

is a sheaf of complete homotopy- E_2 algebras. Dunn additivity gives its locally constant factorization realization in two oriented topological directions. The holomorphic sheaf variable retains the dependence on S .

Theorem 14.1 (Algebraic factorization realization). *The homotopy- E_2 sheaf $\mathcal{P}_S$ determines a complete mixed holomorphic-topological factorization algebra*

$$\mathcal{A}_{S,\mathrm{bulk}}^{\mathrm{alg}} \quad \text{on} \quad \mathbb{R}_{\mathrm{top}}^2 \times S.$$

Its universal boundary category is $\mathrm{Perf}(\mathcal{W}_{S,\hbar})$, and the bulk-boundary morphism is a local equivalence

$$\mathcal{A}_{S,\mathrm{bulk}}^{\mathrm{alg}} \xrightarrow{\sim} Z_{\mathrm{Mor}}^{\mathrm{der}}(\mathrm{Perf}(\mathcal{W}_{S,\hbar})).$$

On a formal Darboux chart this map is the symplectic polyvector-to-Weyl Hochschild comparison. Comparing it with the pointed Hamiltonian centre additionally requires a map between those two polyvector complexes.

Proof. The factorization envelope of a complete homotopy- E_2 algebra is locally constant on oriented two-disks. Sheafwise application to $\mathcal{P}_S$ gives the first assertion. The Fedosov algebroid quantizes the symplectic Poisson tensor on S . Globalized formality identifies its Hochschild cochains with $\mathcal{P}_S$, and Morita invariance identifies Hochschild cochains with the centre of the perfect-module category. The construction is local on S and therefore respects factorization descent. $\square$

14.2 BV REALIZATION

The cyclic Lie sheaf of formal Hamiltonians has the shifted-cotangent action

$$S_{\mathrm{BF}}(A, B) = \int_{\mathbb{R}^2 \times S} \left\langle B, (\mathrm{d} + \bar{\partial})A + \frac{1}{2}[A, A] \right\rangle.$$

A comparison of its classical observables with $\mathcal{A}_{S,\mathrm{bulk}}^{\mathrm{alg}}$ must specify a map from Hamiltonian Chevalley cochains to the symplectic polyvector complex. The finite Lie algebra identification alone does not supply that map. A compatible gauge fixing, extension of graph weights to compactified configurations, and solution of the quantum master equation produce a perturbative factorization realization of the same algebraic bulk-boundary map. The finite jet zero modes carry the exact quantum measure constructed above.

BOOK V

Matrix branes and trace geometry

THE DERIVED COMMUTING STACK

Let $X, Y \in \mathfrak{gl}_N$ and let $\psi \in \mathfrak{gl}_N[1]$. The Koszul differential

$$QX = QY = 0, \quad Q\psi = [X, Y]$$

resolves the commuting equation. The simultaneous conjugation action gives the boundary cdga

$$\mathcal{A}_N = C_{\text{CE}}^*(\mathfrak{gl}_N, \text{Sym}(\mathfrak{gl}_N \oplus \mathfrak{gl}_N \oplus \mathfrak{gl}_N[1])).$$

Proposition 15.1 (Representation of the universal brane). *Substitution of matrices defines a Lie morphism*

$$\rho_N : \mathfrak{h} \longrightarrow \text{Der}(\mathcal{A}_N), \quad f \longmapsto \delta_f,$$

which integrates to an algebra morphism

$$\widehat{U}_\hbar(\mathfrak{h}) \longrightarrow \text{End}(\mathcal{A}_N)[[\hbar]], \quad f \longmapsto \hbar \delta_f.$$

Thus the N -brane stack is a representation of the universal boundary algebra.

Proof. For a polynomial Hamiltonian f , cyclic differentiation gives the matrix vector field

$$\delta_f(X) = -\partial_Y^{\text{cyc}} f(X, Y), \quad \delta_f(Y) = \partial_X^{\text{cyc}} f(X, Y).$$

The cyclic Euler identity implies $\delta_f([X, Y]) = 0$. Hence δ_f commutes with the Koszul differential $Q\psi = [X, Y]$. The same formulas are equivariant for simultaneous conjugation, so δ_f descends to a derivation of the Chevalley–Eilenberg quotient $\mathcal{A}_N$. The trace-moment-map calculation gives

$$[\delta_f, \delta_g] = \delta_{\{f, g\}},$$

first for polynomial Hamiltonians and then continuously for the formal completion. This proves that ρ_N is a continuous Lie morphism. The assignment $f \mapsto \hbar \delta_f$ satisfies

$$[\hbar \delta_f, \hbar \delta_g] = \hbar^2 \delta_{\{f, g\}} = \hbar (\hbar \delta_{\{f, g\}}),$$

which is the Rees relation. The universal property of the completed Rees enveloping algebra gives the displayed morphism. It is the universal boundary action evaluated in the N -dimensional Chan–Paton representation. $\square$

NECKLACE HAMILTONIANS

Let

$$F = \mathbb{C}\langle x, y \rangle, \quad F_{\text{cyc}} = F/[F, F].$$

For a cyclic word f , define cyclic derivatives by cutting the word at every occurrence of the selected letter. Put

$$\partial_f(x) = -\partial_y^{\text{cyc}} f, \quad \partial_f(y) = \partial_x^{\text{cyc}} f.$$

Theorem 16.1 (Necklace Lie algebra). *The bracket*

$$\{f, g\}_{\text{neck}} = [\partial_x^{\text{cyc}} f \partial_y^{\text{cyc}} g - \partial_y^{\text{cyc}} f \partial_x^{\text{cyc}} g]_{\text{cyc}}$$

makes $F_{\text{cyc}}/\mathbb{C}$ a Lie algebra. The assignment $f \mapsto \partial_f$ is a Lie morphism. It preserves the moment map:

$$\partial_f([x, y]) = 0.$$

Proof. The cyclic Euler identity is

$$[x, \partial_x^{\text{cyc}} f] + [y, \partial_y^{\text{cyc}} f] = 0.$$

It gives the moment-map statement. The canonical double Poisson bracket on the free algebra descends to the displayed necklace bracket; its Jacobi identity gives the first assertion. Evaluation on generators gives $[\partial_f, \partial_g] = \partial_{\{f, g\}_{\text{neck}}}$. $\square$

Theorem 16.2 (Trace moment map). *For*

$$J_N(f) = \text{Tr } f(X, Y),$$

one has

$$\{J_N(f), J_N(g)\}_{\mathcal{M}_N} = J_N(\{f, g\}_{\text{neck}}),$$

where $\mathcal{M}_N = \mathfrak{gl}_N^2$ carries $\text{Tr}(dX \wedge dY)$.

Proof. The differential of a trace is the matrix evaluation of the cyclic derivative. Contracting the two differentials with the inverse trace symplectic form gives the necklace formula. $\square$

STABLE TRACES

On the semisimple commuting locus write the joint eigenvalues as (x_i, y_i) . Define

$$p_{ab}^{(N)} = \sum_{i=1}^N x_i^a y_i^b.$$

Theorem 17.1 (Stable multisymmetric Poisson algebra). *The inverse stable trace algebra is the completed polynomial algebra*

$$\mathcal{P}_\infty = \mathbb{C}[[p_{ab} \mid a, b \geq 0, a + b > 0]].$$

Its Poisson bracket is

$$\{p_{ab}, p_{cd}\} = (ad - bc)p_{a+c-1, b+d-1}.$$

Proof. The stable fundamental theorem for multisymmetric functions gives polynomial generators p_{ab} . The bracket follows from $\{x_i, y_j\} = \delta_{ij}$ and the Leibniz rule. $\square$

Remark 17.2. The stable trace algebra is a quotient of the universal closed algebra after representation and scalarization. The universal centre retains Hochschild classes, stabilizer directions, and higher central operations that vanish under the trace.

CHAPTER 18

THE TWO-BRANE CONE

Separate center-of-mass and relative coordinates. Put

$$u = x_1 - x_2, \quad v = y_1 - y_2, \quad E = \frac{1}{2}u^2, \quad F = -\frac{1}{2}v^2, \quad H = -uv.$$

Then

$$H^2 + 4EF = 0.$$

Proposition 18.1 (The A_1 Kirillov–Kostant cone). *With $\{u, v\} = 2$,*

$$\{H, E\} = 4E, \quad \{H, F\} = -4F, \quad \{E, F\} = 2H.$$

After the evident rescaling this is the $\mathfrak{sl}_2$ Lie–Poisson bracket on the nilpotent cone.

Proof. The three identities follow by direct differentiation. The quadratic equation is the determinant-zero equation for the traceless matrix

$$\begin{pmatrix} H & 2E \\ 2F & -H \end{pmatrix}.$$

□

BOOK VI

**Wheel corrections and exact quantum
structure**

THE DUFLO WHEEL

First set the coefficient parameter to 1 in the polynomial Rees algebra. For the resulting ordinary $\mathfrak{sl}_2$ enveloping algebra, take generators

$$[h, e] = 2e, \quad [h, f] = -2f, \quad [e, f] = h.$$

The invariant polynomial and the quadratic Casimir are

$$q = 2ef + \frac{1}{2}h^2, \quad \Omega = ef + fe + \frac{1}{2}h^2.$$

Calculation 19.1 (The zero-point shift). The Duflo map gives

$$\text{Duf}(q) = \Omega + \frac{1}{2}.$$

On the irreducible highest-weight- n module,

$$\left(\Omega + \frac{1}{2}\right)\Big|_{V_n} = \frac{(n+1)^2}{2} \text{id}.$$

Proof. Write $x = ue + vf + wh$. Direct adjoint matrices give $\text{Tr}(\text{ad}_x^2) = 8uv + 8w^2$. Thus the quadratic term of $j^{1/2}(\partial)$ is $(\partial_e \partial_f + \partial_h^2)/6$. Applied to $q = 2ef + h^2/2$, it gives $(2+1)/6 = 1/2$. Higher differential terms annihilate this quadratic polynomial. Symmetrization sends q to Ω . On a highest-weight vector, ef acts by n , fe by zero, and $h^2/2$ by $n^2/2$. Centrality gives the eigenvalue on the entire irreducible module. $\square$

Restoring the Rees parameter gives $\text{Duf}_\hbar(q) = \Omega + \hbar^2/2$. The polynomial calculation permits specialization at 1; arbitrary formal power series do not admit that specialization. Higher wheels assemble into $j^{1/2}$ on higher-degree invariants.

THE LOCAL CENTRE WITH COMPARISON DATA

Assume a compatible jet comparison and a complete homotopy- E_2 refinement as in Theorem 6.1. Put

$$\mathcal{A} = \widehat{U}_h(\mathfrak{h}), \quad \mathcal{A}_{\text{bulk}}^{\text{alg}} = \text{Fact}_{E_2}(\mathcal{P}_\infty).$$

The factorization envelope uses the two oriented topological directions. The formal holomorphic coefficient remains at the Darboux point. Let C_∂ be the category of perfect complete $\mathcal{A}$ -modules in the chosen continuous R -linear Morita theory. Assume its centre is computed by $C_{\text{HH},\text{cont}}^\bullet(\mathcal{A}/R, \mathcal{A})$.

Theorem 20.1 (Local derived-centre comparison). *These data give an equivalence of complete homotopy- E_2 factorization algebras*

$$\mathcal{A}_{\text{bulk}}^{\text{alg}} \xrightarrow{\sim} Z_{\text{Mor},R}^{\text{der}}(C_\partial).$$

At the formal insertion point the map is the selected refinement of Φ_∞ . Every continuous boundary module has the induced derived centre action.

Proof. The compatible jet theorem gives the complete homotopy- E_2 comparison. The factorization envelope preserves this equivalence. The assumed continuous Morita model identifies its target with the centre. A centre element is a derived natural endomorphism of the identity functor, so evaluation gives the action on each boundary module. $\square$

Corollary 20.2 (Global K_3 comparison with descent data). *Let S be a K_3 surface with holomorphic symplectic form. Suppose the local comparisons extend to a sheafwise equivalence with the relative Hochschild centre of $\text{Perf}(\mathcal{W}_{S,\hbar})$. Suppose further that the selected cyclic and mixed refinements extend and satisfy descent in the same completed category. Then derived global sections give the corresponding global centre, cyclic, and open–closed comparisons.*

Proof. Derived global sections preserve equivalences of sheaves. Apply them to the supplied sheafwise maps and their coherence homotopies. $\square$

CYCLIC CLASSES, STABLE TRACES, AND LIE HOMOLOGY

A trace of the unit records the representation dimension. For $B = \mathbb{C}$, one has $\mathrm{Tr}_N(1) = N$, so ordinary traces cannot define a rank-compatible map on all of $HC_o(B)$. A scalar-reduced trace removes this dependence after choosing an augmentation.

Let B be a unital associative $\mathbb{C}$ -algebra. The affine representation scheme $\mathrm{Rep}_N B$ represents unital algebra maps $B \rightarrow \mathrm{Mat}_N(D)$ for commutative $\mathbb{C}$ -algebras D . Put

$$\mathcal{R}_N = \mathcal{O}(\mathrm{Rep}_N B)^{GL_N}, \quad \mathfrak{c}_B = HC_o(B) = B/[B, B],$$

where $[B, B]$ denotes the linear span of commutators. The ordinary trace is the linear map

$$\mathrm{Tr}_N : \mathfrak{c}_B \longrightarrow \mathcal{R}_N, \quad \mathrm{Tr}_N([b])(\rho) = \mathrm{Tr}(\rho(b)).$$

Cyclicity makes it independent of the representative, and conjugation invariance puts its image in $\mathcal{R}_N$.

For Lie and Poisson compatibility, specify a Lie bracket on $\mathfrak{c}_B$, a Poisson bracket on each $\mathcal{R}_N$, and the identities

$$\{\mathrm{Tr}_N \beta, \mathrm{Tr}_N \gamma\}_N = \mathrm{Tr}_N[\beta, \gamma]_{\mathrm{cyc}} \quad (\beta, \gamma \in \mathfrak{c}_B). \quad (2I.1)$$

A Hamiltonian comparison also specifies a Lie map $j : \mathfrak{h} \rightarrow \mathfrak{c}_B$ for this bracket. These are additional data on B . The preceding necklace bracket and trace moment map supply the bracket and trace data for $B = \mathbb{C}\langle x, y \rangle$ with its canonical symplectic bracket. The map j remains an additional comparison. An arbitrary unital algebra does not carry these structures by definition.

For rank stabilization, choose an augmentation $\epsilon : B \rightarrow \mathbb{C}$. It defines the rank-stabilization morphism

$$s_N : \mathrm{Rep}_N B \longrightarrow \mathrm{Rep}_{N+1} B, \quad \rho \longmapsto \rho \oplus \epsilon.$$

Restriction gives $s_N^* : \mathcal{R}_{N+1} \rightarrow \mathcal{R}_N$. Define the reduced cyclic space and trace by

$$\bar{\mathfrak{c}}_B = B/(\mathbb{C}1 + [B, B]), \quad t_N(\bar{b})(\rho) = \mathrm{Tr}(\rho(b)) - N\epsilon(b).$$

Let $\mathfrak{m}_N \subset \mathcal{R}_N$ be the ideal of functions vanishing at the representation $b \mapsto \epsilon(b)1_N$. The completed invariant rings and stable target are

$$\widehat{\mathcal{R}}_N = \varprojlim_{r \geq 1} \mathcal{R}_N / \mathfrak{m}_N^r, \quad \mathcal{R}_\infty^\wedge = \varprojlim_N \widehat{\mathcal{R}}_N.$$

Use the inverse-limit topologies. Write J for the augmentation ideal in $\mathrm{Sym}_{\mathbb{C}} \bar{\mathfrak{c}}_B$ and put

$$\widehat{\mathrm{Sym}} \bar{\mathfrak{c}}_B = \varprojlim_{r \geq 1} \mathrm{Sym}_{\mathbb{C}} \bar{\mathfrak{c}}_B / J^r.$$

The stable matrix Lie algebra is $\mathfrak{gl}_\infty B = \varinjlim_N \mathfrak{gl}_N B$ under block-zero Lie embeddings. For its homology, use trivial coefficients $\mathbb{C}$. Let $V_B = \bigoplus_{n \geq 1} HC_{n-1}(B)$, with $HC_{n-1}(B)$ placed in homological degree n . The notation $\mathrm{Sym}_{\mathrm{gr}} V_B$ denotes the free graded-commutative algebra, with the Koszul sign rule.

Theorem 21.1 (Cyclic traces and stable Lie homology). *Assume the compatible jet comparison and its homotopy Gerstenhaber refinement. For the augmented algebra and completion above, there are maps*

$$\begin{aligned} \mathcal{P}_\infty &\xrightarrow{\Phi_\infty} C_{\text{HH},\text{cont}}^*(\mathcal{A}/R, \mathcal{A}), & \text{Tr}_N : \mathfrak{c}_B &\longrightarrow \mathcal{R}_N, \\ \widehat{\text{Sym}} \bar{\mathfrak{c}}_B &\longrightarrow \mathcal{R}_\infty^\wedge, & \text{Sym}_{\text{gr}} V_B &\xrightarrow{\sim} H_*^{\text{CE}}(\mathfrak{gl}_\infty B; \mathbb{C}). \end{aligned}$$

The first preserves the selected homotopy Gerstenhaber structure. The second is linear and, under the bracket data (21.1), is a Lie map. Under those data, its composite with j preserves the Hamiltonian bracket. The third is a continuous homomorphism of commutative algebras, and the fourth is an isomorphism of graded Hopf algebras. These maps are natural for morphisms preserving the stated data.

Proof. The first arrow is the chosen comparison of Theorem 6.1. Its operadic refinement gives the asserted Gerstenhaber compatibility. The trace formula above constructs the second arrow. Equation (21.1) supplies its Lie compatibility, and the Lie property of j gives the composite assertion.

The augmentation annihilates commutators. Both terms of t_N vanish on a commutator, and their values on a scalar multiple of the unit cancel. Thus t_N is well defined on $\bar{\mathfrak{c}}_B$. At the trivial representation its value is zero, so its image lies in $\mathfrak{m}_N$. Moreover,

$$\begin{aligned} (s_N^* t_{N+1})(\bar{b})(\rho) &= \text{Tr}(\rho(b)) + \epsilon(b) - (N+1)\epsilon(b) \\ &= t_N(\bar{b})(\rho). \end{aligned}$$

The point $N\epsilon$ maps to $(N+1)\epsilon$, so $s_N^*(\mathfrak{m}_{N+1}^r) \subset \mathfrak{m}_N^r$. Hence restriction defines the transition maps between the completed rings. The symmetric-algebra extension of t_N sends J^r into $\mathfrak{m}_N^r$. It therefore induces a ring homomorphism on each corresponding quotient. Taking the inverse limit in r gives a continuous map to $\widehat{\mathcal{R}}_N$. The displayed stabilization identity makes these maps compatible in N . Their inverse limit is the third arrow.

The diagonal Lie map $\mathfrak{gl}_\infty B \rightarrow \mathfrak{gl}_\infty B \oplus \mathfrak{gl}_\infty B$ induces the homology coproduct. Block sum in the opposite direction induces the homology product, using the Künneth isomorphism over $\mathbb{C}$. The Loday–Quillen–Tsygan theorem identifies primitive homology in degree n with $HC_{n-1}(B)$.¹ It identifies the full stable homology with the free graded-commutative Hopf algebra on these primitives, giving the fourth arrow. The trace formulas, the augmentation-preserving rank maps, and stable Lie homology are natural. Naturality of the first arrow requires preservation of the selected jet comparison. $\square$

Scalar reduction need not preserve the supplied Poisson bracket. For example, for $B = \mathbb{C}[x, y]$ with $\{x, y\} = 1$ and $\epsilon(x) = \epsilon(y) = 0$, traces on N points satisfy

$$\{\text{Tr}_N x, \text{Tr}_N y\} = N, \quad t_N(\bar{1}) = 0.$$

A reduced Poisson comparison consequently requires additional compatibility, such as vanishing of ϵ on the selected cyclic brackets, or a restriction to a Hamiltonian subalgebra with that property. Only multiplicativity and continuity are asserted for the third arrow. For $B = \mathbb{C}$, the reduced cyclic space is zero and that arrow is the identity on $\mathbb{C}$.

¹J.-L. Loday and D. Quillen, *Cyclic homology and the Lie algebra homology of matrices*, Commentarii Mathematici Helvetici 59 (1984), 565–591, Theorem 6.2, pp. 584–586.

SYNTHESIS

At each fixed Hamiltonian jet, the Rees enveloping algebra has relative Hochschild complex equivalent to the Chevalley complex with differential $\hbar d_{\text{CE}}$. PBW antisymmetrization proves this over every R_N and in the formal coefficient limit. The Duflo theorem gives the separate multiplicative comparison on invariant functions. Quadratic Moyal Hamiltonians generate the oscillator quotient of the enveloping algebra. Their associative Casimir relation prevents the primitive coproduct from descending.

The Hamiltonian jet limit requires the compatible cochain systems of Theorem 6.1. With its operadic refinement and continuous Morita model, the centre acts on every supplied continuous boundary module M through

$$\mathcal{P}_\infty \xrightarrow{\Phi_\infty} C_{\text{HH,cont}}^\bullet(\mathcal{A}/R, \mathcal{A}) \longrightarrow \text{RHom}_{\mathcal{A}}(M, M).$$

The last arrow evaluates a derived natural endomorphism at M . A matrix-brane realization must construct such a module and compare its centre action with the selected trace functions. A commuting square with scalar representation invariants requires these additional comparison maps. Likewise, Fedosov descent concerns symplectic polyvectors and a Weyl algebroid. Its relation to Hamiltonian Lie cochains requires the specified map between their carriers.

BOOK VII

Technical appendices to the preceding
book

SIGNS IN THE SHIFTED COTANGENT COMPLEX

Let e_a have degree zero and let $e^a[-1]$ have degree -1 . The Chevalley generators dual to them have degrees 1 and 0. Thus

$$C_{\text{CE}}^*(T^{*-}[-1]I) = \text{Sym}(I^\vee[-1] \oplus I).$$

The degree-zero polynomial factor is $\text{Sym } I = \mathcal{O}(I^*)$. The differential on a polynomial p is

$$(d_{\text{CE}}p)(x) = \text{ad}_x p,$$

which is the linear Poisson differential.

A DIRECT JACOBI CHECK FOR HAMILTONIAN JETS

For monomials

$$f = z_1^a z_2^b, \quad g = z_1^c z_2^d,$$

one has

$$\{f, g\} = (ad - bc) z_1^{a+c-1} z_2^{b+d-1}.$$

Substitution of three monomials gives a scalar multiple of

$$\det((a, b), (c, d)) \det((a + c - 1, b + d - 1), (e, f)) + \text{cyclic}.$$

Writing the three coefficients in terms of the 2×2 determinants of the exponent vectors shows pairwise cancellation. The coordinate-free proof is $[\pi, \pi]_{\text{Sch}} = 0$ for $\pi = \partial_{z_1} \wedge \partial_{z_2}$.

BIBLIOGRAPHIC NOTE

The finite affine formality map and the separate symplectic globalization use Kontsevich formality and its Fedosov resolution [13, 26, 51]. The relative Hochschild model is the Cartan–Eilenberg comparison, with both Chevalley terms scaled by $\hbar$. Its finite-base resolution proof is Theorem 4.1. Jet limits and operadic refinements require the additional data stated above. The all-arity centre action is the Deligne–Tamarkin brace action [52, 59]. The global deformation-quantization category is the holomorphic Weyl algebroid. The matrix trace and necklace formulas are the noncommutative symplectic moment-map construction.

BOOK ENTRANCE: ORDERED AND ASSOCIATIVE CHIRAL GEOMETRY

Collision on a complex curve, order on a transverse line, exchange in a plane, and the three corresponding bar constructions.

BOOK VIII

The invariant object

TWO DIRECTIONS OF LOCALITY

26.1 THE LOCAL EXPERIMENT

Let X be a smooth complex algebraic curve and let $I \subset \mathbb{R}$ be an oriented interval. Place two protected observables at (z_1, t_1) and (z_2, t_2) in $X \times I$. The limits $z_1 \rightarrow z_2$ and $t_1 \rightarrow t_2$ carry independent information. The first limit produces a singular distribution along the diagonal of X^2 . The second composes operators in the order selected by the orientation of I .

Principle 26.1 (Two-direction principle). Holomorphic collision belongs to the factorization coefficient on X . Ordered fusion belongs to an internal E_1 multiplication. Exchange belongs to an additional two-dimensional transport structure.

This separation gives five geometric carriers:

carrier	operation
diagonal in X^2	singular chiral operation
oriented interval	ordered fusion and bar construction
oriented plane	braid transport
circle	Hochschild trace
stable curve	modular sewing.

Each later structure is introduced by the datum carried by its row.

26.2 CARRIER AND OPERATION

For the diagonal embedding $\Delta : X \hookrightarrow X^2$ and its complement $j : X^2 \setminus \Delta(X) \hookrightarrow X^2$, a Beilinson–Drinfeld chiral Lie operation is a morphism

$$\mu : j_* j^*(\mathfrak{g} \boxtimes \mathfrak{g}) \longrightarrow \Delta_* \mathfrak{g}$$

in the chiral pseudo-tensor category of right D -modules [5]. The D -module is the carrier. The morphism μ is the operation. A factorization object packages carriers on all powers X^I together with their disjoint-locus structure maps. The complete operator product is the pair consisting of that system and its multiplication maps.

In a formal coordinate z near the diagonal, the pole filtration of $\mathcal{O}_{X^2}(*\Delta)$ has

$$\mathrm{gr}_m^F \mathcal{O}_{X^2}(*\Delta) \cong \mathcal{O}_\Delta(m\Delta) \cong N_{\Delta/X^2}^{\otimes m}, \quad m \geq 1.$$

A principal part of order m therefore has an intrinsic normal symbol. Coordinate changes act triangularly on the filtered object. Residues recover the first symbol; normal jets recover the higher symbols.

THE RAN COEFFICIENT CATEGORY

27.1 THE FIXED SIX-FUNCTOR SETTING

Fix a presentable stable category $\mathcal{D}_X$ of right D -modules on $\mathrm{Ran}(X)$ with the following operations:

- (i) the pointwise tensor product $\otimes^!$;
- (ii) chiral convolution $\otimes^{\mathrm{ch}}$ along disjoint union;
- (iii) base change and projection formulae for the disjoint-union correspondences;
- (iv) geometric realizations and totalizations preserved by the tensor products used below.

The standard renormalized D -module setting of chiral Koszul duality has these operations [31]. Every construction below is formed in this fixed setting.

Let

$$\mathrm{Ran}(X) \times \mathrm{Ran}(X) \xleftarrow{j} (\mathrm{Ran}(X) \times \mathrm{Ran}(X))_{\mathrm{disj}} \xrightarrow{u} \mathrm{Ran}(X)$$

be the disjoint-union correspondence. Define

$$F \otimes^{\mathrm{ch}} G = u_! j^!(F \boxtimes G).$$

The pointwise product is induced by the diagonal of $\mathrm{Ran}(X)$.

Definition 27.1 (Aligned-decomposition projector). For a finite set I , let $\mathrm{Dec}(I)$ be the finite groupoid of ordered decompositions $I = A \sqcup B$ and let

$$a_I : \mathrm{Dec}(I) \longrightarrow \mathrm{Dec}(I) \times \mathrm{Dec}(I), \quad d \longmapsto (d, d)$$

be the diagonal. In the linear species model define

$$e_I = a_{I,!} a_I^!.$$

Thus e_I projects a pair of decompositions onto the summands in which the two decompositions agree. In the six-functor Ran model the same formula denotes extraordinary pullback to the diagonal correspondence followed by the available renormalized pushforward.

Theorem 27.2 (Aligned sector in the finite decomposition model). *Let $\mathcal{S}_I$ be a stable category indexed by ordered decompositions of a finite set I , and assume the direct-sum decomposition*

$$\mathcal{M}_I \simeq \bigoplus_{d_1, d_2 \in \mathrm{Dec}(I)} \mathcal{M}_{d_1, d_2}.$$

The projection

$$e_I : \mathcal{M}_I \longrightarrow \mathcal{M}_I, \quad e_I|_{\mathcal{M}_{d_1, d_2}} = \begin{cases} \mathrm{id}, & d_1 = d_2, \\ 0, & d_1 \neq d_2, \end{cases}$$

is idempotent. Its image is the aligned sector

$$\mathcal{M}_I^{\text{al}} = \bigoplus_{d \in \text{Dec}(I)} \mathcal{M}_{d,d}.$$

The family e_I is natural under bijections and compatible with iterated disjoint union. A mixed distributive law in this model is a natural transformation defined on $\mathcal{M}_I^{\text{al}}$ whose two restrictions to the triple diagonal $d_1 = d_2 = d_3$ agree.

Proof. Idempotence is checked on each direct summand. Naturality follows because a bijection carries an ordered decomposition to an ordered decomposition and preserves equality of the two indices. For three products, both iterated alignments project

$$\bigoplus_{d_1, d_2, d_3} \mathcal{M}_{d_1, d_2, d_3} \quad \text{onto} \quad \bigoplus_d \mathcal{M}_{d, d, d}.$$

The two projectors are therefore equal. This equality is precisely the triple-alignment coherence. $\square$

Theorem 27.3 (Global aligned projector). *In the fixed six-functor Ran category, the common-decomposition inclusion is a finite clopen inclusion on every finite-set chart. The functors*

$$e_I = a_{I,!} a_I^!$$

therefore assemble to a natural idempotent e_{al} on the mixed correspondence object. Its image is the common-decomposition correspondence. The two iterated projectors on three decomposition indices coincide, and Beck–Chevalley transports this equality through every Ran transition map.

Proof. The decomposition index is a finite discrete groupoid. Equality of two decomposition indices selects a union of connected components, so the diagonal inclusion is clopen and its extension-by-zero followed by restriction is an idempotent. A surjection of finite sets pulls equal decompositions to equal decompositions, which gives naturality. For three indices, both iterated projectors select the same clopen union $d_1 = d_2 = d_3$. The six-functor base-change isomorphisms identify the corresponding pull–push composites. $\square$

The aligned image carries the canonical interchange obtained by applying the projection formula to one common disjoint-union correspondence. Locality states that transverse multiplication is a morphism for this factorization coproduct. A strong interchange occurs on the full mixed object exactly when every off-aligned summand vanishes.

27.2 FACTORIZATION COALGEBRAS

A chiral commutative coalgebra is a commutative coalgebra object for $\otimes^{\text{ch}}$. It is factorizing when every coproduct becomes an equivalence on the relevant disjoint locus. Let

$$\text{Fact}_D(X) \subset \text{CoAlg}_{\text{ch}}(\mathcal{D}_X)$$

be the full subcategory of factorizing objects. Chiral Koszul duality identifies the corresponding Lie-theoretic and factorization descriptions on the completed Ran category [31, 43].

Proposition 27.4 (Pointwise closure). *The componentwise tensor product of two factorization objects is a factorization object. Thus $\text{Fact}_D(X)$ is symmetric monoidal under $\otimes^!$ whenever the fixed Ran enhancement is closed under that tensor product.*

Proof. For a decomposition $I = I_1 \sqcup I_2$, restrict to its disjointness locus. The factorization equivalences of F and G give

$$\begin{aligned} j^!(F_I \otimes^! G_I) &\simeq j^! F_I \otimes^! j^! G_I \\ &\simeq (F_{I_1} \boxtimes F_{I_2}) \otimes^! (G_{I_1} \boxtimes G_{I_2}) \\ &\simeq (F_{I_1} \otimes^! G_{I_1}) \boxtimes (F_{I_2} \otimes^! G_{I_2}). \end{aligned}$$

Compatibility under refinement and permutation follows componentwise from the corresponding compatibilities for F and G . $\square$

THE TWO NONCOMMUTATIVE CHIRAL OBJECTS

28.1 TRANSVERSE ORDER

A transversely ordered chiral algebra is an E_1 -algebra in the pointwise coefficient category. Its multiplication composes two operators that occupy the same chiral support and are ordered in an independent real direction. This is the structure carried by a boundary or line defect in a holomorphic–topological theory on $X \times \mathbb{R}$.

Definition 28.1 (Transversely ordered chiral algebra). A transversely ordered chiral algebra is an augmented object

$$\mathcal{A}^\perp \in \text{Alg}_{E_1}^{\text{aug}}(\text{Fact}_D(X), \otimes^!).$$

The superscript $\perp$ records the transverse oriented direction. The shorter phrase *ordered chiral algebra* will always mean this object.

28.2 ASSOCIATIVITY INSIDE CHIRAL CONVOLUTION

Chiral convolution combines disjoint finite subsets before allowing them to collide. Associativity for this product is intrinsic to the curve and has a different source and a different bar construction.

Definition 28.2 (Associative chiral algebra). An associative chiral algebra is an augmented algebra

$$\mathcal{A}_{\text{Ass}}^{\text{ch}} \in \text{Alg}_{E_1}^{\text{aug}}(\mathcal{D}_X, \star^{\text{ch}}).$$

A chiral Lie algebra is a Lie algebra object

$$\mathfrak{g}^{\text{ch}} \in \text{Alg}_{\text{Lie}}(\mathcal{D}_X, \star^{\text{ch}}).$$

The commutator of an associative chiral algebra is a chiral Lie algebra.

The operation $m^\perp : \mathcal{A} \otimes^! \mathcal{A} \rightarrow \mathcal{A}$ composes transverse intervals. The operation $m^{\text{ch}} : \mathcal{A} \star^{\text{ch}} \mathcal{A} \rightarrow \mathcal{A}$ composes chiral configurations. They have different domains, different units, and different deformation complexes.

Theorem 28.3 (Independence of the two associative structures). *Let M be an object for which the split square-zero extensions below exist. A pointwise associative product $\mu_\perp : M \otimes^! M \rightarrow M$ defines a transversely ordered algebra on $\mathbf{1}_\perp \oplus M$, while the reduced chiral-convolution product may independently be zero. A chiral-convolution product $\mu_{\text{ch}} : M \star^{\text{ch}} M \rightarrow M$ defines an associative chiral algebra on $\mathbf{1}_{\text{ch}} \oplus M$, while the reduced pointwise product may independently be zero. Hence neither associative structure determines the other.*

A common carrier equipped with both products is a mixed object after a mixed distributive datum has been supplied and its unit, binary, and triple coherences have been verified.

Proof. Adjoining the unit to an associative product on the augmentation ideal gives a unital split extension. The zero product is associative in either monoidal category. These two constructions vary independently. On a common carrier, compatibility is additional structure because the mixed composites have different a priori factorizations through the two monoidal products. $\square$

Definition 28.4 (Mixed associative chiral datum). A mixed associative chiral datum is a factorization object $\mathcal{A}$ equipped with

- (i) a pointwise multiplication $m^\perp : \mathcal{A} \otimes^\perp \mathcal{A} \rightarrow \mathcal{A}$ and a unit map $\eta_\perp : \mathbf{1}_\perp \rightarrow \mathcal{A}$;
- (ii) a chiral-convolution multiplication $m^{\text{ch}} : \mathcal{A} \star^{\text{ch}} \mathcal{A} \rightarrow \mathcal{A}$ and a unit map $\eta_{\text{ch}} : \mathbf{1}_{\text{ch}} \rightarrow \mathcal{A}$;
- (iii) an aligned mixed distributive law identifying the two composites on $\text{im}(e)$.

The coherence diagrams are the two color-3 faces of the mixed forest complex. A bare symbol $\mathbf{1}$ below denotes the unit of the monoidal category in which the displayed construction is formed.

28.3 FORMAL-DISC REALIZATION

Fix a formal coordinate on a disc. Translation-equivariance turns a chiral-convolution product into a state-field operation $Y(a, z)b$ with coefficients in Laurent series. The following theorem packages the exact relationship with the vertex-algebra language.

Theorem 28.5 (Formal-disc realizations). *Let V be a translation-equivariant state-field system with vacuum and injective state-field map.*

- (i) *If its fields are weakly associative in the sense of Li, then V is a nonlocal vertex algebra.*
- (ii) *If, in addition, V is topologically free over $\mathbb{k}[[\hbar]]$, carries a unitary rational quantum Yang–Baxter operator $S(z)$, and satisfies S -locality and the hexagon compatibility, then V is a quantum vertex algebra.*
- (iii) *A translation-equivariant associative chiral-convolution algebra produces such a state-field system whenever its formal-disc restriction satisfies the reconstruction, lower-truncation, and weak-associativity hypotheses. Conversely, a nonlocal or quantum vertex algebra produces the corresponding formal-disc factorization system by its products of fields.*

Proof. The first assertion is Li’s reconstruction theorem for quasi-compatible fields [53]. The second is the quantum-vertex reconstruction theorem: unitarity gives inverse exchange, the quantum Yang–Baxter equation compares the two three-field exchange paths, and the hexagon relates exchange to the vertex operation [24, 54]. For the third assertion, the formal-disc chiral product supplies the iterated field products; the stated analytic-formal hypotheses are exactly those used by the reconstruction theorems. The converse assigns to a configuration of formal points the iterated field product and uses weak associativity, or braided associativity, for factorization under refinement. $\square$

28.4 THREE SEPARATION LAWS

The complete principal-part filtration, transverse order, and planar exchange form a strict hierarchy of supplied data. The m th normal symbol of a pole is recovered by the m th normal-jet map. An oriented line supplies contractible order chambers and hence E_1 fusion. A plane supplies braid paths and hence exchange. These positive constructions locate each operation in the first geometry that carries it.

TRANSVERSELY ORDERED CHIRAL ALGEBRAS

Definition 29.1 (Ordered chiral algebra). An ordered chiral algebra on X is an augmented E_1 -algebra object

$$A \in \text{Alg}_{E_1}^{\text{aug}}(\text{Fact}_D(X), \otimes^!).$$

Equivalently, A is a factorization object equipped with multiplication maps

$$m_I : A_I \otimes^! A_I \longrightarrow A_I$$

and a unit, compatible with all diagonal transition maps and all disjoint-locus factorization equivalences.

For every decomposition $I = I_1 \sqcup I_2$, locality is the componentwise identity

$$\Delta_{I_1, I_2} \circ m_I = (m_{I_1} \boxtimes m_{I_2}) \circ \sigma \circ (\Delta_{I_1, I_2} \otimes^! \Delta_{I_1, I_2}),$$

where σ reorders the middle two factors. This equation says that ordered fusion commutes with separation of chiral supports.

Theorem 29.2 (Locality recognition). *To give an ordered chiral algebra is equivalent to giving a factorization family A_I , an associative unital multiplication m_I on every component, and the displayed locality identities, compatibly with refinements, permutations, and diagonal transitions.*

Proof. An E_1 -algebra in the symmetric monoidal category $\text{Fact}_D(X)$ is, by definition, an associative multiplication and unit whose structure maps are morphisms in $\text{Fact}_D(X)$. A morphism in $\text{Fact}_D(X)$ is exactly a componentwise morphism compatible with the transition and factorization maps. Expanding this compatibility for a binary decomposition gives the displayed identity; the higher decompositions follow by iteration. $\square$

29.1 PRODUCT FIELD THEORIES

Let $\text{Fact}_{\text{hol,lc}}(X^{\text{an}} \times \mathbb{R})$ denote factorization algebras that are holomorphic along X and locally constant along $\mathbb{R}$. Product additivity gives

$$\text{Fact}_{\text{hol,lc}}(X^{\text{an}} \times \mathbb{R}) \simeq \text{Alg}_{E_1}(\text{HolFact}(X^{\text{an}}))$$

for product-generated objects. Algebraization by regular holonomic Riemann–Hilbert then gives ordered chiral algebras in $\text{Fact}_D(X)$ whenever the coefficient system is regular holonomic and the comparison preserves the required colimits.

Theorem 29.3 (Holomorphic–topological recognition). *Let $\mathbb{F}$ be a product-generated factorization algebra on $X^{\text{an}} \times \mathbb{R}$ that is holomorphic on X and locally constant on $\mathbb{R}$. Then the restriction of $\mathbb{F}$ to a transverse point is an E_1 -algebra in holomorphic factorization algebras on X . A regular-holonomic algebraization produces an ordered chiral algebra on X .*

Proof. An ordered family of disjoint intervals in $\mathbb{R}$ acts on the restriction by factorization. Contractibility of each ordered configuration chamber gives the coherent E_1 structure. Factorization in the X direction is preserved because product embeddings commute. Riemann–Hilbert transports the structure in the regular-holonomic subcategory. $\square$

BLOCK-ORDERED OPERATIONS

The preceding definition is intrinsic. A geometric model arises after one chooses an infinity-operad Ch_X representing the selected chiral factorization theory. The Boardman–Vogt tensor product

$$\text{OrdCh}_X = \text{E}_1 \otimes \text{Ch}_X$$

then represents compatible ordered and chiral operations.

Theorem 30.1 (Operadic exponential law). *Let P and Q be infinity-operads whose Boardman–Vogt tensor product exists, and let $\mathcal{V}$ be a compatible presentable symmetric monoidal infinity-category. Then*

$$\text{Alg}_{P \otimes Q}(\mathcal{V}) \simeq \text{Alg}_P(\text{Alg}_Q(\mathcal{V})).$$

In particular, after a chiral operad Ch_X has been constructed, the tensor product $\text{E}_1 \otimes \text{Ch}_X$ classifies E_1 -algebra objects in Ch_X -algebras.

Proof. The tensor product of infinity-operads is characterized by the displayed universal property [58]. The second statement is the specialization $P = \text{E}_1$ and $Q = \text{Ch}_X$. This formal equivalence applies to each geometric model of Ch_X after that model and its operadic tensor product have been constructed. $\square$

A profile is an ordered list $(I_1, \dots, I_r)$ of finite chiral label sets. Its geometric carrier is $X^{I_1} \times \dots \times X^{I_r}$. Chiral morphisms refine and collide labels inside one block. Transverse morphisms merge adjacent blocks. The tensor-product operad imposes their mixed squares.

30.1 MIXED FORESTS

Fix a cofibrant dg-operad $Q_X \rightarrow \text{Ch}_X$. Apply the functorial $\mathcal{W}$ -construction to $\text{E}_1 \otimes Q_X$. A cell is represented by a levelled forest with transverse and chiral internal edges. Put

$$\text{or}(F) = \det(E_\perp(F)) \otimes \det(E_{\text{ch}}(F)).$$

The differential is the sum of the screen differential and one-edge contractions.

Theorem 30.2 (Two-color forest differential). *Let C_* be the chain complex freely generated by oriented two-color forests, with one determinant-line generator for every transverse edge and one for every chiral edge. Suppose contraction of an edge is functorial on the coefficient system and every codimension-two contraction square commutes before orientations are inserted. With the product orientation, the transverse and chiral contraction operators satisfy*

$$d_\perp^2 = 0, \quad d_{\text{ch}}^2 = 0, \quad d_\perp d_{\text{ch}} + d_{\text{ch}} d_\perp = 0.$$

Thus $d = d_{\text{int}} + d_\perp + d_{\text{ch}}$ squares to zero whenever the internal differential anticommutes with both contraction operators.

Proof. Every term of $d_{\perp}^2$ contracts the same pair of transverse edges in two orders. The underlying coefficient maps agree by functoriality, while exchanging the two transverse determinant generators reverses the sign. The terms cancel in pairs. The same argument proves $d_{\text{ch}}^2 = 0$. For one edge of each color the coefficient maps again form a commuting square and exchanging the two determinant generators contributes the Koszul sign -1 , proving the mixed anticommutator identity. The internal terms vanish by the chain-map assumption. $\square$

This construction supplies a geometric model whenever a concrete Q_X and its screen coefficients are available. The intrinsic definition of ordered chiral algebra remains independent of that choice.

BOOK **IX**

Bar, cobar, and the boundary state

THREE BAR CONSTRUCTIONS

The phrase *the chiral bar* denotes three functors once its monoidal product and algebraic operad have been named.

Definition 3I.1 (Transverse, associative-chiral, and Lie-chiral bars). For augmented objects of the indicated types define

$$\mathrm{Bar}^\perp(\mathcal{A}^\perp) := \mathbf{I} \otimes_{\mathcal{A}^\perp}^{\mathbb{L},!} \mathbf{I},$$

$$\mathrm{Bar}_{\mathrm{Ass}}^{\mathrm{ch}}(\mathcal{A}_{\mathrm{Ass}}^{\mathrm{ch}}) := \mathbf{I} \star_{\mathcal{A}_{\mathrm{Ass}}^{\mathrm{ch}}}^{\mathbb{L}} \mathbf{I},$$

and

$$\mathrm{Bar}_{\mathrm{Lie}}^{\mathrm{ch}}(\mathfrak{g}^{\mathrm{ch}}) := C_*^{\mathrm{ch}}(\mathfrak{g}^{\mathrm{ch}}) = \mathrm{Sym}^{\mathrm{ch}}(\mathfrak{g}^{\mathrm{ch}}[\mathbf{I}])$$

with its chiral Chevalley differential. The first bar is formed with $\otimes^!$, the second with $\star^{\mathrm{ch}}$, and the third is the Lie-to-commutative-coalgebra Koszul functor.

Theorem 3I.2 (Typed bar theorem). *The three bars have the following chain models.*

- (i) $\mathrm{Bar}^\perp(\mathcal{A}^\perp)$ is the tensor coalgebra $T_{\otimes^!}^c(s\bar{\mathcal{A}})$ with coderivation induced by $m^\perp$.
- (ii) $\mathrm{Bar}_{\mathrm{Ass}}^{\mathrm{ch}}(\mathcal{A}_{\mathrm{Ass}}^{\mathrm{ch}})$ is the tensor coalgebra $T_{\star^{\mathrm{ch}}}^c(s\bar{\mathcal{A}})$ with coderivation induced by m^{ch} .
- (iii) $\mathrm{Bar}_{\mathrm{Lie}}^{\mathrm{ch}}(\mathfrak{g}^{\mathrm{ch}})$ is the chiral symmetric coalgebra with differential induced by the chiral Lie bracket.

Each differential squares to zero by the defining relation of its own operad. The outputs live in different coalgebra categories and are compared by a displayed monoidal or PBW morphism.

Proof. Normalize the two simplicial two-sided bars in their respective monoidal categories. Degeneracies remove units, leaving the two tensor coalgebras, and faces contract adjacent factors using the relevant multiplication. The Lie formula is the ordinary operadic bar of the Lie operad internal to chiral convolution. Associativity and Jacobi are exactly the quadratic identities that make the corresponding coderivations square to zero. $\square$

Definition 3I.3 (PBW-admissible chiral Lie algebra). A chiral Lie algebra $\mathfrak{g}^{\mathrm{ch}}$ is PBW-admissible when its chiral enveloping algebra $U^{\mathrm{ch}}(\mathfrak{g})$ has a complete separated PBW filtration, the canonical map

$$\mathrm{Sym}^{\mathrm{ch}}(\mathfrak{g}) \longrightarrow \mathrm{gr}_{\mathrm{PBW}} U^{\mathrm{ch}}(\mathfrak{g})$$

is an equivalence, and the bar filtration is strongly convergent in the chosen nilcomplete Ran category.

Theorem 3I.4 (Filtered chiral PBW comparison). *Let $\mathfrak{g}$ be a chiral Lie algebra in a complete filtered chiral category. Assume:*

- (i) *its chiral enveloping algebra $U^{\mathrm{ch}}(\mathfrak{g})$ exists;*
- (ii) *the PBW filtration is exhaustive, complete, and separated;*

(iii) *the chiral PBW morphism*

$$\mathrm{Sym}^{\mathrm{ch}}(\mathfrak{g}) \longrightarrow \mathrm{gr} U^{\mathrm{ch}}(\mathfrak{g})$$

is an isomorphism;

(iv) *the induced filtrations on the two bar complexes are strongly convergent.*

Then antisymmetrization induces a filtered quasi-isomorphism

$$\mathrm{Bar}_{\mathrm{Ass}}^{\mathrm{ch}}(U^{\mathrm{ch}}(\mathfrak{g})) \simeq \mathrm{Bar}_{\mathrm{Lie}}^{\mathrm{ch}}(\mathfrak{g}).$$

Proof. The associated graded of the associative chiral bar is the Koszul bar of $\mathrm{Sym}^{\mathrm{ch}}(\mathfrak{g})$. The usual Koszul antisymmetrization identifies this coalgebra with the chiral Chevalley coalgebra. Hypotheses (2)–(4) give a complete, separated, strongly convergent comparison spectral sequence. Since its associated-graded map is a quasi-isomorphism, the filtered map is a quasi-isomorphism. This is the PBW lane of chiral Koszul duality [5, 31]. $\square$

Definition 31.5 (Bar-admissible mixed datum). A mixed associative chiral datum is bar-admissible when its aligned distributive law acts on the full bisimplicial two-sided bar and both geometric realizations commute with the tensor products occurring in that bisimplicial object.

Theorem 31.6 (Mixed Fubini theorem). *Let A be a mixed algebra in a presentable stable category. Assume the mixed distributive datum extends to the bisimplicial two-sided bar, and that both monoidal products preserve its geometric realizations separately. Then the two iterated realizations and the diagonal realization are canonically equivalent:*

$$\left| p \mapsto \left| q \mapsto B_{p,q}(A) \right|_{\mathrm{ch}} \right|_{\perp} \simeq \left| \mathrm{diag} B_{\bullet,\bullet}(A) \right| \simeq \left| q \mapsto \left| p \mapsto B_{p,q}(A) \right|_{\perp} \right|_{\mathrm{ch}}.$$

Proof. The distributive datum makes the two simplicial directions commute, so $B_{\bullet,\bullet}(A)$ is a genuine bisimplicial object. Geometric realization is a colimit. Since the two tensor products preserve the relevant colimits, Fubini for colimits identifies both iterated realizations with the realization of the diagonal simplicial object. $\square$

The notation used in the remainder of this part is

$$\mathrm{Bar}_X^{\mathrm{ord}} = \mathrm{Bar}^{\perp}.$$

It always refers to transverse order.

31.1 REALIZATION AND FACTORIZATION

Factorization compares the coefficient at a disjoint configuration with an external tensor product of its coefficients at the separate configurations. This condition is nonlinear. For example, suppose those coefficients are k at one point and at two distinct points, where k is a field. The componentwise sum of two copies has coefficient k^2 at both configurations. Factorization would instead require $k^2 \otimes_k k^2 \cong k^4$ at two points. A componentwise shift also fails: its two-point coefficient is $k[1]$, whereas the tensor product of its singleton coefficients is $k[2]$. Thus an ambient stable coefficient category does not make its factorization objects closed under componentwise sums or shifts.

Geometric realization has a different colimit property. A small indexing ∞ -category J is *sifted* when it is nonempty and its diagonal $J \rightarrow J \times J$ is cofinal. Here cofinal means that, for every diagram D on $J \times J$, the canonical map $\mathrm{colim}_J(D \circ \Delta) \rightarrow \mathrm{colim}_{J \times J} D$ is an equivalence whenever these colimits exist. Iteration gives the same property for $J \rightarrow J^m$, for every finite $m \geq 1$. The simplex category with arrows reversed is sifted, so simplicial geometric realization is a sifted colimit.

Proposition 31.7. *Fix coefficient categories on configuration charts, with specified restriction and collision functors and their coherent identifications. Suppose the coefficient categories admit J -indexed colimits and these functors preserve them. Suppose external tensor products preserve colimits separately in each variable. Let $(F_j)_{j \in J}$ be a diagram of factorization objects and morphisms preserving their factorization equivalences. If J is sifted, the componentwise colimit is a factorization object.*

Proof. Write $F_I = \operatorname{colim}_{j \in J} (F_j)_I$ on a chart indexed by I . The collision maps and their coherences pass to the colimit by hypothesis. For a decomposition $I = I_1 \sqcup \cdots \sqcup I_m$, let u denote restriction to its disjointness locus. Restriction, factorization, cofinality, and separate preservation give

$$\begin{aligned} uF_I &\simeq \operatorname{colim}_{j \in J} u(F_j)_I \\ &\simeq \operatorname{colim}_{j \in J} \bigboxtimes_{a=1}^m (F_j)_{I_a} \\ &\simeq \operatorname{colim}_{(j_1, \dots, j_m) \in J^m} \bigboxtimes_{a=1}^m (F_{j_a})_{I_a} \simeq \bigboxtimes_{a=1}^m F_{I_a}. \end{aligned}$$

Each comparison is natural in the diagram and the decomposition. Refinements and permutations therefore preserve the given coherence diagrams. If an empty-configuration unit is specified, its constant diagram also has that unit as colimit, since a sifted category has contractible groupoid completion. $\square$

For a simplicial two-sided bar, this proposition applies when every simplicial term and every face and degeneracy are already in the specified factorization category. It also requires the actual restriction and collision functors to preserve realization. Normalization may then be computed in an ambient stable category. Its individual shifted summands need not be factorization objects. A coproduct on the normalized complex requires its own compatibility with the factorization equivalences.

THE INTERNAL ORDERED BAR

Let A be an augmented ordered chiral algebra. Assume the restriction and collision functors preserve geometric realizations, and the external and pointwise tensor products preserve them separately. The two-sided simplicial bar has

$$B_r(\mathbf{1}, A, \mathbf{1}) = \mathbf{1} \otimes^! A^{\otimes^! r} \otimes^! \mathbf{1}.$$

Every face and degeneracy is a factorization morphism by the algebra and augmentation axioms. The sifted-colimit proposition therefore constructs its realization $\text{Bar}_X^{\text{ord}}(A)$ as a factorization object.

Theorem 32.1 (Normalized bar). *Choose a symmetric monoidal dg model for the ambient coefficients in which the displayed tensors compute derived tensors and direct-sum normalization computes realization. Write $\tilde{A}$ for the augmentation ideal in that model. The underlying bar complex is*

$$\text{Bar}_X^{\text{ord}}(A) \simeq (T^c(s\tilde{A}), d_{\text{int}} + b_m).$$

Here s decreases cohomological degree by one. The direct sum and its individual shifted summands are formed in the ambient coefficient category. Deconcatenation makes this complex a conilpotent coassociative coalgebra there:

$$\Delta_{\text{dec}}[a_1 | \cdots | a_n] = \sum_{i=0}^n [a_1 | \cdots | a_i] \otimes [a_{i+1} | \cdots | a_n].$$

Proof. Normalization removes words containing a unit and leaves the shifted augmentation ideal in every slot. The faces give the internal differential and adjacent multiplication. The simplicial identities give a square-zero differential. Two successive cuts enumerate the same three subwords in either order. An internal differential or multiplication lies on one side of any surviving cut, with the tensor sign of its prefix. Thus the differential is a coderivation. Every element has finite word support, so sufficiently many reduced cuts give zero. $\square$

32.1 THE BAR COMPARISON ON SEPARATED CONFIGURATIONS

The normalized word decomposition is not itself a decomposition in $\text{Fact}_D(X)$. Its compatibility with factorization is expressed by a map between whole bar complexes. For augmented dg algebras P, Q in the chosen model, define

$$\text{sh} : BP \otimes BQ \longrightarrow B(P \otimes Q)$$

by shuffling the letters $p_i \otimes \mathbf{1}$ and $\mathbf{1} \otimes q_j$, keeping their respective orders. The sign is the Koszul sign for the permutation of the suspended letters. For degree-zero letters,

$$\text{sh}([p] \otimes [q]) = [p \otimes \mathbf{1} | \mathbf{1} \otimes q] - [\mathbf{1} \otimes q | p \otimes \mathbf{1}].$$

Proposition 32.2 (Shuffle and factorization). *The shuffle is a natural quasi-isomorphism and a coalgebra map, where its source has the tensor-product coalgebra structure. For the geometric coefficient models above,*

suppose these maps commute with the specified restriction and collision functors. Then they give factorization equivalences for the whole bar and make deconcatenation a morphism of factorization objects in the derived category.

Proof. In the bar differential of a shuffle, a merge of letters from different factors occurs twice with opposite suspended signs. The equality $(p \otimes 1)(1 \otimes q) = (-1)^{|p||q|}(1 \otimes q)(p \otimes 1)$ cancels these terms. Merges within one factor give the differential of that factor. Internal differentials commute with the same signed permutation. Thus the shuffle is a cochain map.

A cut of a shuffled word determines one cut in each input word, a shuffle of their prefixes, and a shuffle of their suffixes. Conversely these four choices determine exactly that cut shuffle. The signs on both sides come from the same permutation of suspended letters. This proves the coalgebra identity with the tensor-product coproduct.

To verify the quasi-isomorphism, form the bisimplicial object $B_p(k, P, k) \otimes B_q(k, Q, k)$, with k denoting the tensor unit. Its iterated realization is $BP \otimes BQ$. Its diagonal is the simplicial bar of $P \otimes Q$. Cofinality of the simplicial diagonal identifies these realizations. On normalized chains the map from the iterated realization to the diagonal is the shuffle just defined: the monotone paths from $(0, 0)$ to (p, q) triangulate the product of its two simplices. Each path inserts precisely the degeneracies of its shuffle. Adjacent paths have opposite orientations on a shared face, so their signed sum has exactly the product boundary. This is the realization comparison, hence an equivalence.

For $I = I_1 \sqcup I_2$ on the disjointness locus, the given algebra factorization identifies A_I with $A_{I_1} \boxtimes A_{I_2}$. The shuffle therefore gives the comparison $BA_{I_1} \boxtimes BA_{I_2} \rightarrow BA_I$. Its inverse in the derived category is the bar factorization map. The coalgebra identity just proved gives compatibility with deconcatenation. For several blocks, iterated shuffles enumerate the same ordered multishuffles, with the same Koszul signs. This proves refinement coherence. Naturality and the stated functor compatibility give the collision coherence. $\square$

The restriction and collision compatibility in this proposition is a hypothesis on the chosen geometric model. It does not follow from naming an ambient stable category. Conilpotence here refers to finite normalized words in that model. No word-length completion is implicit.

32.2 THREE COPRODUCTS

The theorem system uses three coproducts with distinct geometric meanings:

$$\begin{aligned} \Delta_{\text{fact}} &: \text{separation of chiral support,} \\ \Delta_{\text{dec}} &: \text{cutting an ordered bar word,} \\ \Delta_{\text{Hopf}} &: \text{fusion of representations.} \end{aligned}$$

The first two are present on the transverse bar. The third is additional bialgebra data on an acting algebra.

Proposition 32.3 (Coproduct separation). *Factorization separation is carried by every ordered coefficient. Deconcatenation is generated functorially by the bar construction. A Hopf coproduct is supplied by an independent bialgebra structure. A coalgebra comparison morphism relates the Hopf coproduct to either geometric coproduct whenever a theory identifies their carriers.*

Proof. Factorization separation is part of the coefficient object, and deconcatenation is the canonical cut map on tensor words. A Hopf coproduct is a map from the algebra to a completed tensor square satisfying algebraic coassociativity. Its source, target, and defining universal property differ from the first two, so a relation among them is exactly the data of a comparison morphism. $\square$

32.3 INTERVAL FACTORIZATION HOMOLOGY

For a right $\mathcal{A}$ -module R and a left $\mathcal{A}$ -module L , factorization homology of an interval with endpoint labels gives

$$\int_{([0,1], \{0,1\})} (\mathcal{A}; R, L) \simeq R \otimes_{\mathcal{A}}^{\mathbb{L}} L.$$

This statement holds internally in every presentable monoidal category supporting factorization homology [3, 58]. Taking $R = L = \mathbf{1}$ identifies the interval amplitude with the ordered bar.

COBAR AND TWISTING COCHAINS

Let C be a coaugmented conilpotent differential graded coalgebra in the ambient coefficient model. The ordinary, direct-sum tensor cobar algebra is

$$\mathrm{Cobar}_X^{\mathrm{ord}}(C) = (T(s^{-1}\tilde{C}), d_{\mathrm{int}} + d_{\Delta}).$$

A twisting cochain $\tau : C \rightarrow A$ has degree 1, vanishes on the coaugmentation, and has zero composite with the augmentation of A . It satisfies

$$d\tau + \tau d + m(\tau \otimes \tau)\tilde{\Delta} = 0.$$

The universal twisting cochains are

$$\tau_A : \mathrm{Bar}_X^{\mathrm{ord}}(A) \rightarrow A, \quad \tau_C : C \rightarrow \mathrm{Cobar}_X^{\mathrm{ord}}(C).$$

Theorem 33.1 (Bar–cobar adjunction). *For ordinary augmented dg algebras and coaugmented conilpotent dg coalgebras in the specified coefficient model, tensor cobar is left adjoint to tensor bar. Their mapping set is the set of twisting cochains. An internal adjunction on factorization objects additionally requires both constructions to produce factorization objects and their twisting maps to preserve the geometric data.*

Proof. An algebra map from the tensor algebra $T(s^{-1}\tilde{C})$ is determined by its restriction to $s^{-1}\tilde{C}$. Compatibility with the cobar differential is the Maurer–Cartan equation above. The same equation describes coalgebra maps from C to the cofree conilpotent coalgebra $T^c(s\tilde{A})$. $\square$

COMPLETE BAR-COBAR RECONSTRUCTION

Ordinary tensor cobar uses finite words. Completion changes that carrier. Let k be a commutative ring and $A_{\text{alg}} = \bigoplus_{w \geq 0} A_w$ an augmented algebra in degree zero, with $A_0 = k$, augmentation given by projection onto A_0 , and multiplication adding weights. Put $D = \Omega B(A_{\text{alg}})$ using ordinary direct-sum tensors, and let D_w be its weight- w subcomplex. The complete algebra and its complete resolution are

$$\widehat{A} = \prod_{w \geq 0} A_w, \quad \widehat{D} = \prod_{w \geq 0} D_w.$$

Products of complexes are taken degreewise, with the weight topology.

Theorem 34.1 (Complete reconstruction). *The componentwise counit $\widehat{D} \rightarrow \widehat{A}$ is a quasi-isomorphism of complete differential graded algebras.*

Proof. A weight- w block word has at most w positive-weight entries. Filter by their number. Multiplication lowers this number and splitting preserves it. At a fixed list of $N \geq 2$ entries, the splitting complex is the tensor product of the coefficient module with $N - 1$ copies of $[k \xrightarrow{1} k]$ in degrees $-1, 0$. Changing the first outer cut to an inner gap contracts it. The other differential terms cancel by their prefix signs. This identity survives tensoring with any coefficient module. Subtracting a primitive at the largest entry number terminates, so negative cohomology vanishes. In degree zero, the boundaries impose $c(a)c(b) = c(ab)$, leaving exactly A_w . Taking the degreewise product preserves kernels and images of these differentials, and therefore preserves the computed cohomology. $\square$

For $k[x]$ with weight one on x , the ordinary counit recovers $k[x]$, whereas this weight completion recovers $k[[x]]$. A completed geometric bar-cobar equivalence needs a specified completed tensor product, coalgebra category, and comparison maps compatible with collision and inverse limits. An operadic nilcompletion or a boundary double centralizer is not identified with the ordinary tensor counit by this weight calculation. The coalgebra-unit and geometric duality statements require their own convergence and comparison arguments.

34.1 QUADRATIC RECOGNITION

Let $A = T(V)/(R)$ with $R \subset V \otimes V$. Its quadratic dual coalgebra $A^{\text{q!}}$ is cogenerated by sV with corelations $s^2 R$. The canonical map $A^{\text{q!}} \rightarrow \text{Bar}(A)$ gives the Koszul complex.

Theorem 34.2 (Quadratic Koszul criterion). *Assume positive weight and degreewise finite duals. The following conditions are equivalent:*

- (i) $A^{\text{q!}} \rightarrow \text{Bar}(A)$ is a quasi-isomorphism;
- (ii) $\text{Cobar}(A^{\text{q!}}) \rightarrow A$ is a quasi-isomorphism;
- (iii) the left and right Koszul complexes have homology concentrated in weight zero;
- (iv) $\text{Tor}_i^A(\mathbf{1}, \mathbf{1})$ is concentrated in internal weight i .

Proof. The universal twisting morphism $\mathcal{A}^{\text{q!}} \rightarrow \mathcal{A}$ is Koszul exactly when the associated twisted tensor products resolve the unit. The fundamental theorem of twisting morphisms identifies this condition with the bar and cobar quasi-isomorphisms. Weight purity is the homological form of the same statement [55, 65]. $\square$

MODULES, COMODULES, AND MORITA THEORY

The ordinary tensor counit reconstructs an algebra. Reconstruction of its modules requires the unit and counit of the twisted module functors as well. The following weight conditions make these maps finite in every component.

Let k be a field and let $A_{\text{alg}} = \bigoplus_{w \geq 0} A_w$ be an associative algebra in cohomological degree zero. Assume $A_0 = k$, multiplication adds weights, and augmentation projects onto A_0 . Put $C_{\text{alg}} = B(A_{\text{alg}})$, with its bar weight grading, and set

$$A = \prod_{w \geq 0} A_w, \quad C = \prod_{w \geq 0} (C_{\text{alg}})_w.$$

Products are degree-wise in complexes. The universal twisting map $\tau : C \rightarrow A$ has degree one, sends $[a]$ to a , and vanishes on the empty word and all longer words. The bar differential has the earlier convention $b_2[a|b] = [ab]$.

A complete weight-graded left A -module here means a product $M = \prod_{w \geq w_0} M_w$ of complexes, for some integer w_0 , with differential preserving weight and actions $A_u \otimes M_v \rightarrow M_{u+v}$. A complete weight-graded left C -comodule means such a product N with a counital coassociative coaction $\rho : N \rightarrow C \widehat{\otimes} N$ preserving weight and differential. The completed tensor product is defined componentwise by

$$(C \widehat{\otimes} N)_w = \bigoplus_{u+v=w} (C_{\text{alg}})_u \otimes N_v.$$

This sum is finite since $u \geq 0$ and $v \geq w_0$. Each reduced coaction lowers the weight of its final N -component. It therefore vanishes after more than $w - w_0$ iterations on N_w . These comodules need no chosen coaugmentation.

Write $D_{\text{wt}}(A)$ and $D_{\text{wt}}(C)$ for the categories obtained by inverting maps that are quasi-isomorphisms in every weight, respectively for these modules and comodules. This specifies the weak equivalences; no coderived localization is implicit. Define

$$F(M) = C \widehat{\otimes}_{\tau} M, \quad G(N) = A \widehat{\otimes}_{\tau} N.$$

The first has the coaction induced by Δ_C and the second the left A -action. Writing $\Delta(c) = \sum c_{(1)} \otimes c_{(2)}$ and $\rho(n) = \sum n_{(-1)} \otimes n_{(0)}$, their differentials are

$$\begin{aligned} d_F(c \otimes m) &= d_C c \otimes m + (-1)^{|c|} c \otimes d_M m + \sum (-1)^{|c_{(1)}|} c_{(1)} \otimes \tau(c_{(2)}) m, \\ d_G(a \otimes n) &= a \otimes d_N n - \sum a \tau(n_{(-1)}) \otimes n_{(0)}. \end{aligned}$$

All sums are finite in each weight. The twisting equation, associativity, and coassociativity make both differentials square to zero.

Theorem 35.1 (Koszul module equivalence in positive weights). *These functors induce inverse equivalences*

$$D_{\text{wt}}(A) \simeq D_{\text{wt}}(C).$$

Their counit and unit are

$$\epsilon_M(a \otimes c \otimes m) = a\epsilon_C(c)m, \quad \eta_N(n) = \sum n_{(-1)} \otimes 1 \otimes n_{(0)}.$$

Proof. The differential formulas show that these are cochain maps, respectively A -linear and C -colinear. The twisting terms cancel in the unit equation; the counit equation uses the module action. Coassociativity and the counit identities give the two triangle identities.

The complex $GF(\mathcal{M})$ is the two-sided normalized bar resolution $A \widehat{\otimes} B(A_{\text{alg}}) \widehat{\otimes} M$. For its contraction, use the simplicial boundary convention $\partial = \sum_i (-1)^i d_i$. Multiplying a bar word of length r by $(-1)^r$ translates this convention to the displayed twisting differentials. Writing $\tilde{a} = a - \epsilon_A(a)1$, the extra degeneracy is

$$h(a_0[a_1] \cdots [a_r]m) = 1[\tilde{a}_0[a_1] \cdots [a_r]m].$$

The first face returns the original term minus its scalar first factor. The other faces cancel against applying h after the boundary, including the module-action face. The internal differential of $\mathcal{M}$ cancels by the suspension sign. Thus $\partial h + h\partial = 1 - i\epsilon_M$, with $i(m) = 1[\]m$. This is a linear contraction; i need not be A -linear. It preserves total weight, so the counit is a quasi-isomorphism in every weight.

At fixed total weight, filter $FG(N)$ by the weight of its final N -factor. This filtration is finite. The reduced-coaction term lowers that weight and disappears in the associated graded. The associated graded unit is therefore the inclusion into $B(k, A_{\text{alg}}, A_{\text{alg}}) \otimes N_r$. The rightmost extra degeneracy contracts this augmented bar onto k :

$$h([a_1] \cdots [a_r]a) = (-1)^{r+1}[a_1] \cdots [a_r]\tilde{a}1$$

in the same simplicial convention. The last face gives $1 - i\epsilon$; all other faces cancel in pairs. Consequently the unit is a quasi-isomorphism on each associated graded component, and the finite filtration proves the same for each weight of N .

Both functors preserve weightwise quasi-isomorphisms. Filter by the weight of the final module or comodule component. The twisting terms strictly change that weight in one direction. On associated graded components, the functors tensor complexes of k -vector spaces, which preserves quasi-isomorphisms. The filtration is finite at fixed total weight. Hence F, G descend to the stated localizations, where their unit and counit are isomorphisms. $\square$

This theorem uses an explicit weight decomposition and a lower weight bound on every module and comodule. Derived I -adic completion, with $I = \ker(A \rightarrow k)$, instead uses $\mathcal{M} \rightarrow \varprojlim_p (A/I^p \otimes_A^L M)$. The two completions require a comparison of their filtrations. A coderived category also has a separately specified class of coacyclic complexes. Neither identification follows from the algebra counit alone. For right and left modules, the two-sided bar computes their relative derived tensor product in the same finite weight components. An identification with a geometric derived cotensor product additionally requires the two-sided twisting comparison and its compatibility with the chosen completion and collision maps.

Let $\mathcal{C}$ be a presentable factorization category and let b be a compact factorization generator. The internal endomorphism object

$$A_b = \text{RHom}_{\mathcal{C}}(b, b)$$

has ordered multiplication by composition.

Theorem 35.2 (Factorization Morita theorem). *Assume b generates each local category and the local Morita equivalences commute with factorization descent. Then*

$$\mathcal{C} \simeq \text{Mod}_{A_b}(\text{Fact}_D(X)).$$

Proof. The compact-generator Morita theorem applies on every factorization chart. The descent hypothesis identifies the equivalences on intersections and hence on the Ran colimit. $\square$

DERIVED CENTRE AND BULK ACTION

The ordered Hochschild cochains are

$$Z_{\text{ord}}(A) = \text{RHom}_{A \otimes A^{\text{op}}}(A, A).$$

Theorem 36.1 (Higher Deligne centre). *Let the ambient coefficient category be presentable, stable, symmetric monoidal, closed, and let tensor product preserve geometric realizations separately. Then $Z_{\text{ord}}(A)$ carries a canonical E_2 -algebra structure. It is the universal E_2 algebra acting centrally on A .*

Proof. Hochschild cochains carry the brace operations, and the brace operad is equivalent to chains on little two-disks. This is the Deligne theorem [52, 59]. Universality is the identification of Hochschild cochains with endomorphisms of the identity A -bimodule. $\square$

A boundary condition in a field theory gives a morphism

$$\beta : \text{Obs}_{\text{bulk}}|_{\partial} \longrightarrow Z_{\text{ord}}(A_{\partial}).$$

Locality makes the action central. A boundary condition is open–closed complete on a protected sector when β is an equivalence on that sector.

For a Calabi–Yau category, Hochschild cochains carry the framed E_2 refinement of the cyclic Deligne theorem [10].

CYCLIC CLOSURE AND MODULAR SEWING

A cyclic trace becomes a surface theory after dualizability and nondegeneracy are supplied.

Definition 37.1 (Perfect Calabi–Yau ordered algebra). A perfect Calabi–Yau ordered algebra is a dualizable ordered algebra $\mathcal{A}$ together with a trace

$$\mathrm{tr} : \mathrm{HH}_0(\mathcal{A}) \longrightarrow \mathbf{k}[-d]$$

whose induced pairing $\mathcal{A} \otimes \mathcal{A} \rightarrow \mathbf{k}[-d]$ is nondegenerate.

Theorem 37.2 (Ribbon-graph realization of a Calabi–Yau algebra). *Let $\mathcal{A}$ be a homologically finite, smooth and proper cyclic $\mathbf{A}_\infty$ algebra with a nondegenerate pairing of degree d . Then the cyclic bar construction of $\mathcal{A}$ defines an open topological conformal field theory. Its operations are indexed by chains on moduli spaces of bordered surfaces, equivalently by the corresponding ribbon-graph complexes. The value on the circle is Hochschild homology $\mathrm{HH}_*(\mathcal{A})$.*

Proof. The nondegenerate cyclic pairing identifies outgoing and incoming boundary labels and contracts the two flags of every internal ribbon edge. The $\mathbf{A}_\infty$ identities are exactly the boundary identities for metric ribbon graphs. Smoothness and properness give the dualizability and finite pairing required for all contractions. Costello’s construction then produces the open TCFT and identifies the closed state complex with Hochschild chains [17]. $\square$

Remark 37.3. The trace functional supplies the annular operation. Smoothness, properness, and the nondegenerate Calabi–Yau pairing extend it to the complete ribbon-surface theory of the theorem.

Stable-curve sewing is supplied by a modular operad action. Its contractions form the modular structure carried by the chiral coefficient. The two sewing geometries coexist when their contraction maps supercommute.

BOOK **X**

Exact noncommutative models

FREE AND SQUARE-ZERO PAIRS

Let $M \in \text{Fact}_D(X)$.

$$T_X(M) = \mathbf{1} \oplus M \oplus M^{\otimes 2} \oplus \dots$$

is the free augmented ordered algebra.

Theorem 38.1 (Free–square-zero duality). *Assume exact tensor products and dualizable M . Then*

$$\text{Bar}(T_X(M)) \simeq \mathbf{1} \oplus sM, \quad \text{RHom}_{T_X(M)}(\mathbf{1}, \mathbf{1}) \simeq \mathbf{1} \oplus M^\vee[-1]$$

with square-zero multiplication on the reduced summand. Conversely, for $A = \mathbf{1} \oplus M$ with $M^2 = 0$,

$$\text{Bar}(A) = T^c(sM), \quad A^\dagger \simeq \widehat{T}(M^\vee[-1]).$$

Proof. For the tensor algebra, every word of positive length has a unique last letter. The standard extra degeneracy contracts every bar simplex with an internal cut, leaving the one-letter class. Dualization gives the square-zero algebra. For the square-zero algebra, the multiplication part of the bar differential vanishes, leaving the cofree tensor coalgebra; its continuous dual is the completed tensor algebra. $\square$

THE QUANTUM PLANE

Let

$$A_q = \mathbb{k}\langle x, y \rangle / (yx - qxy), \quad q \in \mathbb{k}^\times.$$

Give x and y weight one.

Theorem 39.1 (PBW basis of the quantum plane). *The monomials $x^a y^b$, $a, b \geq 0$, form a basis of A_q .*

Proof. Use the rewriting rule $yx \mapsto qxy$. Every reduction decreases the inversion number of a word. The leading word yx has distinct initial and terminal letters, so every overlap decomposes into independent reductions. Bergman's diamond argument therefore gives the ordered monomials as normal forms. $\square$

The quadratic dual is

$$A_q^\dagger = \mathbb{k}\langle \xi, \eta \rangle / (\xi^2, \eta^2, \xi\eta + q\eta\xi).$$

Theorem 39.2 (Koszulness). *A_q is Koszul, and its Koszul complex is*

$$0 \rightarrow A_q \otimes \mathbb{k}(\xi\eta) \xrightarrow{d_2} A_q \otimes (\mathbb{k}\xi \oplus \mathbb{k}\eta) \xrightarrow{d_1} A_q \rightarrow \mathbb{k} \rightarrow 0,$$

with

$$d_1(a \otimes \xi + b \otimes \eta) = ax + by, \quad d_2(a \otimes \xi\eta) = ay \otimes \xi - qax \otimes \eta.$$

Proof. Direct substitution gives $d_1 d_2 = ayx - qaxy = 0$. The image of d_1 is the augmentation ideal (x, y) , so the complex is exact at A_q and has cokernel $\mathbb{k}$. We now compute $\ker d_1$.

Write

$$u = \sum_{i,j \geq 0} u_{ij} x^i y^j, \quad v = \sum_{i,j \geq 0} v_{ij} x^i y^j.$$

The relation $y^j x = q^j x y^j$ gives

$$ux + vy = \sum_{a,b \geq 0} (q^b u_{a-1,b} + v_{a,b-1}) x^a y^b,$$

where coefficients with a negative index are zero. Thus $ux + vy = 0$ implies $u_{i,0} = 0$, $v_{0,j} = 0$, and

$$v_{a,b} = -q^{b+1} u_{a-1,b+1}.$$

Define $c = \sum_{i,j \geq 0} u_{i,j+1} x^i y^j$. Then $u = cy$ and, using $x^i y^j x = q^j x^{i+1} y^j$,

$$v = -qcx.$$

Hence $(u, v) = d_2(c)$. Finally, $d_2(c) = 0$ implies $cy = 0$; the PBW basis and $q \neq 0$ show that A_q is a domain, so $c = 0$. The complex is exact, and the quadratic Koszul criterion proves that A_q is Koszul. $\square$

Theorem 39.3 (Yoneda algebra of the quantum plane). *Use the convention that a Yoneda product $\alpha\beta$ means first β and then α . The augmentation module has*

$$\mathrm{Ext}_{A_q}^*(\mathbb{k}, \mathbb{k}) \cong \mathbb{k}\langle \xi, \eta \rangle / (\xi^2, \eta^2, \xi\eta + q\eta\xi) = A_q^!$$

as a bigraded algebra, where $|\xi| = |\eta| = 1$ in cohomological degree and each generator retains its internal weight one.

Proof. Apply $\mathrm{Hom}_{A_q}(-, \mathbb{k})$ to the minimal Koszul resolution in the preceding theorem. Every matrix entry of d_1 and d_2 lies in the augmentation ideal, so the dual differential vanishes. The cohomology has basis

$$1, \quad \xi, \eta, \quad \xi\eta$$

in cohomological degrees 0, 1, 1, 2. Splicing the two degree-one extensions gives the degree-two class. The syzygy $y \otimes \xi - qx \otimes \eta$ identifies the two splicings by $\xi\eta = -q\eta\xi$, while the absence of degree-two x^2 and y^2 syzygies gives $\xi^2 = \eta^2 = 0$. These relations account for the complete four-dimensional cohomology, and therefore present the Yoneda algebra. $\square$

Tensoring A_q with any commutative factorization coefficient produces a concrete ordered chiral algebra with noncommutative transverse multiplication. Its bar, quadratic dual, and derived augmentation boundary are now explicit in generators and relations.

QUIVER PATH ALGEBRAS

Let Q be a finite quiver and $S = \bigoplus_{i \in Q_0} \mathbf{k} e_i$. Assign $M_a \in \text{Fact}_D(X)$ to each arrow $a : i \rightarrow j$ and set

$$M = \bigoplus_{a \in Q_1} e_j M_a e_i, \quad T_S(M) = S \oplus M \oplus M \otimes_S M \oplus \cdots.$$

Words are composable paths.

For the quiver $1 \xrightarrow{a} 2$, the path algebra has basis e_1, e_2, a and relation $a = e_2 a e_1$. The simple modules S_1, S_2 satisfy

$$\text{Ext}^1(S_1, S_2) \cong \mathbf{k}, \quad \text{Ext}^1(S_i, S_i) = 0, \quad \text{Ext}^1(S_2, S_1) = 0.$$

The bar differential records concatenation of paths, and the quadratic dual reverses the arrow with cohomological shift.

Theorem 40.1 (Relative path-algebra bar). *For a tensor path algebra $T_S(M)$ over semisimple S ,*

$$S \otimes_{T_S(M)}^{\mathbb{L}} S \simeq S \oplus sM$$

as an S -bicomodule. A quadratic quotient $T_S(M)/(R)$ is Koszul precisely when its relative quadratic complex is exact away from weight zero.

Proof. The last-arrow contraction is S -bilinear and gives the same extra degeneracy as in the free case. The quotient statement follows from the relative twisting-morphism criterion. $\square$

WEYL AND CLIFFORD DEFORMATIONS

Let V be finite-dimensional and let $\omega : V \otimes V \rightarrow \mathbb{k}$ be alternating. The Weyl algebra is

$$W_{\hbar}(V, \omega) = T(V)[[\hbar]] / \langle uv - vu - \hbar\omega(u, v) \rangle.$$

For a purely odd space and symmetric ω , the same formula with the supercommutator $uv + vu$ gives a Clifford algebra.

Theorem 4I.1 (Filtered PBW theorem). *For alternating ω in the even Weyl case, or symmetric ω in the purely odd Clifford case, the associated graded algebra is $\text{Sym}(V)$ in the Weyl case and $\wedge(V)$ in the Clifford case.*

Proof. Filter by word length. The constant term lowers filtration by two, so the associated graded relation is the commutative or skew-commutative quadratic relation. The cubic ambiguities reduce to the cocycle identity for a central scalar form, which holds because scalars commute. The ordered monomials give the PBW basis. $\square$

The quadratic-linear-constant presentation has a curved Koszul dual. The constant term becomes the curvature functional

$$m_{\circ} : s^2 R \longrightarrow \mathbb{k}[2].$$

This construction is the algebraic origin of the curvature; geometric anomaly curvature enters through a separate comparison morphism.

ASSOCIATIVE CHIRAL ALGEBRAS ON THE FORMAL DISC

Let V be a complete topological vector space with vacuum $\mathbf{1}$ and translation T . A state–field map

$$Y(-, z) : V \longrightarrow \text{End}(V)((z))$$

is a coordinate presentation of an associative chiral-convolution product when its iterated products are expansions of one factorization kernel in the nested regions $|z_1| > |z_2|$ and $|z_2| > |z_1|$.

Proposition 42.1 (Associative convolution from weak associativity). *Suppose $Y(\mathbf{1}, z) = \text{id}$, $Y(a, z)\mathbf{1} \in a + zV[[z]]$, $[T, Y(a, z)] = \partial_z Y(a, z)$, and for every a, b, c there is N such that*

$$(z_1 - z_2)^N Y(a, z_1) Y(b, z_2) c = (z_1 - z_2)^N Y(Y(a, z_1 - z_2) b, z_2) c$$

in the common formal expansion ring. Then the fields define an associative chiral algebra on the formal disc. The converse holds for a vacuum-generated translation-equivariant associative chiral algebra admitting state–field reconstruction.

Proof. The displayed identity is the equality of the two nested restrictions of the three-point chiral product. Vacuum and translation give the unit and coordinate covariance. Factorization on higher configuration spaces is obtained by iterating the equality; coherence follows from associativity of the chiral-convolution product. $\square$

Example 42.2 (Braided current fields). Let $R(z)$ be an invertible solution of the spectral Yang–Baxter equation, normalized by $R^{21}(-z)R(z) = \mathbf{1}$. Generating fields $a_i(z)$ with exchange

$$a_i(z_1) a_j(z_2) = \sum_{k,l} R_{ij}^{kl}(z_1 - z_2) a_l(z_2) a_k(z_1)$$

form a quantum vertex algebra whenever the vacuum, translation, and weak associativity relations hold. The R -matrix supplies the comparison between the two formal expansion regions and records planar exchange in addition to transverse order.

ZAMOLODCHIKOV–FADDEEV EXCHANGE

Let V be finite-dimensional and let $R(u) \in \text{End}(V \otimes V) \otimes \mathbb{k}(u)$ be invertible. Introduce generators $Z_i(u)$ with exchange relation

$$Z_1(u)Z_2(v) = R_{12}(u-v)Z_2(v)Z_1(u).$$

Theorem 43.1 (Yang–Baxter confluence). *The exchange algebra has a consistent normal ordering for three generators precisely when*

$$R_{12}(u-v)R_{13}(u-w)R_{23}(v-w) = R_{23}(v-w)R_{13}(u-w)R_{12}(u-v).$$

Proof. Reverse the word $Z_1(u)Z_2(v)Z_3(w)$. Exchanging positions in the order (12), (13), (23) gives the left side. Exchanging in the order (23), (13), (12) gives the right side. These are the two reductions of the unique length-three overlap. Equality is therefore the diamond condition. $\square$

For the rational permutation solution

$$R(u) = \mathbf{1} - \frac{\hbar P}{u},$$

the Yang–Baxter equation follows from $P_{ij}^2 = \mathbf{1}$ and $P_{12}P_{13}P_{23} = P_{23}P_{13}P_{12}$.

THE RTT YANGIAN OF $\mathfrak{gl}_2$

Let

$$T(u) = 1 + \sum_{r \geq 1} T^{(r)} u^{-r}, \quad R(u) = 1 - \frac{\hbar P}{u}.$$

The RTT Yangian $Y_{\hbar}(\mathfrak{gl}_2)$ is generated by the coefficients $t_{ij}^{(r)}$ subject to

$$R(u-v)T_1(u)T_2(v) = T_2(v)T_1(u)R(u-v).$$

Equivalently,

$$(u-v)[t_{ij}(u), t_{kl}(v)] = \hbar(t_{kj}(u)t_{il}(v) - t_{kj}(v)t_{il}(u)).$$

Proposition 44.1 (Coefficient relations). *Put $t_{ij}^{(0)} = \delta_{ij}$. The RTT identity is equivalent to*

$$[t_{ij}^{(r+1)}, t_{kl}^{(s)}] - [t_{ij}^{(r)}, t_{kl}^{(s+1)}] = \hbar(t_{kj}^{(r)}t_{il}^{(s)} - t_{kj}^{(s)}t_{il}^{(r)}), \quad r, s \geq 0.$$

In particular,

$$[t_{ij}^{(1)}, t_{kl}^{(1)}] = \hbar(\delta_{kj}t_{il}^{(1)} - \delta_{il}t_{kj}^{(1)}),$$

so the first modes form the enveloping algebra of the copy $\hbar\mathfrak{gl}_2$ inside the Yangian.

Proof. Multiply the generating-series relation by $u-v$ and compare the coefficient of $u^{-r-1}v^{-s-1}$. The two terms contributed by multiplication by $u-v$ produce the two commutators on the left, and the two ordered products on the right are the corresponding coefficients of the RTT numerator. Setting $r=0$ gives the displayed first-mode relation. $\square$

Theorem 44.2 (Hopf structure). *The formulas*

$$\Delta(T(u)) = T(u) \otimes T(u), \quad \varepsilon(T(u)) = 1, \quad S(T(u)) = T(u)^{-1}$$

define a Hopf algebra structure in the u^{-1} -adic completion. Coefficientwise,

$$\Delta(t_{ij}^{(r)}) = \sum_{a=1}^2 \sum_{p=0}^r t_{ia}^{(p)} \otimes t_{aj}^{(r-p)}.$$

Proof. Apply the RTT relation in each tensor factor. Matrix multiplication in the auxiliary space gives the RTT relation for $\Delta(T)$. Coassociativity is associativity of matrix multiplication. Expanding the matrix product gives the coefficient formula. The inverse series exists uniquely because $T(u) = 1 + O(u^{-1})$; its defining identities give the antipode axioms. $\square$

The evaluation map

$$\mathrm{ev}_a : T(u) \mapsto 1 + \frac{\hbar E}{u - a}, \quad E = (E_{ij}),$$

lands in $U(\mathfrak{gl}_2)[[u^{-1}]]$ and gives the standard evaluation modules. The PBW theorem identifies the associated graded with $U(\mathfrak{gl}_2[t])$ [27, 60].

This example contains three structures: ordered multiplication in the Yangian, coproduct on tensor products of modules, and exchange by the rational R -matrix.

THE HALL ALGEBRA OF THE A_2 QUIVER

Let $Q : 1 \rightarrow 2$ over $\mathbb{F}_q$. Write S_1, S_2 for the simples and M for the unique non-split extension

$$0 \rightarrow S_2 \rightarrow M \rightarrow S_1 \rightarrow 0.$$

In the twisted Hall algebra, multiplication counts subobjects with Euler-form normalization. Direct counting gives

$$[S_1] * [S_2] = v^{-1}[S_1 \oplus S_2] + v^{-1}[M], \quad [S_2] * [S_1] = [S_1 \oplus S_2], \quad v^2 = q.$$

Hence the root vector

$$E_{12} = E_1 E_2 - v^{-1} E_2 E_1$$

is represented by the indecomposable extension.

Theorem 45.1 (Quantum Serre relation). *The Hall generators satisfy*

$$E_1^2 E_2 - (v + v^{-1}) E_1 E_2 E_1 + E_2 E_1^2 = 0,$$

and the analogous relation with 1 and 2 exchanged.

Proof. Every representation of dimension vector $(2, 1)$ is a linear map $f : \mathbb{F}_q^2 \rightarrow \mathbb{F}_q$ of rank zero or one. Read a word from the top quotient to the bottom quotient. The complete-flag counts are

$\text{rk } f$	112	121	211
0	$q + 1$	$q + 1$	$q + 1$
1	$q + 1$	1	0.

For rank zero every one-dimensional subspace of $\mathbb{F}_q^2$ is admissible, giving $q + 1$ flags in each word. For rank one, the word 121 forces the unique kernel line, the word 211 is incompatible with the final subrepresentation, and the word 112 allows every line at the remaining stage.

The total Euler-form exponents of the three words are respectively $-1, 0, 1$. Their twisted coefficients are therefore $v^{-1}, 1, v$. On the rank-zero stratum the Serre combination has coefficient

$$(q + 1)(v^{-1} - (v + v^{-1}) + v) = 0.$$

On the rank-one stratum it has coefficient

$$v^{-1}(q + 1) - (v + v^{-1}) = v^{-1}(v^2 + 1) - (v + v^{-1}) = 0.$$

The two isomorphism classes exhaust the dimension vector $(2, 1)$, proving the relation. This is Ringel's realization of $U_v^+(\mathfrak{sl}_3)$ [37, 67]. $\square$

MATRIX TRACES AND THE NECKLACE BRACKET

Let $F = \mathbb{k}\langle x, y \rangle$ and $F_{\text{cyc}} = F/[F, F]$. Cyclic derivatives are defined by

$$\partial_x^{\text{cyc}}[a_1 \cdots a_n] = \sum_{a_i=x} a_{i+1} \cdots a_n a_1 \cdots a_{i-1}$$

and similarly for y . Define

$$\{f, g\}_{\text{neck}} = [\partial_x^{\text{cyc}} f \partial_y^{\text{cyc}} g - \partial_y^{\text{cyc}} f \partial_x^{\text{cyc}} g]_{\text{cyc}}.$$

Theorem 46.1 (Trace realization). *For matrices $X, Y \in \mathfrak{gl}_N$ and the canonical symplectic bracket on matrix entries,*

$$\{\text{Tr } f(X, Y), \text{Tr } g(X, Y)\} = \text{Tr} \{f, g\}_{\text{neck}}(X, Y).$$

Proof. Differentiate the trace by moving the varied matrix entry cyclically to the first position. This gives the cyclic derivative. Contract the X derivative of one trace with the Y derivative of the other and subtract the reversed contraction. The matrix indices close into the trace of the cyclic product. $\square$

In the stable commuting-matrix quotient, let

$$p_{ab} = \text{Tr}(X^a Y^b).$$

Then

$$\{p_{ab}, p_{cd}\} = (ad - bc) p_{a+c-1, b+d-1}.$$

For two eigenvalues, the relative invariants

$$A = (x_1 - x_2)^2, \quad B = (x_1 - x_2)(y_1 - y_2), \quad C = (y_1 - y_2)^2$$

satisfy $B^2 = AC$ and carry the standard $\mathfrak{sl}_2$ Poisson bracket after linear rescaling.

BOOK **XI**

The nonlocal ladder

SPECTRAL EXCHANGE AS AN ASSOCIATIVE CHIRAL OPERATION

47.1 THE COEFFICIENT FIELD AND THE EXCHANGE OPERATOR

Let V be a finite-dimensional vector space and let K be a field of meromorphic functions in spectral differences. Fix an invertible operator

$$R(u) \in \text{End}(V \otimes V) \otimes K$$

that satisfies the spectral Yang–Baxter equation

$$R_{12}(u-v)R_{13}(u-w)R_{23}(v-w) = R_{23}(v-w)R_{13}(u-w)R_{12}(u-v). \quad (47.1)$$

The equation states that two routes through the chamber complex of three labelled points give the same transport.

Definition 47.1 (Zamolodchikov–Faddeev algebra). The exchange algebra $\mathcal{Z}_R$ is generated by the components of a current $A(u) \in V \otimes \mathcal{Z}_R$ with relation

$$A_1(u)A_2(v) = R_{12}(u-v)A_2(v)A_1(u). \quad (47.2)$$

A normal word has spectral labels in a fixed chamber $u_1 \succ \cdots \succ u_n$.

Theorem 47.2 (Confluence is Yang–Baxter). *The quadratic rewriting system defined by (47.2) is confluent on words with pairwise distinct spectral labels precisely when (47.1) holds. In that case the normal ordered words form a K -basis of $\mathcal{Z}_R$.*

Proof. Every two-letter inversion is removed by one application of (47.2). The minimal overlap occurs in a word of length three. Moving the leftmost label through the other two gives the coefficient $R_{12}R_{13}R_{23}$, while moving the rightmost label in the opposite sequence gives $R_{23}R_{13}R_{12}$. Equality of the two coefficients is (47.1). Bergman’s diamond lemma then gives the normal-word basis. $\square$

47.2 A DIAGONAL EXCHANGE MODEL

Choose a basis $e_1, \dots, e_N$ of V and functions $q_{ij}(u)$ satisfying

$$q_{ij}(u)q_{ji}(-u) = 1, \quad q_{ii}(u) = 1.$$

The diagonal operator

$$R(u)(e_i \otimes e_j) = q_{ij}(u)e_i \otimes e_j$$

satisfies the Yang–Baxter equation because all three factors are scalars. The generators $a_i(u)$ obey

$$a_i(u)a_j(v) = q_{ij}(u-v)a_j(v)a_i(u).$$

Every ordered monomial is a basis vector. The quantum plane is the constant two-generator specialization $q_{12} = q, q_{21} = q^{-1}$.

Calculation 47.3 (Three-particle coefficient). For $u \succ v \succ w$,

$$a_i(w)a_j(v)a_k(u) = q_{ij}(w-v)q_{ik}(w-u)q_{jk}(v-u)a_k(u)a_j(v)a_i(w).$$

The coefficient is independent of the sequence of adjacent exchanges. This scalar identity is the diagonal Yang–Baxter equation.

Proof. Each inversion of two adjacent letters contributes its scalar exchange factor. Both reduced words for the longest permutation invert the same three pairs. Scalar commutativity makes the products identical. $\square$

RTT AND REFLECTION ALGEBRAS

48.1 THE RTT BIALGEBRA

Let $T(u) = (t_{ij}(u))$ be a matrix of formal currents. The RTT algebra $\mathcal{T}_R$ is defined by

$$R_{12}(u-v)T_1(u)T_2(v) = T_2(v)T_1(u)R_{12}(u-v). \quad (48.1)$$

Theorem 48.1 (RTT coproduct). *The formula*

$$\Delta(T(u)) = T_{13}(u)T_{12}(u), \quad \epsilon(T(u)) = I,$$

defines a coassociative bialgebra structure on $\mathcal{T}_R$.

Proof. Apply (48.1) in each tensor factor. The Yang–Baxter equation moves the three R -operators through the three matrix factors. The resulting identity is the RTT relation for $\Delta(T)$. Matrix multiplication is associative, hence the two iterated coproducts both equal $T_{14}T_{13}T_{12}$. $\square$

For the Yang matrix $R(u) = 1 - \hbar P/u$, expansion at $u = \infty$ gives the RTT Yangian. The coefficient of $u^{-1}v^{-1}$ is

$$[t_{ij}^{(1)}, t_{kl}^{(1)}] = \hbar(\delta_{kj}t_{il}^{(1)} - \delta_{il}t_{kj}^{(1)}),$$

so the first current layer is the Rees enveloping algebra of $\mathfrak{gl}_N$.

48.2 BOUNDARY EXCHANGE

A reflecting boundary reverses the spectral path. For a constant invertible $R \in \text{End}(V^{\otimes 2})$ satisfying Yang–Baxter, define the reflection algebra $\mathcal{K}_R$ by

$$R_{21}K_1R_{12}K_2 = K_2R_{21}K_1R_{12}. \quad (48.2)$$

Theorem 48.2 (Boundary confluence). *The relation (48.2) gives a coherent exchange law for three boundary insertions. The two reductions of every length-three overlap agree by the Yang–Baxter equation and its inverse.*

Proof. A boundary exchange consists of an incoming crossing, reflection, and outgoing crossing. The two reductions of three insertions sweep the same braid in the doubled half-plane. Algebraically, one inserts (48.2) three times and uses Yang–Baxter to pass the middle R -matrix through the outer pair. The remaining factors cancel in inverse pairs. $\square$

The RTT algebra governs bulk line exchange. The reflection algebra governs boundary line exchange. The ZF algebra governs charged fields in a fixed representation. Their common coefficient is the same R -matrix, while their products live in three different endomorphism objects.

QUANTUM VERTEX REALIZATION

49.1 TWO EXPANSION REGIONS

Let V be a complete topological vector space with vacuum $\mathbf{1}$, translation T , state-field map Y , and a unitary rational operator

$$S(z) : V \otimes V \longrightarrow V \otimes V((z)).$$

The two formal regions $|z_1| > |z_2|$ and $|z_2| > |z_1|$ are related by $S(z_1 - z_2)$.

Definition 49.1 (Rational quantum vertex datum). A rational quantum vertex datum satisfies vacuum and translation covariance together with

$$(z_1 - z_2)^N Y(a, z_1) Y(b, z_2) = (z_1 - z_2)^N Y^{21}(S(z_1 - z_2)(a \otimes b); z_2, z_1) \quad (49.1)$$

for a sufficiently large N , and weak associativity in every triple of states.

Theorem 49.2 (Exchange coherence). *Suppose S is unitary,*

$$S^{21}(-z)S(z) = \mathbf{1},$$

and satisfies the spectral Yang–Baxter equation. Then the pairwise continuation maps obtained from (49.1) define a braid-group action on every n -point field product. Weak associativity makes the action compatible with iterated operator products.

Proof. Unitarity identifies a crossing followed by its reverse with the identity. Yang–Baxter identifies the two continuations around every codimension-two triple-collision stratum. These are the defining relations of the braid group. Weak associativity identifies the collision limit in each chamber with the corresponding iterated field. $\square$

When S is the graded flip, the datum is an ordinary local vertex algebra. A nontrivial S retains the ordered chiral product and its exchange monodromy. This is the formal-disc realization of an associative chiral algebra with braiding data in the sense of quantum vertex algebra theory [53, 54].

49.2 THE YANGIAN FIELD

Let $T(u)$ act on a family of evaluation modules. The RTT relation becomes S -locality for the matrix coefficients of $T(u)$. The normalized Yang operator

$$S(u) = f(u) \left(\mathbf{1} - \frac{\hbar P}{u} \right), \quad f(u)f(-u) = \left(\mathbf{1} - \frac{\hbar^2}{u^2} \right)^{-1},$$

is unitary. Its expansion at infinity gives the quantum vertex exchange of Yangian currents. The scalar factor changes normalization and preserves the Yang–Baxter equation.

SHUFFLE REALIZATION AND THE POSITIVE CURRENT ALGEBRA

50.1 PAIRWISE KERNEL

Let $K(z)$ be a rational function and put

$$\text{Sh}_n = \mathbb{C}(z_1, \dots, z_n)^{\mathfrak{S}_n}.$$

For $f \in \text{Sh}_m$ and $g \in \text{Sh}_n$ define

$$(f * g)(z_1, \dots, z_{m+n}) = \text{Sym} \left(f(z_1, \dots, z_m) g(z_{m+1}, \dots, z_{m+n}) \prod_{i \leq m < j} K(z_i - z_j) \right). \quad (50.1)$$

Theorem 50.1 (Associativity of the shuffle product). *The product (50.1) is associative on every subspace closed under the displayed symmetrization and pole conditions.*

Proof. In a triple product, every ordered pair of variables belonging to different blocks contributes exactly one factor $K(z_i - z_j)$. The product of all pairwise factors is independent of the parenthesization. Final symmetrization gives the same function. $\square$

For the tripled Jordan quiver, one uses

$$K(z) = \prod_{a=1}^3 \frac{z + \epsilon_a}{z}, \quad \epsilon_1 + \epsilon_2 + \epsilon_3 = 0.$$

Wheel conditions remove the forbidden triple residues. The resulting shuffle algebra is the positive current algebra that appears in the critical CoHA of $\mathbb{C}^3$ and in the positive affine Yangian of $\mathfrak{gl}_1$ [68, 69].

Calculation 50.2 (Degree one against degree one). For $f(z_1) = z_1^r$ and $g(z_1) = z_1^s$,

$$f * g = z_1^r z_2^s K(z_1 - z_2) + z_2^r z_1^s K(z_2 - z_1).$$

The antisymmetric part records the first current commutator. Its residues at $z_1 - z_2 = -\epsilon_a$ are the three elementary extension channels of the tripled quiver.

Proof. For two variables the symmetrization in the shuffle definition has two terms, which gives the displayed formula. Antisymmetrization exchanges the two chambers. The three numerator factors of the tripled-quiver kernel locate the three elementary shifted extension channels. $\square$

50.2 COPRODUCT AND DOUBLE

Choose a chamber expansion of $K(z)$ at infinity. Splitting the variables into two ordered groups and expanding every cross-kernel in that chamber defines a completed coproduct. A residue pairing between positive and negative shuffle halves gives a continuous Hopf pairing. The completed Drinfeld double is then a quasitriangular current algebra. Product, coproduct, and pairing are separate operations; each is visible in the shuffle formula that defines it.

THE LADDER IN ONE DIAGRAM

The nonlocal structures form the chain

$$\begin{aligned} \text{ordered coefficient} &\longrightarrow \mathcal{Z}_R \longrightarrow \mathcal{T}_R \longrightarrow (V, Y, S), \\ (V, Y, S) &\longrightarrow \text{Sh}^+ \longrightarrow D(\text{Sh}^+). \end{aligned}$$

The arrows retain different data:

Object	Defining operation	Coherence
$\mathcal{Z}_R$	exchange of charged fields	Yang–Baxter equation
$\mathcal{T}_R$	matrix-current product and coproduct	RTT and coassociativity
$\mathcal{K}_R$	boundary exchange	reflection equation
(V, Y, S)	formal field product	S -locality and weak associativity
Sh^+	extension shuffle	pairwise-kernel associativity
$D(\text{Sh}^+)$	positive–negative pairing	quasitriangularity

This ladder gives a concrete generator-and-relation theory of doubly noncommutative chiral observables. The first noncommutativity is ordered composition. The second is exchange between distinct order chambers. Chiral singularity remains the diagonal coefficient throughout.

BOOK **XII**

Field theory and master equations

A FINITE COTANGENT BF MODEL

Let L be a finite-dimensional dg Lie algebra. Its cotangent BV space is

$$\mathcal{E}_L = L[1] \oplus L^*[-2],$$

with the canonical degree -1 symplectic pairing. Write $A \in L[1]$ and $B \in L^*[-2]$. The cotangent action is

$$S_L(A, B) = \langle B, d_L A + \frac{1}{2}[A, A] \rangle.$$

Theorem 52.1 (Cotangent master equation). *The Hamiltonian vector field of S_L is square-zero, and consequently $\{S_L, S_L\} = 0$.*

Proof. The A component is the Maurer–Cartan vector field $d_L A + \frac{1}{2}[A, A]$; the B component is its cotangent lift. The square of the first component is the sum of d_L^2 , the derivation identity for d_L , and the Jacobi identity. The cotangent lift preserves commutators of vector fields, so the full Hamiltonian vector field is square-zero. $\square$

Proposition 52.2 (Coordinate form of the modular term). *Let L be an ordinary finite-dimensional Lie algebra with basis e_a and structure constants $[e_a, e_b] = f_{ab}^c e_c$. Let c^a be the odd coordinates on $L[1]$ and b_a their BV-conjugate coordinates. With*

$$S = \frac{1}{2} b_c f_{ab}^c c^a c^b, \quad \Delta = - \sum_a \frac{\partial}{\partial c^a} \frac{\partial}{\partial b_a},$$

one has

$$\Delta S = \sum_b \text{Tr}(\text{ad}_{e_b}) c^b.$$

Thus the one-loop class of the finite cotangent theory is exactly the modular character.

Proof. Differentiation first in b_a gives $\frac{1}{2} f_{ij}^a c^i c^j$. Odd differentiation in c^a gives $f_{aj}^a c^j$, and the chosen sign in Δ converts this to $f_{ja}^a c^j = \text{Tr}(\text{ad}_{e_j}) c^j$. This is the displayed formula. $\square$

A finite-dimensional dg Lie algebra is unimodular when the supertrace of every adjoint endomorphism vanishes. Every finite-dimensional nilpotent Lie algebra in degree zero is unimodular because each ad_x is nilpotent.

Theorem 52.3 (Finite-dimensional quantum master equation). *Let L be unimodular and let Δ be the BV Laplacian determined by the translation-invariant Berezinian on $\mathcal{E}_L$. Then*

$$\Delta S_L = 0, \quad \frac{1}{2} \{S_L, S_L\} + \hbar \Delta S_L = 0,$$

and equivalently

$$\hbar \Delta(e^{S_L/\hbar}) = 0.$$

The quantum observable differential

$$\partial_{S_L} = \{S_L, -\} + \hbar \Delta$$

satisfies $\partial_{S_L}^2 = 0$.

Proof. The classical master equation gives $\{S_L, S_L\} = 0$. The scalar ΔS_L is the divergence of the Chevalley vector field on $L[1]$. Its linear coefficient is the modular character $x \mapsto \text{str}(\text{ad}_x)$, which vanishes by unimodularity. The BV exponential identity

$$\hbar^2 e^{-S_L/\hbar} \Delta(e^{S_L/\hbar}) = \frac{1}{2} \{S_L, S_L\} + \hbar \Delta S_L$$

proves the exponential form. The standard BV commutator calculation then gives

$$\partial_{S_L}^2 = \left\{ \frac{1}{2} \{S_L, S_L\} + \hbar \Delta S_L, - \right\} = 0.$$

□

Now take the local dg Lie algebra

$$\mathcal{L}_X = \Omega^{\circ,*}(X) \widehat{\otimes} \Omega^*(\mathbb{R}) \otimes \mathbf{I}$$

for a finite-dimensional nilpotent $\mathbf{I}$, and form its shifted cotangent theory with compactly supported density dual. The preceding finite theorem governs every pairing-preserving finite-dimensional deformation retract. The local classical observables are Chevalley cochains and are holomorphic on X and locally constant on $\mathbb{R}$.

Choose the standard interval polarization in which the $L[1]$ variables are fixed at one end and the cotangent variables at the other. Canonical quantization of the boundary linear Poisson algebra gives the associative zero-mode algebra $U(\mathbf{I})$. Gluing the two augmentation boundary states across an interval gives the derived tensor product

$$\mathbf{I} \otimes_{U(\mathbf{I})}^{\mathbb{L}} \mathbf{I} \simeq \text{Bar}(U(\mathbf{I})) \simeq C_*(\mathbf{I}),$$

the Chevalley–Eilenberg chains. A solution of the renormalized quantum master equation in the local BV deformation complex produces the corresponding infinite-dimensional quantum factorization algebra.

BRST AND ORDERED REDUCTION

Let (A, Q) be a dg ordered algebra, with Q a derivation. Extend Q to the bar factorwise. Write b for the multiplication bar differential.

Proposition 53.1 (BRST–bar bicomplex). *With the suspension sign convention,*

$$Q_{\text{Bar}}b + bQ_{\text{Bar}} = 0.$$

Hence $D = Q_{\text{Bar}} + b$ satisfies $D^2 = 0$.

Proof. Apply Q to a bar face that multiplies adjacent entries. The derivation formula splits the result into the two terms obtained by applying Q before that face. The totalization sign makes their sum the anticommutator. $\square$

The filtration by bar length gives

$$E_1 \cong \text{Bar}(H^*(A, Q)) \implies H^*(\text{Bar}(A), D)$$

under boundedness and completeness. Quantum Drinfeld–Sokolov reduction is a current-algebra realization of this bicomplex.

STOKES AND ALGEBRAIC IDENTITIES

Let $C_*(K)$ be the oriented cellular chains of a regular manifold with corners. Suppose each codimension-one face F is assigned a degree-one operator ρ_F on a coefficient complex E , and the assignments are compatible with restrictions to corners.

Theorem 54.1 (Stokes-to-master). *Assume that each ρ_F is a chain map of total degree one, that its restriction to a corner depends on the induced contraction, and that disjoint contractions act with the Koszul sign prescribed by the oriented normal bundle. Then the component of*

$$D^2, \quad D = d_E + d_K + \sum_F \rho_F,$$

supported on a codimension-two corner S is the oriented sum

$$\sum_{F \supset S} \epsilon(F, S) \rho_{F|S} \rho_F.$$

Thus D is a differential precisely when these corner sums vanish.

Proof. Expand D^2 and group terms by support strata. The chain-map condition cancels the mixed internal terms. A term supported on S is obtained by crossing two incident faces. The two orders carry the boundary orientations of the two sides of the oriented normal square. Their signed sum is the displayed expression. $\square$

Associahedra give Stasheff identities. Fulton–MacPherson collision strata give chiral and graph identities after a coefficient map is supplied. Stable-graph boundaries give BV and modular master equations.

TWO EDGE COLORS AND TWO LOOP OPERATORS

Let $G^{\perp, \text{ch}}$ be the completed vector space spanned by stable graphs with transverse and chiral edges. Contracting an edge that joins distinct vertices gives brackets $[-, -]_{\perp}$ and $[-, -]_{\text{ch}}$. Contracting a loop gives operators $\Delta_{\perp}$ and Δ_{ch} . Product orientation gives

$$\Delta_{\perp} \Delta_{\text{ch}} + \Delta_{\text{ch}} \Delta_{\perp} = 0.$$

Theorem 55.1 (Bicolored master equation). *Let $P^{\perp, \text{ch}}$ be a two-colored modular operad whose edge contractions carry the product determinant orientation, and let $\mathfrak{g}_P$ be the completed convolution Lie algebra of its Feynman transform with an endomorphism modular operad. Algebra structures on the transform are Maurer–Cartan elements $\Theta \in \mathfrak{g}_P^1$ satisfying*

$$d\Theta + \frac{1}{2}[\Theta, \Theta]_{\perp} + \frac{1}{2}[\Theta, \Theta]_{\text{ch}} + \hbar_{\perp} \Delta_{\perp} \Theta + \hbar_{\text{ch}} \Delta_{\text{ch}} \Theta = 0.$$

Proof. The Feynman-transform differential contracts one edge. Separating edges give the two brackets and loop edges give the two BV operators. The product orientation makes mixed contractions anticommute. The condition that the structure map commute with the transform differential is exactly the displayed Maurer–Cartan equation [32]. $\square$

TYPED ANOMALIES

An anomaly is a cohomology class in the deformation complex that governs the attempted construction.

construction	complex	class
central extension	local Lie algebra cochains	cocycle
BRST quantization	BRST deformation complex	nilpotence obstruction
BV quantization	local BV cochains	quantum master anomaly
conformal-block family	Atiyah algebra of its determinant line	projective curvature
curved Koszul duality	curved convolution Lie algebra	curvature m_o

A comparison theorem consists of a morphism between two complexes and an equality of the transported classes. This formulation keeps the geometric origin of every scalar visible.

BOOK XIII

Calabi–Yau and automorphic interfaces

CALABI–YAU CATEGORIES

Let C be a smooth linear category with a d -Calabi–Yau structure. Two universal consequences are available.

Theorem 57.1 (Cyclic and shifted outputs). *The Hochschild cochains $CC^*(C)$ form a framed E_2 algebra. The derived moduli stack of objects carries a $(2 - d)$ -shifted symplectic structure whenever the representability hypotheses hold.*

Proof. The first assertion is the cyclic Deligne theorem [10]. The second is the PTVV–Brav–Dyckerhoff construction: the Calabi–Yau trace pairs the tangent complex $R\mathrm{Hom}(E, E)[1]$ with itself in degree $2 - d$ [9, 64]. $\square$

A geometric BV theory or an open factorization category supplies an ordered chiral algebra through the recognition theorem of Chapter 3. This is the bridge from Calabi–Yau geometry to the present volume.

HALL AND BPS ALGEBRAS

A Hall multiplication starts from the extension correspondence

$$\mathcal{M}_\alpha \times \mathcal{M}_\beta \xleftarrow{p} \mathcal{M}_{\alpha,\beta}^{\text{ext}} \xrightarrow{q} \mathcal{M}_{\alpha+\beta}.$$

With a coefficient theory for which p^* and q_* are defined,

$$a * b = q_* p^*(a \boxtimes b).$$

Associativity follows from the stack of two-step filtrations.

For 2-Calabi–Yau categories with good moduli spaces, the BPS algebra is the positive enveloping algebra of a generalized Kac–Moody Lie algebra [22]. For quivers and surface resolutions, preprojective and surface CoHAs recover positive Yangian halves in established cases [25, 68].

The ordered chiral structure enters when Hall multiplication is promoted to a factorization coefficient over a curve or defect parameter. A coproduct, pairing, and completion then produce the corresponding double.

THE IGUSA DENOMINATOR AS A BAR CHARACTER

Let $\mathfrak{g}_\Delta$ be the Gritsenko–Nikulin Borchers–Kac–Moody superalgebra and let $\mathfrak{n}_+$ be its positive nilpotent part. Root spaces are finite-dimensional and the positive cone is locally finite. Fix a proper integral height h that is positive on the effective root cone, and set

$$F^{>H} \mathfrak{n}_+ = \bigoplus_{h(\alpha) > H} \mathfrak{g}_{\Delta, \alpha}, \quad \mathfrak{n}_+^{(H)} = \mathfrak{n}_+ / F^{>H} \mathfrak{n}_+.$$

Height additivity makes $F^{>H} \mathfrak{n}_+$ an ideal; properness makes $\mathfrak{n}_+^{(H)}$ finite-dimensional and nilpotent. Every bar, Chevalley, enveloping, and character formula in this chapter is interpreted coefficientwise in the separated height completion.

Theorem 59.1 (Bar–Chevalley identification). *For each height H there is a natural quasi-isomorphism of conilpotent coalgebras*

$$\text{Bar}(U(\mathfrak{n}_+^{(H)})) \simeq C_*(\mathfrak{n}_+^{(H)}).$$

The quotient maps are compatible with these quasi-isomorphisms, and therefore induce a quasi-isomorphism in the separated height-complete category

$$\text{Bar}_h^\wedge(U(\mathfrak{n}_+)) := \varprojlim_H \text{Bar}(U(\mathfrak{n}_+^{(H)})) \simeq C_{*,h}^\wedge(\mathfrak{n}_+) := \varprojlim_H C_*(\mathfrak{n}_+^{(H)}).$$

The coefficientwise Euler supercharacter of the completed Chevalley complex is

$$\chi C_{*,h}^\wedge(\mathfrak{n}_+) = \prod_{\alpha > 0} (1 - e^{-\alpha})^{\text{sdim } \mathfrak{g}_{\Delta, \alpha}}.$$

Proof. For every H , the PBW filtration on $U(\mathfrak{n}_+^{(H)})$ identifies the associated graded with $\text{Sym}(\mathfrak{n}_+^{(H)})$. The bar complex of the enveloping algebra and the Chevalley complex have the same associated-graded Koszul resolution. The standard antisymmetrization map is a filtered quasi-isomorphism. Naturality for the quotient maps gives the inverse-limit comparison; proper height makes every homogeneous coefficient stabilize. For an even root vector the exterior Chevalley factor contributes $1 - e^{-\alpha}$; for an odd root vector it contributes its inverse. Multiplication over the root spaces gives the product. $\square$

With the positive-root convention $e^{-\rho} = q^{1/2} r^{1/2} s^{1/2}$, the denominator theorem reads

$$e^{-\rho} \chi C_{*,h}^\wedge(\mathfrak{n}_+) = D_\zeta(2\Omega) = 2^{-6} \Delta_\zeta(2\Omega)$$

[38, 40]. This identity is the exact meeting point of ordered bar theory and automorphic products.

FINITE-HEIGHT CURRENTS AND AN EXACT PHYSICAL MODEL

Retain the finite-height quotients $\mathfrak{n}_+^{(H)}$ fixed in the preceding chapter.

The current Lie conformal superalgebra is

$$\text{Cur}(\mathfrak{n}_+^{(H)}) = \mathbb{k}[\partial] \otimes \mathfrak{n}_+^{(H)}, \quad [a_\lambda b] = [a, b].$$

Its universal enveloping vertex superalgebra is V_H . The current generators carry the cocommutative vertex-bialgebra coproduct

$$\Delta(a(-1)\mathbf{1}) = a(-1)\mathbf{1} \otimes \mathbf{1} + \mathbf{1} \otimes a(-1)\mathbf{1}, \quad a \in \mathfrak{n}_+^{(H)},$$

extended uniquely as a vertex-algebra morphism.

Theorem 60.1 (Height-complete current realization). *Let $\mathfrak{n} = \bigoplus_{\alpha > 0} \mathfrak{n}_\alpha$ be a root-graded Lie superalgebra with finite-dimensional root spaces and a proper additive height function. Put*

$$\mathfrak{n}^{(H)} = \mathfrak{n} / \bigoplus_{h(\alpha) > H} \mathfrak{n}_\alpha.$$

Then every $\mathfrak{n}^{(H)}$ is finite-dimensional and nilpotent, the quotient maps form an inverse system, and

$$\widehat{U}_h(\mathfrak{n}) = \varprojlim_H U(\mathfrak{n}^{(H)}), \quad \widehat{C}_{*,h}(\mathfrak{n}) = \varprojlim_H C_*(\mathfrak{n}^{(H)}), \quad \widehat{V}_h(\mathfrak{n}) = \varprojlim_H U^{\text{ch}}(\text{Cur}_X \mathfrak{n}^{(H)})$$

are separated complete algebras, coalgebras, and current algebras. Their associated graded objects are the corresponding completed symmetric objects, and the bar–Chevalley comparison is compatible with the inverse system.

Proof. Properness gives finitely many roots below a fixed height. Additivity implies that the roots of height $> H$ form a Lie ideal and that the quotient is nilpotent. PBW and Chevalley antisymmetrization are natural under quotient maps. Taking inverse limits in each finite total height therefore gives the displayed complete objects and preserves the coefficientwise quasi-isomorphism. $\square$

Apply the finite BF model of Chapter 19 to every $\mathfrak{n}_+^{(H)}$. The inverse system gives a height-complete holomorphic–topological classical field theory with completed boundary algebra and interval complex

$$U_h^\wedge(\mathfrak{n}_+) := \varprojlim_H U(\mathfrak{n}_+^{(H)}), \quad C_{*,h}^\wedge(\mathfrak{n}_+) := \varprojlim_H C_*(\mathfrak{n}_+^{(H)}).$$

This gives an exact algebraic realization, a local classical holomorphic–topological realization, and an exact quantum realization of the finite-dimensional zero-mode sectors of the positive enveloping algebra and its denominator character. The compact $K_3 \times E$ identification is the geometric comparison studied in the companion volumes.

CHARACTER, BRACKET, PARITY, AND ORIENTATION

A character is an Euler invariant. Faithful reconstruction combines it with the bracket, parity decomposition, and orientation line carried by the intended object.

Theorem 6I.1 (Three reconstruction fibres). *The following fibres contain distinct structured objects with the same scalar image.*

- (i) *The PBW-character fibre of a graded vector space with basis x, y, z in degrees $a, b, a + b$ contains both the abelian Lie algebra and the Heisenberg Lie algebra $[x, y] = z$.*
- (ii) *The superdimension-zero fibre contains both the zero super vector space and $\mathbb{k}^{1|1}$.*
- (iii) *The squaring fibre of a line-valued section contains s and $s \otimes \chi$ for every sign local system χ .*

Consequently a faithful reconstruction datum consists respectively of a bracket, a parity-refined dimension, and an oriented square-root line.

Proof. PBW identifies the graded vector-space character of an enveloping algebra with the symmetric-superalgebra character of its underlying graded Lie space, so the two displayed brackets have the same PBW character. The two super vector spaces in the second item both have superdimension zero and have parity dimensions $(0, 0)$ and $(1, 1)$. Finally $\chi^{\otimes 2} \cong \mathbf{1}$, hence

$$(s \otimes \chi)^{\otimes 2} \cong s^{\otimes 2}.$$

The extra structures listed in the conclusion distinguish each pair. □

Recognition Theorem 6I.2 (Minimal PBW recognition). *Let $\phi : \mathfrak{g} \twoheadrightarrow \mathfrak{h}$ be a surjective morphism of positively graded Lie superalgebras with finite-dimensional homogeneous pieces. If the parity-refined Hilbert series of $U(\mathfrak{g})$ and $U(\mathfrak{h})$ agree, then ϕ is an isomorphism. Equality of signed PBW characters suffices when the parity of every primitive degree is prescribed in advance.*

Proof. If the kernel is nonzero, choose its first positive degree. In that degree all decomposable PBW contributions agree by minimality, while the primitive even or odd dimension in $U(\mathfrak{h})$ is strictly smaller. This contradicts equality of parity-refined Hilbert series. With prescribed parity, the formal logarithm of the PBW product recovers the signed primitive dimension inductively in height, and prescribed parity recovers the ordinary dimension. □

EXACT DEPENDENCY THEOREM

Theorem 62.1 (Ordered-chiral closure). *Fix:*

- (i) *a six-functor Ran category and its factorization subcategory;*
- (ii) *a transversely ordered algebra, an associative chiral algebra, or a chiral Lie algebra, as appropriate;*
- (iii) *every mixed distributive, completion, duality, trace, or analytic datum named by the intended construction.*

Then the three bars, their complete cobars, their module/comodule Koszul transforms, the Hochschild centre, the formal-disc realization, the ribbon theory, and the finite-height current realization are defined functorially and satisfy the identities proved in this volume. Every passage between the pointwise and chiral-convolution monoidal products is carried by the displayed mixed distributive datum.

Proof. Each output is one of the functors constructed in the preceding chapters. The source and target monoidal categories are displayed in every definition. The comparison theorems invoke the explicitly supplied distributive, PBW, completion, duality, or trace datum. Composition of these functors gives the asserted functorial system. $\square$

SYNTHESIS

The invariant language has three algebraic axes. Transverse order is an E_1 structure for $\otimes^!$. Associative chirality is an E_1 structure for $\star^{\text{ch}}$. Chiral Lie theory is a Lie structure for $\star^{\text{ch}}$. Their bars are $\text{Bar}^\perp$, $\text{Bar}_{\text{Ass}}^{\text{ch}}$, and $\text{Bar}_{\text{Lie}}^{\text{ch}}$. The aligned projector supplies the exact sector on which a mixed distributive law is defined. Every comparison in the theorem system factors through one of these typed maps.

The constructed operations form the two functorial rows

$$\begin{aligned} \text{HT boundary theory} &\xrightarrow{\text{Obs}_\partial} \text{Alg}_{E_1}(\text{Fact}_D(X)) \xrightarrow{\text{Bar}^{\text{ord}}} \text{Coalg}_{\text{conil}, \text{cpl}}(\text{Fact}_D(X)), \\ \text{Alg}_{E_1}(\text{Fact}_D(X)) &\xrightarrow{Z_{\text{ord}}} \text{Alg}_{E_2}(\text{Fact}_D(X)). \end{aligned}$$

A primitive Lie algebra enters when a separate Chevalley or Hall comparison identifies the bar coalgebra with the Chevalley chains of that Lie algebra. Hall geometry supplies this comparison through extension correspondences. Quantum groups require an additional coproduct and exchange structure. Automorphic forms arise from graded Euler characters after a grading and completion have been fixed. Each arrow therefore preserves the geometric operation named at its source.

BOOK **XIV**

**Technical appendices to the preceding
book**

CHAPTER 64

SIGN CALCULATION FOR THE LENGTH-THREE BAR

For homogeneous $a, b, c \in \vec{A}$, write $[a|b|c] = sa \otimes sb \otimes sc$. With the standard suspension convention,

$$b_m[a|b|c] = (-1)^{|a|}[ab|c] + (-1)^{|a|+|b|+1}[a|bc].$$

Applying b_m again gives

$$b_m^2[a|b|c] = (-1)^{|b|}[(ab)c - a(bc)],$$

which vanishes by associativity.

CHAPTER 65

RATIONAL R -MATRIX VERIFICATION

Let P_{ij} permute tensor factors i and j . Expanding

$$R_{12}(u)R_{13}(u+v)R_{23}(v)$$

and its reversed product shows equality coefficientwise. The cubic coefficient uses

$$P_{12}P_{13}P_{23} = P_{23}P_{13}P_{12},$$

the quadratic coefficients use the symmetric-group relations, and the linear coefficients agree termwise.

CARRIERS AND COMPARISON MAPS

The five-carrier geometry, transverse bar construction, mixed-forest signs, and noncommutative examples use four kinds of categorical data:

- (i) a factorization carrier is paired explicitly with its operation;
- (ii) the Ran category is fixed before its tensor products are used;
- (iii) the block-ordered operad is introduced after a representing chiral operad is chosen;
- (iv) ribbon and stable-curve sewing are derived from nondegenerate Calabi–Yau and modular data.

A comparison between these constructions therefore specifies its source and target categories and the operations preserved by its map.

ENVELOPING BARS AND PLANAR PATH SEQUENCES

The dimension of a bar chain space and the dimension of its homology are different invariants. For an enveloping algebra, the Chevalley comparison gives a Lie-theoretic model for the homology. This makes it possible to compare Motzkin and Riordan path counts with the ordered bar on a specified carrier.

Bar length and Chevalley degree are homological degrees: H_n denotes degree- n homology. The cohomology of the dual Chevalley complex is written H^n . For finite-dimensional weight pieces their dimensions agree.

67.1 CHEVALLEY COMPARISON

Theorem 67.1 (Bar–Chevalley comparison). *Let k be a field of characteristic zero and let $\mathfrak{g}$ be an ordinary Lie algebra over k . Give $U(\mathfrak{g})$ its augmentation killing $\mathfrak{g}$. There is a natural quasi-isomorphism of ordinary conilpotent differential graded coalgebras*

$$\kappa : C_*(\mathfrak{g}) \longrightarrow \text{Bar}^+(U(\mathfrak{g})),$$

where both constructions use direct sums and ordinary tensors over k . It preserves the PBW filtration. If $\mathfrak{g} = \bigoplus_{w \geq 1} \mathfrak{g}_w$ and $[\mathfrak{g}_u, \mathfrak{g}_v] \subseteq \mathfrak{g}_{u+v}$, the map is a quasi-isomorphism in each total weight. It extends to the degreewise product over weights, using completed tensor products by total weight. The completed coalgebras are conilpotent in each weight; no global conilpotence is asserted for infinite sums.

Proof. Put $C_n(\mathfrak{g}) = \Lambda^n \mathfrak{g}$ in homological degree n , with differential

$$d(x_1 \wedge \cdots \wedge x_n) = \sum_{i < j} (-1)^{i+j+1} [x_i, x_j] \wedge x_1 \wedge \cdots \widehat{x_i} \cdots \widehat{x_j} \cdots \wedge x_n.$$

Its coproduct splits the exterior factors between two ordered sublists, with the sign of the corresponding permutation. The ordinary normalized bar differential joins positions $i, i+1$ with sign $(-1)^{i-1}$. Define

$$\kappa(x_1 \wedge \cdots \wedge x_n) = \sum_{\sigma \in S_n} \text{sgn}(\sigma) [x_{\sigma(1)} | \cdots | x_{\sigma(n)}].$$

At degree zero this map preserves the unit. At degree two its differential is $[x_1 x_2 - x_2 x_1] = [[x_1, x_2]]$. In every degree, pair permutations whose merged adjacent entries occur in opposite orders. Their difference is the bracket of those entries, with the displayed exterior sign. This proves the chain-map equation. Cutting an antisymmetrized word amounts to choosing an ordered subset and antisymmetrizing its two subwords. The two procedures give the same permutations and signs, proving the coalgebra identity.

Filter $U(\mathfrak{g})$ by PBW degree, the bar by total PBW degree, and $\Lambda^n \mathfrak{g}$ by degree n . PBW identifies the associated graded algebra with $S(\mathfrak{g})$. The associated graded comparison is the antisymmetrization $\Lambda(\mathfrak{g}) \rightarrow B(S(\mathfrak{g}))$, with zero differential on its source. To verify that it is a quasi-isomorphism, consider the free Koszul resolution $S(V) \otimes \Lambda(V)$ for $V = \mathfrak{g}$. For a basis (x_i) of V , its differential is $\delta = \sum_i x_i \iota_i$, where ι_i contracts the exterior factor x_i . On polynomial degree p and exterior degree q , with $p+q > 0$, the map

$$h = \frac{1}{p+q} \sum_i (x_i \wedge -) \frac{\partial}{\partial x_i}$$

satisfies $\partial h + h \partial = 1$. Only finitely many summands act on each element, even when V is infinite-dimensional. The identity follows by cancellation of cross terms and the Euler identity in the two degrees. Thus this free resolution has augmentation homology k only. The free normalized bar resolution has the same homology by its extra-degeneracy contraction. Antisymmetrization defines an augmentation-preserving comparison between these free resolutions. The simplicial bar sign convention is related to the displayed bar convention by multiplication by $(-1)^n$ in degree n . Free bases permit a reverse comparison and homotopies to be lifted inductively, since every positive-degree cycle in a resolution is a boundary. Tensoring these homotopies with k proves the asserted quasi-isomorphism on associated graded complexes.

Every cycle in the cone of κ has finite PBW filtration degree. A primitive for its highest associated graded part lowers that degree; iteration terminates because the filtration is bounded below. The cone is acyclic, proving the ordinary assertion. For a positive weight grading, PBW degree is at most total weight. The same argument is therefore finite within each weight. Finally, kernels and images of a degree-wise product differential are products of kernels and images, so the componentwise quasi-isomorphisms extend to the stated completion. $\square$

The ordinary comparison does not itself construct a morphism in $\text{FactD}(X)$. For geometric coefficient systems, one must give compatible enveloping and Chevalley constructions on every chart and coherent factorization equivalences for the whole Chevalley complex. On each disjointness locus, with restriction denoted by $j^!$, the required equation is

$$f_{B,I} j^! \kappa_I = (\kappa_{I_1} \boxtimes \kappa_{I_2}) f_{C,I}, \quad I = I_1 \sqcup I_2.$$

Here $f_{C,I}$ and $f_{B,I}$ are the specified Chevalley and bar factorization maps. The bar map may use the signed shuffle comparison under its geometric functor hypotheses. The Chevalley factorization maps, collision compatibility, refinement coherence, and continuity of the completed maps remain additional geometric data to construct or verify. Internal tensor formulas and PBW alone do not provide them.

Calculation 67.2 (Finite-dimensional simple case). For a simple Lie algebra of rank r with exponents $m_1, \dots, m_r$, the Chevalley cohomology is the exterior algebra on primitive classes of degrees $2m_i + 1$. The corresponding homology Poincaré polynomials are

$$\text{Bar}^\perp U(\mathfrak{sl}_2) : 1 + t^3, \quad \text{Bar}^\perp U(\mathfrak{sl}_3) : 1 + t^3 + t^5 + t^8, \quad \text{Bar}^\perp U(\mathfrak{g}_2) : 1 + t^3 + t^{11} + t^{14}.$$

The corresponding Betti sequences for $\mathfrak{sl}_2$ and $\mathfrak{sl}_3$ are 1, 0, 0, 1 and 1, 0, 0, 1, 0, 1, 0, 0, 1. For a finite-dimensional Lie algebra, the Chevalley chain dimensions are $\binom{\dim \mathfrak{g}}{n}$.

For an ordinary augmented algebra A over $\mathbb{C}$, let $I = \ker(A \rightarrow \mathbb{C})$. Its normalized two-sided bar with Chevalley coefficients has homological chain spaces

$$B_0(A) = \mathbb{C}, \quad B_n(A) = I^{\otimes n} \quad (n \geq 1).$$

Here n is bar length, independently of any internal weight or PBW degree. If I has finite dimension d , then $\dim B_n(A) = d^n$; in particular, an augmentation ideal of dimension three gives the chain counts 3^n . In a cohomological convention that places B_n in degree $-n$, its homology is denoted H^{-n} instead of H_n .

For $A = U(\mathfrak{sl}_2)$, PBW gives the basis $e^a h^b f^c$, with $a, b, c \geq 0$, where $[h, e] = 2e$, $[h, f] = -2f$, and $[e, f] = h$. The augmentation ideal has all these basis elements except 1. Consequently $B_n(A)$ is countably infinite-dimensional for every $n \geq 1$. Its total PBW filtration has finite-dimensional associated graded pieces, with generating function

$$\sum_{d \geq 0} \dim \text{gr}_d^{\text{PBW}} B_n(A) z^d = ((1 - z)^{-3} - 1)^n.$$

In particular, 3^n is the dimension in the least possible total PBW degree $d = n$. These least-degree dimensions are 3, 9, 27, 81, 243 for $n = 1, 2, 3, 4, 5$; the full bar chain spaces are infinite-dimensional. The quotient of indecomposables is zero: the perfectness of $\mathfrak{sl}_2$ implies $I/I^2 = 0$.

The Chevalley chain model is different: its degree- n space is $\Lambda^n \mathfrak{sl}_2$, of dimensions 1, 3, 3, 1 in degrees 0, 1, 2, 3. With the convention $d(x \wedge y) = [x, y]$, the map $d : \Lambda^2 \mathfrak{sl}_2 \rightarrow \mathfrak{sl}_2$ sends

$$e \wedge h \mapsto -2e, \quad e \wedge f \mapsto h, \quad h \wedge f \mapsto -2f,$$

and is an isomorphism. The next differential vanishes, since

$$d(e \wedge h \wedge f) = -2e \wedge f - h \wedge h - 2f \wedge e = 0.$$

Together with $d : \mathfrak{sl}_2 \rightarrow \mathbb{C}$ equal to zero, this gives Chevalley homology dimensions 1, 0, 0, 1. The ordinary bar–Chevalley comparison identifies these homology groups with those of $U(\mathfrak{sl}_2)$; it does not identify the two chain spaces.

67.2 THE VIRASORO ROW

The transverse presentation of the Virasoro family is the enveloping algebra of the positive Witt algebra

$$L_1 = \bigoplus_{n \geq 1} \mathbb{C} e_n, \quad [e_i, e_j] = (j - i)e_{i+j},$$

to which Theorem 67.1 applies: L_1 is positively graded with one-dimensional weight spaces, so every weight has finitely many decompositions.

Theorem 67.3 (Pentagonal rigidity). *The bar homology of $U(L_1)$ satisfies*

$$\dim H_n(\text{Bar}^\perp U(L_1)) = 2 \quad \text{for every } n \geq 1,$$

one dimension in each of the weights $(3n^2 - n)/2$ and $(3n^2 + n)/2$. Its graded Euler character is

$$\chi_q(\text{Bar}^\perp U(L_1)) = \prod_{n \geq 1} (1 - q^n), \quad (67.1)$$

and (67.1) is accounted for degree by degree with no residual freedom.

Proof. By Theorem 67.1 the bar of $U(L_1)$ is quasi-isomorphic to $C_*(L_1)$. Each weight subcomplex is finite-dimensional, so its homology has the same dimensions as the weightwise dual cohomology. Goncharova's calculation gives two dimensions in each degree $n \geq 1$, in the weights $(3n^2 \mp n)/2$. Identity (67.1) is the denominator product applied to L_1 , where every root space is one dimensional and even, so $\text{smult}(L_1)_n = 1$. Euler's pentagonal number theorem expands the product as $\sum_{m \in \mathbb{Z}} (-1)^m q^{m(3m-1)/2}$, whose support is exactly $\{(3n^2 \pm n)/2\}$ with coefficient $(-1)^n$ at each of $(3n^2 - n)/2$ and $(3n^2 + n)/2$. Comparing with $\sum_n (-1)^n \dim H_{n,w}$, each weight of the Euler character is saturated by a single class in the degree Goncharova predicts, so no further classes and no cancellation are available. $\square$

For $n = 1, 2, 3, 4$, the nonzero homology groups occupy the weight pairs (1, 2), (5, 7), (12, 15), (22, 26), one dimension in each. The corresponding nonzero coefficients of $\prod_{n \geq 1} (1 - q^n)$ occur at 0, 1, 2, 5, 7, 12, 15, 22, 26 with signs +, −, −, +, +, −, −, +, +.

67.3 PLANAR PATH SEQUENCES

A Motzkin path is a lattice path from $(0, 0)$ to $(n, 0)$, staying at nonnegative height, with steps $(1, 1)$, $(1, 0)$, and $(1, -1)$. Let $M(n)$ count all such paths of length n , and let $R(n)$ count those with no horizontal step at height zero. Both counts include the empty path: $M(0) = R(0) = 1$, while $M(1) = 1$ and $R(1) = 0$. For $n \geq 1$ consider the two sequences

$$a_n = M(n+1) - M(n) = 1, 2, 5, 12, 30, 76, \dots, \quad b_n = R(n+3) = 3, 6, 15, 36, 91, \dots$$

Proposition 67.4 (Motzkin and Riordan identities). *For every $n \geq 0$,*

$$M(n) = R(n) + R(n+1), \quad M(n+1) - M(n) = R(n+2) - R(n).$$

The ordinary generating functions of the two path counts, and hence those of a_n and b_n , belong to $\mathbb{Q}(z, \sqrt{1-2z-3z^2})$.

Proof. Work in the formal power-series ring $\mathbb{Q}[[z]]$, and write $M(z) = \sum_{n \geq 0} M(n)z^n$ and $R(z) = \sum_{n \geq 0} R(n)z^n$. A nonempty Motzkin path either starts with a horizontal step followed by an arbitrary Motzkin path, or has a unique first-return decomposition $UPDQ$, where U and D are the up and down steps, P and Q are arbitrary Motzkin paths, and P is drawn one unit above the horizontal axis. Thus

$$M = 1 + zM + z^2M^2.$$

A nonempty path counted by R has no initial horizontal step. Its first return similarly has the form $UPDQ$, with P arbitrary and Q counted by R . Consequently

$$R = 1 + z^2MR, \quad R = (1 - z^2M)^{-1}.$$

The inverses here exist because their denominators have constant term one. Using $z^2M^2 = M - 1 - zM$, compute

$$(1 + zM)(1 - z^2M) = 1 + zM - z^2M - z^3M^2 = 1 + z.$$

It follows that $(1 + z)R = 1 + zM$. Comparing coefficients of z^{n+1} gives $R(n+1) + R(n) = M(n)$ for all $n \geq 0$. Subtracting consecutive identities gives the difference formula.

Solving the quadratic equation for the formal series with constant term one gives

$$M(z) = \frac{1 - z - \sqrt{1 - 2z - 3z^2}}{2z^2}, \quad R(z) = \frac{1 + zM(z)}{1 + z},$$

where the square root has constant term one. The apparent pole in the formula for M is removable as a formal series. Finite index shifts and differences preserve the indicated quadratic function field, proving the last assertion. These identities are consequences solely of the path decompositions. $\square$

Remark 67.5 (Low-degree homology). Theorem 67.3 gives homology dimensions 2, 2, 2 in degrees 1, 2, 3, whereas a_1, a_2, a_3 are 1, 2, 5. These triples agree only in degree two. The degree-one discrepancy already excludes equality of the full homology dimension sequences.

Theorem 67.6 (Enveloping bars and planar path counts). *Let $B_\bullet(A)$ be the ordinary normalized bar over $\mathbb{C}$, with trivial coefficients and homological degree given by bar length. With $a_n = M(n+1) - M(n)$ and $b_n = R(n+3)$ indexed by $n \geq 1$ as above,*

$$\dim H_1 B_\bullet(U(\mathfrak{sl}_2)) = 0 \neq b_1 = 3, \quad \dim H_1 B_\bullet(U(L_1)) = 2 \neq a_1 = 1,$$

where $L_1 = \bigoplus_{m \geq 1} \mathbb{C}e_m$ and $[e_i, e_j] = (j - i)e_{i+j}$. Hence neither sequence is the bar homology dimension sequence of the corresponding enveloping algebra. Neither is its full bar chain dimension sequence: in both cases B_n is countably infinite-dimensional for every $n \geq 1$.

Proof. For any augmented algebra A with augmentation ideal I , the bar differential out of $B_1 = I$ is zero and the image from $B_2 = I \otimes I$ is I^2 . Therefore

$$H_1 B_\bullet(A) = I/I^2.$$

For an ordinary Lie algebra $\mathfrak{g}$, the inclusion of generators induces an isomorphism

$$\mathfrak{g}/[\mathfrak{g}, \mathfrak{g}] \xrightarrow{\sim} I/I^2, \quad I = \ker(U(\mathfrak{g}) \rightarrow \mathbb{C}).$$

Indeed the generators span I/I^2 , and $[x, y] = xy - yx$ vanishes there. For injectivity, equip $\mathbb{C} \oplus \mathfrak{g}/[\mathfrak{g}, \mathfrak{g}]$ with the square-zero product on its second summand. The quotient map from $\mathfrak{g}$ to that summand is a Lie map into the commutator Lie algebra of this associative algebra. The universal property of $U(\mathfrak{g})$ extends it to an augmented algebra map, which kills I^2 and provides an inverse on the displayed quotient.

The brackets $[h, e] = 2e$, $[h, f] = -2f$, and $[e, f] = h$ show that $\mathfrak{sl}_2$ equals its commutator algebra. For L_1 , every nonzero bracket has weight at least three. Conversely, for $m \geq 3$,

$$e_m = \frac{1}{m-2} [e_1, e_{m-1}].$$

Thus $[L_1, L_1] = \bigoplus_{m \geq 3} \mathbb{C} e_m$, and its abelianization has basis the classes of e_1, e_2 , in weights one and two. This proves both degree-one assertions without using any higher-degree calculation.

PBW gives a countably infinite basis of the augmentation ideal of either enveloping algebra. In the L_1 case the basis consists of nonempty ordered monomials in the countable set $e_1, e_2, \dots$, each monomial having finite length. Finite tensor powers of either augmentation ideal are therefore countably infinite-dimensional. The path counts are finite in every degree, so they cannot count these full chain spaces. $\square$

In particular, a complex whose homology dimensions are b_n cannot be quasi-isomorphic to $B_\bullet(U(\mathfrak{sl}_2))$, and one whose homology dimensions are a_n cannot be quasi-isomorphic to $B_\bullet(U(L_1))$, with the stated degree convention. A discrepancy of chain dimensions alone would not imply this conclusion; the degree-one homology discrepancy does.

The weight grading gives a different, finite comparison. An ordered PBW monomial of weight w specifies a partition of w , so for $n \geq 1$,

$$\sum_{w \geq 0} \dim B_n(U(L_1))_w q^w = \left(\prod_{m \geq 1} (1 - q^m)^{-1} - 1 \right)^n.$$

Each coefficient is finite. Thus a weightwise dimension, a dimension after a specified weight cutoff, and an ungraded dimension are distinct invariants. In a weight completion, dimensions must likewise be interpreted weightwise; the countable algebraic dimensions above refer to the ordinary direct-sum enveloping algebras.

Remark 67.7 (Chiral coefficients and transverse multiplication). Independently of these scalar calculations, a Virasoro chiral coefficient does not determine the transverse bar: a fixed coefficient may be combined with free, square-zero, enveloping, or quantum-plane transverse algebras whose bars differ, while the coefficient and the central charge are unchanged. A chain-level comparison with an ordered bar therefore also requires a specified transverse multiplication and a quasi-isomorphism from the proposed complex to that bar.

THE OBSTRUCTION CONSTITUTION

Each statement below names what is *not* determined, and therefore what must be supplied. Without them the preceding theorems admit false corollaries.

Proposition 68.1 (Where a mixed distributive law can live). *The sector on which a mixed distributive law between $\otimes^!$ and $\star^{\text{ch}}$ is geometrically defined is the common-decomposition sector, the image of the aligned idempotent e_{al} . Off that sector the two products admit no interchange, only the aligned retraction, so no distributive law can be written.*

Proof. A distributive law requires a natural transformation interchanging the two products in a fixed direction, subject to the coherence conditions of each. The canonical comparison in the additive model is the aligned inclusion ι with retraction π , and $\pi\iota = \text{id}$ while $\iota\pi = e_{\text{al}} \neq \text{id}$ in general. A natural transformation subject to both coherence conditions therefore exists only where e_{al} acts as the identity. $\square$

This locates a mixed law without constructing one. No example in the present corpus exhibits one; Zamolodchikov–Faddeev algebras are the natural candidate, since their spectral variable belongs to the chiral direction while their word belongs to the transverse direction.

Theorem 68.2 (No quantum group from dimension loss). *The dimension of the kernel of $V^{\otimes n} \rightarrow \text{Sym}^n V$ determines no R -matrix, associator, or quantum group. That kernel carries no multiplication, coproduct, monodromy, or coherence equation.*

Proof. The kernel is a representation of Σ_n and nothing more. Two algebras with the same underlying graded vector space, hence the same kernel dimensions in every arity, may have different multiplications and inequivalent representation categories; a free algebra and a quadratic quotient with the same Hilbert series is an example. A quantum-group structure is a coproduct together with coherence data, none of which is a function of a dimension. $\square$

Theorem 68.3 (No modular object from topology alone). *The identity $\partial^2 = \circ$ on $\overline{\mathcal{M}}_{g,n}$ assigns no coefficient operations to boundary strata. A modular master element exists only after cyclic contractions and clutching-compatible coefficient maps have been supplied.*

Theorem 68.4 (No bulk from a boundary centre). *The derived centre is terminal among algebraic central actions on the boundary algebra. It does not detect a decoupled bulk sector, so a physical bulk is not determined by the boundary chiral algebra unless a bulk-to-centre comparison is supplied and proved to be an equivalence.*

Proof. Terminality is the universal property of the centre. It is a statement about the boundary category and is insensitive to any sector acting trivially on it, so a bulk with trivial boundary action is invisible. $\square$

Theorem 68.5 (No classification by central charge). *Central charge determines the Virasoro anomaly coefficient. It does not determine the chiral operation, the representation category, the braid monodromy, or the higher-genus conformal blocks. Distinct holomorphic theories at equal central charge exist.*

Theorem 68.6 (No identification of anomalies of different types). *A current-algebra level, a Virasoro central charge, a determinant-line curvature, a BRST ghost-number anomaly, and a Borchers-product weight are invariants in different theories, living in different deformation complexes. They may be related by a theorem in a particular family. They are not one universal scalar, and no identity among them follows from numerical agreement.*

Proof. Each is a class in a distinct complex: an Atiyah-algebra extension class, a Virasoro cocycle, the curvature of a determinant line, a class in H^1 or H^2 of a BRST convolution complex, and a weight in the character lattice of an automorphic form. A map between two of these complexes is additional data; in its absence, equality of numbers is a coincidence of values, not an identification of classes. $\square$

Theorem 68.7 (No unnamed comparison of bars). *An asserted equality $\text{Bar}^\perp(A) = \text{Bar}_{\text{Ass}}^{\text{ch}}(A)$ has no meaning without a functor carrying the pointwise multiplication problem to the chiral-convolution multiplication problem. The two have different underlying gradings, counting interval subdivisions and collision trees respectively. Agreement of Euler characteristics is not a comparison map.*

Theorem 68.8 (A scalar sequence is not homology data). *A sequence of integers becomes bar homology only after a chain complex, a differential, a grading convention, and a rank computation or quasi-isomorphism are specified. Matching initial values, discriminants, or growth rates constructs no such map.*

Proof. Chain-space dimensions and homology dimensions differ by the ranks of the differentials, which are not functions of the chain dimensions. Calculation 67.2 exhibits an algebra whose bar chain dimensions are 3, 9, 27, 81, 243 and whose bar homology dimensions are 1, 0, 0, 1; Proposition 67.4 exhibits two sequences with the same discriminant and growth rate that are transforms of one another and of neither homology. $\square$

Theorem 68.6 together with the central-charge statement retracts any programme treating a tuple of scalars drawn from these sources as a single classifying invariant, and any trace identity asserted among them without a named comparison morphism.

BOOK ENTRANCE: THE IGUSA SQUARE ROOT

The Jacobi seed, Borcherds denominator, root superalgebra, local-wrapped currents, quantum group, Hall model, and Fredholm realization.

BOOK **XV**

The automorphic determinant

THE PROTECTED JACOBI INPUT

69.1 THE K_3 ELLIPTIC GENUS

Let S be a K_3 surface. The right-moving supersymmetric index of a unitary $\mathcal{N} = (4, 4)$ sigma model is the elliptic genus

$$Z_{K_3}(\tau, z) = 2\phi_{0,1}(\tau, z).$$

The normalized weak Jacobi form has weight zero, index one, and Fourier expansion

$$\phi_{0,1}(\tau, z) = \sum_{n \geq 0} \sum_{\ell \in \mathbb{Z}} f(n, \ell) q^n r^\ell.$$

In Eichler–Zagier normalization,

$$\phi_{0,1}(\tau, z) = 4 \sum_{a=2}^4 \frac{\mathcal{I}_a(\tau, z)^2}{\mathcal{I}_a(\tau, 0)^2} \in J_{0,1}^w$$

[29]. The coefficient depends on the discriminant $4n - \ell^2$ and on ℓ modulo 2.

Theorem 69.1 (Exact low Fourier coefficients). *The first four q -rows are*

$$\begin{aligned} \phi_{0,1} = & r^{-1} + 10 + r \\ & + q(10r^{-2} - 64r^{-1} + 108 - 64r + 10r^2) \\ & + q^2(r^{-3} + 108r^{-2} - 513r^{-1} + 808 - 513r + 108r^2 + r^3) \\ & + q^3(-64r^{-3} + 808r^{-2} - 2752r^{-1} + 4016 \\ & \quad - 2752r + 808r^2 - 64r^3) + O(q^4). \end{aligned}$$

In particular,

$$f(0, 0) = 10, \quad \frac{f(0, 0)}{2} = 5.$$

Proof. Insert the theta-series expansions into the preceding quotient formula. Work with $t = q^{1/8}$ and $x = r^{1/2}$; each quotient is an exact formal power series in t with Laurent-polynomial coefficients in x . The fractional powers cancel in the sum. Division through t^{24} gives the displayed rows. The arithmetic calculation is reproduced in Appendix 100. $\square$

69.2 THE VIRTUAL HALF-INDEX

For every (n, ℓ) choose a finite-dimensional super vector space $U_{n,\ell}$ satisfying

$$\text{sdim } U_{n,\ell} = f(n, \ell).$$

The integer $f(n, \ell)$ is its superdimension. In the ordinary Grothendieck group of finite-dimensional super vector spaces, the even and odd dimensions remain separate. The signed integer is their image under superdimension. The determinant depends only on that image.

Lemma 69.2 (Fock character and Berezinian). *Let $V = \bigoplus_{\gamma} V_{\gamma}$ be a locally finite graded super vector space. Then*

$$\mathrm{sch}(\mathrm{Sym}(V_{\bar{0}}) \otimes \bigwedge(V_{\bar{1}})) = \prod_{\gamma} (1 - x^{\gamma})^{-\mathrm{sdim} V_{\gamma}},$$

and

$$\mathrm{sdet}_V(1 - x) = \prod_{\gamma} (1 - x^{\gamma})^{\mathrm{sdim} V_{\gamma}}.$$

Proof. An even one-dimensional summand contributes the geometric series $(1 - x^{\gamma})^{-1}$. An odd one-dimensional summand contributes $1 - x^{\gamma}$. Multiplication over a homogeneous basis proves both identities. $\square$

THE BORCHERDS PRODUCT

70.1 THE EFFECTIVE CHAMBER

Define

$$\begin{aligned} \Gamma_{\text{eff}} = \{ & (n, \ell, m) \in \mathbb{Z}^3 : m > 0, n \geq 0 \} \\ & \cup \{ (n, \ell, 0) : n > 0 \} \cup \{ (0, \ell, 0) : \ell < 0 \}. \end{aligned}$$

The ordering is lexicographic in $(m, n, -\ell)$ and is precisely the ordering selected by the cusp chamber.

For $\gamma = (n, \ell, m)$ put

$$V_\gamma = U_{nm, \ell}, \quad \text{sdim } V_\gamma = f(nm, \ell).$$

Definition 70.1 (Monic virtual determinant). The monic determinant of the half-index is

$$D_5(Z) = q^{1/2} r^{1/2} s^{1/2} \text{sdet}_V(1 - x) = q^{1/2} r^{1/2} s^{1/2} \prod_{\Gamma_{\text{eff}}} (1 - q^n r^\ell s^m)^{f(nm, \ell)}.$$

Theorem 70.2 (Borcherds–Gritsenko–Nikulin product). *The preceding product converges normally on the chosen cusp chamber and extends to a holomorphic genus-two modular form of weight five with a character of order two. It is the monic Igusa form D_5 . The theta-normalized form is*

$$\Delta_5^\theta = 2^6 D_5.$$

Proof. The singular theta lift of the vector-valued form corresponding to $\phi_{0,1}$ produces the product, divisor, weight, and automorphic character. The weight is $f(0, 0)/2 = 5$. The genus-two identification with the Igusa form and the product formula are the Gritsenko–Nikulin theorems [7, 8, 38, 40]. The product of the ten even theta constants has coefficient 2^6 at $q^{1/2} r^{1/2} s^{1/2}$, fixing the scalar. $\square$

70.2 THE THETA CONSTANT NORMALIZATION

Let $\mathfrak{E}$ be the ten even genus-two characteristics. Define

$$\Delta_5^\theta(Z) = \prod_{\mathfrak{m} \in \mathfrak{E}} \mathfrak{g}[\mathfrak{m}](Z, 0).$$

The cusp leading terms of the ten factors contribute six factors of 2 and total exponent $(1/2, 1/2, 1/2)$. Thus

$$[q^{1/2} r^{1/2} s^{1/2}] \Delta_5^\theta = 64.$$

This coefficient selects a canonical scalar normalization of the automorphic section.

Proposition 70.3 (Ten-factor leading-term calculation). *Group the ten even characteristics by the first half-characteristic $a \in (\mathbb{Z}/2)^2$. Their leading contributions in the chosen cusp chamber are*

a	#	scalar	monomial
(0, 0)	4	1	1
(1, 0)	2	2	$q^{1/8}$
(0, 1)	2	2	$s^{1/8}$
(1, 1)	2	2	$q^{1/8} r^{1/4} s^{1/8}$.

Multiplication gives

$$1^4 2^2 2^2 2^2 q^{1/2} r^{1/2} s^{1/2} = 64 q^{1/2} r^{1/2} s^{1/2}.$$

Proof. Insert each characteristic into the genus-two theta series and retain the unique terms of least order in the lexicographic cusp valuation $(m, n, -\ell)$. For $a = (0, 0)$ the zero lattice vector contributes one. For each nonzero a , the two shortest lattice representatives are exchanged by sign and contribute the scalar two. Their quadratic exponents are the entries in the final column. Summing the exponents and multiplying the six factors of two gives the formula. $\square$

Proposition 70.4 (Boundary factor of the Borchers product). *The $m = 0$ part of the monic product is*

$$\prod_{n>0, \ell \in \mathbb{Z}} (1 - q^n r^\ell)^{f(0, \ell)} \prod_{\ell < 0} (1 - r^\ell)^{f(0, \ell)} = (1 - r^{-1}) \prod_{n \geq 1} (1 - q^n r^{-1}) (1 - q^n)^{10} (1 - q^n r).$$

Consequently the first Fourier–Jacobi coefficient is

$$q^{1/2} r^{1/2} (1 - r^{-1}) \prod_{n \geq 1} (1 - q^n r^{-1}) (1 - q^n)^{10} (1 - q^n r) = \eta(\tau)^9 \mathfrak{g}_{\text{II}}(\tau, z),$$

where the odd theta function is fixed by the exact convention

$$\mathfrak{g}_{\text{II}}(\tau, z) = -q^{1/8} r^{-1/2} \prod_{n \geq 1} (1 - q^{n-1} r) (1 - q^n r^{-1}) (1 - q^n).$$

Proof. The q^0 row of $\phi_{0,1}$ has exactly three nonzero coefficients: $f(0, -1) = 1$, $f(0, 0) = 10$, and $f(0, 1) = 1$. Substitution into the $m = 0$ factors gives the first formula. The Jacobi triple product and $\eta(\tau) = q^{1/24} \prod_{n \geq 1} (1 - q^n)$ give the second, including its leading sign and monomial. $\square$

Corollary 70.5 (The scalar square). *The monic weight-ten form is*

$$\chi_{10} = D_5^2 = 2^{-12} \Delta_5^{\theta^2}.$$

Its automorphic character is trivial.

Proof. The identity $\Delta_5^\theta = 2^6 D_5$ gives the scalar factor. The character of D_5 has order two, so its square is trivial. $\square$

DIVISOR, CHARACTER, AND THE MEANING OF THE SQUARE ROOT

The diagonal Humbert divisor is

$$H_1 = \{Z \in \mathbb{H}_2 : z = o\} / \mathrm{Sp}_4(\mathbb{Z}).$$

Theorem 71.1 (Diagonal divisor). *The divisor of D_ς is the $\mathrm{Sp}_4(\mathbb{Z})$ -orbit of H_1 , with multiplicity one. The form transforms by the Maass character ν_ς of order two, and $\chi_{10} = D_\varsigma^2$ is scalar-valued.*

Proof. This is the defining diagonal-divisor property of the Igusa square root and the first row of the Cléry–Gritsenko classification [16, 45, 46]. $\square$

Real roots of the denominator algebra define Weyl reflections. Imaginary roots contribute root spaces and product exponents. This division of labor is intrinsic to a Borchers algebra: the Weyl group is generated by real simple roots, while the imaginary simple roots enter the correction term of the denominator identity.

71.1 AUTOMORPHIC SECTION, ORIENTATION LINE, AND PFAFFIAN OPERATOR

An automorphic square root is a section of a line bundle with character. An orientation is a square root of a determinant line. A Pfaffian operator is a skew family whose regularized Pfaffian is a section of that orientation line. These form a sequence of successive refinements:

$$\begin{aligned} &\text{automorphic section} \longrightarrow \text{determinant-line identification} \\ &\longrightarrow \text{oriented square-root line} \longrightarrow \text{Fredholm family and Pfaffian section.} \end{aligned}$$

Proposition 71.2 (Logical separation of the four levels). *The product theorem constructs the first object in the sequence. A determinant-line comparison constructs the second. A coherent square root of that line constructs the third. A first-order operator requires a symbol, domain, ellipticity or Fredholm condition, and regularization, and thereby produces the fourth refinement.*

Proof. Each stage adds a specified geometric structure. A determinant realization adds an isomorphism from a geometric determinant line to the automorphic line. An oriented realization adds a coherent square-root line and its parity refinement. An operator realization adds a principal symbol, a Fredholm domain, a spectrum, and a regularization whose Pfaffian is the resulting section. $\square$

The title names the monic automorphic section D_ς , characterized by the exact identity $D_\varsigma^2 = \chi_{10}$. Part IV formulates an analytic Pfaffian realization as a further geometric refinement.

BOOK XVI

The denominator algebra

THE LORENTZIAN ROOT LATTICE

Let

$$L = \mathbb{Z}f_2 \oplus \mathbb{Z}f_3 \oplus \mathbb{Z}f_{-2}$$

with bilinear form

$$(f_2, f_{-2}) = -1, \quad (f_3, f_3) = 2,$$

and all remaining basis pairings zero. Its signature is $(2, 1)$. Define the type-II root lattice

$$L_{II} = 2\mathbb{Z}f_2 \oplus \mathbb{Z}f_3 \oplus 2\mathbb{Z}f_{-2} \cong U(4) \oplus \langle 2 \rangle.$$

The Gram-index map is

$$\alpha : \Gamma_{\text{ind}} = \mathbb{Z}^3 \longrightarrow L_{II}, \quad \alpha(n, \ell, m) = 2nf_2 - \ell f_3 + 2mf_{-2}.$$

Then

$$(\alpha(n, \ell, m), \alpha(n', \ell', m')) = -2(2nm' + 2n'm - \ell\ell'),$$

and

$$(\alpha(n, \ell, m), \alpha(n, \ell, m)) = -2(4nm - \ell^2).$$

72.1 REAL SIMPLE ROOTS AND WEYL VECTOR

Put

$$\delta_1 = 2f_2 - f_3, \quad \delta_2 = 2f_{-2} - f_3, \quad \delta_3 = f_3.$$

Proposition 72.1 (Type-II Cartan matrix). *The Gram matrix of the real simple roots is*

$$((\delta_i, \delta_j))_{i,j=1}^3 = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}.$$

The reflections

$$s_i(x) = x - (x, \delta_i)\delta_i$$

are integral involutions preserving L_{II} and its form.

Proof. Substitution into the basis pairing gives the matrix. Since $(\delta_i, \delta_i) = 2$, the reflection formula has integral coefficients. Direct expansion gives $s_i^2 = 1$ and $(s_i x, s_i y) = (x, y)$. $\square$

The Weyl vector is

$$\rho = f_2 - \frac{1}{2}f_3 + f_{-2}, \quad (\rho, \delta_i) = -1.$$

In the Gram-index variables, $e^{-\rho} = q^{1/2}r^{1/2}s^{1/2}$.

THE GRITSENKO–NIKULIN LIE SUPERALGEBRA

The Gritsenko–Nikulin construction specifies a generalized Kac–Moody Lie superalgebra $\mathfrak{g}_{\Delta_5}$, with real simple roots $\delta_1, \delta_2, \delta_3$ and the imaginary correction attached to $\phi_{0,1}$. Its generators and relations supply the bracket whose denominator is compared with the Borchers product.

Theorem 73.1 (Root supermultiplicities and denominator). *On the active positive support,*

$$\text{smult } \mathfrak{g}_{\Delta_5, \alpha(n, \ell, m)} = f(nm, \ell).$$

In the Lorentzian coordinate convention of Gritsenko–Nikulin,

$$\text{den}(\mathfrak{g}_{\Delta_5}) = D_5(2Z) = 2^{-6} \Delta_5^\theta(2Z).$$

Proof. The real roots and Weyl chamber are the type-II case of the Gritsenko–Nikulin construction. The imaginary simple-root correction is chosen so that its Weyl–Kac–Borchers product has exponent $f(nm, \ell)$. The resulting product is the doubled-variable monic Igusa form [38–40]. $\square$

The theorem gives a Lie superalgebra with a bracket, Cartan, Weyl group, root parity, and denominator. The scalar product alone records the signed dimensions of the root spaces; the Gritsenko–Nikulin construction supplies the algebraic operations.

THE ORDERED BAR AS THE DENOMINATOR COMPLEX

Let $\mathfrak{n}_{\Delta_s,+}$ be the positive nilpotent subalgebra of $\mathfrak{g}_{\Delta_s}$. Fix a proper integral height h on the positive root cone and define

$$F^{>H} \mathfrak{n}_{\Delta_s,+} = \bigoplus_{h(\alpha) > H} \mathfrak{g}_{\Delta_s, \alpha}, \quad \mathfrak{n}_{\Delta_s,+}^{(H)} = \mathfrak{n}_{\Delta_s,+} / F^{>H} \mathfrak{n}_{\Delta_s,+}.$$

Height additivity makes $F^{>H} \mathfrak{n}_{\Delta_s,+}$ an ideal. Properness and local finiteness make $\mathfrak{n}_{\Delta_s,+}^{(H)}$ finite-dimensional and nilpotent.

Theorem 74.1 (Height-complete Bar–Chevalley comparison). *For every H , antisymmetrization gives a natural quasi-isomorphism of conilpotent dg coalgebras*

$$\text{Bar}(U(\mathfrak{n}_{\Delta_s,+}^{(H)})) \simeq C_*(\mathfrak{n}_{\Delta_s,+}^{(H)}).$$

The quotient maps preserve these comparisons and induce a quasi-isomorphism in the separated height-complete category

$$\text{Bar}_h^\wedge(U(\mathfrak{n}_{\Delta_s,+})) := \varprojlim_H \text{Bar}(U(\mathfrak{n}_{\Delta_s,+}^{(H)})) \simeq C_{*,h}^\wedge(\mathfrak{n}_{\Delta_s,+}) := \varprojlim_H C_*(\mathfrak{n}_{\Delta_s,+}^{(H)}).$$

Its root-graded Euler supercharacter is

$$\chi_{\text{root}} C_{*,h}^\wedge(\mathfrak{n}_{\Delta_s,+}) = \prod_{\alpha > 0} (1 - e^{-\alpha})^{\text{sdim } \mathfrak{g}_{\Delta_s, \alpha}}.$$

Proof. For each H , the antisymmetrization map sends $\text{Sym}(\mathfrak{n}_{\Delta_s,+}^{(H)}[1])$ to the normalized bar of $U(\mathfrak{n}_{\Delta_s,+}^{(H)})$. The PBW filtrations have the same associated-graded Koszul resolution of $\text{Sym}(\mathfrak{n}_{\Delta_s,+}^{(H)})$, so antisymmetrization is a filtered quasi-isomorphism. Naturality gives a morphism of inverse systems. Every fixed root degree is visible in all sufficiently large height quotients and is unchanged thereafter; hence the inverse-limit map is a quasi-isomorphism coefficientwise. An even root vector becomes odd after the Chevalley shift and contributes $1 - e^{-\alpha}$; an odd root vector becomes even and contributes its reciprocal. Multiplication over the finite-dimensional root spaces gives the product. $\square$

Let $\mathbb{T}_{L_I}$ be the formal root torus and write Z_{root} for its logarithmic coordinate, normalized by

$$e^{-\alpha(n,\ell,m)} = Q^n R^\ell S^m, \quad e^{-\rho} = Q^{1/2} R^{1/2} S^{1/2}.$$

Corollary 74.2 (Bar denominator identity). *On the formal root torus,*

$$e^{-\rho} \chi_{\text{root}} C_{*,h}^\wedge(\mathfrak{n}_{\Delta_s,+}) = D_5(Z_{\text{root}}).$$

The Gritsenko–Nikulin embedding of the root torus into the Siegel period domain sends Z_{root} to $2Z$ [38, 40]. Hence

$$e^{-\rho} \chi_{\text{root}} C_{*,h}^\wedge(\mathfrak{n}_{\Delta_s,+}) = \text{den}(\mathfrak{g}_{\Delta_s}) = D_5(2Z)$$

in Siegel period coordinates.

Proof. Multiply the Euler product in the preceding theorem by the Weyl monomial $e^{-\rho}$. The denominator theorem identifies the result on the root torus, and the stated embedding gives the Siegel-coordinate formula. $\square$

The denominator is therefore the Weyl-shifted Euler supercharacter of the height-complete interval bar of the positive enveloping algebra.

FINITE-HEIGHT CURRENTS

Retain the finite-height quotients $\mathfrak{n}_{\Delta_s,+}^{(H)}$ of the preceding chapter. Define the current Lie conformal superalgebra

$$\text{Cur}(\mathfrak{n}_{\Delta_s,+}^{(H)}) = \mathbb{k}[\partial] \otimes \mathfrak{n}_{\Delta_s,+}^{(H)}, \quad [a_\lambda b] = [a, b],$$

and its universal chiral enveloping algebra

$$V_H = U^{\text{ch}}\left(\text{Cur}_E(\mathfrak{n}_{\Delta_s,+}^{(H)})\right).$$

Theorem 75.1 (Height-complete current envelope). *The quotient maps induce a coefficientwise inverse limit*

$$\widehat{V}_{\Delta_s} = \varprojlim_H V_H.$$

Every V_H is PBW-admissible as an associative chiral algebra, the current generators are primitive for the standard current coproduct, and

$$\text{sch } \widehat{V}_{\Delta_s} = \prod_{j \geq 1} \prod_{\alpha > 0} (1 - t^j e^{-\alpha})^{-\text{sdim } \mathfrak{g}_{\Delta_s, \alpha}}.$$

Moreover,

$$\text{Bar}_{\text{Ass}}^{\text{ch}}(V_H) \simeq \text{Bar}_{\text{Lie}}^{\text{ch}}\left(\text{Cur}_E(\mathfrak{n}_{\Delta_s,+}^{(H)})\right),$$

and these equivalences pass to the separated height completion.

Proof. PBW orders the negative modes $a(-j)$, $j \geq 1$. Even modes contribute symmetric powers and odd modes contribute exterior powers, giving the product. Proper height makes every coefficient stationary. The current coproduct sends $a(z)$ to $a(z) \otimes 1 + 1 \otimes a(z)$. Chiral antisymmetrization identifies the associative-chiral bar of the chiral enveloping algebra with the chiral Chevalley coalgebra; the complete separated PBW filtration is coefficientwise finite. $\square$

75.1 CLASSICAL LOCAL THEORY AND EXACT QUANTUM ZERO MODES

For every H , the finite-dimensional nilpotent Lie superalgebra $\mathfrak{n}_{\Delta_s,+}^{(H)}$ defines a cotangent BF theory on $E \times \mathbb{R}$. Its local dg Lie algebra is

$$\mathcal{L}_H = \Omega^{\circ,*}(E) \widehat{\otimes} \Omega^*(\mathbb{R}) \otimes \mathfrak{n}_{\Delta_s,+}^{(H)}$$

and the classical action is

$$S_H(A, B) = \int_{E \times \mathbb{R}} \Omega_E \wedge \left\langle B, (\bar{\partial} + d_{\mathbb{R}})A + \frac{1}{2}[A, A] \right\rangle.$$

Theorem 75.2 (Finite-height BF realization). *The action S_H satisfies the classical master equation and defines a classical holomorphic–topological factorization algebra. Harmonic reduction along a compact interval produces the finite-dimensional cotangent BV theory of $\mathfrak{n}_{\Delta,+}^{(H)}$. The positive-root height grading makes every even adjoint endomorphism strictly height-increasing and every odd adjoint endomorphism odd; hence the modular supercharacter vanishes and the reduced theory satisfies the quantum master equation.*

Choose the standard complementary Lagrangian boundary polarizations, fixing the $\mathfrak{n}_{\Delta,+}^{(H)}[1]$ coordinate at one endpoint and its cotangent coordinate at the other. The algebraic boundary quantization, the interval amplitude between the two augmentation states, and the holomorphic boundary current envelope are then

$$U(\mathfrak{n}_{\Delta,+}^{(H)}), \quad \mathbf{I} \otimes_{U(\mathfrak{n}_{\Delta,+}^{(H)})} \mathbf{I} \simeq C_*(\mathfrak{n}_{\Delta,+}^{(H)}), \quad V_H.$$

The height quotient maps preserve the polarizations and yield the inverse systems

$$\widehat{U}_h(\mathfrak{n}_{\Delta,+}), \quad C_{*,h}^\wedge(\mathfrak{n}_{\Delta,+}), \quad \widehat{V}_{\Delta,+}.$$

Proof. The classical master equation is the Jacobi identity paired with invariance of the cotangent pairing. Elliptic descent gives the classical factorization algebra. Harmonic reduction along the interval gives a finite-dimensional BV space. If x is even of positive root height, ad_x raises height and is nilpotent on the finite quotient, so its restrictions to the even and odd summands both have trace zero; if x is odd, ad_x is odd and has supertrace zero. Thus the modular character vanishes and the finite BV quantum master equation follows.

The chosen boundary polarization quantizes the linear Kirillov–Kostant boundary bracket to $U(\mathfrak{n}_{\Delta,+}^{(H)})$. Gluing the two augmentation states gives the two-sided derived bar, and the PBW antisymmetrization identifies it with $C_*(\mathfrak{n}_{\Delta,+}^{(H)})$. Holomorphic descendants generate $U^{\text{ch}}(\text{Cur}_E \mathfrak{n}_{\Delta,+}^{(H)}) = V_H$. All maps preserve root height, so they pass coefficientwise to the separated inverse limits. $\square$

BOOK XVII

The compact scalar character

REDUCED CURVE COUNTING ON $K_3 \times E$

Let S be a nonsingular projective K_3 surface and E an elliptic curve. For a primitive class $\beta_h \in H_2(S, \mathbb{Z})$ with $\beta_h^2 = 2h - 2$, let $N_{g,h,d}$ be the reduced disconnected Gromov–Witten invariant of class (β_h, d) with the standard incidence insertion. Define

$$\mathcal{Z}(u, q, s) = \sum_{g,h,d} N_{g,h,d} u^{2g-2} q^{h-1} s^{d-1}.$$

Theorem 76.1 (Primitive Igusa cusp-form theorem). *Under $r = e^{iu}$, the partition function is the Laurent expansion*

$$\mathcal{Z}(u, q, s) = -\frac{1}{\chi_{10}(r, q, s)}.$$

Equivalently, in theta normalization,

$$\mathcal{Z} = -2^{12} \Delta_5^{\theta^{-2}}.$$

Proof. Oberdieck and Pixton identify the primitive reduced partition function with the Laurent expansion of $-\chi_{10}^{-1}$ [62]; their normalization is the monic Borcherds product χ_{10} . The second formula is the identity $\chi_{10} = 2^{-12} \Delta_5^{\theta^2}$. $\square$

The theorem identifies the numerical generating series in its stated primitive sector. The Borcherds root algebra supplies a separate algebraic character to compare with that series.

Corollary 76.2 (Exact character comparison). *Let $j : \mathbb{H}_2 \rightarrow \mathbb{H}_2$ be the half-coordinate map $j(Z) = Z/2$, and let*

$$\text{Fock}(\mathfrak{n}_{\Delta_5,+}) = \text{Sym}(\mathfrak{n}_{\Delta_5,+_{\bar{0}}}) \otimes \bigwedge (\mathfrak{n}_{\Delta_5,+_{\bar{1}}}).$$

Then

$$\mathcal{Z}_{S \times E}^{\text{red}} = -j^* \left(e^{2\rho} \text{sch}(\text{Fock}(\mathfrak{n}_{\Delta_5,+})^{\otimes 2}) \right) = -\chi_{10}^{-1}.$$

Proof. Put

$$P = \prod_{\alpha > 0} (1 - e^{-\alpha})^{\text{sdim } \mathfrak{g}_{\Delta_5, \alpha}}.$$

The denominator theorem gives $\text{den}(\mathfrak{g}_{\Delta_5}) = e^{-\rho} P = D_5(2Z)$, while PBW gives $\text{sch Fock}(\mathfrak{n}_{\Delta_5,+}) = P^{-1}$. Hence

$$e^{2\rho} \text{sch}(\text{Fock}(\mathfrak{n}_{\Delta_5,+})^{\otimes 2}) = \text{den}(\mathfrak{g}_{\Delta_5})^{-2}.$$

Pullback by j sends $\text{den}(\mathfrak{g}_{\Delta_5})$ to $D_5(Z)$, whose square is χ_{10} . The reduced incidence convention supplies the overall minus sign in the Oberdieck–Pixton theorem. $\square$

The equality compares the completed root-Fock supercharacter with the numerical reduced generating series. Its proof uses the denominator identity and the half-coordinate pullback. A map from compact geometric states to this Fock space, preserving operators and their trace, is an additional comparison.

EXTENSIONS OVER THE ELLIPTIC CLASSIFYING STACK

Work on locally noetherian complex schemes with the fppf topology. Let BE be the stack of right E -torsors. For a left E -stack M , an object of $[M/E](T)$ is a right torsor $P \rightarrow T$ and an equivariant map $P \rightarrow M$. Forgetting the map gives $[M/E] \rightarrow BE$. All products of stacks below are two-fibre products.

Proposition 77.1 (The common translation torsor). *There is a natural equivalence*

$$[(M \times N)/E_{\text{diag}}] \simeq [M/E] \times_{BE} [N/E].$$

The equivalences for any finite number of factors commute with regrouping and insertion of the terminal E -stack.

Proof. An object on the right is $(P, m), (Q, n)$ together with a torsor isomorphism $\alpha : P \rightarrow Q$. Send it to $(P, m, n\alpha)$. Conversely, use a common torsor twice and its identity isomorphism. A pair of torsor maps a, b on the right satisfies $b\alpha = \alpha'a$, so b is determined by a . The same construction identifies the maps and their descent isomorphisms. It commutes with base change. For several factors, every regrouping gives one torsor with the same list of equivariant maps, proving all stated coherences. $\square$

For $M = N = E$ with translation, the diagonal quotient is E , with coordinate $y - x$. Both separate quotients are points. Their fibre product over BE retains the relative translation, while their ordinary product loses it.

Let $\mathfrak{M}$ be the stack of coherent sheaves on $X = S \times E$, with families flat over the parameter scheme, and put $\mathfrak{Q} = [\mathfrak{M}/E]$. An object of $\mathfrak{Q}(T)$ is a torsor $P \rightarrow T$ and a flat sheaf on $S \times P$. Fix a polarization and write $\mathfrak{Q}_\alpha$ for a Hilbert-polynomial component. Translation preserves these polynomials. Let $\mathfrak{H}_{\alpha, \beta}$ classify short exact sequences $0 \rightarrow A \rightarrow F \rightarrow B \rightarrow 0$ on a common $S \times P$, with all terms flat over T and polynomials $\alpha, \alpha + \beta, \beta$. Its span over BE is

$$\mathfrak{Q}_\alpha \times_{BE} \mathfrak{Q}_\beta \xleftarrow{p} \mathfrak{H}_{\alpha, \beta} \xrightarrow{q} \mathfrak{Q}_{\alpha+\beta}.$$

The category of spans of fppf stacks over BE has composition by fibre product and tensor product $\times_{BE}$. Its unit object is the identity map of BE .

Theorem 77.2 (Geometric Hall correspondence). *For fixed α, β , q is representable by algebraic spaces and proper. The spans form a coherently associative unital Hall object in the category of spans over BE . Its unit is the zero-sheaf span $BE \leftarrow BE \rightarrow \mathfrak{Q}$.*

Proof. Over a fixed (P, F) , the fibre of q classifies quotients $F \twoheadrightarrow B$ of polynomial β . Its kernel is flat, because F and B are flat. An automorphism fixing the quotient and F is the identity. Trivializing P fppf-locally identifies this fibre with the fixed-polynomial Quot scheme, projective over the parameter scheme by Grothendieck's theorem [41, Theorem 3.2]. The quotient functor descends to an algebraic space, and properness descends with it.

Both composites of three Hall spans classify $\mathfrak{o} \subset A \subset D \subset F$ on the same torsor. One bracketing concatenates inclusions. The other starts with $A \subset F$ and a subobject $B \subset F/A$, and sets $D = F \times_{F/A} B$. These constructions are inverse on objects and isomorphisms. They preserve flatness and commute with base change. For longer flags all regroupings identify the same subobjects and quotients. Their comparison maps agree, proving the pentagon and higher coherences. Finally $\mathfrak{Q}_{\mathfrak{o}} = BE$. The sequences with zero subobject or zero quotient give the unit and its coherences. $\square$

Properness here concerns the full coherent-sheaf stack on a fixed pair of Hilbert-polynomial components. A stability restriction can replace its fibre by an open part of a Quot scheme, which need not be proper. A bounded charge window also needs closure under the extensions and flags used in its operations. A reduced critical-coefficient realization must supply pullback, pushforward, and orientation comparisons on these same flags. Its unit has stabilizer E . A coefficient theory for this quotient must include that nonaffine stabilizer, or work relative to BE , on the unit and every flag.

ADDITIVE PHYSICAL CHARGE AND QUADRATIC AUTOMORPHIC DEGREE

Let

$$\Lambda = \tilde{H}(S, \mathbb{Z})$$

be the Mukai lattice of signature $(4, 20)$. The electric–magnetic charge lattice is

$$\Gamma_{\text{phys}} = \Lambda \oplus \Lambda, \quad c = (Q, P).$$

Its addition is the physical superselection law. Define the Gram map

$$\Pi(Q, P) = \left(\frac{Q^2}{2}, Q \cdot P, \frac{P^2}{2} \right) \in \Gamma_{\text{ind}}.$$

Proposition 78.1 (Polarization identity). *For $c_i = (Q_i, P_i)$,*

$$\Pi(c_1 + c_2) = \Pi(c_1) + \Pi(c_2) + B(c_1, c_2),$$

where

$$B(c_1, c_2) = (Q_1 \cdot Q_2, Q_1 \cdot P_2 + Q_2 \cdot P_1, P_1 \cdot P_2).$$

The cross-term is a symmetric vector-valued two-cocycle:

$$B(c_1, c_2) + B(c_1 + c_2, c_3) = B(c_2, c_3) + B(c_1, c_2 + c_3).$$

Proof. Expand the three quadratic expressions. The cocycle identity is bilinearity of B . □

78.1 THE POLARIZED CENTRAL EXTENSION

Define the abelian central extension

$$0 \longrightarrow \Gamma_{\text{ind}} \longrightarrow \tilde{\Gamma}_{\Pi} \longrightarrow \Gamma_{\text{phys}} \longrightarrow 0, \quad \tilde{\Gamma}_{\Pi} = \Gamma_{\text{phys}} \times \Gamma_{\text{ind}},$$

with multiplication

$$(c, \lambda) \cdot (d, \mu) = (c + d, \lambda + \mu + B(c, d)).$$

The cocycle identity makes this multiplication associative. The polarization identity makes the graph map

$$j_{\Pi} : \Gamma_{\text{phys}} \longrightarrow \tilde{\Gamma}_{\Pi}, \quad c \longmapsto (c, \Pi(c))$$

a homomorphism.

Let $\mathcal{A} = \bigoplus_{c \in \Gamma_{\text{phys}}} \mathcal{A}_c$ be a charge-graded associative algebra, and let $\mathbb{k}[\Gamma_{\text{ind}}]$ have monomials u^λ . Define the polarized central-extension algebra

$$\mathcal{R}_\Pi(\mathcal{A}) = \mathcal{A} \otimes \mathbb{k}[\Gamma_{\text{ind}}]$$

with product

$$(a_c u^\lambda) \star (b_d u^\mu) = a_c b_d u^{\lambda+\mu+B(c,d)}.$$

Give $a_c u^\lambda$ degree $(c, \lambda) \in \tilde{\Gamma}_\Pi$.

Theorem 78.2 (Graph-multiplicative polarization). *The product $\star$ is associative and $\tilde{\Gamma}_\Pi$ -graded. The map*

$$\iota_\Pi : \mathcal{A} \longrightarrow \mathcal{R}_\Pi(\mathcal{A}), \quad a_c \longmapsto a_c u^{\Pi(c)}$$

is an algebra homomorphism whose homogeneous degree is the graph element $j_\Pi(c)$.

Proof. The two parenthesizations differ by

$$B(c, d) + B(c + d, e) - B(d, e) - B(c, d + e),$$

which vanishes by the cocycle identity. The polarization formula gives

$$\iota_\Pi(a_c) \star \iota_\Pi(b_d) = a_c b_d u^{\Pi(c+d)}.$$

Thus the graph of the quadratic Gram map is multiplicative inside the central extension. $\square$

The same construction applies to Lie superalgebras and their enveloping algebras. It retains additive physical charge, Gram degree, and the interaction cross-term in one grading group.

POINT-LOCAL AND WRAPPED SECTORS

Let $p_E : S \times E \rightarrow E$ be projection.

Lemma 79.1 (Positive elliptic degree is wrapped). *Let $C \subset S \times E$ be a proper one-dimensional subscheme whose pushforward $(p_E)_*[C]$ has positive degree. Then $p_E(C) = E$.*

Proof. Properness makes $p_E(C)$ closed. Positive pushforward degree implies that at least one irreducible component of C dominates a one-dimensional closed subset of E . Every one-dimensional closed subset of the irreducible curve E is E itself, so the image of that component, and hence the image of C , is all of E . $\square$

Projection-finite objects define point-supported sectors on $\text{Ran}(E)$. Positive elliptic degree defines wrapped sectors supported over the entire elliptic curve.

Definition 79.2 (The local–wrapped factorization operad). Let Ch_E denote the chosen chiral/factorization operad on E and let Ass denote the ordered associative operad. Define the colored operad Hyb_E with colors

$$l, \quad w_m \quad (m \geq 1)$$

by the following operation objects.

- (i) Operations with local output are

$$\text{Hyb}_E(l^{\otimes k}, l) = \text{Ch}_E(k).$$

- (ii) For $r \geq 1$ and $m = m_1 + \cdots + m_r$, operations with wrapped output are

$$\text{Hyb}_E(l^{\otimes k}, w_{m_1}, \dots, w_{m_r}; w_m) = \text{Ch}_E(k) \otimes \text{Ass}(r).$$

All other operation objects are initial.

Operadic composition uses chiral insertion and disjoint union in the first factor and ordered substitution in the second. The equality

$$(\gamma_1 \circ \gamma_2) \otimes (a_1 \circ a_2) = (\gamma_1 \otimes a_1) \circ (\gamma_2 \otimes a_2)$$

is the mixed interchange law. The arities $(k, r) = (k, 0), (0, 2), (k, 1)$ recover, respectively, local factorization, wrapped multiplication, and the action of local insertions on a wrapped sector.

Definition 79.3 (Hybrid local–wrapped algebra). A hybrid local–wrapped algebra over E is an algebra over Hyb_E . It consists of a factorization algebra $\mathbb{F}_{\text{loc}}$, a height- or charge-complete family $\mathcal{W} = \prod_{m>0} \mathcal{W}_m$, ordered products

$$\mu_{m_1, \dots, m_r} : \mathcal{W}_{m_1} \otimes \cdots \otimes \mathcal{W}_{m_r} \longrightarrow \mathcal{W}_{m_1 + \cdots + m_r},$$

and factorization-compatible actions of the local operation objects $\text{Ch}_E(k)$ on every wrapped product. The operad axioms state associativity, local factorization, and compatibility of the two operations.

A compact Hall theory of $S \times E$ produces a correspondence-valued Hyb_E -algebra when disjoint-support direct sums, wrapped extensions, and the mixed extension correspondences satisfy the same base-change identities.

BOOK XVIII

The Borcherds algebra and its quantum
group

THE LOCAL-WRAPPED ROOT ALGEBRA

The positive cone is graded by $\gamma = (n, \ell, m)$, with elliptic degree m . Set

$$\mathfrak{n}_{\Delta_s, +_{\text{loc}}} = \bigoplus_{\substack{\alpha > 0 \\ m(\alpha) = 0}} \mathfrak{g}_{\Delta_s, \alpha}, \quad \mathfrak{n}_{\Delta_s, +_{\text{wr}}} = \bigoplus_{\substack{\alpha > 0 \\ m(\alpha) > 0}} \mathfrak{g}_{\Delta_s, \alpha}.$$

Proposition 80.1 (Semidirect local–wrapped decomposition). *The space $\mathfrak{n}_{\Delta_s, +_{\text{loc}}}$ is a Lie subsuperalgebra, $\mathfrak{n}_{\Delta_s, +_{\text{wr}}}$ is a Lie ideal, and*

$$\mathfrak{n}_{\Delta_s, +} = \mathfrak{n}_{\Delta_s, +_{\text{loc}}} \ltimes \mathfrak{n}_{\Delta_s, +_{\text{wr}}}.$$

The proper height filtration preserves both summands.

Proof. The bracket adds root degrees. The sum of two roots of elliptic degree zero again has degree zero, and addition of a nonnegative degree to a positive degree remains positive. Height is additive on every nonzero bracket. $\square$

Define the complete enveloping Hopf algebra and current envelope

$$\mathbb{U}_{\Delta_s} = \widehat{U}_h(\mathfrak{n}_{\Delta_s, +}), \quad \mathbb{V}_{\Delta_s, E} := \widehat{V}_{\Delta_s} = \varprojlim_H U^{\text{ch}}(\text{Cur}_E(\mathfrak{n}_{\Delta_s, +}^{(H)})).$$

The adjoint action of $\mathfrak{n}_{\Delta_s, +_{\text{loc}}}$ on $\mathfrak{n}_{\Delta_s, +_{\text{wr}}}$ extends by derivations to $\widehat{U}_h(\mathfrak{n}_{\Delta_s, +_{\text{wr}}})$. Let $\#$ denote the corresponding completed smash product.

Definition 80.2 (Current-generated local–wrapped operad). The colored operad $\text{Hyb}_{E, \Delta}^{\text{cur}}$ is generated by the chiral current operations of $\text{Cur}_E(\mathfrak{n}_{\Delta_s, +_{\text{loc}}})$, the ordered multiplications of $\widehat{U}_h(\mathfrak{n}_{\Delta_s, +_{\text{wr}}})$, and the adjoint derivation action of the former on the latter, modulo the chiral Jacobi, associative, derivation, and semidirect-product relations.

Theorem 80.3 (Local–wrapped Borchers decomposition). *Let $\mathfrak{n}_{\Delta_s, +_{\text{loc}}}$ be the sum of positive root spaces of elliptic degree $m = 0$ and let $\mathfrak{n}_{\Delta_s, +_{\text{wr}}}$ be the sum of those with $m > 0$. Then*

$$\mathfrak{n}_{\Delta_s, +} = \mathfrak{n}_{\Delta_s, +_{\text{loc}}} \ltimes \mathfrak{n}_{\Delta_s, +_{\text{wr}}},$$

where $\mathfrak{n}_{\Delta_s, +_{\text{loc}}}$ is a Lie subsuperalgebra and $\mathfrak{n}_{\Delta_s, +_{\text{wr}}}$ is a Lie ideal. For the proper-height completion,

$$\widehat{U}_h(\mathfrak{n}_{\Delta_s, +}) \cong \widehat{U}_h(\mathfrak{n}_{\Delta_s, +_{\text{wr}}}) \# \widehat{U}_h(\mathfrak{n}_{\Delta_s, +_{\text{loc}}}),$$

and

$$\widehat{C}_{*, h}(\mathfrak{n}_{\Delta_s, +}) \simeq \text{Tot } C_*(\mathfrak{n}_{\Delta_s, +_{\text{loc}}}; \widehat{C}_{*, h}(\mathfrak{n}_{\Delta_s, +_{\text{wr}}}).$$

The primitive Lie superalgebra of the completed enveloping algebra is the height completion $\widehat{\mathfrak{n}_{\Delta_s, +_h}} = \varprojlim_H \mathfrak{n}_{\Delta_s, +}^{(H)}$, and the Weyl-shifted Euler supercharacter of the completed Chevalley complex is $D_s(2Z)$.

Proof. Root addition adds elliptic degree. Thus the $m = 0$ summand is closed under the bracket, and every bracket with an $m > 0$ root again has $m > 0$. PBW orders wrapped generators before local generators and gives the completed smash product. The Chevalley complex of a semidirect product is the Hochschild–Serre total complex. Primitive elements of a completed enveloping algebra are recovered height by height. The denominator theorem supplies the last character identity. $\square$

Remark 80.4. The theorem constructs a Borchers root algebra with local and wrapped colors. The word Hall is reserved for an algebra obtained from geometric extension correspondences.

THE FORMAL IGUSA QUANTUM GROUP

Let $\mathfrak{b}_+$ be the positive Borchers Borel half. Form the restricted graded dual $\mathfrak{b}_- := \mathfrak{b}_+^{\vee, \text{gr}}$ with bracket and cobracket transported by the invariant pairing, identify the two Cartan copies along their common radical quotient, and complete by a proper root height. The completed Drinfeld double of this paired Borel datum is denoted $\mathfrak{g}_{\Delta_s}^{\text{pair}}$. Its positive and negative halves, diagonal Cartan, invariant form, and canonical tensor form a height-admissible Manin triple and hence a quasitriangular Lie superbialgebra.

Definition 8I.1 (Height-admissible Lie superbialgebra). A root-graded Lie superbialgebra is height-admissible when every root space is finite-dimensional, every bounded-height region contains finitely many roots, and the bracket, cobracket, invariant tensor, and all iterated operations are coefficientwise finite in the height completion.

Proposition 8I.2 (Height admissibility). *The paired Gritsenko–Nikulin Borchers Lie superbialgebra $\mathfrak{g}_{\Delta_s}^{\text{pair}}$ is height-admissible.*

Proof. The Borchers root spaces are finite-dimensional and the chosen height is proper. The bracket and cobracket preserve total root degree. In a bounded total height, finitely many decompositions of a root occur. The canonical tensor is therefore a coefficientwise finite sum in every height quotient. $\square$

Theorem 8I.3 (Quantized Borchers superalgebra). *Choose the symmetrizable Borchers–Cartan superdatum and multiplicity index set that present $\mathfrak{g}_{\Delta_s}$. Hong’s construction gives a quantized enveloping Hopf superalgebra*

$$U_q(\mathfrak{g}_{\Delta_s})$$

over $\mathbb{Q}(q)$, generated by its Cartan elements and the real and imaginary simple-root generators, with triangular decomposition

$$U_q(\mathfrak{g}_{\Delta_s}) \cong U_q^-(\mathfrak{g}_{\Delta_s}) \otimes U_q^0(\mathfrak{g}_{\Delta_s}) \otimes U_q^+(\mathfrak{g}_{\Delta_s}).$$

Its PBW root spaces have the same graded ranks as the classical Borchers superalgebra. The canonical pairing of the two Borel halves yields a universal R -matrix in the root-height completion of

$$U_q^+(\mathfrak{g}_{\Delta_s}) \widehat{\otimes} U_q^-(\mathfrak{g}_{\Delta_s}).$$

Consequently the quantization preserves the root-multiplicity data that enter $D_s(2Z)$.

Proof. The Gritsenko–Nikulin algebra is a symmetrizable Borchers–Kac–Moody Lie superalgebra. Apply the generators-and-relations construction of the quantized Borchers superalgebra. The nondegenerate bilinear form, triangular decomposition, centre, and universal R -matrix are established by Hong [44]. The PBW theorem identifies the associated graded root spaces with their classical counterparts. Since the denominator exponents are the classical signed root multiplicities, they are unchanged by the flat root-graded deformation. $\square$

Remark 81.4. The quantum group is attached to the Borchers generalized Cartan superdatum. A geometric Hall quantum group is identified with it through a Hall coproduct, pairing, completion, and primitive comparison.

A HOLOMORPHIC–TOPOLOGICAL BF REALIZATION

For every height bound H , let $\mathfrak{n}_{\Delta,+}^{(H)}$ be the finite-dimensional nilpotent quotient. On $E \times \mathbb{R}$ form the shifted cotangent theory of

$$\mathcal{L}_H = \Omega^{\circ,*}(E) \widehat{\otimes} \Omega^*(\mathbb{R}) \otimes \mathfrak{n}_{\Delta,+}^{(H)}$$

with action

$$S_H(A, B) = \int_{E \times \mathbb{R}} \Omega_E \wedge \left\langle B, (\bar{\partial} + d_{\mathbb{R}})A + \frac{1}{2}[A, A] \right\rangle.$$

Theorem 82.1 (Classical local and exact zero-mode realization). *For every H :*

- (i) S_H satisfies the classical master equation and defines a classical holomorphic–topological factorization algebra;
- (ii) harmonic reduction along a compact interval gives the finite-dimensional cotangent BV theory of $\mathfrak{n}_{\Delta,+}^{(H)}$, whose positive-root grading makes its modular supercharacter zero and hence gives the quantum master equation;
- (iii) after choosing the complementary Lagrangian boundary polarizations, the boundary algebra is $U(\mathfrak{n}_{\Delta,+}^{(H)})$, the interval amplitude is $\mathbf{1} \otimes_{U(\mathfrak{n}_{\Delta,+}^{(H)})}^{\mathbb{L}} \mathbf{1} \simeq C_*(\mathfrak{n}_{\Delta,+}^{(H)})$, and the holomorphic boundary current envelope is V_H .

The transition maps yield a pro-object with boundary algebra $\widehat{U}_h(\mathfrak{n}_{\Delta,+})$, interval complex $C_{*,h}^{\wedge}(\mathfrak{n}_{\Delta,+})$, and current envelope $\widehat{V}_{\Delta}$.

Proof. The classical master equation is the Jacobi identity and invariance of the cotangent pairing. The classical observables satisfy factorization descent. Harmonic reduction is finite-dimensional. An even adjoint operator raises positive-root height and is nilpotent on the finite quotient; an odd adjoint operator is odd. Both have supertrace zero, so the modular term in the finite BV Laplacian vanishes. The chosen polarization quantizes the boundary Lie–Poisson bracket to the enveloping algebra; the two augmentation states give the two-sided bar, identified with Chevalley chains by PBW. Holomorphic descendants give the current algebra. All structure maps preserve height. $\square$

The theorem gives the precise field-theoretic realization used in this volume: a classical local theory together with an exact quantum protected zero-mode sector.

THE DETERMINANT AND FREDHOLM SQUARE-ROOT OPERATORS

Let

$$\mathcal{V}_H = \bigoplus_{\substack{\alpha > 0 \\ h(\alpha) \leq H}} \mathfrak{g}_{\Delta_s, \alpha}$$

and define

$$\mathcal{D}_H|_{\mathfrak{g}_{\Delta_s, \alpha}} = (1 - e^{-\alpha}) \text{id}.$$

Theorem 83.1 (Universal determinant section). *The finite-height Berezinian is*

$$\text{Ber}(\mathcal{D}_H) = \prod_{\substack{\alpha > 0 \\ h(\alpha) \leq H}} (1 - e^{-\alpha})^{\text{sdim } \mathfrak{g}_{\Delta_s, \alpha}}.$$

The coefficientwise limit satisfies

$$e^{-\rho} \text{Ber}(\mathcal{D}_\infty) = D_5(2Z).$$

After pullback by $j(Z) = Z/2$,

$$j^*(e^{-\rho} \text{Ber}(\mathcal{D}_\infty)) = D_5(Z), \quad D_5(Z)^2 = \chi_{10}(Z).$$

Proof. The Berezinian of a diagonal endomorphism is the product of its eigenvalues with exponent equal to the superdimension of the eigenspace. Proper height makes every coefficient stationary. The denominator theorem supplies the Weyl line and the root-to-Siegel coordinate map. $\square$

83.1 TRACE-CLASS REALIZATION ON THE ROOT HILBERT SUPERSPACE

Choose homogeneous orthonormal bases of the finite-dimensional root spaces and set

$$\mathcal{H}_\Delta = \widehat{\bigoplus_{\alpha > 0} \mathfrak{g}_{\Delta_s, \alpha}}$$

with its induced $\mathbb{Z}/2$ grading. Let $\mathfrak{C}_{\text{abs}}$ be the subchamber on which

$$\sum_{\alpha > 0} \dim \mathfrak{g}_{\Delta_s, \alpha} \left| e^{-\alpha(Z)} \right| < \infty.$$

For $Z \in \mathfrak{C}_{\text{abs}}$, define the trace-class diagonal operator

$$K_Z|_{\mathfrak{g}_{\Delta_s, \alpha}} = e^{-\alpha(Z)} \text{id}, \quad D_Z = 1 - K_Z.$$

On each parity summand form the skew doubling

$$\mathbb{A}_{Z, \bar{\epsilon}} = \begin{pmatrix} 0 & D_{Z, \bar{\epsilon}} \\ -D_{Z, \bar{\epsilon}}^t & 0 \end{pmatrix}.$$

Orient every finite-height truncation by the fixed root order and normalize its Pfaffian to 1 at $K_Z = 0$. The oriented Fredholm super-Pfaffian is the stabilized ratio

$$\mathrm{sPf}_F(\mathbb{A}_Z) = \frac{\mathrm{Pf}_F(\mathbb{A}_{Z,\bar{0}})}{\mathrm{Pf}_F(\mathbb{A}_{Z,\bar{1}})}.$$

Theorem 83.2 (Polarized Fredholm realization). *Choose the protected parity refinement and the root-space Hilbert norms above. For every Z in the absolute-convergence chamber, the diagonal operator $K_Z|_{\mathfrak{g}_{\Delta,\alpha}} = e^{-\alpha(Z)} \mathrm{id}$ is trace class on the even and odd root Hilbert spaces. For $D_Z = 1 - K_Z$,*

$$\mathrm{sdet}_F(D_Z) = \frac{\det_F(D_Z|_{\bar{0}})}{\det_F(D_Z|_{\bar{1}})} = \prod_{\alpha > 0} (1 - e^{-\alpha(Z)})^{\mathrm{sdim} \mathfrak{g}_{\Delta,\alpha}}.$$

After choosing the standard polarization of $\mathcal{H}_\Delta \oplus \mathcal{H}_\Delta^$ and its orientation, the paritywise skew block operators $\mathbb{A}_{Z,\bar{0}}$ and $\mathbb{A}_{Z,\bar{1}}$ have an oriented Fredholm super-Pfaffian*

$$\mathrm{sPf}_F(\mathbb{A}_Z) = \frac{\mathrm{Pf}_F(\mathbb{A}_{Z,\bar{0}})}{\mathrm{Pf}_F(\mathbb{A}_{Z,\bar{1}})} = \mathrm{sdet}_F(D_Z).$$

Hence

$$e^{-\rho} \mathrm{sdet}_F(D_Z) = D_\zeta(2Z).$$

Proof. The defining summability condition for $\mathfrak{C}_{\mathrm{abs}}$ is precisely the trace norm of the diagonal operator, summed in both parities. The Fredholm determinant of $1 - K_Z$ is the convergent product over eigenvalues; taking the even/odd ratio gives the superdeterminant. On each finite root-height truncation the Pfaffian of the oriented skew doubling equals the determinant of D_Z . Trace-norm convergence of the truncations passes this identity to the Fredholm limit. The final equality is the Gritsenko–Nikulin denominator formula. $\square$

Remark 83.3. The polarization and orientation in this theorem belong to the root Hilbert space. A compact moduli-space orientation is a separate line bundle and is compared with this one by a specified intertwiner.

This constructs the square-root operator on the root Hilbert superspace where its stated summability condition holds. The nonemptiness estimate below concerns the minimal virtual superspace, whose dimensions are $|\mathrm{sdim} \mathfrak{g}_{\Delta,\alpha}|$. Applying that estimate to the full root superspace requires a bound on both parity dimensions. A compact geometric realization requires an operator intertwiner and a compatible map of determinant lines.

WHAT THE AUTOMORPHIC CHARACTER DETERMINES

Theorem 84.1 (Denominator data and reconstruction data). *The Borchers product determines:*

- (i) *the signed root multiplicity $\text{sdim } \mathfrak{g}_{\Delta, \alpha}$ in every active root degree;*
- (ii) *the Weyl vector and denominator character;*
- (iii) *the divisor and automorphic multiplier of the section.*

A Lie bracket is supplied by the Borchers–Kac–Moody presentation, parity dimensions by a parity-refined root datum, Hall multiplication by extension correspondences, and an oriented square root by a square-root line with monodromy. These reconstruction data refine the scalar product independently.

Proof. The first three items are read from the exponent, prefactor, and automorphy law. Independence is witnessed by explicit pairs: two brackets on the same graded vector space have the same PBW character; an even–odd pair has signed dimension zero; distinct Hall correspondences may have the same Euler product; and twisting a square-root line by a sign local system preserves its square. Hence each refined object is reconstructed from the additional datum named in the statement. $\square$

Recognition Theorem 84.2 (Complete primitive recognition). *Let $\mathfrak{p}$ be a positive graded Lie superalgebra and let $\Psi : \mathfrak{n}_{\Delta, +} \rightarrow \mathfrak{p}$ be a graded Lie map. The map is an isomorphism if either:*

- (i) *the parity-refined Hilbert series of $U(\mathfrak{n}_{\Delta, +})$ and $U(\mathfrak{p})$ agree; or*
- (ii) *each primitive degree has prescribed pure parity and the signed PBW characters agree.*

The induced height completions, Chevalley complexes, and current envelopes are then isomorphic.

Proof. Choose the first height containing a homogeneous kernel vector. All decomposable PBW terms in that height are determined by lower heights and agree. The primitive parity dimension therefore drops, contradicting the first hypothesis. In the second case the logarithm of the PBW product recovers signed primitive dimensions inductively, and prescribed parity recovers ordinary dimensions. Naturality of completion, Chevalley chains, and current envelopes gives the final assertion. $\square$

BOOK **XIX**

**The universal Borcherds Hall and
three-Calabi–Yau models**

FINITE SIMPLE-ROOT WINDOWS

Let I_Δ be the multiset of simple roots of the Gritsenko–Nikulin Borchers superalgebra. A repeated imaginary simple root is written once for each copy. The parity map is

$$p : I_\Delta \longrightarrow \mathbb{Z}/2.$$

Choose a proper height function

$$h : I_\Delta \longrightarrow \mathbb{Z}_{>0}$$

whose sublevel sets are finite. Put

$$I_H = \{i \in I_\Delta : h(i) \leq H\}.$$

The restriction of the invariant form gives a finite symmetric Borchers–Cartan matrix

$$\begin{aligned} A_H &= (a_{ij})_{i,j \in I_H}, \\ a_{ii} &= 2 && \text{for real } i, \\ a_{ii} &\leq 0 && \text{for imaginary } i, \\ a_{ij} &\leq 0 && \text{for } i \neq j. \end{aligned}$$

The evenness of the root lattice makes every imaginary diagonal entry even.

Construction 85.1 (The Igusa Borchers quiver). Order I_H . For $i < j$ place $-a_{ij}$ arrows from i to j . At an imaginary vertex i place

$$g_i = 1 - \frac{a_{ii}}{2}$$

loops. A real vertex has loop number zero. Denote the resulting quiver by $Q_{\Delta,H}$.

Proposition 85.2 (Euler form). *The symmetrized Euler form of $Q_{\Delta,H}$ is A_H :*

$$\langle e_i, e_j \rangle + \langle e_j, e_i \rangle = a_{ij}.$$

Proof. For $i \neq j$, the total number of arrows between the two vertices is $-a_{ij}$, so the symmetrized Euler entry is a_{ij} . At i the value is $2 - 2g_i = a_{ii}$. $\square$

The inclusions $I_H \subset I_{H+1}$ identify $Q_{\Delta,H}$ with a full subquiver of $Q_{\Delta,H+1}$. Representations supported on the old vertices form an exact extension-closed subcategory. This gives compatible Hall embeddings.

THE REAL-ROOT SEED WINDOW

The three real simple roots are

$$\delta_1 = 2f_2 - f_3, \quad \delta_2 = 2f_{-2} - f_3, \quad \delta_3 = f_3,$$

with Gram matrix

$$A_{\text{re}} = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}.$$

Choose the order $\delta_1 < \delta_2 < \delta_3$. The real-root quiver Q_{re} has two arrows from vertex i to vertex j for each $i < j$.

Proposition 86.1 (Real-root Hall seed). *The symmetrized Euler form of Q_{re} is A_{re} . The Hall composition algebra generated by its three simple representations is the positive quantum Kac–Moody algebra with Cartan matrix A_{re} .*

Proof. The diagonal Euler entries are 2. The total number of arrows between distinct vertices is two, hence the symmetrized off-diagonal entries are -2 . The Ringel–Green Hall theorem identifies the composition algebra with the positive quantum algebra defined by that symmetric Cartan matrix. $\square$

The three vertices already contain the Weyl group and the reflective divisor. Imaginary vertices add the automorphic correction. The finite-height tower therefore begins with a completely explicit hyperbolic Hall algebra and enlarges it by one finite collection of imaginary simple roots at each stage.

86.1 THE FIRST GINZBURG DIFFERENTIAL

Let $a_{ij}^{(1)}, a_{ij}^{(2)} : i \rightarrow j$ be the two arrows for $i < j$. Add reverse arrows $(a_{ij}^{(r)})^* : j \rightarrow i$ of degree -1 and loops t_i of degree -2 . The canonical three-Calabi–Yau completion has

$$da_{ij}^{(r)} = 0, \quad d(a_{ij}^{(r)})^* = 0,$$

and

$$dt_i = \sum_{a:s(a)=i} aa^* - \sum_{a:t(a)=i} a^*a. \quad (86.1)$$

The identity $d^2 = 0$ follows immediately. Formula (86.1) is the derived moment-map equation for the doubled real-root quiver.

Calculation 86.2 (Vertex one). At vertex 1,

$$dt_1 = \sum_{r=1}^2 \left(a_{12}^{(r)} (a_{12}^{(r)})^* + a_{13}^{(r)} (a_{13}^{(r)})^* \right).$$

At vertex 3,

$$dt_3 = - \sum_{r=1}^2 \left((a_{13}^{(r)})^* a_{13}^{(r)} + (a_{23}^{(r)})^* a_{23}^{(r)} \right).$$

The sum over vertices is $\sum_a (aa^* - a^*a)$. It vanishes in the cyclic quotient and after matrix trace, but generally not in the path algebra.

Proof. The outgoing arrows at vertex 1 have targets 2 and 3. The incoming arrows at vertex 3 have sources 1 and 2. The displayed formulas follow from these incidence lists. Each arrow contributes one commutator to the sum over vertices. The trace of a commutator vanishes, giving the asserted trace identity. $\square$

HALL REALIZATION OF THE POSITIVE ALGEBRA

Let $\mathcal{A}_H = \text{Rep}_{\mathbb{F}_q}(Q_{\Delta,H})$. Its twisted Hall product is

$$[M] * [N] = v^{\langle \dim M, \dim N \rangle} \sum_{[L]} |\{N' \subset L : N' \cong N, L/N' \cong M\}| [L], \quad v^2 = q.$$

The class $[o]$ is the unit: the required subobject is unique on either side. Associativity follows by counting flags $N_1 \subset N_2 \subset L$ in two orders. The twist factors agree because the Euler form is bilinear:

$$\langle \alpha, \beta \rangle + \langle \alpha + \beta, \gamma \rangle = \langle \beta, \gamma \rangle + \langle \alpha, \beta + \gamma \rangle.$$

Thus this normalization defines the unital twisted Hall algebra on isomorphism classes.

Theorem 87.1 (Finite-window composition algebra). *At each vertex i of $Q_{\Delta,H}$, choose the one-dimensional representation S_i supported at i , with every loop acting by zero. Use the generic composition algebra generated by these classes across finite fields, with the twisted subobject-count multiplication above. After extending coefficients from $\mathbb{Z}[v, v^{-1}]$ to $\mathbb{C}(v)$, this is the positive quantum generalized Kac–Moody algebra for the even Borcherds–Cartan matrix A_H . The parameter in Kang–Schiffmann is $v_{\text{KS}} = v^{-1}$. Their specified semisimple perverse-sheaf construction on nilpotent representation varieties gives the corresponding canonical basis.*

Proof. This is the selected-simple composition construction of Kang–Schiffmann, Section 2.3 and Theorem 2.2, with one selected zero-loop simple at each vertex. Their geometric realization uses its split graded Grothendieck group over $\mathbb{Z}[v_{\text{KS}}, v_{\text{KS}}^{-1}]$ (Theorem 4.1), with the stated inversion of parameter [48]. The theorem does not use every simple object of the full representation category. A vertex with c loops has q^c one-dimensional simple representations over $\mathbb{F}_q$, whereas only the chosen zero-loop representation is the generator here. $\square$

A semi-derived Hall realization of a full quantum group additionally specifies two-periodic complexes, the invertible acyclic classes and Cartan reduction [2, 57]. Its coproduct, pairing and completion are separate from the positive composition algebra used here.

87.1 PARITY AND THE QUANTIZED SUPERDATUM

The Gritsenko–Nikulin denominator algebra carries a parity map on its simple-root multiset. It is additive on the free lattice of labelled simple generators. Projection to the Lorentzian degree can identify contributions of different parity. Both parity dimensions must therefore be retained in each projected root space. Hong’s construction applied to this Borcherds superdatum gives the positive Borel

$$U_q^+(\mathfrak{g}_{\Delta,}^{(H)})$$

and the full quantized Borcherds superalgebra $U_q(\mathfrak{g}_{\Delta,}^{(H)})$.

Theorem 87.2 (Ordinary and super presentations). *The ordinary quiver Hall presentation and Hong's quantum superalgebra use the same specified Cartan matrix and labelled simple generators. Their relation ideals and parity conventions are different inputs. Equality of their Cartan gradings alone does not identify their PBW characters.*

Proof. The two presentations are those of the preceding Hall construction and Hong's superalgebra construction. The difference already occurs for a single isotropic generator. The enveloping algebra of a one-dimensional even abelian Lie algebra is $\mathbb{C}[x]$. Declaring its generator odd only as a graded vector space leaves a nonzero vector x^2 in degree two. The enveloping algebra of a one-dimensional odd abelian Lie superalgebra is $\wedge(\xi)$, since its defining relation is $2\xi^2 = [\xi, \xi] = 0$. Its degree-two component is zero. Thus parity assignment to an ordinary PBW basis does not produce the super relation ideal or its character. A comparison must respect these relations degree by degree. $\square$

THE THREE-CALABI-YAU COMPLETION

Let $kQ_{\Delta,H}$ be the path algebra. Its algebraic three-Calabi-Yau completion is

$$\Gamma_{\Delta,H} = \Pi_3(kQ_{\Delta,H}).$$

Write J for the ideal generated by all arrows in its graded quiver. Its completed version is the separate topological dg algebra $\widehat{\Gamma}_{\Delta,H} = \varprojlim_n \Gamma_{\Delta,H}/J^n$. The algebraic Calabi-Yau theorem is used for $\Gamma_{\Delta,H}$; a topological Calabi-Yau assertion for $\widehat{\Gamma}_{\Delta,H}$ requires its completed bimodule category. This is the Ginzburg model with zero potential: a degree-minus-one reverse arrow for every arrow and a degree-minus-two loop at each vertex. The loop differential is the signed sum of incident arrow-reverse-arrow products.

Theorem 88.1 (Borcherds-Ginzburg projections). *For each finite quiver $Q_{\Delta,H}$, its undeformed three-Calabi-Yau completion $\Gamma_{\Delta,H}$ is the Ginzburg dg algebra with zero potential. If $H \leq H'$ and $Q_{\Delta,H}$ is the full subquiver on a subset of vertices, there is a surjective dg algebra map*

$$\pi_{H'H} : \Gamma_{\Delta,H'} \longrightarrow \Gamma_{\Delta,H}.$$

It kills the added vertices, every arrow incident to them, the corresponding reverse arrows, and the added degree-minus-two loops. These maps compose strictly and extend continuously to the completed algebras.

Proof. The finite Calabi-Yau assertion is the Ginzburg model of Keller's completion theorem [33, 49]. We concatenate paths from left to right, so aa^* is based at the source of a . Original and reverse arrows have zero differential. For the degree-minus-two loop t_i , the differential is

$$dt_i = \sum_{s(a)=i} aa^* - \sum_{t(a)=i} a^*a.$$

The projection deletes exactly the terms involving an added vertex, leaving the same sum for the full subquiver. It therefore commutes with the differential. It preserves path length and extends to the completed path algebra. Successive projections kill the union of their omitted vertices and arrows, proving strict composition. $\square$

The evident inclusion of graded quivers does not give the reverse dg map. A single old vertex has loop differential zero. Adding an arrow $a : i \rightarrow j$ changes that differential to aa^* at i . The map fixing the old loop therefore fails to commute with the differential.

Write $\Gamma_{\Delta} = (\Gamma_{\Delta,H}, \pi_{H'H})_H$ for this pro-system of dg algebras. It is separate from the ordinary Hall embeddings obtained by extension by zero on representations of a full subquiver. A direct system of Calabi-Yau dg categories compatible with those Hall embeddings requires additional dg functors. It does not follow from the inclusion of the displayed Ginzburg generators.

The quantum group retains its separately specified root-height completion. Its transition maps must preserve the Cartan generators, coproduct, and pairing. Quotienting the positive nilpotent Lie algebra by height and truncating a full quantum group are different operations.

HALL STATES AND PROTECTED SUPERSTATES

The representation varieties of $Q_{\Delta,H}$ carry the perverse-sheaf complexes used in the composition-algebra construction. The full equivariant derived category is a larger carrier. Already for one real vertex and dimension one, its quotient is $B\mathbb{G}_m$, with $H^*(B\mathbb{G}_m; \mathbb{Q}) = \mathbb{Q}[c_1]$, where $|c_1| = 2$. Its total equivariant cohomology is infinite-dimensional. Finite root multiplicities therefore refer to the specified composition algebra, not to total equivariant cohomology of this stack.

Definition 89.1 (Igusa Hall category). For every dimension vector $\mathbf{d}$, let

$$\mathfrak{R}_{H,\mathbf{d}} = [\mathrm{Rep}_{\mathbf{d}}(Q_{\Delta,H})/G_{\mathbf{d}}].$$

The finite-height Hall category is the direct sum of the equivariant constructible derived categories on the $\mathfrak{R}_{H,\mathbf{d}}$, equipped with Hall convolution along the stack of short exact sequences. Denote it by $\mathcal{B}_{\Delta,H}$ and set

$$\mathcal{B}_{\Delta} = \varinjlim_H \mathcal{B}_{\Delta,H}.$$

Independently, the protected one-particle superspace is the root-graded superspace

$$\mathcal{H}_{\mathrm{ip},H} = \bigoplus_{\alpha > 0} \mathfrak{g}_{\Delta,\alpha}^{(H)},$$

with the parity of the Borchers superalgebra. Its multiparticle completion is the PBW/Fock superspace of $U(\mathfrak{n}_{\Delta,+}^{(H)})$.

Theorem 89.2 (Hall and protected-state comparison). *The composition subalgebra in the finite-window Hall theorem is the ordinary positive quantum generalized Kac–Moody algebra. This statement does not identify the Grothendieck group of the full category $\mathcal{B}_{\Delta,H}$ with that composition algebra. The protected one-particle superspace has signed root multiplicities*

$$\mathrm{sdim} \mathcal{H}_{\mathrm{ip},H,\alpha} = \mathrm{sdim} \mathfrak{g}_{\Delta,\alpha}^{(H)}.$$

Its height-complete PBW character is

$$\mathrm{sch} \mathrm{Fock}(\mathfrak{n}_{\Delta,+}) = \prod_{\alpha > 0} (1 - e^{-\alpha})^{-\mathrm{sdim} \mathfrak{g}_{\Delta,\alpha}},$$

and its Weyl-shifted inverse is $D_5(2Z)$. The composition canonical basis and the protected root presentation use the specified simple-root indexing and Cartan grading; the parity refinement belongs to the Borchers superdatum.

Proof. The composition-algebra assertion is the finite-window theorem. The protected assertion is PBW for the Borchers Lie superalgebra. Proper height passes both constructions to the limit. The Gritsenko–Nikulin denominator theorem gives the final product identity. $\square$

EXTENSION DETERMINANTS AND CRITICAL ORIENTATIONS

Let $\mathcal{C}$ be a three-Calabi–Yau dg category over $\mathbb{C}$, with its chosen trace. For families of objects A, B over a base T , let $C(A, B)$ denote the relative derived Hom complex on T . Suppose these complexes are perfect, and that the trace gives base-change-compatible duality

$$C(B, A) \simeq C(A, B)^\vee[-3].$$

Put $L(A, B) = \det C(A, B)$ and $D(A) = L(A, A)$. The determinant convention is $\det C = \bigotimes_i (\det C^i)^{(-1)^i}$, with symmetry sign specified by $\chi(C) \bmod 2$.

Theorem 90.1 (Extension determinant). *For an exact triangle $A \rightarrow F \rightarrow B \rightarrow A[1]$, there is a base-change-compatible isomorphism*

$$D(F) \simeq D(A) \otimes D(B) \otimes L(A, B)^{\otimes 2}.$$

The determinant comparisons obtained by refining a flag agree, with the stated determinant symmetry signs.

Proof. Apply derived Hom to the triangle in each variable. Determinant additivity gives

$$D(F) \simeq L(B, A) \otimes D(A) \otimes D(B) \otimes L(A, B).$$

Dualization and the odd shift each invert the determinant line. Thus Calabi–Yau duality identifies $L(B, A)$ with $L(A, B)$. Reorder by determinant symmetry to obtain the formula. For a full flag, all iterated decompositions identify with the same ordered tensor product of $C(A_i, A_j)$, for its successive quotients A_i . Determinant additivity for refined filtrations proves coherence. Each operation commutes with base change. $\square$

A square-root orientation consists of lines o_A and isomorphisms $o_A^{\otimes 2} \simeq D(A)$. Extension compatibility requires maps

$$o_A \otimes o_B \otimes L(A, B) \longrightarrow o_F$$

whose squares are the determinant comparisons. Choosing coordinates on a smooth critical chart trivializes its ambient canonical line. Hall convolution requires transition maps compatible with the cross line and the zero object. On three-step flags, the two orientation maps can differ by an element of μ_2 , even when their squares agree. That sign must be trivialized coherently.

Example 90.2 (A nontrivial cross line). Let X be a smooth projective Calabi–Yau threefold, with a chosen volume form, and let $x \in X$. On $X \times T$, where $T = \mathbb{P}^1$, take

$$A = \mathcal{O}_X \boxtimes \mathcal{O}_T(1), \quad B = \mathcal{O}_x \boxtimes \mathcal{O}_T.$$

Then $C(A, B) = \mathcal{O}_T(-1)[0]$, and relative Serre duality gives $C(B, A) = \mathcal{O}_T(1)[-3]$. Thus $L(A, B) = L(B, A) = \mathcal{O}_T(-1)$. The self-Hom complexes of A and B have constant determinant lines, since the factors $\mathcal{O}_T(1)$ cancel in the first self-Hom. Nevertheless

$$D(A \oplus B) = D(A) \otimes D(B) \otimes \mathcal{O}_T(-2).$$

The orientation cross factor is $\mathcal{O}_T(-1)$. A sign change cannot replace this line of nonzero degree.

Corollary 90.3 (Oriented critical coefficients). *Suppose the chosen critical charts have square-root orientation maps on overlaps and extension flags, with unit and flag coherences. Suppose their vanishing-cycle comparison maps preserve these data. Then the local vanishing-cycle complexes descend to the specified oriented coefficient object. Their Hall convolution is associative whenever its pullback and pushforward maps are defined and satisfy base change on the flag diagrams.*

Proof. The overlap cocycle identifies the local coefficient complexes by descent. On a three-step flag, orientation coherence identifies the two Thom–Sebastiani coefficient maps. Proper or compact-support base change, on the stipulated domain, identifies the two composites of convolution. The zero flag gives the unit equation. $\square$

For the quiver charts, the remaining construction is an equivariant system of square roots and coefficient maps on the actual representation and extension stacks, compatible with changes of height. For compact $S \times E$, the reduced complexes and the common elliptic torsor must also be retained. A coordinate volume on an affine chart specifies neither descent system by itself.

THE COLORED ELLIPTIC RAN SPACE

The elliptic degree is a global charge carried by the whole configuration. Define

$$\mathrm{Ran}^{\mathrm{wr}}(E) = \bigsqcup_{m \geq 0} \mathrm{Ran}(E)_m$$

with monoidal operation

$$(I, m) \sqcup (J, n) = (I \sqcup J, m + n)$$

on disjoint supports. The component $m = 0$ is point-local. The components $m > 0$ are wrapped.

Let

$$\mathfrak{n}_{\Delta, +} = \bigoplus_{m \geq 0} \mathfrak{n}_{\Delta, +}[m]$$

be the elliptic-degree grading. The root bracket obeys

$$[\mathfrak{n}_{\Delta, +}[m], \mathfrak{n}_{\Delta, +}[n]] \subset \mathfrak{n}_{\Delta, +}[m + n].$$

Construction 9I.1 (Hybrid factorization algebra). Form the chiral Chevalley factorization coalgebra

$$\mathcal{F}_{\Delta, E}^{\mathrm{hyb}} = C_*^{\mathrm{ch}}\left(\mathrm{Cur}_E(\mathfrak{n}_{\Delta, +})\right).$$

Retain the elliptic-degree grading on every Chevalley word and place its degree- m summand on $\mathrm{Ran}(E)_m$. Disjoint union adds the degree labels.

Theorem 9I.2 (Local–wrapped factorization). $\mathcal{F}_{\Delta, E}^{\mathrm{hyb}}$ is a graded factorization coalgebra on $\mathrm{Ran}^{\mathrm{wr}}(E)$. Its degree-zero color is the current theory of $\mathfrak{n}_{\Delta, +}[0]$. Every degree- m color is a module for the degree-zero color. The factorization product of degrees m and n lands in degree $m + n$. Its chiral primitives are $\mathrm{Cur}_E(\mathfrak{n}_{\Delta, +})$, and constant modes recover $\mathfrak{n}_{\Delta, +}$.

Proof. The chiral Chevalley construction is a factorization coalgebra for every chiral Lie algebra. The bracket preserves elliptic-degree addition, so its differential and coproduct preserve the graded Ran placement. The primitive theorem for the Chevalley coalgebra recovers the current Lie algebra. $\square$

This construction retains the nonnegative root-degree label on a point-supported current theory. It does not replace a sheaf projecting onto the whole elliptic curve by a point-supported sheaf. The compact wrapped comparison must map its common-torsor extension correspondences to these graded current operations, preserving support and charge. The bracket on the present carrier is the Borchers bracket.

PHYSICAL CHARGE AND GRAM DEGREE

Let Γ_{phys} be an additive physical charge lattice and let

$$\Pi(c) = \left(\frac{Q^2}{2}, Q \cdot P, \frac{P^2}{2} \right)$$

be the Gram map. Its polarization is

$$B(c, d) = \Pi(c + d) - \Pi(c) - \Pi(d).$$

Define the central extension

$$\tilde{\Gamma}_{\Pi} = \Gamma_{\text{phys}} \times \Gamma_{\text{ind}}$$

with multiplication

$$(c, \lambda)(d, \mu) = (c + d, \lambda + \mu + B(c, d)).$$

Theorem 92.1 (Graph homomorphism). *The map*

$$j_{\Pi} : \Gamma_{\text{phys}} \longrightarrow \tilde{\Gamma}_{\Pi}, \quad c \longmapsto (c, \Pi(c))$$

is a group homomorphism. For every Γ_{phys} -graded algebra A , the Rees lift

$$a_c \longmapsto a_c u^{\Pi(c)}$$

turns multiplication into an additive $\tilde{\Gamma}_{\Pi}$ -grading with cocycle $u^{-B(c, d)}$.

Proof. The defining multiplication gives

$$j_{\Pi}(c)j_{\Pi}(d) = (c + d, \Pi(c) + \Pi(d) + B(c, d)) = j_{\Pi}(c + d).$$

The algebra identity follows immediately. □

The universal Igusa category is graded directly by the Lorentzian root lattice. A geometric $K_3 \times E$ theory enters through a charge functor to $\tilde{\Gamma}_{\Pi}$.

THE UNIVERSAL PROTECTED FIELD THEORY

For each H take cotangent BF theory with gauge Lie superalgebra $\mathfrak{n}_{\Delta,+}^{(H)}$ on $\mathbb{R}_{\text{top}}^2 \times E$. Its fields are

$$\Omega^\bullet(\mathbb{R}^2) \widehat{\otimes} \Omega^{\circ,\bullet}(E) \widehat{\otimes} \left(\mathfrak{n}_{\Delta,+}^{(H)}[1] \oplus (\mathfrak{n}_{\Delta,+}^{(H)})^\vee[2] \right).$$

The action is

$$S_H = \int_{\mathbb{R}^2 \times E} \langle B, (d + \bar{\partial})A + \frac{1}{2}[A, A] \rangle.$$

Theorem 93.1 (Classical local theory and exact quantum harmonic sector). *For every finite height H the shifted-cotangent action defines a classical mixed holomorphic–topological factorization algebra on $\mathbb{R}_{\text{top}}^2 \times E$. Harmonic reduction on a compact product cell gives the finite-dimensional cotangent BV theory of $\mathfrak{n}_{\Delta,+}^{(H)}$, and this reduced theory satisfies the quantum master equation. Complementary Lagrangian boundary polarizations give the boundary enveloping algebra $U(\mathfrak{n}_{\Delta,+}^{(H)})$, the two-sided bar complex $C_*(\mathfrak{n}_{\Delta,+}^{(H)})$, and the current envelope on E . The proper-height inverse system is coefficientwise finite and defines a complete classical local theory together with an exact quantum harmonic sector.*

Proof. The invariant cotangent pairing and the Jacobi identity give the classical master equation and factorization descent. Harmonic reduction leaves a finite positive nilpotent Lie superalgebra. Every even adjoint endomorphism raises proper height and is strictly upper triangular; every odd adjoint endomorphism has vanishing supertrace. The reduced BV divergence therefore vanishes. Canonical quantization in the selected polarization gives the enveloping algebra, and the augmentation boundary states give the two-sided bar, identified with Chevalley chains by PBW. Each map preserves height, so the inverse system is defined coefficientwise. $\square$

ABSOLUTE CONVERGENCE OF THE VIRTUAL ROOT OPERATOR

For each positive root α , let V_α be the minimal parity realization of the signed multiplicity: it is purely even of dimension $\text{sdim } \mathfrak{g}_{\Delta, \alpha}$ when this integer is nonnegative and purely odd of dimension $-\text{sdim } \mathfrak{g}_{\Delta, \alpha}$ when it is negative. Set

$$\mathcal{V}_\Delta = \widehat{\bigoplus_{\alpha > 0} V_\alpha}.$$

This virtual root superspace is the canonical linear carrier of the denominator character; the Lie bracket remains carried by $\mathfrak{g}_{\Delta, \alpha}$.

Let $h(\alpha)$ be the proper height. Jacobi-form coefficient estimates give constants $C_0, C_1 > 0$ such that

$$\dim V_\alpha = \left| \text{sdim } \mathfrak{g}_{\Delta, \alpha} \right| \leq C_0 \exp(C_1 h(\alpha)) \quad (94.1)$$

for every positive root. The number of Gram triples of height H is polynomial in H .

Theorem 94.1 (Nonempty trace-class chamber). *Let Z_0 lie in the interior of the positive Weyl chamber. For sufficiently large $t > 0$,*

$$\sum_{\alpha > 0} \dim V_\alpha \left| e^{-\alpha(tZ_0)} \right| < \infty.$$

Hence the absolute-convergence chamber contains a conical tail in every interior direction.

Proof. Interior positivity gives a constant $c(Z_0) > 0$ with $\Re \alpha(Z_0) \geq c(Z_0)h(\alpha)$. Combine this with (94.1). The contribution of roots of height H is bounded by a polynomial in H times

$$\exp((C_1 - tc(Z_0))H).$$

The series converges when $tc(Z_0) > C_1$. □

Corollary 94.2 (Trace-class root Hamiltonian). *On the conical tail, the diagonal operator*

$$K_Z|_{V_\alpha} = e^{-\alpha(Z)} \text{id}$$

is trace class on both parity summands. The Fredholm product is holomorphic and equals the absolutely convergent Borchers product.

Proof. The trace norm is the summable series of the preceding theorem, separated by parity. Standard trace-class determinant theory gives holomorphy, and diagonalization gives the product over root eigenvalues. □

THE VIRTUAL ROOT FREDHOLM OPERATOR

Let

$$\mathcal{H}_\Delta^{\text{vir}} = \widehat{\bigoplus_{\alpha > 0} V_\alpha}$$

with an orthonormal basis in each minimal parity realization. This is the virtual root Hilbert space used in the preceding absolute-convergence theorem. In the absolute-convergence chamber define

$$K_Z|_{V_\alpha} = e^{-\alpha(Z)} \text{id}, \quad D_Z = 1 - K_Z.$$

Theorem 95.1 (Oriented Fredholm square root). *The operator K_Z is trace class on the even and odd summands of the virtual root Hilbert space. Its Fredholm Berezinian is*

$$\text{sdet}_F(D_Z) = \prod_{\alpha > 0} (1 - e^{-\alpha(Z)})^{\text{sdim } \mathfrak{g}_{\Delta, \alpha}}.$$

Choose an order of the positive roots, first by height and then lexicographically. This choice orients every finite-height root space. The skew doubling

$$\mathbb{A}_Z = \begin{pmatrix} 0 & D_Z \\ -D_Z^t & 0 \end{pmatrix}$$

has an oriented Fredholm super-Pfaffian satisfying

$$\text{sPf}_F(\mathbb{A}_Z) = \text{sdet}_F(D_Z).$$

Consequently

$$e^{-\rho} \text{sPf}_F(\mathbb{A}_Z) = D_\zeta(2Z).$$

Proof. Absolute convergence of the Borchers product is the trace-norm summability condition. The determinant formula follows from the spectrum. At finite height the Pfaffian of the oriented skew doubling is the determinant. Trace-norm convergence passes the identity to the limit. The denominator theorem supplies the Weyl shift. $\square$

95.1 AUTOMORPHIC ORIENTATION

The product defines a section of the automorphic line $\mathcal{L}^\zeta \otimes \nu_\zeta$. On the absolute-convergence chamber, the oriented Fredholm Pfaffian is a trivialization of this line. Analytic continuation uses the automorphy factor of D_ζ and therefore glues the chamberwise Fredholm models with transition character ν_ζ .

Corollary 95.2 (Orientation character). *The automorphic orientation line of the root Dirac family is the sign local system ν_ζ . On every real simple reflection it has value -1 .*

Proof. The Fredholm product equals D_ζ on the convergence chamber, and D_ζ transforms with automorphy factor ν_ζ . This transformation law is the gluing rule for the automorphic orientation line. A real simple root occurs with multiplicity one, so reflection reverses its oriented normal line. $\square$

THE UNIVERSAL IGUSA PROTECTED EFFECTIVE THEORY

Definition 96.1 (Universal Igusa completion). The protected completion selected by the K_3 half-index is the quintuple

$$\mathfrak{R}_\Delta = \left(\Gamma_\Delta, \mathcal{B}_\Delta, \mathcal{H}_{\text{ip}}, \mathcal{F}_{\Delta, E}^{\text{hyb}}, \mathbb{A}_Z \right).$$

Its components are the pro-system of finite three-Calabi–Yau completions, the ordinary quiver Hall category, the Borchers one-particle Lie superalgebra, the local–wrapped current factorization algebra, and the virtual root Dirac family.

Theorem 96.2 (Igusa protected completion). *The quintuple $\mathfrak{R}_\Delta$ satisfies:*

- (i) *its one-particle Lie superalgebra is $\mathfrak{g}_{\Delta, \zeta}$;*
- (ii) *its ordinary Hall presentation and Hong’s quantum supergroup retain the stated Cartan grading; comparison of their relation ideals and parity-refined PBW spaces requires an additional map;*
- (iii) *its chiral primitive algebra on E is the local–wrapped current algebra of $\mathfrak{g}_{\Delta, \zeta}$;*
- (iv) *its raw multiparticle supercharacter is P^{-1} , where $e^{-\rho} P = \text{Den}_\Delta$; equivalently $e^\rho \text{sch Fock}(\mathfrak{n}_{\Delta, +}) = \text{Den}_\Delta^{-1}$;*
- (v) *the half-coordinate pullback of the doubled Weyl-normalized character is $-\chi_{10}^{-1}$ in the Oberdieck–Pixton normalization;*
- (vi) *its oriented first-order determinant is Den_Δ and its orientation monodromy is ν_ζ .*

Proof. The first three statements are the Borchers root, quiver Hall, Hong quantum-supergroup, Ginzburg, and colored-Ran theorems. PBW gives the fourth. The identity $\chi_{10} = D_\zeta^{-2}$ and the established reduced scalar theorem give the fifth. The Fredholm theorem gives the sixth. $\square$

The tuple retains algebraic constructions associated with the Igusa presentation and their scalar characters. A geometric type-IIA or topological-string realization must construct compatible functors from its oriented extension category to these targets. Primitive states and extension amplitudes are inputs to those functors. Their action on morphisms, convolution, orientations, and chiral operations must also be specified and proved compatible.

THE PRIMITIVE COMPARISON AND ITS EXTENSIONS

Let $\mathfrak{p}$ be a positive-root-graded Lie superalgebra with finite-dimensional homogeneous pieces and a proper positive height. A primitive realization of the Igusa presentation is a parity-preserving homogeneous map from its labelled simple-root spaces to $\mathfrak{p}$, satisfying the defining relations.

Theorem 97.1 (Universal presentation map). *Let $\mathfrak{p}$ be a positive-root-graded Lie superalgebra over $\mathbb{C}$, with finite-dimensional homogeneous pieces. Suppose a homogeneous parity-preserving map from every simple-root generating space of $\mathfrak{n}_{\Delta,+}$ to $\mathfrak{p}$ satisfies the defining Borchers–Serre relations. It extends uniquely to a graded Lie-superalgebra map*

$$f : \mathfrak{n}_{\Delta,+} \longrightarrow \mathfrak{p}.$$

It induces continuous maps on height completions, enveloping algebras, Chevalley chains, and level-zero currents. The map f is an isomorphism if and only if it is surjective and the root-graded PBW series agree in each parity.

Proof. Extend the chosen maps to the free Lie superalgebra on the simple-root spaces. The relation hypothesis makes the defining ideal lie in its kernel, giving f . The generators prove uniqueness. On enveloping algebras the map sends $x_1 \cdots x_j$ to $f(x_1) \cdots f(x_j)$. On Chevalley chains it is the graded symmetric map on the shifted generators. It commutes with the Chevalley differential because f preserves brackets. The current map is $\partial^j x \mapsto \partial^j f(x)$, preserving $[x_\lambda y] = [x, y]$. Root degree preservation gives compatible maps at each height and hence on the stated limits.

If f is surjective, $U(f)$ is surjective. Proper positive height makes each root-graded PBW component finite-dimensional. Equality of its even and odd dimensions makes $U(f)$ bijective in every component. PBW embeds each Lie superalgebra into its enveloping algebra, so f is injective. The converse follows by applying the constructions to its inverse. $\square$

A geometric comparison also requires functors on the actual Hall and critical categories, a compatible coproduct and Hopf pairing, and maps of determinant lines and operator domains. These categories and lines are not the domain of the primitive Lie map. The additional maps must be constructed and checked against that map. They are not consequences of its universal property.

THE EIGHT DIAGONAL-DIVISOR FORMS

Cléry and Gritsenko classify eight genus-two modular forms whose divisor is the diagonal with order one [16]. Gritsenko–Nikulin and the CHL denominator constructions give the sourced reflective presentations used for the corresponding rows [34–36, 38–40]. Every row has a minimal virtual product superspace obtained from its Fourier exponents, together with a Fock determinant and a diagonal Fredholm operator. A sourced denominator presentation refines this product model by its Borchers–Cartan bracket, Weyl group, and quantum supergroup. The half-integral rows are formed on their multiplier covers.

form	group	weight	character order
Δ_5	$\mathrm{Sp}_4(\mathbb{Z})$	5	2
Δ_2	Γ_2	2	4
Δ_1	Γ_3	1	6
$\Delta_{1/2}$	Γ_4	1/2	8
∇_3	$\Gamma_0^{(2)}(2)$	3	2
∇_2	$\Gamma_0^{(2)}(3)$	2	2
$\nabla_{3/2}$	$\Gamma_0^{(2)}(4)$	3/2	4
Q_1	$\Gamma_2(2)$	1	4

Theorem 98.1 (Eight-row algebraic realization). *Each diagonal-divisor row has a proper-height virtual root superspace, a super-Fock determinant, a colored abelian current theory, and a trace-class Fredholm denominator operator whose character is the corresponding automorphic product. For every row equipped with its reflective denominator presentation, the finite-window Borchers quiver, ordinary Hall category, three-Calabi–Yau completion, and parity-refined quantum supergroup refine this product model and recover the same denominator character.*

Proof. Local finiteness of the product exponents gives the graded superspace and its proper-height completion. The Fock and Fredholm constructions depend only on these signed multiplicities and reproduce the product coefficientwise. A reflective denominator presentation supplies a symmetrizable Borchers superdatum; the quiver, Hall, Ginzburg, current, and Hong constructions then apply to that datum and preserve its PBW denominator. $\square$

ALGEBRAIC AND PHYSICAL CLOSURE

Theorem 99.1 (Universal Igusa realization theorem). *The constructions have the following inputs and outputs. An arrow labelled by a construction records its dependence on that input:*

$$\begin{array}{llll}
 \phi_{o,I} \xrightarrow{\text{Borch}} D_5, & \phi_{o,I} \xrightarrow{\text{root datum}} \mathfrak{g}_{\Delta_5}, & A_{\Delta} \xrightarrow{\text{quiver}} Q_{\Delta,H} \xrightarrow{\text{Hall}} \mathcal{B}_{\Delta}, \\
 kQ_{\Delta,H} \xrightarrow{\Pi_3} \Gamma_{\Delta,H}, & \mathfrak{g}_{\Delta_5} \xrightarrow{\text{Cur}_E} \mathcal{F}_{\Delta,E}^{\text{hyb}}, & \mathfrak{g}_{\Delta_5} \xrightarrow{\text{Hong}} \widehat{U}_q(\mathfrak{g}_{\Delta_5}), \\
 \mathfrak{g}_{\Delta_5} \xrightarrow{\text{root spectrum}} \mathbb{A}_Z, & e^{-\rho} \text{sPf}_F(\mathbb{A}_Z) = \text{Den}_{\Delta}(Z) = D_5(2Z), \\
 \mathcal{Z}_{K_3 \times E}^{\text{red}} = -(j^* \text{Den}_{\Delta})^{-2} = -\chi_{10}^{-1}.
 \end{array}$$

The system contains parity-refined one-particle spaces, an ordinary Hall extension product, the Borchers root bracket, a local-wrapped collision law, a quantum supergroup, an automorphic orientation line, a first-order Fredholm determinant, and the reduced scalar square.

Proof. The Borchers lift, Gritsenko–Nikulin presentation, and primitive reduced curve-counting theorem give the scalar identities. The enveloping and Chevalley maps are induced by the presentation. The quiver Hall algebra and finite Calabi–Yau completion have their respective quiver and path-algebra inputs. Their common grading specifies the degrees a comparison must preserve. A map between those constructions requires the additional functors, coefficient maps, and pairings described above. $\square$

BOOK **XX**

**Technical appendices to the preceding
book**

EXACT THETA-SERIES CALCULATION

Set $t = q^{1/8}$ and $x = r^{1/2}$. Then

$$\begin{aligned}\mathfrak{g}_2(\tau, z) &= \sum_{n \in \mathbb{Z}} t^{4n^2+4n+1} x^{2n+1}, \\ \mathfrak{g}_3(\tau, z) &= \sum_{n \in \mathbb{Z}} t^{4n^2} x^{2n}, \\ \mathfrak{g}_4(\tau, z) &= \sum_{n \in \mathbb{Z}} (-1)^n t^{4n^2} x^{2n}.\end{aligned}$$

Squaring numerator and denominator, dividing in $\mathbb{Q}[x^{\pm 1}][[t]]$, and summing the three quotients gives the coefficient table in Chapter 1. The calculation uses integer convolution and formal-series inversion; fractional powers cancel before conversion from (t, x) to (q, r) .

THE ROOT REFLECTION MATRICES

In the basis (f_2, f_3, f_{-2}) , the form is

$$G = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 2 & 0 \\ -1 & 0 & 0 \end{pmatrix}.$$

The three reflection matrices are

$$S_1 = \begin{pmatrix} 1 & 4 & 4 \\ 0 & -1 & -2 \\ 0 & 0 & 1 \end{pmatrix}, \quad S_2 = \begin{pmatrix} 1 & 0 & 0 \\ -2 & -1 & 0 \\ 4 & 4 & 1 \end{pmatrix}, \quad S_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Direct multiplication gives

$$S_i^2 = I, \quad S_i^\top G S_i = G.$$

AUTOMORPHIC, HOMOLOGICAL, AND HALL COMPARISONS

The half-index determinant, theta normalization, Borchers denominator, reduced scalar theorem, charge Gram map, and geometric Hall comparison datum belong to the following constructions:

- (i) the automorphic square root is completed entirely within the Borchers product theorem;
- (ii) the BKM bracket comes from the Gritsenko–Nikulin algebra;
- (iii) the bar–Chevalley comparison gives the denominator a homological source;
- (iv) the current and BF constructions give an exact first-order algebraic model;
- (v) orientation data refine determinant lines, while an operator Pfaffian requires an actual Fredholm family;
- (vi) Weyl reflections are attached to real roots, and imaginary roots enter through multiplicities;
- (vii) the compact Hall comparison uses additive Mukai charge, the polarized central extension, and hybrid local–wrapped sectors.

BOOK ENTRANCE: CALABI–YAU QUANTUM GROUPS

An extension of geometric objects defines a correspondence. Its action on cohomology depends on the coefficient theory and the allowed pushforward. Tensor products and exchange require a coproduct and compatible intertwiners. A trace on a specified representation then produces an amplitude. The reconstruction problem is to retain these operations through the comparisons between Calabi–Yau categories, boundary fields, Hall algebras, and quantum groups.

BOOK **XXI**

**Categorical and field-theoretic
foundations**

CALABI–YAU CATEGORIES

103.1 SMOOTHNESS, PROPERNESS, AND THE CALABI–YAU CLASS

Let C be a small stable $\mathbb{k}$ -linear category. Smoothness means that the diagonal bimodule is perfect. Properness means that the morphism complexes are perfect over $\mathbb{k}$. A d -Calabi–Yau structure is a nondegenerate negative-cyclic class

$$\theta_{\text{CY}} \in HC_d^-(C)$$

whose underlying Hochschild trace identifies the diagonal bimodule with its dual shifted by d .

For $E \in C$, the tangent complex of the derived moduli stack of objects is

$$T_E \mathcal{M}_C \simeq \text{RHom}_C(E, E)[1].$$

The Calabi–Yau trace pairs two tangent vectors by composition and trace.

Theorem 103.1 (The two universal categorical outputs). *Let C be a smooth proper d -Calabi–Yau category.*

- (i) *The Hochschild cochains $CC^*(C)$ carry a framed E_2 -algebra structure.*
- (ii) *Whenever the moduli stack of objects is representable in the PTVV setting, $\mathcal{M}_C$ carries a $(2-d)$ -shifted symplectic form.*

Proof. The cyclic Deligne theorem upgrades the brace E_2 structure on Hochschild cochains by the circle action supplied by the cyclic trace [10]. The shifted form at E is

$$\text{RHom}(E, E)[1]^{\otimes 2} \xrightarrow{\circ} \text{RHom}(E, E)[2] \xrightarrow{\text{Tr}} \mathbb{k}[2-d].$$

Nondegeneracy is the Calabi–Yau duality isomorphism, and closedness is the negative-cyclic lift [9, 64]. □

The integer d controls the shift of the moduli form. The Hochschild operation remains framed E_2 . These structures meet through deformation theory while retaining their separate operadic degrees.

LOCAL BV THEORIES AND FACTORIZATION OBSERVABLES

104.1 CYCLIC ELLIPTIC LOCAL LIE ALGEBRAS

Let X be a complex manifold. A classical local field theory is specified by a sheaf $\mathcal{L}$ of elliptic L_∞ algebras, a local invariant pairing, and an action functional S satisfying the classical master equation. For an open set $U \subset X$, the classical observables are

$$\text{Obs}^{\text{cl}}(U) = C_{\text{CE,cont}}^*(\mathcal{L}(U)).$$

Here $\mathcal{L}(U)$ consists of unrestricted smooth fields with its nuclear smooth-section topology. Continuous cochains use the completed symmetric powers of its continuous dual, with the completed tensor products of the elliptic-observable construction [18, 19]. The dual consists of compactly supported distributions. For $U \subset V$, restriction of fields $\mathcal{L}(V) \rightarrow \mathcal{L}(U)$ induces the covariant map on observables. For pairwise disjoint $U_i \subset V$, restrict to $\bigoplus_i \mathcal{L}(U_i)$ and multiply the pulled-back cochains. These are the prefactorization maps.

Theorem 104.1 (Factorization observables). *An elliptic classical BV theory produces a factorization algebra Obs^{cl} . A renormalized quantization satisfying the quantum master equation produces a factorization algebra Obs^{q} over $\mathbb{k}[[\hbar]]$.*

Proof. Restriction is a morphism of local L_∞ algebras, so its contravariant continuous Chevalley map commutes with the differential. For disjoint opens, the completed cochains of the finite direct sum are the completed tensor product of the cochains. This constructs the displayed maps and their composition identities. The elliptic descent theorem in this completed coefficient category supplies Weiss descent [18, 19]. In the quantum theory, renormalized time-ordered products define the structure maps, and the quantum master equation makes their differential square-zero and compatible with nested ultraviolet contractions [18, 19]. $\square$

104.2 BOUNDARY AND DEFECT RESTRICTION

Let $Y = C \times \mathbb{R}_{\geq 0}$, with C a complex curve, and let a boundary condition be given at $C \times \{0\}$. Boundary observables are holomorphic along C and ordered along the inward normal line. A compact boundary object b in the boundary factorization category gives

$$A_b = \text{RHom}(b, b).$$

Theorem 104.2 (Ordered boundary algebra). *When boundary observables are holomorphic on C and locally constant along the oriented normal direction, A_b is an ordered chiral algebra*

$$A_b \in \text{Alg}_{\text{E}_1}(\text{Fact}_D(C), \otimes^!).$$

If b is a local compact generator and the Morita equivalences obey factorization descent, the boundary category is equivalent to A_b -modules in factorization coefficients.

Proof. Composition of boundary endomorphisms gives the internal E_1 multiplication. Holomorphic collision gives the factorization coefficient on C . The componentwise locality identity makes multiplication compatible with every disjoint-support factorization map. Local compact-generator Morita theory descends over the Ran diagrams. These are the recognition and Morita theorems of Volume I. $\square$

THREE CURVE-LEVEL ALGEBRAIC OUTPUTS

Let C be a smooth curve and let $\mathcal{D}_C$ be the fixed Ran category. The same geometric source can produce three algebraic outputs, and their distinction is functorial.

Definition 105.1 (Curve-level faces). A transverse face is

$$\mathcal{A}^\perp \in \text{Alg}_{\mathcal{E}_1}(\text{Fact}_D(C), \otimes^\perp).$$

An associative chiral face is

$$\mathcal{A}_{\text{Ass}}^{\text{ch}} \in \text{Alg}_{\mathcal{E}_1}(\mathcal{D}_C, \star^{\text{ch}}).$$

A chiral Lie face is

$$\mathfrak{g}^{\text{ch}} \in \text{Alg}_{\mathcal{L}\text{ie}}(\mathcal{D}_C, \star^{\text{ch}}).$$

Their bars are $\text{Bar}^\perp$, $\text{Bar}_{\text{Ass}}^{\text{ch}}$, and $\text{Bar}_{\text{Lie}}^{\text{ch}}$.

Theorem 105.2 (Source separation). *A boundary normal direction determines the transverse face. A chiral-convolution multiplication determines the associative chiral face. A Lie bracket for chiral convolution determines the Lie face. The commutator functor sends the associative chiral face to the Lie face. A chiral enveloping algebra $U^{\text{ch}}(\mathfrak{g})$ satisfies*

$$\text{Bar}_{\text{Ass}}^{\text{ch}}(U^{\text{ch}}(\mathfrak{g})) \simeq \text{Bar}_{\text{Lie}}^{\text{ch}}(\mathfrak{g})$$

on the nilcomplete PBW locus. A mixed source joins the transverse and associative chiral faces through an aligned distributive law.

Proof. Each statement is the corresponding operadic restriction or adjunction in the named monoidal category. The PBW equivalence is the enveloping lane of chiral Koszul duality. The mixed statement is the distributive-law theorem of Volume I. $\square$

This theorem replaces every dimension-based dispatch of an E_n level. The geometry of the source selects the available directions; the operadic level records the operations actually constructed.

HOLOMORPHIC CHERN–SIMONS THEORY

Let X be a complex threefold with holomorphic volume form Ω_X , and let $\mathfrak{a}$ be a quadratic Lie algebra. The BV fields are

$$\mathcal{E}_X = \Omega_X^{\circ,*}(X, \mathfrak{a})[1],$$

with degree -1 pairing

$$\omega(A, B) = \int_X \Omega_X \wedge \langle A, B \rangle.$$

The action is

$$S_{\text{hCS}}(A) = \int_X \Omega_X \wedge \left(\frac{1}{2} \langle A, \bar{\partial} A \rangle + \frac{1}{6} \langle A, [A, A] \rangle \right).$$

Theorem 106.1 (Classical hCS factorization algebra). *The action S_{hCS} satisfies the classical master equation. Its classical observables form a holomorphic factorization algebra on X .*

Proof. The Hamiltonian vector field is the Maurer–Cartan differential $\bar{\partial} A + \frac{1}{2} [A, A]$. Its square is the Dolbeault square, the derivation identity, and the Jacobi identity. The elliptic Dolbeault complex then gives factorization observables by the preceding theorem. $\square$

A perturbative quantization is controlled by the local BV deformation complex. In complex dimension three, its one-loop gauge anomaly occupies the degree-four invariant-polynomial summand; mixed and gravitational components enter when the background fields carry the corresponding couplings. A quantization is a trivialization of the complete local anomaly class, compatible with restriction and factorization [20].

PUSHFORWARD TO A CURVE

Let $p : X \rightarrow C$ be a holomorphic map to a complex curve. For a factorization algebra $\mathbb{F}$ on X , define

$$(p_*\mathbb{F})(U) = \mathbb{F}(p^{-1}U).$$

Theorem 107.1 (Factorization pushforward). *The assignment $p_*\mathbb{F}$ is a factorization algebra on C . Suppose in addition that p is proper and locally trivial in a topology compatible with the factorization theory, that $\mathbb{F}$ is locally constant in the fiber directions, and that product-chart excision and base change hold. Then on every trivializing open $U \subset C$ with fiber F_p there is a natural equivalence*

$$(p_*\mathbb{F})(U) \simeq \int_{F_p} \mathbb{F}|_{F_p \times U},$$

compatible on overlaps. If the resulting object is holomorphic and regular holonomic along C , Riemann–Hilbert algebraization gives a factorization D -module on C .

Proof. Disjoint opens in C have disjoint inverse images, so the factorization products push forward. Weiss descent is inherited from inverse images of Weiss covers. On a product chart, local constancy in the fiber and factorization excision identify the value with fiberwise factorization homology. The base-change hypothesis glues these local equivalences. Regular-holonomic Riemann–Hilbert transports the holomorphic factorization object to the algebraic D -module setting. $\square$

Thus a curve shadow is attached to an actual map and an actual field theory. Different fibrations and boundary conditions produce different ordered chiral algebras from the same categorical source.

THE DERIVED CENTRE AND THE PHYSICAL BULK

For an ordered boundary algebra A , define

$$Z_{\text{ord}}(A) = \text{RHom}_{A \otimes A^{\text{op}}}(A, A).$$

It is a canonical E_2 algebra. A bulk theory acting on the boundary gives

$$\beta : \text{Obs}_{\text{bulk}}|_{\partial} \longrightarrow Z_{\text{ord}}(A).$$

Definition 108.1 (Open–closed completeness). A boundary condition is open–closed complete on a protected sector when β is an equivalence on that sector.

The definition turns the phrase “bulk equals centre” into a comparison theorem. The centre exists algebraically for every ordered boundary algebra; the physical bulk is identified with it by the open–closed map.

BOOK XXII

Hall and BPS algebras

EXTENSION CORRESPONDENCES

Let $\mathcal{M} = \coprod_{\gamma \in \Gamma} \mathcal{M}_\gamma$ be a moduli stack in an exact or derived category. Let $\mathcal{M}_{\alpha, \beta}^{\text{ext}}$ classify extensions

$$0 \longrightarrow E_\beta \longrightarrow E \longrightarrow E_\alpha \longrightarrow 0.$$

The correspondence is

$$\mathcal{M}_\alpha \times \mathcal{M}_\beta \xleftarrow{p} \mathcal{M}_{\alpha, \beta}^{\text{ext}} \xrightarrow{q} \mathcal{M}_{\alpha + \beta}.$$

Theorem 109.1 (Hall associativity). *Assume a coefficient theory in which p^* and q_* are defined and satisfy base change for the two-step filtration stack. Then*

$$a * b = q_* p^*(a \boxtimes b)$$

defines an associative product.

Proof. The two parenthesizations are the two projections from the stack of filtrations $0 \subset E_1 \subset E_2 \subset E_3$. Base change identifies both composites with pullback to this stack followed by pushforward to E_3 . $\square$

For critical CoHAs, the coefficient theory includes vanishing cycles and a coherent orientation. For symplectic CoHAs, it includes the selected equivariant Borel–Moore theory. Completion is chosen so that each output degree receives a convergent sum.

ORIENTATION DATA AND THE THREEFOLD HALL PRODUCT

For compact Calabi–Yau threefolds, natural orientation data on moduli of coherent sheaves and perfect complexes is constructed by Joyce and Upmeyer [47]. A general cohomological Hall algebra for a 3-Calabi–Yau category uses strong orientation data, which carries the coherent multiplicativity required by the attractor and extension correspondences [50].

Theorem 110.1 (Threefold Hall algebra from strong orientation). *Let $\mathcal{C}$ be a 3-Calabi–Yau category satisfying the geometric hypotheses of Kinjo–Park–Safronov, and assume strong orientation data. Then its cohomological Donaldson–Thomas complexes carry an associative cohomological Hall product. The product is the parabolic-induction correspondence supplied by the attractor Lagrangian correspondence.*

Proof. Kinjo–Park–Safronov prove the functoriality of the Donaldson–Thomas perverse sheaf under the attractor Lagrangian correspondence and use strong orientation to define the parabolic-induction morphism. Their base-change and associativity theorem gives the CoHA [50]. $\square$

Definition 110.2 (Reduced $K_3 \times E$ Hall datum). A reduced $K_3 \times E$ Hall datum is a strong-oriented threefold Hall datum together with descent along the elliptic translation action, a reduced deformation theory, and a separated completion in physical charge. A full quantum-group datum additionally includes a coproduct, Cartan extension, negative half, and nondegenerate continuous Hopf pairing.

THE GEOMETRIC E -EQUIVARIANT HALL SOURCE

Let S be a projective K_3 surface, let E be an elliptic curve, and put

$$X = S \times E.$$

Write $\mathrm{Coh}_{\leq 1}(X)$ for coherent sheaves whose support has dimension at most one. This is an extension-closed abelian category. Its numerical charge group is denoted Γ_X , and the degree of the projected one-cycle on E is the additive function

$$m : \Gamma_X \longrightarrow \mathbb{Z}_{\geq 0}.$$

For $\gamma \in \Gamma_X$, let $\mathcal{M}_\gamma$ be the moduli stack of objects of charge γ . Translation on the elliptic factor gives an action of E on every $\mathcal{M}_\gamma$ and on every extension stack.

Oberdieck and Shen construct the E -equivariant Grothendieck ring of stacks and its Hall convolution. In the present notation the completed algebra is

$$\mathcal{H}_X^{E, \mathrm{mot}} = \widehat{\bigoplus_{\gamma \in \Gamma_X} K_o^E(\mathrm{St}/\mathcal{M}_\gamma)},$$

where the completion is taken in effective curve class and Euler characteristic.

Theorem III.1 (Geometric reduced Hall algebra). *The extension correspondence in $\mathrm{Coh}_{\leq 1}(X)$ defines an associative E -equivariant motivic Hall product on $\mathcal{H}_X^{E, \mathrm{mot}}$. The Oberdieck–Shen reduced integration morphism*

$$I_X^E : \mathcal{H}_X^{E, \mathrm{mot}} \longrightarrow \widehat{\mathbb{Q}[\Gamma_X][\epsilon]/(\epsilon^2)}$$

is multiplicative on the regular subalgebra on which their integration theorem is defined. The coefficient of ϵ is the reduced contribution of the positive-dimensional E -orbits.

Proof. The stack of short exact sequences is E -equivariant because translation preserves exactness and numerical charge. The stack of two-step filtrations identifies the two composites of three Hall products, so equivariant base change proves associativity. The integration statement is the E -equivariant motivic Hall theorem of Oberdieck–Shen [63]. Their stabilizer stratification separates fixed contributions from moving elliptic orbits and packages the latter in the square-zero parameter ϵ . $\square$

LOCAL AND WRAPPED GEOMETRY INSIDE THE HALL ALGEBRA

Define the completed degree-zero and positive-degree pieces

$$\mathcal{H}_X^{\text{loc}} = \widehat{\bigoplus_{m(\gamma)=0} K_o^E(\text{St}/\mathcal{M}_\gamma)}, \quad \mathcal{H}_X^{\text{wr}} = \widehat{\bigoplus_{m(\gamma)>0} K_o^E(\text{St}/\mathcal{M}_\gamma)}.$$

A pure one-dimensional support of degree zero projects to a finite subset of E . A support of positive degree projects onto E . The first sector is point-local on the elliptic curve; the second is wrapped.

Theorem II.2.1 (Geometric local–wrapped decomposition). *Hall multiplication respects elliptic degree:*

$$\mathcal{H}_{m_1} * \mathcal{H}_{m_2} \subseteq \mathcal{H}_{m_1+m_2}.$$

Consequently $\mathcal{H}_X^{\text{loc}}$ is a subalgebra, $\mathcal{H}_X^{\text{wr}}$ is a closed two-sided ideal, and

$$\mathcal{H}_X^{E, \text{mot}} \cong \mathcal{H}_X^{\text{loc}} \ltimes \mathcal{H}_X^{\text{wr}}$$

as a complete vector space with multiplication determined by the Hall correspondences. The reduced integration morphism preserves this decomposition.

Proof. The cycle of the middle term of a short exact sequence is the sum of the cycles of its subobject and quotient. Projection to E therefore adds degrees. Degree zero is closed under addition, and positive degree absorbs addition by a nonnegative degree. The same grading is present on the target group algebra of I_X^E . $\square$

This theorem supplies the geometric version of the local–wrapped splitting constructed from the Borchers root algebra in the companion volume. The two constructions have the same additive elliptic degree and become comparable after a primitive root framing is chosen.

COHOMOLOGICAL REFINEMENT AND THE GRAM CHARACTER

A cohomological refinement uses the Donaldson–Thomas sheaf on the derived moduli stack. The datum required for the convolution is the strong orientation coherence of Kinjo–Park–Safronov. A strong orientation enhancement of the natural Joyce–Upmeyer orientation therefore produces

$$\mathcal{H}_X^{E, \text{coh}} = \widehat{\bigoplus}_{\gamma \in \Gamma_X} H_*^{\text{BM}, E}(\mathcal{M}_\gamma, \Phi_\gamma),$$

with its cohomological Hall product.

Theorem II.3.1 (Strong-oriented cohomological lift). *For every strong orientation enhancement on the E -equivariant derived moduli problem, the attractor correspondence constructs an associative cohomological Hall algebra $\mathcal{H}_X^{E, \text{coh}}$. Elliptic degree again gives a subalgebra in degree zero and a closed ideal in positive degree. Forgetting the cohomological coefficient system recovers the corresponding motivic Hall correspondences.*

Proof. The parabolic-induction theorem of Kinjo–Park–Safronov gives the cohomological product from the strong-oriented attractor Lagrangian correspondence [50]. Numerical charge and elliptic degree are constant on connected components of every correspondence, so the proof of the preceding decomposition applies verbatim. $\square$

For the dyonic charge lattice

$$\Gamma_{\text{phys}} = \widetilde{H}(S, \mathbb{Z}) \oplus \widetilde{H}(S, \mathbb{Z}),$$

write

$$\Pi(Q, P) = \left(\frac{Q^2}{2}, Q \cdot P, \frac{P^2}{2} \right).$$

To compare sheaf charges with this lattice, choose an additive map

$$c : \Gamma_X \longrightarrow \Gamma_{\text{phys}}.$$

Put $B_\Pi(a, b) = \Pi(a + b) - \Pi(a) - \Pi(b)$ and give $\widetilde{\Gamma}_{\text{phys}} = \Gamma_{\text{phys}} \times \mathbb{Z}^3$ the group law

$$(a, u)(b, v) = (a + b, u + v + B_\Pi(a, b)).$$

Then $j_\Pi(a) = (a, \Pi(a))$ is a homomorphism. Assume $j_\Pi c$ preserves the selected charge completions. On the regular subalgebra $\mathcal{H}_{\text{reg}}$ of the reduced integration theorem, define

$$I_{X, \Pi, c}^E = (j_\Pi c)_* I_X^E : \mathcal{H}_{\text{reg}} \longrightarrow \widehat{\mathbb{Q}[\widetilde{\Gamma}_{\text{phys}}][\epsilon]/(\epsilon^2)}.$$

Write $I_{X, \Pi}^E$ for this map when c has been fixed.

Proposition 113.2 (Polarized reduced character). *The map $I_{X,\Pi,c}^E$ is multiplicative. Its coefficients are the reduced invariants supplied by I_X^E , regraded by $j_\Pi c$. If $g \in \mathbf{1} + F^1 H$ is group-like in a complete filtered Hall bialgebra and the logarithm converges, then $\log g$ is primitive.*

Proof. Expanding $\Pi(a + b + d)$ gives

$$B_\Pi(a, b) + B_\Pi(a + b, d) = B_\Pi(b, d) + B_\Pi(a, b + d).$$

This proves associativity of the group law. The definition gives $j_\Pi(a + b) = j_\Pi(a)j_\Pi(b)$. Composing its group-algebra map with reduced integration proves multiplicativity on $\mathcal{H}_{\text{reg}}$. For the last assertion, the two factors in $\Delta g = (g \hat{\otimes} \mathbf{1})(\mathbf{1} \hat{\otimes} g)$ commute. Taking the convergent logarithm gives $\Delta \log g = \log g \hat{\otimes} \mathbf{1} + \mathbf{1} \hat{\otimes} \log g$. $\square$

The primitive reduced Gromov–Witten theorem supplies the separate character $-\chi_{\text{io}}^{-1}$ in classes primitive in the K3 factor [62]. Comparing this character with the Hall character requires the same charge map, sector, and normalization. A primitive root framing must also preserve the extension bracket and parity of the chosen BPS states.

BPS PRIMITIVES

A connected graded bialgebra has primitive Lie algebra

$$\mathrm{Prim}(H) = \{x : \Delta x = x \otimes 1 + 1 \otimes x\}.$$

In cohomological Donaldson–Thomas theory, the BPS sheaf isolates the primitive part of the Hall algebra.

Theorem II.4.1 (Two-Calabi–Yau BPS algebra). *For the geometric two-Calabi–Yau categories covered by the BPS-algebra theorem, the BPS algebra is the positive enveloping algebra of a generalized Kac–Moody Lie algebra. This scope includes semistable sheaves on $K3$ and other symplectic surfaces under the stated good-moduli hypotheses.*

Proof. The theorem is the Davison–Hennecart–Schlegel Mejia construction [22]. The simple BPS sheaves give the simple-root spaces, extension correspondences give the bracket, and the integrality/PBW theorem identifies the Hall algebra with the positive enveloping algebra. $\square$

This result supplies a global $K3$ generalized Kac–Moody algebra at the two-Calabi–Yau Hall level. Yangian structures arise from additional equivariant correspondences and coproducts in the surface and quiver models described below.

THE INTEGRAL HALL–BORCHERDS AMALGAM

Let $\mathfrak{p}_X^+$ be a complete charge-graded BPS Lie superalgebra supplied by a geometric Hall source, and let $\mathfrak{n}_{\Delta,+}$ be the positive Igusa Borchers Lie superalgebra. The polarized charge extension places both gradings over one abelian Cartan algebra $\mathfrak{h}_\Pi$. Form the two complete Borels

$$\mathfrak{b}_X = \mathfrak{h}_\Pi \ltimes \mathfrak{p}_X^+, \quad \mathfrak{b}_\Delta = \mathfrak{h}_\Pi \ltimes \mathfrak{n}_{\Delta,+}.$$

The Cartan acts through the physical charge on the first Borel and through Gram degree on the second.

Definition 115.1 (Integral protected Lie algebra). The integral Hall–Borchers algebra is the completed amalgamated coproduct

$$\mathfrak{b}_{X \boxtimes \Delta} = \mathfrak{b}_X \widehat{\Pi}_{\mathfrak{h}_\Pi} \mathfrak{b}_\Delta$$

in complete charge-filtered dg Lie superalgebras.

Theorem 115.2 (Hall–Borchers amalgamation). *The completed amalgam exists and represents pairs of continuous Lie morphisms from $\mathfrak{b}_X$ and $\mathfrak{b}_\Delta$ that agree on $\mathfrak{h}_\Pi$. The Cartan projections of the two Borels induce canonical retractions*

$$r_X : \mathfrak{b}_{X \boxtimes \Delta} \longrightarrow \mathfrak{b}_X, \quad r_\Delta : \mathfrak{b}_{X \boxtimes \Delta} \longrightarrow \mathfrak{b}_\Delta.$$

Its completed enveloping algebra is the amalgamated free product

$$\widehat{U}(\mathfrak{b}_{X \boxtimes \Delta}) \cong \widehat{U}(\mathfrak{b}_X) \widehat{*}_{\widehat{U}(\mathfrak{h}_\Pi)} \widehat{U}(\mathfrak{b}_\Delta).$$

The geometric Hall operations survive under r_X , while the Igusa bracket, PBW character, and automorphic determinant survive under r_Δ .

Proof. Construct the ordinary Lie coproduct from the free product of the two Borels and divide by the ideal identifying their two Cartan copies. Complete by total positive charge and proper root height. Every fixed filtered degree uses finitely many Lie words, so the completion is separated and has the stated mapping property. The projection $\mathfrak{b}_X \rightarrow \mathfrak{h}_\Pi \rightarrow \mathfrak{b}_\Delta$ together with the identity on $\mathfrak{b}_\Delta$ gives r_Δ by universality; the second retraction is symmetric. The enveloping functor preserves coproducts of Lie algebras, and PBW identifies the completed algebraic filtration with the displayed completed free product. $\square$

Definition 115.3 (Comparison ideal). A primitive comparison $\varphi : \mathfrak{n}_{\Delta,+} \rightarrow \mathfrak{p}_X^+$ preserving Cartan degree defines the closed ideal

$$\mathcal{J}_\varphi = \overline{\langle i_\Delta(x) - i_X(\varphi(x)) \mid x \in \mathfrak{n}_{\Delta,+} \rangle} \subset \mathfrak{b}_{X \boxtimes \Delta}.$$

The diagonal realization is

$$\mathfrak{b}_{X \simeq \Delta, \varphi} = \mathfrak{b}_{X \boxtimes \Delta} / \mathcal{J}_\varphi.$$

Proposition 115.4 (Graph quotient of a primitive comparison). *Let $\varphi : \mathfrak{b}_\Delta \rightarrow \mathfrak{b}_X$ be a continuous Lie morphism restricting to the identity on $\mathfrak{h}_\Pi$. Use a completion in which the amalgam and closed quotient have their stated universal properties. Then*

$$\mathfrak{b}_{X \boxtimes \Delta} / \mathcal{J}_\varphi \cong \mathfrak{b}_X.$$

The two factor maps become id and φ . If φ is an isomorphism, the quotient is also isomorphic to $\mathfrak{b}_\Delta$.

Proof. The maps id and φ agree on $\mathfrak{h}_\Pi$. They give $q_\varphi : \mathfrak{b}_{X \boxtimes \Delta} \rightarrow \mathfrak{b}_X$, which kills the generators of $\mathcal{J}_\varphi$. The inclusion of $\mathfrak{b}_X$ gives a map s in the other direction. The composite $q_\varphi s$ is the identity. The other composite is the identity on the geometric factor. On the Borchers factor, the defining relation identifies x with $s\varphi(x)$, so it is also the identity there. The two factor images topologically generate the quotient. Continuity proves that the maps are inverse on the completion. $\square$

The graph quotient implements a specified primitive comparison. Constructing that comparison requires geometric simple states whose extension brackets satisfy the Borchers relations. The quotient formula alone produces neither those states nor their orientation data.

COPRODUCTS, PAIRINGS, AND DOUBLES

A positive Hall algebra becomes a quantum group after additional structure is supplied. Let H^+ be a complete graded bialgebra, let H° be a Cartan algebra, and let H^- be the opposite or dual half. Let

$$\langle -, - \rangle : H^+ \widehat{\otimes} H^- \longrightarrow \mathbb{k}$$

be a nondegenerate Hopf pairing.

Definition II6.1 (Completed Hall double). The completed Drinfeld double is the topological algebra

$$\widehat{D}(H^+) = H^- \widehat{\otimes} H^\circ \widehat{\otimes} H^+$$

with internal products in the three factors and cross-relations determined by the Hopf pairing. Its representation category is monoidal by the coproduct and braided by the universal R -matrix whenever the canonical tensor converges.

Theorem II6.2 (Double construction). *Let H^+ and H^- be paired complete graded bialgebras, let H° be a Cartan bialgebra acting by the stated weight grading, and let*

$$\langle -, - \rangle : H^+ \widehat{\otimes} H^- \longrightarrow \mathbb{k}$$

be a continuous nondegenerate Hopf pairing. Assume the triangular multiplication map from $H^- \widehat{\otimes} H^\circ \widehat{\otimes} H^+$ is separated and complete. Then the pairing cross-relations determine a unique completed double with this triangular decomposition. On every continuous module category on which the canonical tensors converge, the completed canonical element gives the universal R -matrix.

Proof. The Hopf pairing defines the cross-relations between the paired halves and makes the three canonical embeddings bialgebra morphisms. The Hopf identities identify the two reductions of every triple product, proving associativity. Separated triangular decomposition gives uniqueness. Finite-degree canonical tensors form a compatible inverse system; convergence on the selected module category yields the quasitriangular element [28, 30]. $\square$

Theorem II6.3 (Quantum Hall–Borcherds amalgam). *Let H_X and H_Δ be complete Hopf algebras containing the same complete Cartan Hopf algebra H° , with continuous Hopf retractions onto H° . Their completed amalgamated free product*

$$H_{X \boxtimes \Delta} = H_X \widehat{*}_{H^\circ} H_\Delta$$

has a unique complete Hopf structure extending the two coproducts, counits, and antipodes. It represents pairs of continuous Hopf morphisms agreeing on H° and has canonical Hopf retractions to both factors. A mixed universal R -matrix is equivalent to a continuous cross-pairing satisfying the two hexagon identities; the intrinsic R -matrices of the two retracts remain defined independently.

Proof. The algebraic amalgamated free product exists because complete filtered algebras are obtained degreewise from finite quotients. The two coproducts agree on H° , so the universal property extends them to a coproduct on the free product. Coassociativity is checked on the two generating Hopf subalgebras. The same argument extends the counit and antipode. The Hopf retractions are induced by the factor retractions onto H° . A quasitriangular structure extending both factors requires the additional commutation data between the two generating halves; writing this data as a cross-pairing turns the quasitriangular and Yang–Baxter equations into the two hexagon identities. $\square$

The deconcatenation coproduct on a bar complex and the representation-theoretic coproduct on a quantum group occupy different structures. A geometric comparison between them is a theorem about a selected model.

ORDERED CHIRAL BIALGEBRAS

A tensor product of representations requires compatible actions on threefold products and the unit. Retain the factorization coefficient category $(\text{Fact}_D(C), \otimes^!)$ on the smooth complex curve C . Fix a category $\widehat{\mathcal{V}}_C$ of complete filtered objects with specified presentations $F \simeq \varprojlim_r F_r$, where $F_r \in \text{Fact}_D(C)$. Use derived limits in the fixed Ran enhancement and define

$$F \widehat{\otimes}^! G = \varprojlim_{r,s} (F_r \otimes^! G_s).$$

Assume these limits remain factorizing and give $\widehat{\mathcal{V}}_C$ a symmetric monoidal structure with unit $\mathbf{1}$. Its associativity, symmetry, and unit equivalences must preserve the presentations and the Ran structure. Morphisms are continuous maps compatible with these presentations. These are hypotheses on the chosen completion.

Definition 117.1 (Ordered chiral bialgebra). An ordered chiral bialgebra is a counital E_1 -coalgebra object in complete ordered chiral algebras:

$$A \in \text{CoAlg}_{E_1}(\text{Alg}_{E_1}(\widehat{\mathcal{V}}_C, \widehat{\otimes}^!)).$$

It has continuous multiplication m and unit η , and continuous morphisms of unital ordered chiral algebras

$$\Delta : A \longrightarrow A \widehat{\otimes}^! A, \quad \epsilon : A \longrightarrow \mathbf{1}.$$

The counit is the chosen augmentation. Write τ for the coefficient symmetry, including Koszul signs. Suppressing ambient associativity and unit equivalences, compatibility reads

$$\begin{aligned} \Delta m &= (m \widehat{\otimes}^! m)(\text{id} \widehat{\otimes}^! \tau \widehat{\otimes}^! \text{id})(\Delta \widehat{\otimes}^! \Delta), \\ \Delta \eta &= \eta \widehat{\otimes}^! \eta, \quad \epsilon m = \epsilon \widehat{\otimes}^! \epsilon, \quad \epsilon \eta = \text{id}_{\mathbf{1}}. \end{aligned}$$

Coassociativity and the counit identities read

$$\begin{aligned} (\Delta \widehat{\otimes}^! \text{id})\Delta &= (\text{id} \widehat{\otimes}^! \Delta)\Delta, \\ (\epsilon \widehat{\otimes}^! \text{id})\Delta &= \text{id}_A = (\text{id} \widehat{\otimes}^! \epsilon)\Delta. \end{aligned}$$

For a homotopy-coherent presentation, these identities include the higher coherences of the displayed coalgebra object. Every operation and coherence preserves diagonal transitions and disjoint-locus factorization maps. An antipode is a continuous convolution inverse $S : A \rightarrow A$ of id_A :

$$m(S \widehat{\otimes}^! \text{id})\Delta = \eta \epsilon = m(\text{id} \widehat{\otimes}^! S)\Delta.$$

A bialgebra with an antipode is a Hopf object in this coefficient category.

A continuous left A -module is an object $M \in \widehat{\mathcal{V}}_C$ with a unital E_1 action $\rho_M : A \widehat{\otimes}^! M \rightarrow M$. Its action and coherences are morphisms in these factorization coefficients. Write $\text{Mod}_A(\widehat{\mathcal{V}}_C)$ for their category.

Theorem 117.2 (Representation fusion). *An ordered chiral bialgebra makes $\text{Mod}_A(\widehat{\mathcal{V}}_C)$ monoidal. Its product is $M \widehat{\otimes}^! N$ with action*

$$\begin{aligned} A \widehat{\otimes}^! (M \widehat{\otimes}^! N) &\xrightarrow{\Delta \widehat{\otimes}^! \text{id}} (A \widehat{\otimes}^! A) \widehat{\otimes}^! (M \widehat{\otimes}^! N) \\ &\xrightarrow{\text{id} \widehat{\otimes}^! \tau_{A,M} \widehat{\otimes}^! \text{id}} (A \widehat{\otimes}^! M) \widehat{\otimes}^! (A \widehat{\otimes}^! N) \\ &\xrightarrow{\rho_M \widehat{\otimes}^! \rho_N} M \widehat{\otimes}^! N. \end{aligned}$$

The unit is $\mathbf{1}$ with action induced by ϵ . The ambient associator and unit equivalences are module equivalences.

Suppose a continuous universal R -matrix is also given. It is an invertible morphism $R : \mathbf{1} \rightarrow A \widehat{\otimes}^! A$ for the tensor-product algebra, satisfying

$$\begin{aligned} R\Delta(a) &= \Delta^{\text{op}}(a)R, & \Delta^{\text{op}} &= \tau_{A,A}\Delta, \\ (\Delta \widehat{\otimes}^! \text{id})(R) &= R_{13}R_{23}, & (\text{id} \widehat{\otimes}^! \Delta)(R) &= R_{13}R_{12}, \\ (\epsilon \widehat{\otimes}^! \text{id})(R) &= \eta = (\text{id} \widehat{\otimes}^! \epsilon)(R). \end{aligned}$$

The first equation is an identity of morphisms from A to its completed tensor square. Leg notation uses the unit and symmetry of $\widehat{\mathcal{V}}_C$. In a homotopy-coherent presentation, require compatible lifts of these identities. Then $c_{M,N} = \tau_{M,N} \circ R_{M,N}$ is a braiding, where $R_{M,N}$ acts through the external tensor-product module action.

Proof. The symmetry combines the independent module actions into an $A \widehat{\otimes}^! A$ -action. Restriction along the unital algebra map Δ gives the displayed action. For three modules, the two actions use $(\Delta \widehat{\otimes}^! \text{id})\Delta$ and $(\text{id} \widehat{\otimes}^! \Delta)\Delta$. Coassociativity makes the ambient associator A -linear. The counit identities make the ambient unit equivalences A -linear. The coalgebra coherences give the pentagon and triangle coherences. The completion hypotheses keep these maps continuous and factorizing. Quasitriangularity makes $c_{M,N}$ an intertwiner and invertibility makes it an equivalence. The two coproduct identities for R give the hexagons. Its counit identities give compatibility with the unit. $\square$

The tensor in this construction is the completed transverse tensor $\widehat{\otimes}^!$ on factorization coefficients. A chiral-convolution or Hall coproduct requires its own construction and a comparison in the specified category. A spectral version additionally requires a difference coordinate, a Laurent-series completion, and compatible domains for iterated expansions. Its identities must hold in those domains before defining module actions.

Example 117.3 (A counital algebra map without coassociativity). In ordinary complex vector spaces, let $A = \mathbb{C}[x]$ and put

$$\Delta(x) = x \otimes \mathbf{1} + \mathbf{1} \otimes x + x \otimes x^2, \quad \epsilon(x) = 0.$$

Polynomial substitution extends these assignments to unital algebra maps. Writing $F(u, v) = u + v + uv^2$, the equations $F(0, v) = v$ and $F(u, 0) = u$ prove both counit identities. But

$$\begin{aligned} F(F(u, v), w) - F(u, F(v, w)) &= -uv^2w^4 - uv^2w^2 - 2uvw^3 - 2uvvw \\ &= -uvw(\mathbf{1} + w^2)(vw + 2) \neq 0. \end{aligned}$$

On three regular modules, the two actions differ already on $1 \otimes 1 \otimes 1$. Thus the canonical tensor associator is not $\mathcal{A}$ -linear. This ordinary example shows why multiplicativity and the counit identities must be supplemented by coassociativity. It makes no assertion about a geometric realization in factorization coefficients.

BOOK XXIII

Exact model cases

THE A_2 HALL ALGEBRA

For the quiver $1 \rightarrow 2$ over $\mathbb{F}_q$, the twisted Hall algebra has generators E_1, E_2 and $v^2 = q$. Counting subrepresentations gives

$$E_1^2 E_2 - (v + v^{-1}) E_1 E_2 E_1 + E_2 E_1^2 = 0.$$

For a representation of dimension vector $(2, 1)$, the arrow has rank zero or one. The relevant flag counts are

rank	112	121	211
0	$q + 1$	$q + 1$	$q + 1$
1	$q + 1$	1	0.

After the Hall twists $v^{-1}, 1, v$, both rows give zero. This is the elementary geometric origin of the quantum Serre relation [37, 67].

THE TRIPLED JORDAN QUIVER AND $\mathbb{C}^3$

Let Q have one vertex and loops x, y, z , with potential

$$W = \text{Tr } x[y, z].$$

The Jacobi algebra is $\mathbb{C}[x, y, z]$. The critical CoHA uses the vanishing cycles of $\text{Tr } W$ on the representation spaces. Its positive currents are generated by the one-vertex dimension sectors.

Theorem II9.1 (Affine-Yangian positive half). *After the standard equivariant localization and completion, the critical CoHA of the tripled Jordan quiver is identified with the positive half of the affine Yangian of $\mathfrak{gl}_1$. The independently defined affine Yangian carries its standard Cartan and negative current halves; the positive-half comparison embeds the CoHA into that triangular presentation.*

Proof. The CoHA product becomes the shuffle product under localization. Dimensional reduction identifies the tripled-quiver critical theory with the equivariant surface CoHA, and Davison's affine-BPS theorem identifies its completed positive algebra with the strictly positive affine Yangian. The shuffle kernel satisfies the affine-Yangian current relations with equivariant parameters $\epsilon_1, \epsilon_2, \epsilon_3$ and Calabi–Yau relation $\epsilon_1 + \epsilon_2 + \epsilon_3 = 0$. PBW and current comparison identify the positive algebra. The independently presented full affine Yangian is the triangular algebra generated by its negative currents, Cartan currents, and this positive half [21, 23, 68, 69]. $\square$

The vacuum representation produces a $W_{1+\infty}$ -type vertex algebra through an evaluation homomorphism. The CoHA, its double, and the vacuum algebra occupy successive levels: positive algebra, paired quantum group, and distinguished representation.

SURFACE COHAS AND ADE YANGIANS

Let $S_\Gamma \rightarrow \mathbb{C}^2/\Gamma$ be the minimal resolution for a finite subgroup $\Gamma \subset \mathrm{SL}_2(\mathbb{C})$. The exceptional curves have ADE intersection matrix. Moduli of sheaves supported on these curves carry convolution correspondences.

Theorem 120.1 (Curve-supported surface CoHA). *The completed positive CoHA of one-dimensional sheaves supported on the exceptional curves is identified with a nonstandard positive half of the affine Yangian of the corresponding ADE Lie algebra.*

Proof. The theorem is the curve-supported surface construction of Diaconescu–Porta–Sala–Schiffmann–Vasserot [25]. The simple curve sheaves give the simple generators, Hecke correspondences give the current operators, and the completed PBW comparison gives the affine Yangian presentation. $\square$

This theorem is a local K_3 model whenever an ADE configuration occurs on a K_3 surface. The global K_3 surface adds compact curve classes and monodromy beyond the local root lattice.

COULOMB BRANCHES AND SHIFTED YANGIANS

For a three-dimensional $\mathcal{N} = 4$ gauge theory, the BFN Coulomb branch is defined by equivariant Borel–Moore homology of an affine-Grassmannian correspondence. Its convolution algebra quantizes the Coulomb branch [II]. In quiver gauge theories, shifted Yangians and their truncations provide presentations of these quantizations. Boundary conditions select the shift and truncation data.

The BFN construction supplies a second geometric source of Yangians. Its grading and coproduct arise from loop rotation and convolution, while the surface CoHA source arises from sheaf extensions. A comparison between the two is a model-specific equivalence.

BOOK XXIV

K₃ symmetries

THE MUKAI LATTICE

For a K3 surface S , the Mukai lattice is

$$\tilde{H}(S, \mathbb{Z}) = H^0(S, \mathbb{Z}) \oplus H^2(S, \mathbb{Z}) \oplus H^4(S, \mathbb{Z})$$

with pairing

$$((r, c, s), (r', c', s')) = c \cdot c' - rs' - r's.$$

It is even unimodular of signature $(4, 20)$ and rank 24.

THE NAKAJIMA HEISENBERG ALGEBRA

Let

$$\mathbb{H}_S = \bigoplus_{n \geq 0} H^*(S^{[n]}).$$

For $m \in \mathbb{Z} \setminus \{0\}$ and $\alpha \in H^*(S)$, the Nakajima correspondence gives an operator $p_m(\alpha)$.

Theorem 123.1 (K_3 Heisenberg action). *The operators satisfy*

$$[p_m(\alpha), p_n(\beta)] = m \delta_{m+n,0} \int_S \alpha \cup \beta \text{ id}.$$

They generate the rank-twenty-four Heisenberg algebra acting irreducibly on $\mathbb{H}_S$ with vacuum $1 \in H^(S^{[0]})$.*

Proof. The commutator is the excess intersection of the two incidence correspondences. Opposite creation and annihilation correspondences meet along the diagonal with multiplicity m and intersection pairing $\int_S \alpha \beta$. Correspondences of the same sign commute. The vacuum construction gives the Fock representation [61]. $\square$

This action is global, compact, and abelian at the Lie level. It is an exact K_3 symmetry and supplies the rank-twenty-four Fock sector of any broader K_3 algebra.

THE ORTHOGONAL LLV ALGEBRA

For a compact hyperkähler manifold, every Lefschetz class gives an $\mathfrak{sl}_2$ triple on cohomology. The Lie algebra generated by all such triples is the Looijenga–Lunts–Verbitsky algebra.

Theorem 12.4.1 (Orthogonal LLV symmetries). *For a $K3$ surface S , the LLV algebra acting on $H^*(S)$ is*

$$\mathfrak{g}_{\text{LLV}}(S) \cong \mathfrak{so}(\widetilde{H}(S), (-, -)_{\text{Muk}}) \cong \mathfrak{so}(4, 20).$$

For an irreducible hyperkähler manifold M of $K3^{[n]}$ type with $n \geq 2$, one has $b_2(M) = 23$ and

$$\mathfrak{g}_{\text{LLV}}(M) \cong \mathfrak{so}(4, 21).$$

Moduli spaces of stable objects on S carry their own LLV actions; Mukai and monodromy correspondences relate these actions to the $K3$ lattice.

Proof. The Looijenga–Lunts–Verbitsky theorem identifies the Lie algebra generated by all Lefschetz triples with the orthogonal algebra of the Mukai extension $H^2(M) \oplus U$. For S , the extension has signature $(4, 20)$. For $K3^{[n]}$ type with $n \geq 2$, the Beauville–Bogomolov lattice has signature $(3, 20)$, so adjoining the hyperbolic plane gives signature $(4, 21)$ [56, 70]. $\square$

The LLV algebra is a classical cohomological symmetry. A Yangian deformation of the full orthogonal action consists of two further structures: a Drinfeld current presentation and a compatible spectral coproduct.

LOCAL ADE YANGIANS

An ADE configuration of (-2) -curves on S gives a root sublattice of $H^2(S, \mathbb{Z})$. Sheaves supported on the configuration produce the surface CoHA of the preceding part and hence a completed positive affine ADE Yangian. Nakajima quiver varieties supply compatible Yangian actions on their equivariant cohomology.

Thus K_3 geometry contains local nonabelian Yangian sectors indexed by ADE enhancements of the Picard lattice. Picard monodromy transports the ADE root sublattices over the corresponding Noether–Lefschetz loci. A family of sheaf moduli and Hecke correspondences compatible with that monodromy transports the Yangian actions themselves.

THE K_3 SYMMETRY TRICHOTOMY

The three exact structures are

Structure	Carrier	Operation
Mukai Heisenberg	$\bigoplus_n H^*(S^{[n]})$	creation and annihilation
LLV $\mathfrak{so}(4, 20)$	$H^*(S)$; separately $\mathfrak{so}(4, 21)$ on $K_3^{[n]}$ type	Lefschetz Lie action
Local ADE Yangian	curve-supported equivariant moduli	Hecke and current correspondences

A global K_3 quantum algebra is a system that contains these three actions with explicit comparison homomorphisms. The two-Calabi–Yau BPS theorem supplies a global generalized Kac–Moody positive algebra. A full Yangian or double is specified by a compatible coproduct, spectral parameter, negative half, Cartan part, pairing, and completion.

K₃ CORRESPONDENCES ON FIXED COEFFICIENTS

Fix rational cohomology and a collection $\mathcal{M}_S$ of smooth projective moduli spaces on a projective K₃ surface S . Include the Hilbert schemes $S^{[n]}$ and close the collection under finite products. For $M, N \in \mathcal{M}_S$, define the graded kernel space

$$K(M, N) = H^*(M \times N, \mathbb{Q})[2 \dim_{\mathbb{C}} M].$$

A kernel of degree r acts from $H^j(M)$ to $H^{j+r}(N)$ by

$$T_\alpha(v) = (p_N)_*(p_M^*v \cup \alpha).$$

The shift in $K(M, N)$ accounts for integration over M . All pushforwards in this construction are proper.

Proposition 127.1 (Kernel composition). *The product*

$$\beta \circ \alpha = (p_{13})_*(p_{12}^*\alpha \cup p_{23}^*\beta)$$

defines an associative graded category with diagonal identities. The kernel action identifies $K(M, N)$ with $\text{Hom}^(H^*(M), H^*(N))$.*

Proof. The Künneth isomorphism and Poincaré duality identify $H^*(M \times N)[2 \dim M]$ with $H^*(M)^\vee \otimes H^*(N)$. Under this identification, T_α is the associated linear map. Integration over the middle factor contracts its dual pair, so $T_{\beta \circ \alpha} = T_\beta T_\alpha$. Faithfulness then proves associativity and the diagonal identity. External products give tensor products with their Koszul signs. $\square$

A geometric symmetry is specified by selected kernels in this category. Nakajima incidence kernels connect different Hilbert schemes. Lefschetz multiplication and its adjoint act on each individual cohomology space. Using every cohomology kernel would instead give all linear maps; the selected geometric kernels carry the representation-theoretic content.

THE ALGEBRA GENERATED BY GEOMETRIC ACTIONS

Put $V_S = \bigoplus_{M \in \mathcal{M}_S} H^*(M, \mathbb{Q})$. Choose a set of homogeneous operators T_λ on V_S , each sending a vector with finite support to a vector with finite support. The Nakajima operators on the Hilbert-scheme summand satisfy this condition: each changes the point number by a fixed integer. Individual Lefschetz operators, extended by zero on other summands, also satisfy it. Let F be the free unital graded algebra on symbols t_λ .

Proposition 128.1 (Geometric action algebra). *The map*

$$\rho : F \longrightarrow \text{End}_{\mathbb{Q}}(V_S), \quad t_\lambda \longmapsto T_\lambda$$

defines a faithful action of $A_S = F/\ker \rho$. For every algebra B , maps $A_S \rightarrow B$ are exactly maps $F \rightarrow B$ that kill $\ker \rho$. Products and commutators of the selected kernels are their convolution products and commutators on the relevant moduli spaces.

Proof. Finite words in the selected operators preserve finite support, so ρ is defined by composition. The quotient acts faithfully by definition of its kernel. The factorization property is the universal property of an algebra quotient. The convolution assertion follows from Proposition 127.1. $\square$

An equivariant ADE construction uses a separate coefficient field, for example $\text{Frac } H_T^*(\text{pt}, \mathbb{Q})$, and its prescribed localization and proper pushforwards. Joining it to A_S requires maps between the actual state spaces that intertwine those operators and their coefficients. A completion of A_S requires a specified topology and continuity of every generator. The indefinite Mukai norm alone supplies neither condition. The desired common K3 quantum algebra must construct these local-to-global actions and their mixed convolution relations.

TENSOR RECONSTRUCTION AND GEOMETRIC KERNELS

Let $\mathcal{C}$ be an essentially small rigid linear monoidal category and let $\omega : \mathcal{C} \rightarrow \text{Vect}_{\text{fd}}(\mathbb{Q})$ be a strong monoidal functor. A choice of geometric kernels specifies the morphisms of $\mathcal{C}$; it is part of the input. Define

$$H = \left(\bigoplus_{M \in \mathcal{C}} \omega(M)^* \otimes \omega(M) \right) / \langle [\lambda \circ \omega(f) \otimes v]_M - [\lambda \otimes \omega(f)v]_N \rangle,$$

where $f : M \rightarrow N$, $v \in \omega(M)$, and $\lambda \in \omega(N)^*$.

Proposition 129.1 (Matrix-coefficient coalgebra). *For a basis (e_i) with dual basis (e^i) , the formulas*

$$\Delta[\lambda \otimes v]_M = \sum_i [\lambda \otimes e_i]_M \otimes [e^i \otimes v]_M, \quad \epsilon[\lambda \otimes v]_M = \lambda(v)$$

are basis-independent and make H a coalgebra. Tensor product gives its multiplication, and duals give its antipode. Each $\omega(M)$ is a comodule by

$$v \mapsto \sum_i e_i \otimes [e^i \otimes v]_M.$$

Every selected kernel is a comodule morphism.

Proof. The identity tensor $\sum_i e_i \otimes e^i$ is basis-independent. Inserting it twice proves coassociativity; evaluation proves both counit identities. The defining relations express naturality under f and are preserved by these formulas. The strong monoidal constraints identify the coefficient of $M \otimes N$ with the product of coefficients of M and N . Their pentagon and unit identities give the bialgebra axioms. The evaluation and coevaluation identities for duals give the antipode identities. The same relations prove that $\omega(f)$ intertwines the coactions. $\square$

The linear dual of H represents linear natural endomorphisms of ω . Tensor-compatible natural automorphisms are the corresponding multiplicative, unit-preserving families, not all elements of this dual. An equivalence with a comodule category requires the additional exactness and reconstruction hypotheses stated in the Tannaka theorem below. The action algebra A_S acts by selected kernels; the coend records symmetries commuting with those kernels.

BOOK XXV

The exact $K_3 \times E$ protected completion

THE PROTECTED INPUT

The K_3 elliptic genus is $2\phi_{0,1}$. The Borchers half determines the weight-five denominator D_5 and its Borchers Lie superalgebra $\mathfrak{g}_\Delta$. Fix its positive root spaces with their parity and proper height. The positive algebra and its root Fock space are defined from this Lie superalgebra, independently of a compact Hall realization.

For each finite zero-potential quiver window $Q_{\Delta,H}$, form the Ginzburg algebra $\Gamma_{\Delta,H}$. Restriction to full subquivers gives the projections of Proposition 136.1. Write Γ_Δ for this specified inverse system. A non-degenerate continuous Calabi–Yau structure on a completed limit is additional data. An ordinary Hall category $\mathcal{B}_\Delta$ must carry its chosen quiver coefficients and extension correspondences. Its identification with a root presentation requires the corresponding Hall theorem on that precise datum.

Definition 130.1 (Root-datum realizations). A protected root-datum realization records

$$\mathfrak{P}(K_3 \times E) = (\Gamma_\Delta, \mathcal{B}_\Delta, \mathcal{H}_{\text{ip}}, \mathcal{F}_{\Delta,E}^{\text{hyb}}, \widehat{U}_q(\mathfrak{g}_\Delta), \mathbb{A}_Z).$$

Here $\mathcal{H}_{\text{ip}} = \bigoplus_{\alpha > 0} (\mathfrak{g}_\Delta)_\alpha$ with its parity. The current-factorization, paired quantum-group, and Fredholm entries retain their own coefficient categories, completions, and defining hypotheses. Maps among these entries are specified realization maps, not consequences of using the same root labels.

The enveloping and Chevalley constructions are functorial under the primitive maps of Theorem 135.1. The root Fock trace is computed from the actual graded root spaces. A Fredholm realization adds a family of operators on a specified domain and the convergence and orientation data identifying its Pfaffian with the denominator. These are the analytic inputs to the square-root comparison below.

Definition 130.2 (Integral protected realization). Let $\mathfrak{p}_X^+$ be the BPS primitive Lie algebra of a selected strong-oriented cohomological lift of the geometric E -equivariant Hall source. The integral protected realization is

$$\mathfrak{P}_{\text{int}}(X, \Delta) = \left(\mathcal{H}_X^{E, \text{mot}}, \mathfrak{P}(K_3 \times E), \mathfrak{b}_{X \boxtimes \Delta}, r_X, r_\Delta, I_{X, \Pi}^E \right).$$

At the motivic level the first, second, and character components are defined. A strong orientation enhancement supplies $\mathfrak{p}_X^+$ and hence the amalgamated primitive component.

Proposition 130.3 (Primitive graph comparison). *Suppose the two primitive Borels, their common Cartan action, and their completed amalgam are supplied as above. Their Cartan retractions induce retractions from the amalgam to each factor. For each specified primitive map φ preserving Cartan action, its graph quotient is canonically the geometric Borel.*

Proof. The identity on one Borel and the Cartan projection of the other agree on the common Cartan, so the amalgam gives the retraction. The quotient statement is Proposition 115.4. $\square$

The motivic integration map and primitive comparison are different maps. Strong orientation gives the coefficient coherence of a cohomological Hall construction under its stated geometric hypotheses. Extracting a primitive Lie algebra additionally requires the appropriate BPS or coproduct construction.

CYCLIC OPEN-STRING OPERATIONS

Let A be a finite-dimensional, uncurved cyclic A_∞ algebra over $\mathbb{C}$, with a nondegenerate pairing of degree -3 . Use the cohomological suspension $s : A \rightarrow A[1]$ of degree -1 . The suspended operations $b_n : (A[1])^{\otimes n} \rightarrow A[1]$ have degree one. Our sign convention is the coderivation convention $b^2 = 0$ on the tensor coalgebra, with the Koszul rule for moving homogeneous inputs. Write ω for the resulting degree -1 symplectic form on $A[1]$. Fix the Hamiltonian convention $\iota_{X_F} \omega = dF$ and $[X_F, X_G] = X_{\{F, G\}}$.

For the universal degree-zero coordinate x on $A[1]$, put

$$S(x) = \sum_{n \geq 1} \frac{1}{n+1} \omega(x, b_n(x^{\otimes n})).$$

Cyclic invariance is read with the suspended Koszul signs. All functions are formal power series at the origin. An equivalent unsuspended convention writes this as $\frac{1}{2} \langle a, m_1 a \rangle + \sum_{n \geq 2} \frac{1}{n+1} \langle a, m_n(a^{\otimes n}) \rangle$.

Proposition 13I.1 (Cyclic master equation). *The action satisfies $\{S, S\} = 0$.*

Proof. Cyclicity makes the $n+1$ derivatives of the arity- $n+1$ term equal, with the suspended Koszul signs. Thus $X_S(x) = \sum_{n \geq 1} b_n(x^{\otimes n})$. The coderivation identity implies $X_S^2 = 0$. The Hamiltonian identity gives $X_{\{S, S\}} = 2X_S^2 = 0$. Nondegeneracy of ω makes $\{S, S\}$ constant. Uncurvedness removes b_0 , so S starts in order two and its bracket has no constant term. The constant is therefore zero. $\square$

A finite-height Ginzburg realization requires such a cyclic model on the selected proper objects before this scalar action is applied. A Hall realization additionally requires a moduli problem with critical coefficient complexes. For an extension $A_o \rightarrow F \rightarrow B_o$, put $L(A_o, B_o) = \det R \operatorname{Hom}(A_o, B_o)$. Three-Calabi–Yau duality gives

$$R \operatorname{Hom}(B_o, A_o) \simeq R \operatorname{Hom}(A_o, B_o)^\vee[-3].$$

Applying the determinant functor to the two derived Hom triangles gives

$$\det R \operatorname{Hom}(F, F) \cong \det R \operatorname{Hom}(A_o, A_o) \otimes \det R \operatorname{Hom}(B_o, B_o) \otimes L(A_o, B_o)^{\otimes 2}.$$

Indeed, the four subquotients are the two diagonal Hom complexes and the two cross complexes. Dualization and the odd shift cancel on their determinant lines, so the two cross determinants coincide. Orientation lines require square-root isomorphisms along this formula, compatible on triple flags and with any quotient descent. The cyclic master equation does not select these isomorphisms.

AN ALGEBRAIC OPEN–CLOSED COMPARISON

A finite-dimensional ordinary Lie algebra $\mathbf{I}$ gives a precise coefficient model for an enveloping-algebra centre. Let $U = U(\mathbf{I})$ over $\mathbb{C}$. The adjoint module U^{ad} has action $x \cdot u = xu - ux$. Its Chevalley cochains are

$$C^n(\mathbf{I}, U^{\text{ad}}) = \text{Hom}_{\mathbb{C}}(\Lambda^n \mathbf{I}, U),$$

with differential

$$\begin{aligned} (df)(x_1, \dots, x_{n+1}) &= \sum_i (-1)^{i+1} [x_i, f(x_1, \widehat{x}_i, \dots, x_{n+1})] \\ &\quad + \sum_{i < j} (-1)^{i+j} f([x_i, x_j], x_1, \widehat{x}_i, \widehat{x}_j, \dots, x_{n+1}). \end{aligned}$$

The second line places the bracket first and retains the order of the remaining arguments.

Theorem 132.1 (Hochschild–Chevalley comparison). *Restriction followed by antisymmetrization defines a quasi-isomorphism*

$$C_{\text{HH}}^{\bullet}(U, U) \longrightarrow C^{\bullet}(\mathbf{I}, U^{\text{ad}}), \quad \phi \longmapsto \left[(x_1, \dots, x_n) \mapsto \sum_{\sigma \in S_n} \text{sgn}(\sigma) \phi(x_{\sigma(1)}, \dots, x_{\sigma(n)}) \right].$$

This is a comparison of ordinary cochain complexes.

Proof. The Chevalley bimodule resolution has terms $K_n = U \otimes \Lambda^n \mathbf{I} \otimes U$. Its differential has the left-minus-right multiplication terms and the bracket terms with the signs dual to the displayed differential. Filter K_n by total PBW degree, counting each exterior generator as degree one. The multiplication terms preserve this degree and bracket terms lower it. The associated graded is the Koszul resolution of the diagonal in $\text{Sym}(\mathbf{I}) \otimes \text{Sym}(\mathbf{I})$, with regular sequence $x_i \otimes 1 - 1 \otimes x_i$. That Koszul complex is exact away from degree zero. The filtration is bounded below and exhaustive, and each element has finite filtration degree, so lifting a leading term proves exactness of $K_{\bullet}$ by induction. Its augmentation has image U . Thus $K_{\bullet}$ is a free bimodule resolution.

Antisymmetrization gives a map from $K_{\bullet}$ to the ordinary bar resolution, lifting the identity of U . Adjacent products in its bar differential pair to commutators; $xy - yx = [x, y]$ gives exactly the bracket terms of $K_{\bullet}$. The outer products give its left-minus-right terms. A comparison between projective resolutions lifting the identity is a homotopy equivalence. Applying $\text{Hom}_{U \otimes U^{\text{op}}}(-, U)$ gives the stated formula and quasi-isomorphism. $\square$

This is the classical Cartan–Eilenberg comparison. For a finite Lie superalgebra, the same construction uses the super-exterior powers and permutation Koszul signs. Odd generators produce symmetric exterior directions, so the resolution can be unbounded; its algebraic direct sums and cochain products remain part of the convention. The PBW and super-Koszul proof applies with those conventions.

A physical mixed holomorphic–topological realization requires a map from its independently defined bulk observables to this cochain object. The theorem specifies the adjoint coefficients and the actual algebraic comparison. An E_2 refinement, a current-factorization realization, and compatibility with root-height transitions require maps preserving their additional operations. A quotient of Lie algebras alone does not provide a covariant map of Hochschild centres.

THE EXACT SCALAR AND FIRST-ORDER OBSERVABLES

Set

$$\mathrm{Den}_\Delta(Z) = D_5(2Z), \quad j(Z) = Z/2, \quad j^* \mathrm{Den}_\Delta = D_5.$$

The one-particle root spaces give

$$\mathrm{sch Fock}(\mathfrak{g}_{\Delta_5,+}) = \prod_{\alpha > 0} (1 - e^{-\alpha})^{-\mathrm{sdim} \mathfrak{g}_{\Delta_5,\alpha}}.$$

The oriented root operator gives

$$e^{-\rho} \mathrm{sPf}_F(\mathbb{A}_Z) = D_5(2Z).$$

The primitive reduced curve-counting theorem gives

$$\mathcal{Z}_{K_3 \times E}^{\mathrm{red}} = -(j^* \mathrm{Den}_\Delta)^{-2} = -\chi_{\mathrm{io}}^{-1}.$$

Theorem 133.1 (Protected square-root theorem). *Assume the stated root-space Fredholm comparison with its orientation, Weyl shift, and convergence domain. Within that realization, the denominator section Den_Δ is simultaneously:*

- (i) *the Weyl denominator of $\mathfrak{g}_{\Delta_5}$;*
- (ii) *the Euler character of the bar–Chevalley complex;*
- (iii) *the oriented Fredholm Pfaffian of $\mathbb{A}_Z$;*
- (iv) *the PBW determinant of the parity-refined root superspace $\mathcal{H}_{\mathrm{ip}}$;*
- (v) *the automorphic denominator whose half-coordinate pullback squares to the reduced scalar denominator.*

All five identifications preserve root degree and Weyl transport.

Proof. The first two are the denominator and bar–Chevalley theorems. The third is the Fredholm theorem. PBW for the Borchers Lie superalgebra identifies the root superspace determinant, giving the fourth. The half-coordinate identity $j^* \mathrm{Den}_\Delta = D_5$, the theta normalization, and the reduced curve-counting theorem give the fifth. $\square$

PHYSICAL MEANING

The root-datum realizations retain several distinct constructions. The finite Ginzburg algebras and their projections form an inverse system. A completed three-Calabi–Yau sector requires a continuous nondegenerate Calabi–Yau structure and compatible cyclic models on the selected proper objects. The quiver Hall category retains its chosen coefficients and extension correspondences. A Hall theorem on that datum must identify its root presentation. The Borchers root superspace defines the root-graded Fock space and its supercharacter. Its identification with physical one-particle states requires a parity-preserving graded comparison intertwining the primitive actions.

A realization on the elliptic curve must construct the current object with its local and wrapped supports, collision operations, and comparison maps. The quantum Borchers algebra, with its specified pairing and completion, acts as a line-operator symmetry only through a representation preserving fusion and exchange. The root operator gives the Fredholm product on its specified convergence domain. A physical determinant interpretation requires an operator comparison and an orientation-preserving identification of the determinant lines and sections. The Fock character gives the scalar square. An operator trace additionally requires its state space and trace domain.

Simple-root states satisfying the defining relations give a primitive Lie map. The explicit enveloping and Chevalley extensions preserve products and brackets. A compact string realization must additionally construct its brane functor, Hall orientation maps, paired quantum coproduct, and trace-line comparison.

PRIMITIVE COMPARISONS AND THEIR ALGEBRAIC EXTENSIONS

Let $\mathfrak{n}_\Delta$ be the positive Borchers Lie superalgebra. Choose homogeneous parity-preserving images of its simple generators in a charge-graded Lie superalgebra $\mathfrak{p}$. Assume these images satisfy every defining relation.

Theorem 135.1 (Extension of a primitive comparison). *The chosen images determine a unique Lie-superalgebra map $f : \mathfrak{n}_\Delta \rightarrow \mathfrak{p}$. It induces*

$$U(f)(x_1 \cdots x_r) = f(x_1) \cdots f(x_r), \quad C_*(f) = \text{Sym}^c(f[1]).$$

These are respectively an enveloping-algebra map and a Chevalley chain-coalgebra map. Filtration-preserving maps extend to the specified inverse-limit completions. If f is surjective and both algebras have finite charge components and proper positive height, equality of their parity-refined PBW characters implies that f and $U(f)$ are isomorphisms.

Proof. The defining presentation gives f . Its tensor-algebra extension carries $xy - (-1)^{|x||y|}yx - [x, y]$ to the corresponding relation. It therefore descends to $U(f)$. The shifted symmetric-coalgebra map commutes with the Chevalley differential because f preserves brackets. Compatible maps on filtration quotients give maps on their inverse limits. Surjectivity of f gives surjectivity of $U(f)$. Character equality makes each finite charge component an isomorphism, separately in both parities. The PBW embeddings then imply injectivity of f . $\square$

At level zero the current Lie conformal algebra is $\mathbb{C}[\partial] \otimes \mathfrak{n}_\Delta$ with $[x_\lambda y] = [x, y]$ on constant generators. The map $\partial^j x \mapsto \partial^j f(x)$ preserves this bracket and sesquilinearity. A nonzero central level additionally requires preservation of the specified invariant pairing. Hall convolution, Calabi–Yau structures, quantum coproducts, and determinant lines require their own comparison data.

FINITE QUIVERS AND CALABI–YAU COMPLETIONS

For a finite quiver Q with zero potential, let $\Gamma(Q, \circ)$ be its path-length-completed Ginzburg algebra. Path products are read from left to right: if $a : i \rightarrow j$ and $b : j \rightarrow k$, then $ab : i \rightarrow k$. For each arrow $a : i \rightarrow j$, add $a^* : j \rightarrow i$ of degree -1 . At each vertex add a loop t_i of degree -2 . Original arrows have degree zero, and

$$da = da^* = \circ, \quad dt_i = e_i \left(\sum_a [a, a^*] \right) e_i.$$

Proposition 136.1 (Restriction to a full subquiver). *If $Q_\circ$ is a full subquiver, retaining its generators and killing all other vertices, arrows, reverse arrows, and loops defines a continuous dg-algebra projection*

$$\Gamma(Q, \circ) \longrightarrow \Gamma(Q_\circ, \circ).$$

The inclusion of retained graded generators need not commute with the differential.

Proof. The projection respects source and target idempotents and extends multiplicatively. At a retained vertex, every term involving a discarded arrow vanishes. The other terms are precisely dt_i in the smaller algebra. All arrow differentials vanish, so the projection is a chain map. It preserves the path-length filtration and is continuous. For the final assertion, start with one vertex i and no arrows, where $dt_i = \circ$. Add a vertex j and an arrow $a : i \rightarrow j$. In the larger algebra $dt_i = aa^*$, a nonzero length-two path. Thus the graded inclusion is not a dg map. $\square$

Ginzburg completion supplies a three-Calabi–Yau algebra for each finite quiver in its stated setting. The displayed projections specify an inverse system for compatible zero-potential windows. A Calabi–Yau structure on a completed limit requires its own nondegeneracy and continuity argument. Comparisons with Hall categories and quantized Borchers algebras must also preserve orientations, root relations, and the paired Hopf data.

BOOK XXVI

Quantum groups, exchange, and
representation theory

FROM POSITIVE HALF TO FULL QUANTUM GROUP

Suppose the compact or local positive algebra H^+ is paired continuously with a negative half H^- , carries a completed coproduct, and admits Cartan data and a separated triangular completion. The completed double

$$\mathcal{U} = \widehat{D}(H^+)$$

has triangular decomposition $H^- \widehat{\otimes} H^0 \widehat{\otimes} H^+$. Its universal R -matrix acts on continuous tensor products of modules. The classical limit is a Lie bialgebra whose cobracket is the first-order term of the coproduct asymmetry.

For $\mathbb{C}^3$, H^+ is the positive affine Yangian. For local ADE surfaces, it is the completed positive affine ADE Yangian. For the Igusa algebra, the paired Borel construction of the preceding part supplies a height-complete formal quantum group; a compact Hall realization maps to it through primitive, coproduct, and pairing comparisons.

TANNAKA RECONSTRUCTION

Let $\mathcal{L}$ be an essentially small rigid $\mathbb{k}$ -linear abelian monoidal category with finite-dimensional morphism spaces, finite-length objects, and exact tensor product. Let

$$\omega : \mathcal{L} \rightarrow \mathbf{Vect}_{\text{fd}}$$

be a faithful exact strong monoidal functor. Assume the coend

$$H = \int^{M \in \mathcal{L}} \omega(M)^* \otimes \omega(M)$$

exists and that the canonical comparison functor to finite-dimensional H -comodules is comonadic. Tensor product makes H a bialgebra, and rigidity supplies its antipode. A braiding gives a coquasitriangular form; a meromorphic braiding gives its spectral version.

Theorem 138.1 (Rigid Tannaka reconstruction). *Under the preceding finite and comonadic hypotheses, the comparison functor is a monoidal equivalence*

$$\mathcal{L} \simeq \mathbf{Comod}_{\text{fd}}(H).$$

Braiding transports to the universal coquasitriangular form on H .

Proof. Matrix coefficients define the universal H -coaction on every $\omega(M)$. Tensor product and duality give the bialgebra and antipode structures. The Barr–Beck comonadic theorem identifies $\mathcal{L}$ with coalgebras for the induced comonad, and the coend identifies that comonad with $H \otimes -$. The hexagon identities transport to the coquasitriangular identities. $\square$

This reconstruction explains why a line-operator category, a fiber functor, and exchange together produce a quantum group. The ordered boundary algebra provides ordered composition. A compatible bialgebra coproduct supplies tensor products of its modules, and exchange supplies additional braiding data.

KZ AND RATIONAL EXCHANGE

For a simple Lie algebra $\mathfrak{g}$, the Knizhnik–Zamolodchikov connection is

$$\nabla = d - \frac{\hbar}{2\pi i} \sum_{i < j} \Omega_{ij} d \log(z_i - z_j).$$

Flatness follows from the infinitesimal braid relations. Drinfeld–Kohno identifies its monodromy tensor category with the corresponding Drinfeld–Jimbo category after a choice of associator.

Rational difference exchange instead gives the Yangian R -matrix

$$R(u) = 1 - \frac{\hbar P}{u}$$

in the fundamental representation. The Yang–Baxter equation is the confluence identity for three exchanges. The KZ and Yangian constructions share the geometry of configuration spaces while using different flat connections.

THE CENTRE AND THE DOUBLE

The Hochschild centre $Z_{\text{ord}}(\mathcal{A})$ controls central operations on the ordered algebra in its specified coefficient category. To form a Drinfeld centre of modules, first specify a monoidal module category. For an ordered chiral bialgebra satisfying Definition 117.1, use

$$\mathcal{M}_A = \text{Mod}_A(\widehat{\mathcal{V}}_C)$$

with the monoidal product of Theorem 117.2. An object of $Z(\mathcal{M}_A)$ is a module M together with natural equivalences

$$c_{M,N} : M \widehat{\otimes}^! N \xrightarrow{\sim} N \widehat{\otimes}^! M \quad (N \in \mathcal{M}_A)$$

satisfying the half-braiding hexagon and unit identities, with their coherences. This categorical centre is braided. A comparison with the Hochschild centre must specify a realization functor, the source and target of the comparison, and the operations it preserves. Dualizability alone does not supply such a comparison. Derived loop-space descriptions apply in the geometric categories and under the hypotheses of their respective centre theorems [6].

The centre and the double answer complementary questions. The centre classifies central actions in the chosen category. The double reconstructs a quantum group from paired positive and negative halves. A field-theory comparison requires explicit realizations of the relevant line and bulk operations and a map compatible with both constructions.

BOOK **XXVII**

***A compact six-dimensional protected
source***

HOLOMORPHIC COTANGENT BF THEORY ON $K_3 \times E$

Let S be a K_3 surface with holomorphic symplectic form σ_S , let E be an elliptic curve with holomorphic form dz , and put

$$X = S \times E, \quad \Omega_X = \sigma_S \wedge dz.$$

For a finite proper-height quotient $\mathfrak{n}_{\Delta,+}^{(H)}$ of the positive Igusa Borchers algebra, take fields

$$A \in \Omega^{\circ,\bullet}(X, \mathfrak{n}_{\Delta,+}^{(H)})[1], \quad B \in \Omega^{\circ,\bullet}(X, (\mathfrak{n}_{\Delta,+}^{(H)})^\vee)[1].$$

The invariant evaluation pairing and Ω_X define the BV action

$$S_H(A, B) = \int_X \Omega_X \wedge \langle B, \bar{\partial} A + \frac{1}{2}[A, A] \rangle. \quad (141.1)$$

This is the holomorphic cotangent theory of the dg Lie algebra of $\mathfrak{n}_{\Delta,+}^{(H)}$ -valued Dolbeault forms.

Theorem 141.1 (Classical compact-background source). *The action (141.1) satisfies the classical master equation and defines a holomorphic factorization algebra on the compact Calabi–Yau threefold X .*

Proof. The shifted cotangent pairing is invariant. The classical master equation is the Jacobi identity of $\mathfrak{n}_{\Delta,+}^{(H)}$ together with $\bar{\partial}^2 = 0$ and the derivation property of $\bar{\partial}$. Local observables of an elliptic cyclic L_∞ algebra form a factorization algebra by the standard BV construction. $\square$

TREE EXACTNESS AND THE QUANTUM MASTER EQUATION

Every interaction vertex in (141.1) has one B -leg and two A -legs. The propagator pairs an A -leg with a B -leg.

Lemma 142.1 (Loop bound). *A connected Feynman graph of the cotangent theory has loop number at most one.*

Proof. A graph with V interaction vertices has exactly V internal or external B -half-edges. Every internal edge consumes one of them, hence $E \leq V$. The loop number is $E - V + 1 \leq 1$. $\square$

The positive algebra is graded by strictly positive root height. For homogeneous $x \in \mathfrak{n}_{\Delta,+}^{(H)}$, the operator ad_x raises height. Its coadjoint operator lowers dual height. A product of adjoint operators labelled by positive roots has total nonzero height shift and therefore has zero trace.

Proposition 142.2 (Vanishing of one-loop Lie weights). *For every finite positive-height quotient, the Lie weight of each one-loop graph in the cotangent theory vanishes.*

Proof. A one-loop graph has $E = V$. Every B -leg is consumed by an internal edge, so its external legs are A -legs. Its Lie weight is a cyclic trace of adjoint operators, or the dual trace with the conventionally equivalent signs. Each external label has positive root height. The product therefore strictly raises height on the finite-dimensional adjoint representation and has zero trace. The same argument applies to the dual representation. $\square$

This vanishing removes the one-loop Lie coefficients in a perturbative construction with the stated propagator and vertex types. A renormalized quantum factorization algebra still requires admissible analytic kernels, local extensions, the quantum master equation, and compatible height transitions on the chosen observable spaces. These are the remaining quantum realization data.

CLASSICAL REDUCTION ALONG K_3

Choose a Kähler metric on S and normalize a closed harmonic $(0, 2)$ -form η by $\int_S \sigma_S \wedge \eta = 1$. Let $K = \mathbb{C}[\eta]/(\eta^2)$ with $|\eta| = 2$ and zero differential. There is a unital dg-algebra map

$$i : K \longrightarrow \Omega^{0,*}(S), \quad \eta \longmapsto \eta.$$

Its image is closed under multiplication because $(0, 4)$ -forms vanish. The Dolbeault cohomology of a K_3 surface is one-dimensional in degrees zero and two and zero in degree one, so i is a quasi-isomorphism. It preserves the trace defined by integration against σ_S .

Let $\mathfrak{l}$ be a finite-dimensional Lie algebra, and write $\mathfrak{d} = \mathfrak{l} \ltimes \mathfrak{l}^*$ for its cotangent Lie algebra, with invariant evaluation pairing. For an open $U \subset E$, tensor with $\Omega^{0,*}(U)$, using the completed smooth tensor product in the S direction. The field comparison is

$$K \otimes \Omega^{0,*}(U) \otimes \mathfrak{d} \xrightarrow{i \otimes \text{id}} \Omega^{0,*}(S \times U) \otimes \mathfrak{d}.$$

The bracket is multiplication of forms followed by the Lie bracket, with Koszul signs.

Proposition 143.1 (Classical field comparison). *The displayed map is a trace-preserving dg Lie quasi-isomorphism, compatible with restriction in U . Its induced minimal operations in the K_3 factor have no higher products.*

Proof. Multiplicativity of i proves bracket compatibility. Hodge projection p and $h = \bar{\partial}^* G$ on the compact surface satisfy $pi = \text{id}$ and $\bar{\partial}h + h\bar{\partial} = \text{id} - ip$. Tensoring these continuous operators with forms on U gives the same contraction, with the tensor-product signs. Thus $i \otimes \text{id}$ is a quasi-isomorphism. Restriction acts only on the U factor and commutes with it. The normalized trace on K is the restriction of the Dolbeault trace. Choose the contraction with $hi = 0$. In every higher transferred tree, the first product of inputs from $i(K)$ remains in $i(K)$ and the next internal homotopy kills it. Thus every higher transferred product vanishes. $\square$

The coefficient algebra K retains both the degree-zero and degree-two K_3 modes. The proposition compares classical field complexes and their brackets. Constructing an equivalence of quantum observable factorization algebras requires compatible quantization, support conventions, and the induced maps on observable completions. The ordinary Hochschild comparison of Theorem 132.1 is a separate algebraic calculation, not that quantum field map.

PROTECTED STATES AND THE IGUSA TRACE

Let

$$\mathcal{H}_{\text{ip}} = \widehat{\bigoplus_{\alpha > 0} \mathfrak{g}_{\Delta, \alpha}}$$

with its Borchers parity. The multiparticle state space is the completed super-Fock space

$$\mathcal{H}_{\text{Fock}} = \widehat{\text{Sym}}(\mathcal{H}_{\text{ip}, \bar{0}}) \otimes \widehat{\Lambda}(\mathcal{H}_{\text{ip}, \bar{1}}).$$

For Z in the absolute-convergence chamber, let e^{-Hz} act by $e^{-\alpha(Z)}$ on the root space of degree α .

Theorem 144.1 (Protected Hamiltonian trace). *The supertrace is*

$$\text{str}_{\mathcal{H}_{\text{Fock}}}(e^{-Hz}) = \prod_{\alpha > 0} (1 - e^{-\alpha(Z)})^{-\text{sdim } \mathfrak{g}_{\Delta, \alpha}}.$$

After multiplication by the inverse Weyl line, its reciprocal is $\text{Den}_{\Delta}(Z) = D_{\mathfrak{s}}(2Z)$. After the half-coordinate pullback and doubling, the disconnected protected scalar is $-(j^ \text{Den}_{\Delta})^{-2} = -\chi_{\text{io}}^{-1}$ in the Oberdieck–Pixton normalization.*

Proof. An even oscillator contributes $(1 - e^{-\alpha})^{-1}$ and an odd oscillator contributes $1 - e^{-\alpha}$. Multiplication over the root basis gives the product. The denominator identity and the reduced curve-counting theorem supply the final two equalities. $\square$

THE COMPACT-SOURCE COMPARISON

The compact threefold provides the classical cotangent field theory and the dg Lie comparison of Proposition 143.1. The root datum separately gives the Borchers algebra, its enveloping algebra, and the root-graded Fock space. The latter has the trace computed above. Proposition 142.2 supplies the finite one-loop Lie cancellation, while Theorem 132.1 computes the ordinary algebraic centre model with adjoint coefficients.

Problem 145.1 (Geometric compact comparison). Construct objects E_i representing the simple branes and maps on their derived morphism complexes that preserve units, composition, and higher composition homotopies. Realize these maps on extension stacks with the reduced critical coefficients and orientation isomorphisms on triple flags. Construct compatible tensor and unit constraints on the resulting boundary functor. Compare the chosen quantum bulk action with the independently defined boundary centre, including support, collisions, and completion. Finally construct a trace and a map of its value line to the automorphic line in the same charge normalization.

A primitive map satisfying the root relations gives the enveloping and Chevalley maps of Theorem 135.1. A strong monoidal equivalence transports categorical centres. Neither construction determines the brane functor or its cyclic trace. These are the additional maps required for compact geometric reconstruction.

BOOK XXVIII

Physics of the comparison maps

HOLOMORPHIC–TOPOLOGICAL BOUNDARY THEORIES

A three-dimensional holomorphic–topological theory on $C \times \mathbb{R}$ has local observables that are holomorphic along C and locally constant along $\mathbb{R}$. A boundary produces an ordered chiral algebra. A second topological direction permits exchange and gives an E_2 enhancement or a meromorphic braided line category. The bulk acts through the derived centre.

The finite cotangent BF model gives the exact local calculation:

$$\mathrm{Bar}(U(\mathbf{I})) \simeq C_*(\mathbf{I}).$$

For $\mathbf{I} = \mathfrak{n}_{\Delta,+}^{(H)}$, this calculation realizes every finite-height Igusa denominator sector.

SIX-DIMENSIONAL hCS ON $K_3 \times E$

Choose a holomorphic volume form $\Omega_S \wedge dz_E$. The hCS theory has classical fields

$$\Omega^{0,*}(S \times E, \mathfrak{a})[1]$$

and action S_{hCS} . Its factorization observables are native to the threefold. Pushforward along p_E gives a factorization algebra on E whose coefficient complex contains the Dolbeault modes of S .

A protected reduction to a chiral algebra consists of a deformation retract of the K_3 Dolbeault complex, compatibility of the transferred L_∞ operations with the trace, a boundary or defect selecting ordered fusion, and a quantization trivializing the local anomaly. These data produce the ordered chiral algebra to which the Hall/BPS comparison may be applied.

The Hodge equality $\chi(\mathcal{O}_{S \times E}) = 0$ controls one compact supertrace. The gauge and gravitational anomalies are classes in the local BV deformation complex and are evaluated by their own characteristic forms.

OMEGA BACKGROUNDS

Let a torus T act holomorphically on the local geometry and preserve the selected supercharge. For a generator V of the action, the equivariant differential is

$$Q_\Omega = Q + \iota_V, \quad Q_\Omega^2 = \mathcal{L}_V.$$

After localization, fixed loci produce deformation parameters and often a spectral R -matrix. Toric three-folds and ADE surface resolutions carry such actions. A deformation family equipped with monodromy-compatible moduli correspondences and wall-crossing equivalences transports the localized algebra from algebraic or equivariant K3 fibers to the selected generic fiber.

ANOMALIES AS TYPED CLASSES

The principal anomaly classes are

construction	deformation complex
hCS quantization	local BV cochains
critical CoHA orientation	Picard/gerbe theory of the critical stack
Hall coproduct	correspondence and localization complex
projective conformal blocks	Atiyah algebra of the determinant line
automorphic multiplier	character group of the modular cover.

A comparison theorem is a morphism between the corresponding complexes together with equality of the transported classes. This formulation turns numerical coincidences into geometric identities precisely when an intertwining map exists.

BOOK **XXIX**

**Automorphic characters and scalar
shadows**

BORCHERDS PRODUCTS AS CHARACTERS

Let $\mathfrak{n}_+$ be a locally finite positive Lie superalgebra. The Chevalley Euler character is

$$\chi^{C_*}(\mathfrak{n}_+) = \prod_{\alpha > 0} (1 - e^{-\alpha})^{\text{sdim } \mathfrak{g}_\alpha}.$$

A Borchers product becomes the Weyl-shifted character of $C_*(\mathfrak{n}_+)$. This construction produces a character of an existing algebra. Hall geometry supplies a second algebra, and a primitive comparison theorem identifies the two.

THE UNIVERSAL BORCHERDS WEIGHT FORMULA

For a weight-zero weak Jacobi form

$$\varphi(\tau, z) = \sum c(n, \ell) q^n r^\ell,$$

the Borcherds lift has weight

$$\text{wt}(\text{Borch}(\varphi)) = \frac{c(o, o)}{2}.$$

For the K_3 half-index, $c(o, o) = 10$ and the weight is 5. This formula lives entirely on the automorphic layer. Hodge Euler characteristics, Heisenberg ranks, central charges, and BV anomalies are evaluated by their own traces.

THE GRITSENKO–CLÉRY ATLAS

The eight diagonal-divisor forms are

$$\Delta_5, \Delta_2, \Delta_1, \Delta_{1/2}, \nabla_3, \nabla_2, \nabla_{3/2}, Q_1$$

with weights

$$5, 2, 1, \frac{1}{2}, 3, 2, \frac{3}{2}, 1.$$

Each form is an automorphic product on its own paramodular or Hecke congruence group with its own character or multiplier [16]. A geometric realization is specified row by row by a charge lattice, Hall problem, trace, and primitive algebra comparison.

SCALARIZATION HIERARCHY

The constructions descend through four levels:

$$\begin{aligned} \text{factorization or Hall object} &\longrightarrow \text{primitive Lie algebra} \\ &\longrightarrow \text{PBW character} \longrightarrow \text{scalar partition function.} \end{aligned}$$

Each arrow retains the quotient of structure displayed at its target. The first object contains supports, extension maps, and brackets. The second retains the infinitesimal algebra. The third retains signed graded dimensions. The fourth adds a trace and normalization. Upward reconstruction is supplied by the corresponding geometric operations and comparison morphisms.

BOOK **XXX**

The unified theorem system

THE FOUNDATIONAL THEOREM

Definition 154.1 (Complete Calabi–Yau quantum-group datum). A complete comparison datum specifies independent categorical, field, boundary, Hall, Hopf, and trace objects together with:

- (i) PTVV-representable moduli for the smooth proper Calabi–Yau category;
- (ii) a renormalized BV quantization satisfying the quantum master equation and a local compact factorization generator b ;
- (iii) a boundary reduction or proper locally trivial map to a curve satisfying factorization pushforward, together with the supplied curve-level faces and aligned distributive law;
- (iv) a strong-oriented Hall coefficient theory whose pull–push maps satisfy base change and whose positive algebra is separated and complete;
- (v) paired negative and Cartan halves, a completed coproduct, a continuous nondegenerate Hopf pairing, and a separated triangular completion on which the canonical tensors converge;
- (vi) compatible open–closed and Hall–boundary morphisms;
- (vii) a parity-refined primitive comparison for every automorphic identification.

Theorem 154.2 (Calabi–Yau quantum-group assembly). *Let a complete datum consist of:*

- (i) *a Calabi–Yau category and a representable moduli-of-objects problem;*
- (ii) *a quantized cyclic elliptic BV theory and a boundary or defect generator;*
- (iii) *a Hall extension correspondence with coefficient functor, base-change theorem, strong orientation data, and separated charge completion;*
- (iv) *paired positive and negative bialgebras, Cartan data, a nondegenerate continuous Hopf pairing, and a completed double;*
- (v) *every curve-level distributive, PBW, and analytic comparison used below.*

Then the datum determines functorially:

- (i) *framed E_2 Hochschild cochains and the shifted-symplectic moduli structure;*
- (ii) *the three curve-level algebraic faces and their three bars;*
- (iii) *the positive Hall/BPS algebra and its primitive Lie algebra;*
- (iv) *the completed quantum double and braided representation category;*
- (v) *every supplied comparison among categorical, field-theoretic, Hall, current, quantum, and automorphic realizations;*
- (vi) *the Weyl-shifted PBW/Chevalley character of any parity-refined primitive Lie-superalgebra identification.*

Proof. Cyclic Deligne and shifted symplectic geometry give the first output [9, 10, 64]. Boundary factorization gives the curve faces, and Volume I gives their bars and centres. The Hall correspondence and its base-change theorem give the associative product; strong orientation supplies the coefficient coherence required in the 3-Calabi–Yau setting [50]. The paired bialgebra data give the double and universal R -matrix. The comparison maps commute by hypothesis. PBW and Chevalley antisymmetrization give the final character. Each construction is natural under a morphism preserving the displayed data. $\square$

The local quiver, equivariant surface, K_3 correspondence, and Igusa Borchers constructions have the carriers stated above. A geometric $K_3 \times E$ comparison requires the primitive map and determinant-line map of Problem 145.1.

LANDSCAPE OF ESTABLISHED REALIZATIONS

Geometry	Positive algebra	Full or centred object	Character
A_2 quiver	$U_v^+(\mathfrak{sl}_3)$	$U_v(\mathfrak{sl}_3)$	PBW root product
$\mathbb{C}^3$	$Y^+(\widehat{\mathfrak{gl}}_1)$	$Y(\widehat{\mathfrak{gl}}_1)$	plane-partition and Fock traces
$\widetilde{\mathbb{C}^2}/\Gamma$	$Y^+(\widehat{\mathfrak{gl}}_\Gamma)$	paired completion when the negative half and coproduct are supplied	surface Fock traces
K_3	$U(\mathfrak{g}_{\text{BPS}}^+)$	Heisenberg and LLV actions	K_3 Hall characters
$K_3 \times E$	Borcherds root envelope; geometric reduced character	$U_q(\mathfrak{g}_{\Delta_3})$ and the root-current model and classical BF theory	$-\chi_{\text{io}}^{-1}$

The A_2 and $\mathbb{C}^3$ rows provide complete standard quantum-group models. The ADE row provides an established completed positive half together with its geometric current actions; its paired full double is formed from the negative half, Cartan data, coproduct, continuous Hopf pairing, and triangular completion stated in the table. The K_3 row provides global two-Calabi–Yau BPS and cohomological symmetries. The final row records independently constructed root, character, and classical field objects. Their geometric Hall realization requires the comparison in Problem 145.1.

CONSTRUCTED COMPARISONS AND THEIR GEOMETRIC EXTENSIONS

The Hall flag argument constructs associativity once its coefficient and orientation maps are defined. The tensor-action formula constructs representation fusion from a coassociative counital coproduct. The matrix-coefficient formulas construct the coend of a supplied fibre functor. The primitive presentation gives enveloping and Chevalley maps. The classical cyclic and K_3 field calculations retain their actual differentials, traces, and brackets.

The quantum field realization requires admissible quantization and observable completions. The compact Hall realization requires the brane and orientation maps of Problem 145.1. The Fredholm and automorphic comparisons require their analytic domains and value-line isomorphisms. These constructions preserve the full geometric reconstruction problem beyond its finite algebraic and classical calculations.

ACTIONS AND COMPARISON MAPS

For a specified boundary algebra $\mathcal{A}_b$, a coherent bulk action is a map

$$\beta : \text{Obs}_{\text{bulk}}|_{\partial} \longrightarrow Z_{\text{ord}}(\mathcal{A}_b).$$

The target is the derived endomorphism object of the diagonal bimodule in the chosen factorization coefficient category. A Hall construction instead starts with

$$\mathcal{M}_{\alpha} \times \mathcal{M}_{\beta} \xleftarrow{p} \mathcal{E}_{\alpha,\beta} \xrightarrow{q} \mathcal{M}_{\alpha+\beta}.$$

Critical coefficients, orientations, and admissible pull–push maps give its product. Framed moduli supply a representation after their action correspondences and triple-flag identities have been constructed. A paired Hopf structure gives $D(H^+, H^0, H^-)$. Its coproduct defines tensor products of continuous modules, and a convergent universal R -matrix gives exchange.

A strong monoidal equivalence between this module category and a boundary category transports categorical centres. It does not identify that categorical centre with the object $Z_{\text{ord}}(\mathcal{A}_b)$ without a realization theorem. A trace on a specified representation then gives an amplitude. Its comparison with an automorphic section requires a map of value lines with the same normalization.

COMPARISON DATA FOR CALABI–YAU CONSTRUCTIONS

Comparisons among Calabi–Yau categories, holomorphic field theory, chiral algebras, Hall algebras, quantum groups, and automorphic products require the following structures:

- (i) for a smooth proper d -Calabi–Yau category with its nondegenerate negative-cyclic class, the categorical outputs are framed E_2 Hochschild cochains and, when the moduli stack of objects is representable in the PTVV setting, $(2 - d)$ -shifted symplectic moduli; the shift d does not change the Hochschild operadic level to E_n ;
- (ii) a geometric BV theory and boundary category produce factorization observables and transverse ordered chiral algebras; associative chiral convolution is recorded as a separate supplied structure;
- (iii) pushforward to a curve uses an actual map $p : X \rightarrow C$; formal cycle-specialization notation alone specifies no pushforward;
- (iv) extension stacks, coefficient sheaves, orientations, and completions define the Hall algebra;
- (v) coproduct, Cartan, Hopf pairing, and topology define the double;
- (vi) K_3 Heisenberg, LLV orthogonal symmetry, local ADE Yangians, and global BPS generalized Kac–Moody algebras occupy distinct strata joined by explicit comparison maps where constructed;
- (vii) the graph of the $K_3 \times E$ Gram map becomes multiplicative in the polarized central extension;
- (viii) point-local and wrapped sectors form a hybrid factorization problem;
- (ix) the Igusa form is the Weyl-shifted PBW/Chevalley character of $\mathfrak{g}_{\Delta_5}$, and compact Hall geometry maps to it through the primitive comparison;
- (x) anomaly identities are transported through explicit maps between deformation complexes.

The categorical, geometric, Hall, and automorphic operations thus have separate carriers and comparison maps.

BOOK XXXI

Technical appendices to the preceding
book

TYPED DATA TABLE

Object	Minimal data	Primary operation
CY category	nondegenerate negative-cyclic trace	Hochschild and shifted-symplectic structures
Factorization algebra	local observables and Weiss descent	disjoint-open product
Transversely ordered chiral algebra	factorization coefficient with an internal E_1 structure for $\otimes^!$	ordered fusion
Associative chiral algebra	an E_1 structure for chiral convolution $\star^{\text{ch}}$	nonlocal chiral composition
CoHA	extension correspondence and coefficient theory	pull–push product
Bialgebra	compatible product and coassociative counital coproduct	fusion of representations
Double	positive and negative halves, Cartan data, Hopf pairing	universal R -matrix
Borcherds product	Jacobi or vector-valued modular input	graded character

CHAPTER 160

THE THREE K_3 BRACKETS

The Heisenberg bracket is the excess intersection of two Nakajima incidence kernels. The LLV bracket is the commutator of diagonal Lefschetz kernels and their adjoints. The local Yangian bracket is the convolution commutator of ADE Hecke kernels with spectral equivariance. In $\text{Corr}_{K_3}(S)$ all three are compositions of geometric kernels. Their mixed commutators are the corresponding triple-intersection classes. The action algebra and Tannaka coend retain these classes while preserving the three geometric constructions.

EXACT ARITHMETIC CHECKS

The companion verification calculation establishes:

- (i) the low Fourier rows of $\phi_{0,1}$ through q^3 by exact theta-series division;
- (ii) the vector-valued cocycle identity for the Gram polarization;
- (iii) the rational Yang–Baxter equation on $(\mathbb{C}^2)^{\otimes 3}$;
- (iv) the A_2 Hall flag counts at ranks zero and one;
- (v) the type-II Gram matrix and reflection identities.

Each calculation uses integer or rational arithmetic and appears as a worked formula in the relevant chapter.

COHERENT BOUNDARY ACTIONS AND BV FIELDS

Boundary operations determine a deformation problem once their interface and coherence maps are supplied. A local realization gives a shifted cotangent BV theory. Comparing it with a physical bulk requires a map of fields, actions, and coherent boundary operations.

BOOK XXXII

The full mixed boundary datum

THREE PRODUCTS AND ONE INTERFACE

Holomorphic collisions and transverse composition act on different coefficient objects. A common boundary theory must specify their interaction on an interface. This interaction is part of the input to the following constructions.

Let X be a smooth complex curve and let $\mathcal{D}_X = D\text{-Mod}(\text{Ran } X)$ in a fixed six-functor enhancement. Write $\otimes^!$ for the pointwise product and $\star^{\text{ch}}$ for chiral convolution.

Definition 162.1 (Complete mixed boundary datum). A complete mixed boundary datum on X is a tuple

$$\mathbb{A} = (L, \mathcal{A}^{\text{ch}}, \mathcal{A}^\perp, K, \iota, \lambda)$$

with the following components.

- (i) L is a complete chiral Lie algebra.
- (ii) $\mathcal{A}^{\text{ch}}$ is a complete associative algebra for $\star^{\text{ch}}$.
- (iii) $\mathcal{A}^\perp$ is a complete E_1 algebra in $(\text{Fact}_D(X), \otimes^!)$.
- (iv) K is an $(\mathcal{A}^{\text{ch}}, \mathcal{A}^\perp)$ interface bimodule.
- (v) $\iota : L \rightarrow (\mathcal{A}^{\text{ch}})_{\text{Lie}}$ is a chiral Lie morphism.
- (vi) λ is the aligned mixed distributive law relating the two module actions on K .

Every structure map is continuous for the chosen weight filtrations. The tuple includes the module actions, the mixed coherence homotopies, and their identities in the chosen enhancement. When a reduced bar is used, the relevant augmentations are supplied as additional data. Existence of this tuple for a physical boundary theory is a separate construction.

The three native bars are

$$\text{Bar}_{\text{Lie}}^{\text{ch}}(L), \quad \text{Bar}_{\text{Ass}}^{\text{ch}}(\mathcal{A}^{\text{ch}}), \quad \text{Bar}^\perp(\mathcal{A}^\perp).$$

The interface has a fourth bar. Its degree- r terms are alternating words in the two associative colours with one marked interface letter. Its differential contracts adjacent compatible letters.

Construction 162.2 (Interface bar). Let $\overline{\mathcal{A}}^{\text{ch}}$ and $\overline{\mathcal{A}}^\perp$ be the augmentation ideals. Set

$$\text{Bar}^{\text{mix}}(\mathbb{A}; K) = \bigoplus_{p, q \geq 0} (s\overline{\mathcal{A}}^{\text{ch}})^{\otimes p} \otimes sK \otimes (s\overline{\mathcal{A}}^\perp)^{\otimes q}.$$

The differential is the sum of the internal differential, the two bar differentials, the left action on K , the right action on K , and the mixed terms supplied by λ .

Theorem 162.3 (Four-colour differential). *The interface differential squares to zero. Its codimension-two terms are exactly the associative relations in the two colours, the two module relations, and the mixed distributive relation.*

Proof. Filter by total word length. Every component of the square contracts two adjacent edges. Disjoint contractions cancel by the determinant-line sign. Nested contractions in one colour cancel by associativity. A contraction meeting the interface cancels by the corresponding module axiom. The remaining square has one edge of each colour and is the aligned distributive square. The datum λ identifies its two composites. $\square$

COMPATIBLE MODULES AND THE MIXED CENTRE

An operation on a boundary theory must act on boundary-changing morphisms as well as on objects. Natural transformations retain this compatibility, and derived natural transformations retain its higher homotopies.

A compatible module is a triple $(M^{\text{ch}}, M^{\perp}, \kappa)$ with an $\mathcal{A}^{\text{ch}}$ action, an $\mathcal{A}^{\perp}$ action, and an interface comparison over K . Let $\text{Mod}_{\mathbb{A}}^{\text{mix}}$ be their stable category.

Definition 163.1 (Mixed derived centre). The mixed derived centre is

$$Z_{\text{mix}}(\mathbb{A}) = \text{End}\left(\text{id}_{\text{Mod}_{\mathbb{A}}^{\text{mix}}}\right).$$

It is computed by the endomorphisms of the diagonal compatible bimodule.

Theorem 163.2 (Universal interior action). $Z_{\text{mix}}(\mathbb{A})$ carries an internal E_2 algebra structure. Every action already defined as a coherent natural endomorphism of the identity determines a point of this centre. The space of its factorizations through the representing endomorphism object is contractible. Morita-equivalent complete mixed data have equivalent centres.

Proof. Endomorphisms of the identity functor form the higher centre of a module category. Composition and insertion of natural transformations give the brace operations and hence the E_2 structure. A supplied coherent natural action defines an endomorphism of the identity. A physical bulk insertion determines such an action only after its action on modules, morphisms, and higher compositions has been constructed. The factorization is the universal property of the endomorphism object. Morita equivalence identifies the module categories and their identity functors. $\square$

RELATIVE INDUCTION FROM A CHIRAL ALGEBRA

Let V be a chiral algebra with conformal vacuum. Its algebraic collar $\text{Coll}(V)$ is the free transverse E_1 extension generated by one boundary copy of V , modulo the chiral relations already present in V . Weight completion gives $\widehat{\text{Coll}}(V)$.

Theorem 164.1 (Universal collar). *For every complete transversely ordered chiral algebra B and every chiral morphism $V \rightarrow B$, there is a unique continuous E_1 morphism*

$$\widehat{\text{Coll}}(V) \longrightarrow B$$

extending the chiral map. The construction is left adjoint to restriction from transverse ordered algebras to chiral coefficients.

Proof. The algebraic collar is the relative operadic induction from the chiral operad to the mixed operad. Relative induction is left adjoint to restriction. Completion preserves the adjunction on positive complete objects because every map is determined weight by weight. $\square$

When relative induction supplies a regular interface and a continuous aligned distributive law, these data define a mixed envelope with $L = V_{\text{Lie}}$, $A^{\text{ch}} = U^{\text{ch}}(L)$, and $A^\perp = \widehat{\text{Coll}}(V)$. Constructing that interface law in the selected geometric coefficient category is part of the envelope construction.

BOOK XXXIII

The deformation algebra and its BV envelope

THE CENTRE-EXTENDED DEFORMATION ALGEBRA

Varying a boundary product changes its compatibility with the interface and the central action. The deformation complex must therefore include all these operations. A local field realization is further data needed to turn their deformation equations into a BV action.

Fix a cofibrant coloured operad $\mathcal{P}_{\text{mix}}$ presenting the structures of a complete mixed datum and its central action. Let $\mathcal{P}_{\text{mix}}^!$ be its bar cooperad. The convolution complex

$$\text{Def}_{\mathbb{A}}^{\text{full}} = \text{Hom}_{\mathbb{S}}(\mathcal{P}_{\text{mix}}^!, \text{End}_{\mathbb{A}})$$

contains the chiral operation, both associative products, the interface, and the centre action.

Theorem 165.1 (Complete deformation L_{∞} algebra). *The convolution complex has a canonical complete L_{∞} structure. Its Maurer–Cartan points are complete mixed boundary structures together with a coherent central action. Twisting by the point defined by $\mathbb{A}$ gives the tangent algebra of deformations of the full datum.*

Proof. The infinitesimal decomposition maps of the bar cooperad insert collections of operations into one another. Composition in the endomorphism operad turns these decompositions into the convolution brackets. The cooperad identities imply the L_{∞} identities. A Maurer–Cartan element is precisely an operad morphism from a cofibrant resolution of $\mathcal{P}_{\text{mix}}$ to $\text{End}_{\mathbb{A}}$. Twisting linearizes this mapping space at the selected structure. $\square$

165.1 FIELD EQUATIONS AND BIANCHI IDENTITIES

Let α be a degree-one field in the localized deformation algebra. Its curvature is

$$\mathcal{F}(\alpha) = \sum_{n \geq 1} \frac{1}{n!} \ell_n(\alpha^{\otimes n}).$$

The linearization at α is

$$d_{\alpha}x = \sum_{n \geq 0} \frac{1}{n!} \ell_{n+1}(\alpha^{\otimes n}, x).$$

Theorem 165.2 (Universal Bianchi identity). *For every α ,*

$$d_{\alpha}\mathcal{F}(\alpha) = 0.$$

At a Maurer–Cartan point, $d_{\alpha}^2 = 0$.

Proof. Both identities are the exponential form of the L_{∞} Jacobi relations. Expand the coderivation square on the cofree cocommutative coalgebra and evaluate it on e^{α} and $e^{\alpha}x$. $\square$

SHIFTED COTANGENT TRANSGRESSION

Let $\mathbf{I}$ be a complete elliptic L_∞ algebra on a three-manifold $\mathcal{M}$. Its shifted cotangent theory has fields

$$\mathcal{E} = T^*[-1]\mathbf{I}[1] = \mathbf{I}[1] \oplus \mathbf{I}^\vee[-2].$$

The canonical action is

$$S(\alpha, \beta) = \langle \beta, \mathcal{F}(\alpha) \rangle.$$

Theorem i66.1 (Universal BV envelope). *The action satisfies the classical master equation. Its Euler–Lagrange equations are*

$$\mathcal{F}(\alpha) = 0, \quad d_\alpha^\vee \beta = 0.$$

The tangent complex at a solution is the cotangent complex of the Maurer–Cartan moduli problem.

Proof. The Hamiltonian vector field of S is the cotangent lift of the homological vector field defining the L_∞ structure. The square of that vector field is zero, hence $\{S, S\} = 0$. Variation in β gives the Maurer–Cartan equation. Variation in α gives the dual linearized equation. $\square$

Definition i66.2 (Local realization and chiral BV envelope). A local realization of the complete mixed datum consists of an elliptic local L_∞ algebra $\mathbf{I}_\mathbb{A}^{\text{full}}$ on $\mathcal{M} = X^{\text{an}} \times \mathbb{R}$, a specified continuous filtered L_∞ quasi-isomorphism

$$\Gamma(\mathcal{M}, \mathbf{I}_\mathbb{A}^{\text{full}}) \longrightarrow \text{Def}_\mathbb{A}^{\text{full}},$$

and the local pairing needed for cotangent transgression. Here Γ denotes the chosen complex of global sections with its declared topology. The comparison includes compatibility with the specified boundary operations. The dual fields use the density-valued continuous dual, with compact support or boundary conditions that make the integrated pairing and integration by parts defined.

For a supplied local realization, define

$$\mathbf{BEnv}_3^{\text{full}}(\mathbb{A}) = T^*[-1] \text{MC}(\mathbf{I}_\mathbb{A}^{\text{full}}).$$

Its action is the cotangent action displayed above. Constructing a local realization from an abstract mixed datum remains an additional existence problem.

INTRINSIC ALGEBRAIC QUANTIZATION

Let $\mathfrak{X}$ be a finite perfect derived formal moduli problem with a homological orientation μ : the corresponding Berezinian is invariant under the homological vector field. Functions on its shifted cotangent are completed polyvectors:

$$O(T^*[-1]\mathfrak{X}) = \widehat{\mathrm{PV}}(\mathfrak{X}) = \prod_{p \geq 0} \Gamma(\mathfrak{X}, \wedge^p T_{\mathfrak{X}})[p].$$

The homological orientation defines the divergence operator

$$\Delta_\mu : \widehat{\mathrm{PV}}(\mathfrak{X}) \longrightarrow \widehat{\mathrm{PV}}(\mathfrak{X})[1].$$

Theorem 167.1 (Oriented BV complex). *The operator Δ_μ has order two and satisfies*

$$\begin{aligned} \Delta_\mu^2 &= 0, & [d_{\mathfrak{X}}, \Delta_\mu] &= 0, \\ \Delta_\mu(fg) &= \Delta_\mu(f)g + (-1)^{|f|}f\Delta_\mu(g) + (-1)^{|f|}\{f, g\}. \end{aligned}$$

Consequently

$$\mathbf{QHdg}(\mathfrak{X}, \mu) = \left(\widehat{\mathrm{PV}}(\mathfrak{X})[[\hbar]], d_{\mathfrak{X}} + \hbar\Delta_\mu \right)$$

is a square-zero quantum BV complex.

Proof. In oriented Darboux coordinates, Δ_μ is the sum of paired second derivatives. Their graded commutation gives $\Delta_\mu^2 = 0$. Invariance of the Berezinian under the homological vector field gives $[d_{\mathfrak{X}}, \Delta_\mu] = 0$. Expanding the operator on a product gives the bracket as the cross term. $\square$

The orientation identifies polyvectors with differential forms by contraction. Under this identification,

$$\iota_\mu \Delta_\mu = d_{\mathrm{dR}} \iota_\mu.$$

Hence the BV complex is the Hodge-completed derived de Rham complex with differential

$$d_{\mathfrak{X}} + \hbar d_{\mathrm{dR}}.$$

The Rees parameter records Hodge degree.

Definition 167.2 (Intrinsic quantum complex of a mixed datum). For a complete mixed datum $\mathbb{A}$ equipped with compatible homological orientations on its finite perfect charts, define $\mathbf{QHdg}_3^{\mathrm{full}}(\mathbb{A})$ by applying the preceding BV complex to finite perfect charts of

$$\mathrm{MC}(\mathbf{I}_{\mathbb{A}}^{\mathrm{full}})$$

and taking the separated inverse limit over the chart and weight filtrations.

Theorem 167.3 (Quantum master equation). *A degree-zero action $S = S_0 + \hbar S_1 + \cdots$ defines a quantum state $e^{S/\hbar} \mu$ precisely when*

$$dS + \frac{1}{2}\{S, S\} + \hbar \Delta_\mu S = 0.$$

The equation is equivalent to

$$(d + \hbar \Delta_\mu) e^{S/\hbar} = 0.$$

Proof. Apply the second-order product formula to the exponential and collect powers of $\hbar$. □

PERTURBATIVE FACTORIZATION REALIZATION

The algebraic quantum differential does not specify a propagator or a space of singular kernels. Fix a local realization, elliptic gauge-fixing data, support and boundary conditions, and a renormalization prescription. A heat-kernel propagator is defined away from the diagonal. Extending its graph integrals to collisions requires counterterms and compatibility with the boundary conditions.

Definition 168.1 (Factorization realization data). A factorization realization consists of renormalized local observable complexes, continuous disjoint-insertion maps, and coherent homotopies for their compositions. It includes extensions of graph weights to the required collision strata, their Stokes identities, and a quantum master equation in the permitted local functional complex. Weiss descent is a further required property of these actual complexes. A boundary realization also includes a comparison with the supplied mixed datum, preserving its two products, Lie operation, interface, and mixed law.

Under these hypotheses the disjoint-insertion maps and composition homotopies define the stated factorization object. The boundary comparison identifies its boundary operations with those of $\mathbb{A}$. The construction problem is to produce these data from the physical fields and action. A graphwise identity supplies the corresponding composition equation, whereas descent and boundary identification require their own maps and proofs.

BOOK XXXIV

Boundary duality, exchange, and reduction

THE KOSZUL–MORITA SQUARE

A change from modules to comodules must transport the interface as well as each algebra. The mixed bar supplies the comparison object. Exchange then asks for motion in an additional topological direction, while BRST reduction asks whether the differential respects every transported operation.

Let P be a dualizable boundary state for $A^\perp$. Its boundary dual is

$$A_P^\dagger = \mathrm{RHom}_{A^\perp}(P, P).$$

The ordinary bar construction gives explicit twisted tensor functors. Their extension to the mixed datum must also transport the interface and its geometric operations.

Let k be a field and let $A_{\mathrm{alg}} = \bigoplus_{w \geq 0} A_w$ be an augmented associative algebra in cohomological degree zero. Assume $A_0 = k$, augmentation projects onto A_0 , and multiplication adds weights. Set $A = \prod_{w \geq 0} A_w$ and $C = \prod_{w \geq 0} B(A_{\mathrm{alg}})_w$, with degreewise products and the weight topology. The bar suspension lowers cohomological degree by one, with $b_2[a|b] = [ab]$. The degree-one twisting map $\tau : C \rightarrow A$ sends $[a]$ to a and vanishes on the empty word and longer words.

A module or comodule has the form $\prod_{w \geq w_0} M_w$, with differential preserving weight, for some integer w_0 . Actions and coactions preserve weight. Comodule coactions are counital and coassociative, and use the tensor product

$$(C \widehat{\otimes} M)_w = \bigoplus_{u+v=w} B(A_{\mathrm{alg}})_u \otimes M_v.$$

The same convention defines $A \widehat{\otimes} M$. These sums are finite in each weight. Repeated reduced coactions vanish in weight w after more than $w - w_0$ steps. Invert quasi-isomorphisms in each weight to obtain $D_{\mathrm{wt}}(A)$ and $D_{\mathrm{wt}}(C)$.

Proposition 169.1 (Ordinary twisted tensor equivalence). *The functors*

$$F(M) = C \widehat{\otimes}_\tau M, \quad G(N) = A \widehat{\otimes}_\tau N$$

induce inverse equivalences $D_{\mathrm{wt}}(A) \simeq D_{\mathrm{wt}}(C)$. Writing $\Delta c = \sum c_{(1)} \otimes c_{(2)}$ and $\rho n = \sum n_{(-1)} \otimes n_{(0)}$, their differentials are

$$\begin{aligned} d_F(c \otimes m) &= d_C c \otimes m + (-1)^{|c|} c \otimes d_M m + \sum (-1)^{|c_{(1)}|} c_{(1)} \otimes \tau(c_{(2)}) m, \\ d_G(a \otimes n) &= a \otimes d_N n - \sum a \tau(n_{(-1)}) \otimes n_{(0)}. \end{aligned}$$

The counit and unit are

$$\epsilon_M(a \otimes c \otimes m) = a \epsilon_C(c) m, \quad \eta_N(n) = \sum n_{(-1)} \otimes 1 \otimes n_{(0)}.$$

Proof. The twisting equation makes both differentials square to zero. The action, coaction, and counit identities show that the displayed maps are chain maps and satisfy the triangle identities. For $GF(M)$, the extra degeneracy

of the two-sided bar inserts $1[\bar{a}_0|a_1|\cdots|a_r]m$ into $a_0[a_1|\cdots|a_r]m$, where $\bar{a}_0 = a_0 - \epsilon_A(a_0)1$. With simplicial boundary $\sum_i (-1)^i d_i$, it contracts the augmented bar. Multiplying length- r words by $(-1)^r$ gives the displayed differential convention. This contraction preserves weight, so the counit is a weightwise quasi-isomorphism. For $FG(N)$, filter each weight by the weight of the final N -factor. The reduced coaction lowers that weight. The associated graded unit is the contracted right-sided bar $k \otimes N_v \rightarrow B(k, A_{\text{alg}}, A_{\text{alg}}) \otimes N_v$. The filtration is finite in each total weight, so the unit is also a quasi-isomorphism. The same finite filtrations show that F, G preserve weightwise quasi-isomorphisms, since tensoring complexes over k preserves them. Thus the unit and counit give the asserted inverse equivalences. $\square$

For a right module R and a left module L with the same lower-weight bounds, the two-sided bar has underlying weight-completed complex

$$\prod_n \bigoplus_{u+v_1+\cdots+v_r+w=n} R_u \otimes s\bar{A}_{v_1} \otimes \cdots \otimes s\bar{A}_{v_r} \otimes L_w, \quad v_i > 0,$$

where the sum also runs over $r \geq 0$ and $\bar{A} = \ker(A_{\text{alg}} \rightarrow k)$. Internal differentials and the two endpoint actions and adjacent products form its bar differential. Its realization computes $R \otimes_A^L L$ in these weight-graded categories. Comparing it with a derived cotensor product requires a two-sided twisting comparison on the chosen comodule category and completion. Weight completion here is not identified with derived augmentation-adic completion.

Proposition 169.2 (Centre transport under a mixed equivalence). *Suppose an equivalence between the compatible mixed module categories is supplied, with coherent compatibility with all four bar objects and the interface operations. It identifies the derived endomorphism objects of their identity functors.*

Proof. Conjugate a natural endomorphism by the equivalence and a quasi-inverse. The unit and counit homotopies give inverse maps on derived endomorphism objects and preserve composition and its coherent operations. $\square$

Constructing such a mixed equivalence requires the restriction, collision, interface, and distributive-law comparisons for the actual geometric coefficients. Its unit and counit must respect those maps and the chosen inverse limits. A diagonal bimodule comparison must produce an actual compatible bicomodule before tensor products can be identified with geometric cotensor products. Transport of mixed Hochschild chains and traces further requires their cyclic comparison and the trace domains used in their definition. These are additional mixed Koszul–Morita constructions. The ordinary weight-graded equivalence supplies its stated one-colour functors, without identifying the other mixed carriers or their physical traces.

PLANAR EXCHANGE AND TANNAKA RECONSTRUCTION

A second topological direction gives paths exchanging parallel defects. Their homotopy classes form braid groups. A rigid meromorphic line category $\mathcal{L}$ with a faithful fibre functor ω has coend

$$\mathcal{O}(\mathbb{G}) = \int^{M \in \mathcal{L}} \omega(M)^* \otimes \omega(M).$$

Theorem 170.1 (Quantum symmetry from exchange). *The coend is a coquasitriangular bialgebra. Its continuous dual acts on every line object. The meromorphic braiding gives the universal R -matrix on completed matrix coefficients. A spectral rational braiding yields an RTT presentation.*

Proof. Tensor product in $\mathcal{L}$ gives the coproduct on matrix coefficients. Evaluation gives the counit. Rigidity gives the antipode on the Hopf envelope. The braiding defines the coquasitriangular pairing, and the hexagon identities give the Yang–Baxter equation. Writing this naturality on a generating fibre produces the RTT relation. $\square$

BRST REDUCTION

Let Q be a square-zero derivation of a mixed datum. Extend Q factorwise to every bar word. Since Q is a derivation for each structure map, it anticommutes with every bar differential.

Theorem 171.1 (BRST–bar bicomplex). *The total differential*

$$D = Q + b_{\text{ch}} + b_{\perp} + b_{\text{mix}}$$

satisfies $D^2 = 0$. A bounded-below complete filtration gives a convergent spectral sequence

$$E_1 = H_Q(\text{Bar}^{\text{mix}}(\mathbb{A}; K)) \implies H_D(\text{Bar}^{\text{mix}}(\mathbb{A}; K)).$$

When Q -cohomology is concentrated and tensor-exact, the target is the mixed bar of the reduced datum.

Proof. The square vanishes by the derivation identity and the four-colour bar theorem. The filtration by BRST degree gives the spectral sequence. Concentration and tensor exactness identify the E_1 differential with the bar differential of the reduced operations. $\square$

BOOK **XXXV**

Conformal covariance and Einstein completion

PROJECTIVE CONNECTIONS AND THE VIRASORO TANGENT COMPLEX

A conformal boundary algebra transforms under changes of coordinates. Projective connections record its central term. Three-dimensional gravity supplies independently defined connection and coframe fields whose boundary transformation law can be compared with this algebraic structure.

Let $\mathcal{P}_c(X)$ be the formal moduli problem of projective connections with central parameter c . At a projective connection p , its tangent complex is locally

$$\Omega^{\circ, \bullet}(X, T_X) \xrightarrow{\mathcal{D}_{p,c}} \Omega^{\circ, \bullet}(X, K_X^{\otimes 2}),$$

where

$$\mathcal{D}_{p,c}(\epsilon) = \epsilon \partial p + 2(\partial \epsilon) p + \frac{c}{12} \partial^3 \epsilon.$$

Theorem 172.1 (Projective covariance). *A conformal map $\text{Vir}_c \rightarrow A$ defines a formal family of mixed structures over $\mathcal{P}_c(X)$. Its tangent algebra is the semidirect product of the Virasoro projective complex with the deformation algebra relative to a fixed projective connection.*

Proof. The conformal vector acts on fields by infinitesimal coordinate change. The central term is the affine cocycle in the transformation law of a projective connection. The action therefore gives a map from the projective tangent algebra to derivations of the mixed datum. The tangent algebra of the associated family is the semidirect product. $\square$

THREE-DIMENSIONAL GRAVITY AS A FLAT CARTAN THEORY

Let M be an oriented three-manifold, let $\Lambda = -\ell^{-2}$, and let $\mathfrak{g}_\Lambda$ have basis J_a, P_a with

$$[J_a, J_b] = \epsilon_{ab}{}^c J_c, \quad [J_a, P_b] = \epsilon_{ab}{}^c P_c, \quad [P_a, P_b] = \ell^{-2} \epsilon_{ab}{}^c J_c.$$

The invariant pairing is $\langle J_a, P_b \rangle = \eta_{ab}$ and vanishes on the two diagonal summands. A Cartan connection is

$$\mathcal{A} = \omega^a J_a + e^a P_a.$$

Its curvature is

$$F_{\mathcal{A}} = \left(R^a + \frac{1}{2\ell^2} \epsilon^a{}_{bc} e^b \wedge e^c \right) J_a + T^a P_a.$$

Theorem 173.1 (Cartan–Einstein equivalence). *On the locus where e is invertible, $F_{\mathcal{A}} = 0$ is equivalent to*

$$T^a = 0, \quad G_{\mu\nu} + \Lambda g_{\mu\nu} = 0, \quad g_{\mu\nu} = \eta_{ab} e_\mu^a e_\nu^b.$$

The Chern–Simons action for $\mathcal{A}$ equals the Einstein–Hilbert action with cosmological constant, up to its boundary term and normalization.

Proof. The P component of the flatness equation is the torsion equation. Invertibility of e gives the Levi–Civita spin connection. The J component states that the curvature two-form has constant sectional curvature $-\ell^{-2}$. In three dimensions the Riemann tensor is determined by the Ricci tensor, so this is the Einstein equation. Expanding the Chern–Simons three-form with the off-diagonal invariant pairing gives $e \wedge R + \ell^{-2} e^3$ plus an exact form [1, 71]. $\square$

FOUR EQUIVALENT LINEAR COMPLEXES

Fix a constant-curvature solution (e, ω) . There are four deformation complexes.

- (i) The flat adjoint complex $\Omega^\bullet(M, \mathfrak{g}_\Lambda)$ with differential $d_{\mathcal{A}}$.
- (ii) The Einstein–Cartan complex on $(\partial e, \partial \omega)$ with linearized torsion and curvature.
- (iii) The metric Einstein complex on $h_{\mu\nu}$ with linearized Einstein operator.
- (iv) The Calabi complex resolving local Killing fields by the Killing, linearized curvature, and Bianchi operators.

Theorem 174.1 (Chain equivalence of gravitational presentations). *An invertible torsion-free dreibein gives chain maps among the four complexes. These maps are quasi-isomorphisms on every geodesically convex open set. Their cohomology describes infinitesimal isometries, physical linearized fields, and Bianchi identities in the four languages.*

Proof. The dreibein identifies internal translations with tangent vectors and internal Lorentz rotations with skew endomorphisms. Splitting $d_{\mathcal{A}}$ in this basis gives the linearized torsion and curvature operators. Eliminating $\partial \omega$ by the linearized torsion equation gives the metric Einstein operator. The symbol sequence is the Calabi symbol sequence and is exact on constant-curvature space. The Poincaré lemma for the flat adjoint connection and exactness of the symbol homotopies give local quasi-isomorphisms. $\square$

BROWN–HENNEAUX AND THE PROJECTIVE BOUNDARY

Write the AdS connection as

$$A^\pm = \omega \pm \ell^{-1} e.$$

The Einstein action is the difference of two $\mathfrak{sl}_2$ Chern–Simons actions with level

$$k = \frac{\ell}{4G}.$$

Theorem 175.1 (Brown–Henneaux central charge). *The classical asymptotic Poisson algebra of each chiral Chern–Simons sector has Virasoro central charge*

$$c = 6k = \frac{3\ell}{2G}.$$

The projective connection carried by the boundary stress tensor is the projective variable of the preceding conformal family.

Proof. Hamiltonian reduction of the boundary $\mathfrak{sl}_2$ current Poisson algebra gives the classical Virasoro bracket. Its central term is $6k$. Substitution of the gravitational level gives the second equality. The coadjoint transformation of the boundary stress tensor is the projective-connection transformation law [12]. $\square$

EINSTEIN COMPLETION OF A CONFORMAL PAIR

Let A_L and A_R be conformal mixed boundary data with equal Brown–Henneaux central charge. Their projective families map to the same $\mathcal{P}_c(X)$. Define their relative field algebra by the homotopy fibre product over the projective tangent algebra.

Definition 176.1 (Candidate Einstein BV completion). Suppose the relative algebra and a continuous L_∞ action of the Cartan algebra have been supplied, together with their local realization. The associated cotangent candidate is

$$\mathbf{EinBV}_3(A_L, A_R) = T^*[-1] \text{MC}(\mathfrak{g}_{\text{grav}} \ltimes_{\text{Ham}} \mathfrak{m}_{A_L, A_R}).$$

Here $\mathfrak{g}_{\text{grav}}$ is the Cartan deformation algebra and $\mathfrak{m}_{A_L, A_R}$ is the relative mixed deformation algebra after projective matching.

Theorem 176.2 (Cotangent action for a supplied coupled algebra). *Suppose projective matching and a continuous L_∞ action have constructed the displayed complete semidirect product. Suppose also that its local realization and integrated dual pairing are defined. Its shifted cotangent has action*

$$S(\alpha, \beta) = \langle \beta, \mathcal{F}(\alpha) \rangle$$

and satisfies the classical master equation. Its Bianchi identity is the L_∞ identity of the coupled algebra.

Proof. The supplied semidirect product has a square-zero homological vector field. Its cotangent lift is Hamiltonian for the displayed action and remains square-zero. This proves the classical master equation and the Bianchi identity by the preceding cotangent construction. $\square$

The gravitational summand of this cotangent action is the cotangent theory of Cartan deformations. It introduces independent dual fields. To identify it with Einstein BV theory requires a comparison preserving the BV pairing and the action, including boundary terms, on the invertible-coframe sector. The flatness equations alone do not provide that comparison. A physical coupled completion must also construct the matter action and its equivariant moment map. The intended Einstein reconstruction therefore retains these field and action comparisons as existence problems.

HIGHER-SPIN COMPLETION

Replace $\mathfrak{sl}_2$ by $\mathfrak{sl}_N$ and choose its principal $\mathfrak{sl}_2$ embedding. The adjoint representation decomposes as

$$\mathfrak{sl}_N \cong \bigoplus_{s=2}^N V_{2s-2}, \quad \sum_{s=2}^N (2s-1) = N^2 - 1.$$

Theorem 177.1 (Principal higher-spin spectrum). *The principal embedding gives one massless gauge field of each integer spin $2, 3, \dots, N$. Classical Drinfeld–Sokolov reduction of the boundary affine Poisson algebra gives the principal W_N algebra. A pair of principal W_N boundary data therefore has an $\mathfrak{sl}_N \oplus \mathfrak{sl}_N$ Einstein–Cartan completion with the same spin spectrum.*

Proof. The irreducible $\mathfrak{sl}_2$ module V_{2s-2} corresponds to a frame-like field of spin s . The dimension identity exhausts the adjoint. Principal Drinfeld–Sokolov reduction produces one generating current in each conformal weight $2, \dots, N$. The boundary currents and bulk gauge fields therefore match representation by representation [14, 42]. $\square$

BOOK XXXVI

Exact gravitational complexes and quantum tests

THE FOUR COMPLEXES IN COORDINATES

The Cartan flatness equations and the metric Einstein equations describe the same classical solutions on the invertible-coframe locus. Their infinitesimal complexes make the comparison explicit. Quantum determinants additionally depend on gauge fixing, boundary conditions, and determinant-line data.

178.1 FLAT ADJOINT COMPLEX

At a flat Cartan connection $\mathcal{A}$, the first complex is

$$\Omega^0(M, \mathfrak{g}_\Lambda) \xrightarrow{d_{\mathcal{A}}} \Omega^1(M, \mathfrak{g}_\Lambda) \xrightarrow{d_{\mathcal{A}}} \Omega^2(M, \mathfrak{g}_\Lambda) \xrightarrow{d_{\mathcal{A}}} \Omega^3(M, \mathfrak{g}_\Lambda).$$

Degree zero is the gauge parameter, degree one is the fluctuation, degree two is the linearized curvature, and degree three is the Bianchi identity.

178.2 EINSTEIN–CARTAN COMPLEX

Write a gauge parameter as (ξ^a, λ^a) and a fluctuation as $(u^a, v^a) = (\delta e^a, \delta \omega^a)$. The first differential is

$$\begin{aligned} u^a &= D_\omega \xi^a + \epsilon^a_{bc} e^b \lambda^c, \\ v^a &= D_\omega \lambda^a + \ell^{-2} \epsilon^a_{bc} e^b \xi^c. \end{aligned}$$

The second differential is

$$\begin{aligned} \delta T^a &= D_\omega u^a + \epsilon^a_{bc} v^b \wedge e^c, \\ \delta F^a &= D_\omega v^a + \ell^{-2} \epsilon^a_{bc} e^b \wedge u^c. \end{aligned}$$

The third differential is the linearized pair of Cartan Bianchi identities.

Proposition 178.1 (Chain identity). *The composite of the first two Einstein–Cartan differentials vanishes on a constant-curvature torsion-free background.*

Proof. Apply D_ω^2 to ξ and λ . Replace the background curvature by $-\ell^{-2}e \wedge e$ and use $D_\omega e = 0$. The curvature terms cancel the explicit ℓ^{-2} terms. The remaining expressions are Jacobi identities for $\epsilon_{ab}{}^c$. $\square$

178.3 METRIC COMPLEX

The dreibein identifies the translation parameter with a vector field. Modulo Lorentz rotations,

$$h_{\mu\nu} = 2\eta_{ab} e^a_{(\mu} u^b_{\nu)}.$$

The metric differential begins with the Killing operator

$$K(\zeta)_{\mu\nu} = \nabla_\mu \zeta_\nu + \nabla_\nu \zeta_\mu.$$

The linearized cosmological Einstein operator is

$$(E_\Lambda h)_{\mu\nu} = \frac{1}{2} \left(-\nabla^2 h_{\mu\nu} - \nabla_\mu \nabla_\nu h + \nabla_\mu \nabla^\rho h_{\rho\nu} + \nabla_\nu \nabla^\rho h_{\rho\mu} \right. \\ \left. - g_{\mu\nu} (\nabla^\rho \nabla^\sigma h_{\rho\sigma} - \nabla^2 h) \right) - \Lambda (h_{\mu\nu} - g_{\mu\nu} h).$$

It satisfies $E_\Lambda K = 0$ and $\nabla^\mu (E_\Lambda h)_{\mu\nu} = 0$.

Theorem 178.2 (Explicit metric reduction). *Solving the linearized torsion equation for $v = \delta\omega$ and substituting into the linearized curvature equation gives $E_\Lambda h = 0$. The kernel of the map from (u, v) to h is the infinitesimal Lorentz action.*

Proof. The torsion equation is algebraic in the antisymmetric part of the covariant derivative of u and determines v . The symmetric part of u is $h/2$. Substitution into $\delta F = 0$ yields the standard variation of the Ricci tensor and the displayed Einstein operator. An antisymmetric $u_{\mu\nu}$ is exactly a Lorentz rotation of the dreibein. $\square$

178.4 CALABI COMPLEX

On a constant-curvature manifold, the Calabi complex begins

$$\Gamma(TM) \xrightarrow{K} \Gamma(S^2 T^* M) \xrightarrow{R'} \Gamma(\mathcal{R}) \xrightarrow{B'} \Gamma(\mathcal{B}),$$

where R' is the linearized Riemann tensor and B' is the linearized algebraic and differential Bianchi operator. In three dimensions the Riemann tensor is equivalent to the Einstein tensor, so the metric and Calabi complexes are locally quasi-isomorphic.

BOUNDARY BFV THEORY AND THE VIRASORO MOMENT MAP

The variation of Chern–Simons theory on a manifold with boundary gives the boundary symplectic form

$$\Omega_{\partial} = \frac{k}{4\pi} \int_{\partial M} \langle \delta A \wedge \delta A \rangle.$$

A chiral polarization selects $A_{\bar{z}}$ as coordinate. The residual moment map is the flatness constraint.

Theorem 179.1 (Boundary current algebra). *Canonical quantization of the boundary symplectic form gives the affine current algebra at level k . Drinfeld–Sokolov reduction of the principal $\mathfrak{sl}_2$ constraint gives the Virasoro moment map*

$$T(z) = \frac{1}{2(k + h^{\vee})} : J^a J_a : (z) + \text{improvement},$$

with the classical Brown–Henneaux limit $c = 6k$.

Proof. The inverse of Ω_{∂} is the current Poisson bracket. Quantization gives the affine central extension. The BRST complex of the principal constraint eliminates the gauge directions and retains the improved quadratic current. Its classical central term is $6k$ in the gravitational normalization. $\square$

MATTER STRESS TENSOR AND THE EINSTEIN MOMENT MAP

Let $\mathfrak{m}$ be the relative field algebra of a conformal pair. Projective covariance gives a Hamiltonian action of vector fields and therefore a moment map

$$\mu_{\text{matt}} : \mathfrak{m} \longrightarrow \Omega^{\circ, \bullet}(X, K_X^{\otimes 2})[-1].$$

Its classical component is the stress tensor.

Theorem 180.1 (Conservation from the master equation). *The coupled BV action*

$$S_{\text{tot}} = S_{\text{grav}} + S_{\text{matt}} + \langle h, \mu_{\text{matt}} \rangle$$

satisfies the classical master equation precisely when the matter moment map is equivariant. The component of that equation linear in the diffeomorphism ghost is

$$\nabla^\mu T_{\mu\nu} = 0.$$

Proof. The master bracket of the coupling with itself is the defect of equivariance. The bracket with the gravitational action is the infinitesimal diffeomorphism of the moment map. Their cancellation is the Hamiltonian equivariance identity. Pairing the identity with the diffeomorphism ghost and integrating by parts gives stress-tensor conservation. $\square$

ONE-LOOP GRAVITY AND ANALYTIC TORSION

Expand Chern–Simons theory around an acyclic flat connection $\mathcal{A}$. Gauge fixing gives the Laplacians of the flat adjoint de Rham complex.

Theorem 181.1 (One-loop determinant). *The absolute value of the one-loop partition function is the square root of the Ray–Singer torsion of the flat adjoint bundle:*

$$|Z_{1-loop}(\mathcal{A})| = \tau_{RS}(M, \text{ad } \mathcal{A})^{1/2}.$$

The phase is the eta-invariant contribution fixed by the framing convention.

Proof. Bosonic and ghost Gaussian integrals give alternating determinants of the Laplacians in the flat de Rham complex. Their product is the analytic torsion. The odd signature operator supplies the determinant phase. This is the standard stationary-phase calculation for Chern–Simons theory [66, 71]. $\square$

The same determinant complex is obtained from the Einstein–Cartan, metric, and Calabi presentations through their chain equivalences. One-loop equality is therefore a consequence of quasi-isomorphism together with compatible determinant-line orientations.

BTZ, CARDY, AND THE AREA LAW

For a unitary modular boundary theory with left and right central charges c_L, c_R , the asymptotic entropy is

$$S_{\text{Cardy}} = 2\pi \sqrt{\frac{c_L}{6} \left(L_o - \frac{c_L}{24} \right)} + 2\pi \sqrt{\frac{c_R}{6} \left(\bar{L}_o - \frac{c_R}{24} \right)}.$$

For the BTZ black hole,

$$L_o - \frac{c}{24} = \frac{(r_+ + r_-)^2}{16G\ell}, \quad \bar{L}_o - \frac{c}{24} = \frac{(r_+ - r_-)^2}{16G\ell}, \quad c = \frac{3\ell}{2G}.$$

Theorem 182.1 (Cardy–Bekenstein equality). *Substitution gives*

$$S_{\text{Cardy}} = \frac{2\pi r_+}{4G}.$$

This is the Bekenstein–Hawking entropy of the BTZ horizon.

Proof. Each square root becomes $\pi(r_+ \pm r_-)/(4G)$. Their sum is $2\pi r_+/(4G)$. The modular asymptotic is Cardy's theorem, and the geometric charges are the Chern–Simons holonomies of the BTZ solution [4, 15]. $\square$

HIGHER-SPIN HOLONOMY AND BOUNDARY CHARGES

In $\mathfrak{sl}_N$ Chern–Simons gravity, a smooth Euclidean saddle is selected by the conjugacy class of the thermal holonomy. The invariant polynomials

$$\mathrm{Tr}(\mathrm{Hol}^s), \quad 2 \leq s \leq N,$$

fix the spin- s charges. Drinfeld–Sokolov reduction identifies the same charges with zero modes of the W_N currents.

Theorem 183.1 (Bulk–boundary charge equality). *For a principal $\mathfrak{sl}_N$ connection in Drinfeld–Sokolov gauge, the invariant polynomials of the spatial holonomy equal the classical W_N zero-mode Hamiltonians. Their Poisson brackets are the classical W_N brackets.*

Proof. Drinfeld–Sokolov gauge writes the connection as the principal raising operator plus lowest-weight components. The coefficients of the characteristic polynomial are the invariant polynomials and are polynomial functions of those components. Hamiltonian reduction gives the W_N Poisson structure on the same functions. $\square$

BOOK XXXVII

**Anomalies, Hall sources, and automorphic
completion**

FOUR NATIVE ANOMALY CLASSES

A local anomaly, a determinant-line curvature, and an arity-zero bar term belong to different complexes. A cancellation in one complex becomes a geometric cancellation only through a comparison preserving the relevant class and its primitive. Hall and automorphic constructions require their coefficient and orientation maps for the same reason.

The construction carries four curvature classes.

- (i) A local BV anomaly lies in the degree-one local deformation complex.
- (ii) A projective anomaly is the curvature of the determinant line over parameter space.
- (iii) A framing anomaly is the transgression of a characteristic class to a three-manifold.
- (iv) A curved Koszul term is the arity-zero component of a curved coalgebra.

Theorem 184.1 (Curvature-killing extension). *Let κ be a closed central degree-two class in the deformation complex of a theory. Adjoining a degree-one generator u with $du = \kappa$ produces a central extension in which the lifted curvature vanishes. Tensoring with the inverse invertible field theory gives the geometric realization of the same cancellation.*

Proof. The extended differential squares to zero because $d\kappa = 0$ and κ is central. The new Maurer–Cartan equation contains the compensating field u . An invertible theory with curvature $-\kappa$ adds the opposite class under tensor product. $\square$

HALL AND CALABI–YAU SOURCES

Let $\mathcal{C}$ be a Calabi–Yau category with a Hall moduli problem. An oriented extension correspondence gives a Hall algebra H^+ . A factorization pushforward to the boundary curve gives a chiral-convolution algebra. Relative induction supplies the remaining colours.

Proposition 185.1 (Transport under a supplied mixed Morita equivalence). *Suppose a Hall construction supplies a complete mixed boundary datum and a Morita equivalence of its compatible module category with that of $\mathbb{A}$. The equivalence identifies their derived centres. A comparison of local deformation algebras, BV pairings, and actions is additional data for a BV comparison. Quantum line and amplitude comparisons additionally require the monoidal and cyclic structures used to define them.*

Proof. Conjugating natural transformations by the equivalence and its inverse identifies the endomorphisms of the two identity functors. The unit and counit homotopies provide the inverse comparison and preserve composition. This proves the centre comparison. It does not construct the additional local, symplectic, or cyclic maps. $\square$

An algebra map alone does not induce the required forward centre map. For example, the homomorphism $\mathbb{C}[x] \rightarrow M_2(\mathbb{C})$ with $x \mapsto \text{diag}(0, 1)$ sends the central element x to a noncentral matrix. Its commutator with the matrix unit E_{12} is $-E_{12}$. Thus even the degree-zero condition fails for a centre map extending the given algebra map. Hall multiplication maps must be supplemented by an actual coherent module comparison before centre transport can be applied.

For $X = K_3 \times E$, suppose a geometric Hall source supplies the complete charge-graded BPS Lie superalgebra and Cartan action required in Volume IV. Its semidirect product with the common Cartan defines $\mathfrak{b}_X$. Volume IV then constructs the completed Lie amalgam with the Igusa Borchers Borel $\mathfrak{b}_\Delta$. Applying the completed enveloping functor gives the cocommutative Hopf algebra

$$H_{X \boxtimes \Delta} = \widehat{U}(\mathfrak{b}_X) \widehat{*}_{\widehat{U}(\mathfrak{b}_\Pi)} \widehat{U}(\mathfrak{b}_\Delta).$$

The two Hopf retractions preserve the geometric and automorphic sectors separately. A q -deformed mixed Hopf algebra is supplied by the quantum amalgam theorem when a geometric Hall coproduct and pairing are present.

185.1 MIXED REALIZATION OF THE AMALGAM

The Hopf retractions above are maps of their stated algebraic objects. A mixed realization must construct chiral and transverse operations, an interface, and a compatible distributive law. To realize the retractions, it must then supply maps preserving those operations and the chosen completions.

A BV retraction requires further maps between the local deformation algebras, together with compatible pairings and actions. The counterexample to forward centre functoriality prevents deducing these maps from the Hopf retractions alone. The intended geometric–automorphic theory asks for this mixed realization and for a primitive comparison quotient on its actual boundary operators. Its interaction

ideal is the kernel of that constructed quotient, rather than an ideal determined by equality of scalar characters.

IGUSA ROOT STATES AND THE MIXED-BOUNDARY COMPARISON

The Gritsenko–Nikulin presentation defines the root Lie superalgebra $\mathfrak{g}_{\Delta_5}$ and its positive subalgebra $\mathfrak{n}_+$. Let $d_\alpha = \text{sdim}(\mathfrak{g}_{\Delta_5})_\alpha$, and fix the positive cone, its proper height, and Weyl vector ρ . Put

$$P = \prod_{\alpha > 0} (1 - e^{-\alpha})^{d_\alpha}, \quad \text{Den}_\Delta = e^{-\rho} P = D_5(2Z), \quad j(Z) = Z/2.$$

The products are first interpreted in the positive-height completion.

Theorem 186.1 (Root character and reduced scalar). *The root Fock space $\text{Sym}(\mathfrak{n}_+)$, formed with super symmetry, has supercharacter P^{-1} . Consequently*

$$-j^*(e^{2\rho} \text{sch}(\text{Sym}(\mathfrak{n}_+)^{\otimes 2})) = -(j^* \text{Den}_\Delta)^{-2} = -\chi_{10}^{-1}.$$

For a nonsingular projective K_3 surface S and an elliptic curve E , this scalar also has an enumerative interpretation. Let $N_{g,h,d}$ be the reduced disconnected Gromov–Witten invariant in class (β_h, d) , where β_h is nonzero and primitive with square $2h - 2$. Use the incidence insertion $\beta_h^\vee \otimes [\text{point}]$, with $\langle \beta_h, \beta_h^\vee \rangle = 1$. Then the series

$$\mathcal{Z}(u, q, s) = \sum_{g,h,d \geq 0} N_{g,h,d} u^{2g-2} q^{h-1} s^{d-1}$$

is the Laurent expansion of $-\chi_{10}(r, q, s)^{-1}$ under $r = e^{iu}$, by [62, Theorem 1].

Proof. An even root vector contributes $(1 - e^{-\alpha})^{-1}$ to the supercharacter of its symmetric algebra. An odd root vector contributes $1 - e^{-\alpha}$, because its symmetric algebra in super vector spaces is an exterior algebra. Multiplying the two contributions gives P^{-1} . Proper height makes every coefficient calculation finite. The Weyl factor gives $e^{2\rho} P^{-2} = \text{Den}_\Delta^{-2}$. Pullback by j gives $D_5(Z)^{-2} = \chi_{10}(Z)^{-1}$. Oberdieck–Pixton, Theorem 1, identifies the separately defined reduced numerical generating series with its negative [62]. This last theorem is an enumerative input; the character calculation supplies the root side. $\square$

A primitive realization starts with parity-preserving root classes in a charge-graded Lie superalgebra $\mathfrak{p}$ satisfying the defining relations. Require the classes to preserve charge. The free Lie presentation gives an even charge-preserving map $f : \mathfrak{n}_+ \rightarrow \mathfrak{p}$.

Applying f to tensor words constructs $U(f)$. With trivial coefficients, use Chevalley chains $C_*^{\text{CE}}(\mathfrak{a}) = \text{Sym}(\mathfrak{a}[1])$, where the homological suspension raises degree by one and the differential contracts pairs by the bracket. Applying the suspension of f to every factor gives a chain map because f preserves brackets.

For the level-zero Lie conformal current, define $\text{Cur}(\mathfrak{a}) = \mathbb{C}[\partial] \otimes \mathfrak{a}$, with ∂ even. Its generator bracket is $[x_\lambda y] = [x, y]$. Extend it by $[\partial x_\lambda y] = -\lambda[x_\lambda y]$ and $[x_\lambda \partial y] = (\partial + \lambda)[x_\lambda y]$. The formula $\partial^j x \mapsto \partial^j f(x)$ defines $\text{Cur}(f)$. Charge preservation makes these maps compatible with the specified proper-height quotients and their inverse limits.

A complete mixed boundary realization requires further geometric maps. The compact source consists of sheaves and extension flags on a common elliptic torsor. Its critical realization must supply reduced coefficient complexes, orientation cross-line maps, unit and flag coherences, and the admissible pull–push operations. A chiral comparison must preserve actual support and moving-insertion operations. A quantum-double comparison must preserve coproducts, pairings and the domain of exchange operators. The finite Calabi–Yau quiver completions require their specified dg maps; ordinary full-subquiver inclusions do not in general give those maps.

The mixed datum to which a BV envelope is applied must therefore include these operations and compatible maps between them. Identifying the primitive algebra of that envelope requires an actual map of its deformation complex to the root complex. A determinant comparison requires a skew perfect family, a square-root line and section, and an isomorphism to the automorphic line preserving transport and normalization. An operator-trace interpretation further requires its state space and trace domain. Constructing this datum and these comparisons is the Igusa mixed-boundary problem. The root character identity specifies the scalar that their final evaluation must recover.

THE CARRIER LATTICE

The complete theory uses the carrier set

$$\mathfrak{C} = \{\text{diag}, \text{ch}, \perp, \text{int}, \text{ex}, \text{cyc}, \text{mod}, \text{BV}, \text{Ein}\}.$$

For a subset $S \subset \mathfrak{C}$, let $\text{Str}_S(\mathbb{A})$ be the homotopy limit of all mapping spaces attached to incidence cells supported in S .

Theorem 187.1 (Carrier descent). *When S and T form a relation-excise pair, the square*

$$\begin{array}{ccc} \text{Str}_{S \cup T}(\mathbb{A}) & \longrightarrow & \text{Str}_S(\mathbb{A}) \\ \downarrow & & \downarrow \\ \text{Str}_T(\mathbb{A}) & \longrightarrow & \text{Str}_{S \cap T}(\mathbb{A}) \end{array}$$

is a homotopy pullback. Thus a global structure is assembled from its carrier restrictions together with their common interface.

Proof. Relation-excision identifies the incidence category of $S \cup T$ with the pushout of the incidence categories for S and T over their intersection. Homotopy limits send this pushout to the displayed pullback. $\square$

Definition 187.2 (Relation-complete carrier cover). Let $U \subseteq \mathfrak{C}$ and let $\mathcal{U} = \{S_i\}_{i \in I}$ be a finite family with $U = \bigcup_i S_i$. The family is *relation-complete* when the canonical functor from the Čech homotopy colimit

$$\text{hocolim}_{\emptyset \neq J \subseteq I} \mathcal{I}_{\cap_{j \in J} S_j} \longrightarrow \mathcal{I}_U$$

is an equivalence. Equivalently, every incidence cell and every relation among its faces is contained in one member of the cover, with all overlaps recorded by the intersections.

Theorem 187.3 (Power-set coherence). *The assignment*

$$S \longmapsto \text{Str}_S(\mathbb{A})$$

is a contravariant functor on the full power set of $\mathfrak{C}$. For every relation-complete carrier cover $\mathcal{U} = \{S_i\}_{i \in I}$ of U , the restriction map is an equivalence

$$\text{Str}_U(\mathbb{A}) \xrightarrow{\sim} \text{holim}_{\emptyset \neq J \subseteq I} \text{Str}_{\cap_{j \in J} S_j}(\mathbb{A}).$$

Thus every subtheory, every intersection of subtheories, and the complete theory are values of one descent object.

Proof. An inclusion $S \subseteq T$ induces the full inclusion $\mathcal{I}_S \subseteq \mathcal{I}_T$ and therefore the restriction map on homotopy limits. Composition of inclusions gives functoriality. A relation-complete cover presents $\mathcal{I}_U$ as the displayed homotopy colimit of its intersection diagram. Mapping that colimit into the diagram of structure spaces converts it into the stated homotopy limit. The resulting equivalence is natural under refinement of the carrier cover. $\square$

Definition 187.4 (Exact-support deformation complex). For $S \subseteq \mathfrak{C}$, let $\text{Def}_{\leq S}$ be the subcomplex generated by incidence cells supported in S , and set

$$\text{Def}_{=S} = \text{cofib} \left(\text{hocolim}_{T \subsetneq S} \text{Def}_{\leq T} \longrightarrow \text{Def}_{\leq S} \right).$$

Thus $\text{Def}_{=S}$ retains the deformation cells whose minimal carrier support is exactly S .

Theorem 187.5 (Carrier-support spectral sequence). *Let $\text{Def}_{\mathbb{A}}^{\text{full}}$ be the cellular convolution complex of the complete incidence diagram. Filter it by the cardinality of minimal carrier support. The finite filtration gives a convergent spectral sequence*

$$E_1^{p,q} = \bigoplus_{\substack{S \subseteq \mathfrak{C} \\ |S|=p}} H^{p+q}(\text{Def}_{=S}) \implies H^{p+q}(\text{Def}_{\mathbb{A}}^{\text{full}}).$$

A class on the S -summand is a deformation or curvature class whose first complete geometric carrier is S . The first differential is the signed incidence map between exact-support quotients.

Proof. The support-cardinality filtration is finite because $\mathfrak{C}$ is finite. Its p th associated-graded quotient is the direct sum of the exact-support complexes $\text{Def}_{=S}$ with $|S| = p$. The spectral sequence of a finite filtered complex gives the displayed first page and converges strongly to the cohomology of the total convolution complex. The cellular determinant orientation gives the incidence signs. $\square$

SIXTEEN STRUCTURAL LAWS

The theory is governed by the following laws.

- (i) A geometric carrier determines the type of its operation.
- (ii) The diagonal stores the complete principal-part filtration.
- (iii) The interval stores ordered fusion.
- (iv) The plane stores exchange.
- (v) The circle stores trace.
- (vi) The stable curve stores modular sewing.
- (vii) The transverse, associative-chiral, and chiral-Lie bars are distinct.
- (viii) Chiral PBW compares the last two on enveloping objects.
- (ix) The interface bar records the mixed distributive law.
- (x) The derived centre is the universal interior action.
- (xi) The deformation algebra is the tangent algebra of the centre-extended datum.
- (xii) Maurer–Cartan curvature is the field equation.
- (xiii) The L_∞ identity is the Bianchi identity.
- (xiv) Shifted cotangent is the universal classical BV envelope.
- (xv) Hodge-completed derived de Rham theory is the intrinsic algebraic quantum complex of the oriented formal moduli problem.
- (xvi) A gravitational realization requires projective matching, a coupled local field algebra, and a comparison preserving the physical BV action and pairing.

COHERENT ACTIONS AND GRAVITATIONAL RECONSTRUCTION

A supplied mixed boundary datum defines compatible modules and their derived natural endomorphisms. A specified coloured operad organizes deformations of that datum together with its central action. These are constructions relative to the full operations and their coherence maps. The centre alone does not recover the datum from which the deformation algebra is formed.

A supplied local realization of the deformation algebra admits shifted cotangent transgression. With compatible invariant orientations, the finite perfect charts also carry the algebraic BV differential. The physical bulk is defined independently by its fields, action, gauge symmetry, boundary conditions, and quantization. Reconstruction requires a coherent action map from its observables to an independently defined geometric centre, followed by an equivalence theorem on the stated sector.

For gravity, the invertible coframe gives the classical Cartan–Einstein correspondence. The cotangent construction supplies a formal comparison theory with extra dual fields. Its identification with gravitational BV fields requires an action-preserving and pairing-preserving map. Quantum boundary currents, a continuum measure, global holonomy sectors, allowed fillings, and compatible states remain necessary for the quantum and global comparison.

Hall and Calabi–Yau geometry can supply boundary operations when their correspondences, coefficients, orientations, and comparison maps are constructed. Exchange additionally uses a monoidal category and a fibre functor. Cyclic pairings and sewing determine amplitudes only on their stated domains. These comparisons are the mathematical steps by which local boundary operations could reconstruct physical and geometric theories.

BOOK **XXXVIII**

**Technical appendices to the preceding
book**

THE CHERN–SIMONS NORMALIZATION

The gravitational charge calculation uses the action normalization. The mixed bar calculation uses the determinant orientation of its edges. The following formulas specify these conventions.

With invariant pairing $\langle J_a, P_b \rangle = \eta_{ab}$, the Chern–Simons action is

$$S_{\text{CS}}[\mathcal{A}] = \frac{k}{4\pi} \int_M \langle \mathcal{A} \wedge d\mathcal{A} + \frac{2}{3} \mathcal{A} \wedge \mathcal{A} \wedge \mathcal{A} \rangle.$$

For $A^\pm = \omega \pm \ell^{-1}e$ one finds

$$S_{\text{CS}}[A^+] - S_{\text{CS}}[A^-] = \frac{k}{2\pi\ell} \int_M \epsilon_{abc} e^a \wedge \left(R^{bc} + \frac{1}{3\ell^2} e^b \wedge e^c \right) + \int_{\partial M} (\cdots).$$

The equality with the Einstein–Hilbert normalization gives $k = \ell/(4G)$.

CHAPTER 191

THE PRINCIPAL EMBEDDING COUNT

The exponents of $\mathfrak{sl}_N$ are $1, 2, \dots, N-1$. Under the principal $\mathfrak{sl}_2$ embedding,

$$\mathfrak{sl}_N = \bigoplus_{m=1}^{N-1} V_{2m}.$$

The dimensions satisfy

$$\sum_{m=1}^{N-1} (2m+1) = (N-1)(N+1) = N^2 - 1.$$

Writing $s = m+1$ gives one field at every spin $s = 2, \dots, N$.

LOW-ARITY MIXED SIGNS

Let $x|k|y$ be a suspended interface word. The two length-lowering contractions are

$$b_L[x|k|y] = (-1)^{|x|}(xk)|y, \quad b_R[x|k|y] = (-1)^{|x|+|k|}x|(ky).$$

Applying the remaining contraction gives the two composites $(xk)y$ and $x(ky)$ with opposite determinant-line signs. Their sum vanishes by the interface associativity relation. A transverse and a chiral contraction acquire the product orientation

$$\det(E_{\text{ch}} \oplus E_{\perp}) = \det(E_{\text{ch}}) \otimes \det(E_{\perp}),$$

so reversing their order contributes the Koszul sign and yields anticommutation.

BIBLIOGRAPHY

- [1] Achucarro Achucarro and Paul K. Townsend, *A chern–simons action for three-dimensional anti-de sitter supergravity theories*, Physics Letters B **180** (1986), no. 1–2, 89–92.
- [2] Jonathan D. Axtell and Kyu-Hwan Lee, *Quantum generalized kac–moody algebras via hall algebras of complexes*, 2019.
- [3] David Ayala, John Francis, and Hiro Lee Tanaka, *Factorization homology of stratified spaces*, Selecta Mathematica **23** (2017), 293–362, available at [1409.0848](#).
- [4] Máximo Bañados, Claudio Teitelboim, and Jorge Zanelli, *The black hole in three-dimensional space-time*, Physical Review Letters **69** (1992), no. 13, 1849–1851.
- [5] Alexander Beilinson and Vladimir Drinfeld, *Chiral algebras*, American Mathematical Society Colloquium Publications, vol. 51, American Mathematical Society, 2004.
- [6] David Ben-Zvi, John Francis, and David Nadler, *Integral transforms and drinfeld centers in derived algebraic geometry*, Journal of the American Mathematical Society **23** (2010), no. 4, 909–966, available at [0805.0157](#).
- [7] Richard E. Borcherds, *Automorphic forms on $o_{3+2,2}(\mathbb{R})$ and infinite products*, Inventiones Mathematicae **120** (1995), 161–213.
- [8] ———, *Automorphic forms with singularities on grassmannians*, Inventiones Mathematicae **132** (1998), 491–562, available at [alg-geom/9609022](#).
- [9] Christopher Brav and Tobias Dyckerhoff, *Relative calabi–yau structures ii: shifted lagrangians in the moduli of objects*, Selecta Mathematica (N.S.) **27** (2021), no. 4, 63, available at [1812.11913](#).
- [10] Christopher Brav and Nick Rozenblyum, *The cyclic deligne conjecture and calabi–yau structures*, 2023.
- [11] Alexander Braverman, Michael Finkelberg, and Hiraku Nakajima, *Towards a mathematical definition of coulomb branches of 3-dimensional $n = 4$ gauge theories, ii*, Advances in Theoretical and Mathematical Physics **22** (2018), no. 5, 1071–1147, available at [1601.03586](#).
- [12] J. David Brown and Marc Henneaux, *Central charges in the canonical realization of asymptotic symmetries: An example from three-dimensional gravity*, Communications in Mathematical Physics **104** (1986), no. 2, 207–226.
- [13] Damien Calaque, Vasily Dolgushev, and Gilles Halbout, *Formality theorems for hochschild chains in the lie algebroid setting*, Journal für die reine und angewandte Mathematik **612** (2007), 81–127, available at [math/0504372](#).
- [14] Andrea Campoleoni, Stefan Fredenhagen, Stefan Pfenninger, and Stefan Theisen, *Asymptotic symmetries of three-dimensional gravity coupled to higher-spin fields*, Journal of High Energy Physics **2010** (2010), no. 11, 007, available at [1008.4744](#).
- [15] John L. Cardy, *Operator content of two-dimensional conformally invariant theories*, Nuclear Physics B **270** (1986), 186–204.
- [16] Fabien Cléry and Valery Gritsenko, *The siegel modular forms of genus 2 with the simplest divisor*, Proceedings of the London Mathematical Society **102** (2011), no. 6, 1024–1052, available at [0812.3962](#).
- [17] Kevin Costello, *Topological conformal field theories and calabi–yau categories*, Advances in Mathematics **210** (2007), 165–214.
- [18] Kevin Costello and Owen Gwilliam, *Factorization algebras in quantum field theory, volume 1*, New Mathematical Monographs, vol. 31, Cambridge University Press, 2017.
- [19] ———, *Factorization algebras in quantum field theory, volume 2*, New Mathematical Monographs, vol. 41, Cambridge University Press, 2021.
- [20] Kevin Costello and Si Li, *Anomaly cancellation in the topological string*, 2019.
- [21] Ben Davison, *Affine bps algebras, w algebras, and the cohomological hall algebra of $\mathfrak{a}^2$* , Homotopical methods in geometry and physics, 2026, pp. 163–199.

- [22] Ben Davison, Lucien Hennecart, and Sebastian Schlegel Mejia, *Bps algebras and generalised kac–moody algebras from 2-calabi–yau categories*, 2023.
- [23] Ben Davison and Sven Meinhardt, *Cohomological donaldson–thomas theory of a quiver with potential and quantum enveloping algebras*, *Inventiones Mathematicae* **221** (2020), 777–871, available at [1601.02479](#).
- [24] Alberto De Sole, Matteo Gardini, and Victor G. Kac, *On the structure of quantum vertex algebras*, 2019.
- [25] Duiliu-Emanuel Diaconescu, Mauro Porta, Francesco Sala, Olivier Schiffmann, and Eric Vasserot, *Cohomological hall algebras of one-dimensional sheaves on surfaces and yangians*, 2026.
- [26] Vasilij Dolgushev, *Covariant and equivariant formality theorems*, *Advances in Mathematics* **191** (2005), no. 1, 147–177, available at [math/0307212](#).
- [27] Vladimir Drinfeld, *Hopf algebras and the quantum yang–baxter equation*, *Soviet Mathematics Doklady* **32** (1985), 254–258.
- [28] ———, *Quantum groups*, *Proceedings of the international congress of mathematicians, berkeley 1986, 1987*, pp. 798–820.
- [29] Martin Eichler and Don Zagier, *The theory of jacobi forms*, *Progress in Mathematics*, vol. 55, Birkhäuser, 1985.
- [30] Pavel Etingof and Olivier Schiffmann, *Lectures on quantum groups*, International Press, 1998.
- [31] John Francis and Dennis Gaitsgory, *Chiral koszul duality*, *Selecta Mathematica* **18** (2012), no. 1, 27–87, available at [1103.5803](#).
- [32] Ezra Getzler and Mikhail Kapranov, *Modular operads*, *Compositio Mathematica* **110** (1998), 65–126.
- [33] Victor Ginzburg, *Calabi–yau algebras*, arXiv preprint (2006), available at [math/0612139](#).
- [34] Suresh Govindarajan, *Borcherds–kac–moody lie superalgebras from counting twisted CHL dyons*, *Journal of High Energy Physics* **2011** (2011), no. 5, 089, available at [1006.3472](#).
- [35] Suresh Govindarajan and K. Gopala Krishna, *Generalized kac–moody algebras from CHL dyons*, *Journal of High Energy Physics* **2009** (2009), no. 4, 032, available at [0807.4451](#).
- [36] Suresh Govindarajan and Sutapa Samanta, *Borcherds–kac–moody lie superalgebras from counting twisted CHL dyons—ii*, *Nuclear Physics B* **948** (2019), 114770, available at [1905.06083](#).
- [37] James A. Green, *Hall algebras, hereditary algebras and quantum groups*, *Inventiones Mathematicae* **120** (1995), 361–377.
- [38] Valery Gritsenko and Viacheslav Nikulin, *Siegel automorphic form corrections of some lorentzian kac–moody lie algebras*, *American Journal of Mathematics* **119** (1997), no. 1, 181–224, available at [alg-geom/9504006](#).
- [39] ———, *Automorphic forms and lorentzian kac–moody algebras. part i*, *International J. Math.* **9** (1998), no. 2, 153–199, available at [alg-geom/9610022](#).
- [40] ———, *Automorphic forms and lorentzian kac–moody algebras. part ii*, *International J. Math.* **9** (1998), no. 2, 201–275, available at [alg-geom/9611028](#).
- [41] Alexander Grothendieck, *Techniques de construction et théorèmes d’existence en géométrie algébrique IV: les schémas de Hilbert*, *Séminaire bourbaki*, 1961, pp. 249–276.
- [42] Marc Henneaux and Soo-Jong Rey, *Nonlinear w_∞ as asymptotic symmetry of three-dimensional higher spin anti-de sitter gravity*, *Journal of High Energy Physics* **2010** (2010), no. 12, 007, available at [1008.4579](#).
- [43] Gijs Heuts, *Koszul duality and a conjecture of francis–gaitsgory*, 2024.
- [44] Jin Hong, *Center and universal r-matrix for quantized borcherds superalgebras*, *Journal of Mathematical Physics* **40** (1999), no. 6, 3123–3145, available at [math/9810115](#).
- [45] Jun-ichi Igusa, *On siegel modular forms of genus two*, *American Journal of Mathematics* **84** (1962), no. 1, 175–200.
- [46] ———, *On siegel modular forms of genus two (ii)*, *American Journal of Mathematics* **86** (1964), no. 2, 392–412.
- [47] Dominic Joyce and Markus Upmeyer, *Orientation data for moduli spaces of coherent sheaves over calabi–yau 3-folds*, *Advances in Mathematics* **381** (2021), 107627, available at [2001.00113](#).
- [48] Seok-Jin Kang and Olivier Schiffmann, *Canonical bases for quantum generalized kac–moody algebras*, *Advances in Mathematics* **200** (2006), no. 2, 455–478, available at [math/0311089](#).
- [49] Bernhard Keller, *Deformed calabi–yau completions*, *Journal für die reine und angewandte Mathematik* **654** (2011), 125–180, available at [0908.3499](#).

- [50] Tasuki Kinjo, Hyeonjun Park, and Pavel Safronov, *Cohomological hall algebras for 3-calabi–yau categories*, 2024.
- [51] Maxim Kontsevich, *Deformation quantization of poisson manifolds*, Letters in Mathematical Physics **66** (2003), 157–216, available at [q-alg/9709040](https://arxiv.org/abs/q-alg/9709040).
- [52] Maxim Kontsevich and Yan Soibelman, *Deformations of algebras over operads and the deligne conjecture*, 2000.
- [53] Haisheng Li, *Nonlocal vertex algebras generated by formal vertex operators*, 2005.
- [54] ———, *Quantum vertex algebras and their ϕ -coordinated quasi modules*, 2009.
- [55] Jean-Louis Loday and Bruno Vallette, *Algebraic operads*, Grundlehren der mathematischen Wissenschaften, vol. 346, Springer, 2012.
- [56] Eduard Looijenga and Valery Lunts, *A lie algebra attached to a projective variety*, Inventiones Mathematicae **129** (1997), 361–412.
- [57] Ming Lu, *Quantum borchers–bozec algebras via semi-derived ringel–hall algebras*, 2021.
- [58] Jacob Lurie, *Higher algebra*, 2017. Available from the author’s webpage.
- [59] James E. McClure and Jeffrey H. Smith, *A solution of deligne’s hochschild cohomology conjecture*, Contemporary Mathematics **293** (2002), 153–193, available at [math/9910126](https://arxiv.org/abs/math/9910126).
- [60] Alexander Molev, *Yangians and classical lie algebras*, Mathematical Surveys and Monographs, vol. 143, American Mathematical Society, 2007.
- [61] Hiraku Nakajima, *Heisenberg algebra and hilbert schemes of points on projective surfaces*, Annals of Mathematics **145** (1997), no. 2, 379–388.
- [62] Georg Oberdieck and Aaron Pixton, *Holomorphic anomaly equations and the igusa cusp form conjecture*, Inventiones Mathematicae **213** (2018), no. 2, 507–587, available at [1706.10100](https://arxiv.org/abs/1706.10100).
- [63] Georg Oberdieck and Junliang Shen, *Reduced donaldson–thomas invariants and the ring of dual numbers*, Proceedings of the London Mathematical Society **118** (2019), no. 1, 191–220, available at [1612.03102](https://arxiv.org/abs/1612.03102).
- [64] Tony Pantev, Bertrand Toën, Michel Vaquié, and Gabriele Vezzosi, *Shifted symplectic structures*, Publications Mathématiques de l’IHÉS **117** (2013), 271–328, available at [1111.3209](https://arxiv.org/abs/1111.3209).
- [65] Stewart Priddy, *Koszul resolutions*, Transactions of the American Mathematical Society **152** (1970), 39–60.
- [66] Daniel B. Ray and Isadore M. Singer, *r-torsion and the laplacian on riemannian manifolds*, Advances in Mathematics **7** (1971), no. 2, 145–210.
- [67] Claus Michael Ringel, *Hall algebras and quantum groups*, Inventiones Mathematicae **101** (1990), 583–591.
- [68] Olivier Schiffmann and Eric Vasserot, *Cohomological hall algebras of quivers and yangians*, 2023.
- [69] Alexander Tsybaliuk, *The affine yangian of $\mathfrak{gl}_1$ revisited*, Advances in Mathematics **304** (2017), 583–645, available at [1404.5240](https://arxiv.org/abs/1404.5240).
- [70] Mikhail Verbitsky, *Cohomology of compact hyper-kähler manifolds and its applications*, Geometric and Functional Analysis **6** (1996), 601–611.
- [71] Edward Witten, *(2 + 1)-dimensional gravity as an exactly soluble system*, Nuclear Physics B **311** (1988), no. 1, 46–78.